

B1 Insulated Cables



Finite Element Analysis for Cable Rating Calculations

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WG B1.87

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Executive summary

This Technical Brochure addresses the computation of cable circuits current ratings using the Finite Element Method (FEM). FEM itself is not new, it has been in use for well over half a century now and is considered to be an essential tool for the modern day engineer in a wide range of technical disciplines. There has long been interest in applying FEM to high voltage cable problems, with pioneering work done prior to the year 2000. At this time it was only possible (and desirable) to apply FEM to a very limited subset of cases, typically only for cables at the highest voltages. Engineers conducting such models were highly skilled and experienced, with analysis often being conducted by research institutes within TSOs or major cable suppliers, using computing hardware that was state-of-the-art for the time. The software was often highly specialised, even bespoke to the application, making it beyond the reach of the majority of engineers in a standard cable project.

The continued growth in interest in the use of FEM for cable ratings led to the publication of an IEC Technical Report in 2003, representing one of the first times that formal guidance and recommendations towards best practice had been collected together. Although highly relevant at the time of its publication, the B1 community expressed a desire for enhanced and updated guidance to be provided, leading to the launch of a Task Force (B1.84) and subsequently Working Group (B1.87) to prepare this Technical Brochure.

What has changed since 2003 that made this new guidance desirable? From the perspective of our ability to rapidly deploy FEM on cable problems, there have been enormous developments in both software and hardware. Commercial software packages with much more intuitive user interfaces, powerful libraries of tools such as solvers and meshing algorithms and the ability to easily develop models mean that the level of training required to build models has reduced and the development time is much shorter. Computing hardware has moved forward massively, resulting in standard desktop computers being able to handle problems that were previously beyond the capability of all but the most powerful mainframes or servers. These factors combined with a need to reduce conservatism and make cable circuits ever more economically competitive has led to FEM being very widely deployed for cable system analysis. What was once a niche subject that would only be of interest to a very small group of specialists is now widely used in the design of cable systems around the world. Therefore the time was right to produce a comprehensive set of recommendations for use by the B1 community.

The structure of this Technical Brochure is informed by the needs of different user groups, and it is highly likely that many readers will not need to follow it page by page. Chapter 1 provides an introduction to the scope of the TB, summarising the Terms of Reference and demonstrating how different user groups can get the most value from the content. Chapter 2 presents a literature review of relevant developments, primarily from CIGRE, Jicable and IEEE publications. It is not intended to be a fully comprehensive review of all works in the public domain, hence the focus on a subset that will be most accessible to the cable community.

The WG is keen to emphasise that FEM is not always the best tool for the job, and that standard analytical calculations still have an important place in the toolbox. Chapter 3 assesses the Value Proposition of FEM from a cable rating perspective and provides some high level recommendations on when FEM should be used and for what reasons. This leads on to Chapter 4, which provides guidance to Clients; these could be customers of the cable suppliers, for example TSOs or project developers. The aim is to provide an overview of what to expect when an FEM study is commissioned, what input data should be provided, how to get the most value from the work and to set out some norms for the contents of reports.

Chapter 5 presents an extensive set of Guidance Points covering all aspects of the finite element modelling process, from the initial steps to define and set up a model through to recommendations on how to implement more complex models. These Guidance Points should not be seen as a step by step instruction; the nature of FEM means that there will be unique features to every model, and it is

impossible to provide a fool-proof recipe for construction of models. The guidance is informed by the collective experience of the working group members, supplemented with evidence from the technical literature which was summarised in Chapter 2. Verification that the model is performing as intended is a vitally important step, and often a step which does not receive sufficient attention. Chapter 6 sets out recommendations on how the verification should be conducted and gives insight into what should be reported to the client and which records should be retained by the engineer conducting the modelling work.

The TB is completed by a set of six case studies which cover a wide variety of different installation types. The intention is to provide insight into the effects of common modelling choices that must be made and to highlight where benefit can be found from simplifying assumptions. The case studies also provide an insight into how the guidance points can be applied. Readers should note that sufficient information is provided to allow the cases to be replicated. However, due to variations in meshing and solver settings mean that an exact match to the results should not be expected.

While the WG has made every effort to provide comprehensive guidance, the sheer number of cases in which FEM might be used for cable rating calculations means that some gaps might exist. As the industry wide experience of applying FEM to cables continues to grow, we fully expect new publications to become available that demonstrate further improvements or enhancements. While it remains unlikely that the use of FEM will displace standardised analytical calculations for the purpose of cable system design, it will continue to be an essential part of the cable engineers toolbox.

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1. Introduction

This Chapter provides an overview of the rationale for the formation of WG B1.87 and explains the scope of work which has been addressed. The structure of the TB is outlined, and guidance is given on how different user groups can gain the most benefit from the material contained within it.

The Finite Element Analysis (FEA) is the simulation of any given physical phenomenon, described by partial differential equations, using the numerical technique called the Finite Element Method (FEM).

1.1 Scope of the Working Group

The work of B1.87 was initiated by the findings of Task Force B1.84. The scope of B1.84 included two separate IEC documents, IEC TR 62095 (addressing finite element modelling of cables) and IEC 60853 (addressing transient temperature rise calculations for cables). This TB deals only with the first topic; that of IEC 60853 is being addressed in the parallel WG B1.91. It should be noted that the number '84' has not been used for either of the follow on working groups to avoid confusion around the scope.

1.1.1 Task Force B1.84

During the August 2020 meeting of SC B1, a proposal was submitted by the Swedish National Committee. The text of the proposal is included below.

The IEC 60853 (cyclic and emergency current rating of cable) is not very up to date, and rather difficult to read and apply. Some methods and derivations in IEC 60853 are not traceable anymore. Any future work on future updates will not become easier as time passes by.

The IEC TR 62095 (Finite element method - current rating) is not very up-to-date either. Both documents are in need of an update.

It is suggested to check with IEC whether help of B1 is needed, or whether IEC plans for a revision themselves. If then necessary, work should start within B1 that can be implemented in these future-update IEC documents.

In response to this proposal, TF B1.84 was established and duly completed the following scope of work:

- Reviewed the current state of IEC TR 62095 and established that it did not fully meet the requirements of the modern B1 user community.
- Reviewed other CIGRE B1 work that relates to establish what had already been published by CIGRE, and to identify any gaps.
- Established the status of IEC TR 62095 with IEC TC20 WG19 (Current rating and short-circuit limits of cables), and agreed that CIGRE B1 could provide valuable support to WG19 by undertaking work on these topics.
- Determined that a WG was required and produced the Terms of Reference.

1.1.2 Scope of B1.87

The formal scope of this TB, as laid out in the Terms of Reference for WG B1.87, is as follows.

1. Conduct a review of published papers and reports on the topic of finite element analysis for cable ratings.
2. Evaluate the information needed by different user groups (utilities/developers, cable suppliers, contractors, consultants, researchers etc.).
3. Develop guidance for 'purchasers' on how to correctly specify FE modelling for cable rating purposes (inputs needed, definition of cases to be solved, quality control, contents of reports).
4. Provide guidance to FE users on:
 - a. When FE is needed, and when analytical methods will suffice.
 - b. Correct model set up for common cable rating cases, including assessment of 2D vs 3D models.
 - c. Means of verifying that a model has been implemented correctly and is not adversely affected by any modelling assumptions.
 - d. Handling the calculation of cable electrical losses, including the advantages and

disadvantages of multiple-physics modelling vs a simpler analytical/FE approach.

e. Implementation of transient/variable loading scenarios.

f. Handling of convective and radiative heat transfer, including the use of empirically derived convection coefficients.

5. Provision of a small number of worked examples which are sufficiently detailed to be replicated by independent users.
6. By solving the same case using different tools, meshes etc., define the expected 'tolerance' around the true answer.

It is important to note that the Scope does not include elementary discussions on the finite element method; the guidance provided to users will assume general competence in the use of commercially available software tools. The purpose of the TB will not be to teach the reader how to use FE tools, but instead to focus on application to cable rating problems. The guidance provided to 'purchasers' of rating studies is to be produced in a manner that can be understood without specific knowledge of FE tools.

1.2 Relation to IEC TR 62095

As described in Section 1.1, the TF B1.84 starting point was the IEC TR 62095. It was noted that the report has not been significantly updated in many years and may not fully meet the needs of modern users. Much of the guidance in the document was written at a time when FE studies were undertaken only by a small group of experienced analysis engineers, or by researchers. Many such users would need to create their own numerical codes to build and solve the models; although some software tools were commercially available, they often required a high degree of training and were not always easy to use. It was comparatively rare for FE studies to be commissioned in cable projects, and the computational power required to solve such models restricted the cable rating cases where FE was used.

In the 2020's, the situation is very different. User-friendly software tools are widely available, and standard laptop computers are capable of solving models that in the late 2000's had to be solved using a dedicated server computer. It is now increasingly common for purchasers of cable systems to request that parts of rating studies be performed using FE, most notably in situations where IEC 60287 does not easily allow the problem to be solved. This means that the number of users solving cable ratings via FE is much higher, and their specific experience is (on average) less than would have been the case 15 years ago. On this basis, the type of guidance needed is now different to that offered by IEC TR 62095. While much of the information contained there is still valid, there are many gaps which could be filled by new work from CIGRE. The work completed by the Task Force has identified that there are no current plans within IEC for a major update of this technical report, although some minor changes may be made after 2023. Consultation with users from the countries represented by the TF has concluded that the existing TR 62095 does not sufficiently meet the needs of many users.

A comprehensive review of the IEC TR 62095 is presented in Chapter 2. WG B1.87 was fortunate to be joined by the original author of the document, Prof George Anders, who has supported the group throughout as an Invited Expert.

1.2.1 Feedback Received on IEC TR 62095

The members of TF B1.84 discussed the existing IEC TR 62095 technical report and the application of the finite element method in the context of thermal rating of power cables. Thereby, several issues were identified by the group, which are briefly discussed below:

Examples: One of the biggest gaps in the existing IEC TR 62095 concerns the provision of the examples. The given examples are not well defined so that crucial information which would be needed to duplicate and test the given examples are missing. This information contains type and definition of boundary conditions, input values, convection coefficients and so forth. Furthermore, the problem specification is not completely clear.

Model Verification: The users consulted agreed that there was a need for advice on model verification steps or methods, similar to CIGRE TB 880. This would allow the user to check the results or partial results of the finite element method, as these are usually easy to obtain and can reveal an incorrectly defined simulation at an early stage and therefore, avoid incorrect simulation results. The steps in the verification should be explained separately to the examples themselves.

Best Practice: The group agrees that some form of best practice guidance is needed for the use of the most common cable assessment studies. This should show users of IEC TR 62095 the basic structure of common cable rating simulation problems and also give them the opportunity to test their own simulation tools (software, settings, parameters, etc.) using common examples. This is particularly valuable as there are a large number of solvers, programmes and other simulation tools and it would not be possible to deal with each one in a single document.

Representation of Cable Losses: Another issue is the erroneous representation of electromagnetic phenomena by simplified finite element models. There are several simplifications in the application of the finite element method, for example the representation of segmented stranded conductors as solid conductors or the missing representation of the twisting of shield and armour in 2D models, which lead to incorrect results in electromagnetic simulations. These can lead to incorrect thermal ratings, which is why guidelines for the correct representation of certain cable components should be given here, as these issues can be overcome by sophisticated implementation, e.g., 2.5D models, coil domains etc.

Sensitivity Studies: Another identified issue is the lack of some sensitivity analysis of the impact of different solvers, mesh strategies and calculation tools, especially in the context of multiple-physics simulations. This is considered essential as it allows users to check if the rating of the cable is heavily dependent upon assumptions or model parameters.

Time Step Calculations: Also identified as a relevant point is the treatment of solver settings and solver errors. In the simulation of thermal cable ratings, many domains of different physics and time parameters sometimes come together, which means that a problem must be solved by several different solver types. Since an electromagnetic solver in the frequency domain is very different from a thermal solver in the time domain, there are special requirements for the mesh as well as for the solver settings. This can lead to various problems such as a lack of convergence, increased calculation time, etc. Therefore, this topic should also be addressed.

When to Use FE: Users felt that it should be clearly indicated in which areas the finite element method performs better or might be more convenient than IEC 60287/60853 and in which cases there is no performance advantage. It should also be clearly shown where the limitations of the finite element method are in the context of thermal rating of power cables and in which situations complex, elaborate simulations might not bring any significant added value. This should help to avoid unnecessarily complex models and also contribute to keeping the expectations regarding the possibilities of the finite element method in the thermal rating of power cables within a realistic framework.

1.2.2 Links to IEC TC 20 WG 19

In order to avoid duplication of work, or undertaking work which would not provide value to both the IEC and CIGRE communities, TF B1.84 engaged with Mr Roberto Gaspari, the Convenor of IEC TC 20 WG19 (Current rating and short-circuit limits of cables). In addition, the following members of TF B1.84 were also Members of WG19:

- Prof George Anders, Canada
- Hajo Eckstein, Germany
- Frédéric Lesur, France
- Dr James Pilgrim, United Kingdom

The summary below identifies the topics which were discussed during the TF.

IEC Work Plan: The current version of IEC TR 62095 at the time of the TF had a stability date of 2023, meaning that no changes could be published prior to this time. WG19 planned to introduce a new document after the stability date had passed, a version of which has already been approved within the WG19 (although not yet finalised through the voting process). There is no plan at this time for major changes in the next revision.

Development Areas: Mr Gaspari felt that there were opportunities to provide more guidance in some areas, for example explaining when CFD might be useful and when it is not, along with discussion on the simplification of models and the use of sensitivity checks.

Broader FE issues: the topic of a more general TB on the application of FE to cable systems was discussed. The shared opinion was that IEC/CIGRE should not teach engineers how to use software tools, but there could be a place for guidance on technical issues that need to be addressed. There might be some merit in the production of a TB which covered the topic of FE modelling not just from

the perspective of the software user, but also from the perspective of a client requesting studies from a supplier or consultant.

Overall, there was mutual agreement that some improvement on the present IEC TR 62095 was possible, but there was a need to determine what is within the remit of IEC 62095 and what would be better as a separate CIGRE TB. This is particularly true with respect to the amount of explanation and guidance that can be provided in an informative TB, in comparison to within a Technical Report by IEC.

1.3 Structure of the TB

The TB is structured into 8 Chapters, of which this is the first. Chapter 2 presents a review of the technical literature related to finite element analysis, including not only work done in CIGRE but also a thorough review of work published in conferences such as Jicable and peer reviewed journals such as the IEEE Transactions. It should be noted that the review is not intended to be comprehensive, and many worthwhile papers are not included. Instead the aim is to provide the reader with a starting point for further research and learning on the topic.

Chapter 3 introduces the Value Proposition for Finite Element modelling of cables to obtain current ratings. While FE is an extremely powerful tool, it is not the intention that it should replace standard based calculations. The aim is to highlight the places where there is true value to the use of FE, thus allowing both clients and cable system designers alike to determine when it is worthwhile to use FE tools. There is also a focus on defining when it is NOT appropriate or necessary to use FE. This information is presented in an accessible way, which does not require the reader to be a master of FE tools.

Having identified when FE can provide real value to a cable project, Chapter 4 then presents a short set of guidance for clients. The intention is to describe in a concise way how a finite element based cable rating study should be requested, with a summary of the key information that would need to be made available by the client to the party conducting the modelling work. While the guidance presented is deliberately not a comprehensive checklist of items, it will add clarity concerning what information and input data should be made available during the procurement stage and as input to the work of the modelling team. In order to set realistic expectations on both sides, this chapter also defines what information should be provided back to the client as a "minimum viable report". As with Chapter 3, it is not necessary to be a user of FE tools to follow the guidance provided here.

Chapter 5 consists of the largest body of work in the TB and provides Modelling Guidance to users of FE tools. These users could be cable system designers within a cable manufacturing business, engineers working for an EPC contractor, independent engineering consultants or researchers working on cable system simulations. The guidance presented has been derived not just from the collective experience of the group, several of whom have published peer reviewed papers on the topic, but also from the process of setting up and solving the series of Case Studies that are presented in Chapter 7. As can be expected, the group did not immediately agree 100% with each others ideas. This sparked a very productive series of technical exchanges, enabling us to identify the most appropriate guidance on a wide range of modelling issues that can be encountered during cable rating studies.

For any cable rating study to have value, it is essential that the results can be verified. Unlike the earlier work of WG B1.56 in TB 880, it is not possible for two engineers using different software tools to obtain exactly the same answer. Therefore, the process of "checking" results becomes much more complex than for a simpler IEC 60287 based study. Chapter 6 provides an overview of how the outputs of the modelling process can be verified. This includes recommendations on "in process" checks that can be made, appropriate sensitivity studies to carry out, and mechanisms by which some complexity can be removed in order to sanity check the model. It is after all essential that a cable rating calculation obeys the laws of physics!

Chapter 7 presents a series of six different case studies that have been modelled in detail by multiple members of the WG. The agreed process for modelling each scenario is clearly described, and a limited set of results is presented to show the importance of certain features of each model. The intention is not for users to attempt to exactly match the answers, although sufficient information is provided that it should be possible to reproduce the results. Instead, the aim is to provide practical demonstrations of how each case can be modelled, with a wide range of cable and installation types being assessed.

1.4 How to best use the TB

This TB will be used in different ways by different user groups within the cable sector. For a quick start guide:

Clients / Purchasers: if you are reading this TB as you would like guidance on how to procure a high quality cable rating study, then you have come to the right place! It is recommended that you read Chapters 3 and 4 in their entirety. You should also make the contractor aware of this document, and that you will be using the guidance provided in it (see Section 1.5). The other parts of the TB may spark your imagination but are not essential reading.

Cable System Designers: if you are reading this TB as you are designing a cable system with some complex thermal sections, and you think that FE may be needed, then this is soon to be your favourite CIGRE TB. The WG suggest that you focus your attention on Chapters 5 and 6, as well as a quick refresher on Chapter 3. You may also wish to look at the specific Case Studies in Chapter 7 that are relevant to your case. Chapter 2 may also provide inspiration on other sources of guidance within the wider engineering literature.

Researchers: several of the authors of this TB are or were researchers working on cable system simulations. They recommend that you read all of it, and if you are new to the subject then you may wish to reproduce some of the Case Study models in Chapter 7. If you publish a paper which presents a new and better way of modelling the cable system than the one we have described, then please remember to reference this TB in your work. The field of cable system simulation is advancing quickly, and the WG has no doubt that there will come a time when some of our recommendations can be updated.

1.5 How to reference this TB in a technical specification

To ensure that the recommendations presented in this CIGRE Technical Brochure benefit your power cable project, it is proposed to use the following wording in a technical specification for a power cable project:

"All current rating calculations shall be performed with a verified current rating calculation tool and on the basis of IEC 60287. Where CIGRE TB 880 provides additional guidance compared to IEC 60287, this shall be followed. If the CIGRE TB 880 provides a different interpretation of the IEC calculations, it is suggested to follow the TB 880 guidance. Where finite element modelling is used, the guidance of CIGRE TB 963 shall be followed. In the event that an alternative modelling approach that contradicts the guidance in CIGRE TB 963 is considered beneficial, a clear technical justification should be given."

Evidence could be requested to show which calculation tool and version has been used, and to show how the tool aligns with the guidance provided in this TB, for example in the form of a verification report. This will provide clarity on the performance of the tool and subsequently the quality of the calculated current rating.

2. Literature review

2.1 The state of finite element analysis

The development of the numerical approach which today is known as the finite element method (FEM), has its roots in mechanical engineering. Alexander Hrennikoff and Richard Courant, who are often called the fathers of the finite element method, as well as John Argyris, Ray Clough (name giver to the finite element method), M.J. Turner and Olgierd Zienkiewicz, who are also referred to as the pioneers of the finite element method, all were working in structural analysis. For example, some of the early problems in the 1950s and 1960s that triggered the development of FEM were mechanical stress analysis for airplane components or the investigation of the effect of earthquakes on buildings and other structures.

The first paper found by the WG to use FEM for cable rating was published in 1973 [1]. Ever since, finite element has evolved and is used throughout a wide variety of engineering and scientific fields. A good summary on today's role of this approach can be found in [2].

FEM has become the computational workhorse for engineering design analysis and scientific modelling of a wide range of physical processes, including material and structural mechanics, fluid flow and heat conduction, various biological processes for medical diagnosis and surgery planning, electromagnetics and semi-conductor circuit and chip design and analysis, additive manufacturing, i.e. virtually every conceivable problem that can be described by partial differential equations (PDEs).

Over the 60 years since its invention, the FEM has become an extremely highly valued and applied calculation tool whose validity has been confirmed in countless tests. For example, FEM simulations are already replacing vehicle crash tests in some standards, are used for the design of fire protection systems through the calculation of fire behaviour or are used as proof for magnetic field exposures in complex energy systems such as transformers, substations, power cable joint-bays and so on.

2.2 The early developments in FEA/FEM for cable ratings

Following these developments, the cable industry has been using FEA/FEM for cable ratings since the 80s and 90s. The review included in this TB is limited to publications written in English. Therefore, early Jicable publications are not considered here. However, it should be noted that as early as 1987, Kladas, Pichon and Razek published a paper on numerical models based on the finite element method for calculating the electromagnetic and thermal behaviour of power cables and that Pichon and Razek published another paper on thermal modelling of transient conditions in power cables in 1987 [3], [4].

In their paper published in 1999, Anders et al. describe how to solve the heat conduction problem for complex power cable arrangements, e.g., crossings and thermally unfavourable environments [5]. The use of the finite element method is also recommended, as it allows a more accurate representation of mutual heating effects, an accurate representation of the boundaries and brings advantages for transient analysis. In addition, the following cases are mentioned in which the analytical method might fail:

- cables in touching formation or very close proximity
- when an isothermal boundary cannot be assumed
- when the medium surrounding the cable is composed of several materials

Furthermore, the paper briefly describes the concept of the finite element method for solving the heat conduction problem. The paper also includes the original example which was later incorporated into IEC TR 62095 as example 2 [6].

Also in 1999, P. Argaut et al. published a paper in which they used finite element method to calculate magnetic shielding techniques for power cables [7]. The variations and dimensions of the metallic plates relative to the flat-laid cable arrangement are given, but there is no information on other simulation parameters, such as boundary conditions or material parameters.

At Jicable 2003, J. Leduc et al. published a paper in which finite element analysis was used in combination with a DTS system to perform dynamic ratings of four 120 kV power cables and 10 overhead lines [8]. The models were validated on a laboratory test setup with varying loads over a period of 60 hours in which the FEM model was in excellent agreement with the measurements.

However, no information is given on how the cable is modelled or how the finite element analysis was set up.

An IEC Technical Report (IEC TR 62095 [6]) was published in 2003 with the aim of providing an overview on the application of FEA for cable ratings. The following section reviews this report in more detail. As the purpose of this TB is primarily to build upon the recommendations in the IEC TR 62095, the WG has taken the decision not to include a detailed review of the academic literature related to the use of FEM in cable applications prior to 2003.

2.3 What information does IEC TR 62095 provide?

The IEC TR 62095 consists of four chapters and an appendix. These are:

- Chapter 1: Introduction
- Chapter 2: Overview of the finite element method
- Chapter 3: Practical considerations when applying the finite element method for cable rating calculations.
- Chapter 4: Examples of application of the finite element method for cable rating calculations
- Annex A: Development of equations

all of which were reviewed by the working group and are summarised in the following.

2.3.1 Introduction

2.3.1.1 General

- The FEM is used as one of the numerical approaches to overcome the simplifying assumptions made in the IEC 60287 [9] or IEC 60853 [10]. This is the case for large cable diameters and cables located close to ground surface (cable external diameter > laying depth/10).
- FEM is used as a method to solve the heat conduction equations.
- FEM could be used to compute the temperature distribution in the soil surrounding the cable.

2.3.1.2 Field of application

FEM allows for more accurate modelling of the region's boundaries, e.g.:

- A convective boundary at the earth surface
- Constant heat flux circular boundaries for heat or water pipes in the vicinity of the cables
- An isothermal boundary at the water level at the bottom of a trench

as well as the calculation of:

1. Temperature rise in groups of cables
2. Unclear isothermal boundaries /medium surrounding cable composed of material with different thermal resistivities
3. Transient analysis for complex cable arrangements
4. Complex installations such as groups of different cable designs in tunnels or monopiles, crossings of several cable circuits, and so on

Temperature rise in groups of cables

In analytical methods, the case of a group of cables is often dealt with on the basis of the restricted application of superposition, where the assumption is made that the presence of another cable does not disturb the heat flux. Such a procedure is reasonably correct when the cables are separated from each other. However, when this is not the case, for example for cables in touching formation, the temperature rise caused by simultaneous operation of all cables should be considered. The finite element method offers such possibility.

Unclear isothermal boundaries / medium surrounding cable composed of material with different thermal resistivities

IEC 60287 calculations are not precise enough when isothermal boundaries are unclear or when the medium surrounding the cable is composed of materials with different thermal resistivities.

Numerical approaches, such as the finite element method, permit more accurate modelling of the region's boundaries. Thus, when an isothermal boundary cannot be assumed, for example, for cables installed in shallow troughs, the finite element method provides a suitable tool for the thermal analysis.

In actual cable installations there are often several soil characteristics present. The non-uniform soil conditions are handled easily by the finite element method.

Transient analysis between internal and external circuits of a cable

In IEC 60853 calculations, separate computations are performed for the internal and the external parts of the cable. Here, again, the finite element method offers a solution with minimal simplifying assumptions.

Complex cable installations

There are no known analytical solutions for multiple cable circuits carrying different loads in either ventilated or unventilated tunnels or vaults. Cable crossing can be solved analytically for very simple installations for two circuits in a uniform soil where the hot spot location can be identified. In almost all complex installations, the FEM offers a solution to cable rating problems.

2.3.1.3 Information obtained from numerical methods

IEC TR 62095 states that the most important additional information obtained from the numerical method is the resulting temperature distribution (temperature field) within the cable and its surroundings, rather than just evaluating the temperatures at the connection points of the thermal equivalent circuit diagram, such as conductor or cable surface, as is the case with the analytical method.

The WG note that analytical methods can be used to calculate temperature distributions although IEC 60287 calculations do not fully describe the method.

2.3.1.4 Alternative methods

At the end of the chapter "General" in IEC TR 62095 an overview of alternative numerical methods and where they are applicable is given. These methods are:

- Finite difference methods
- Boundary element methods
- The superposition method
- Conformal transformation

2.3.2 Overview of the finite element method

The chapter 2 of IEC TR 62095 gives a conceptual introduction to the finite element method in a frame of cable rating calculations. At the beginning of the chapter, FEM is introduced as a numerical technique for solving partial differential equations (PDEs). In the first paragraph it is also stated that

- FEM for cable rating is usually about solving the heat conduction problem, but may also include heat and mass transfer (i.e., heat transfer by other mechanisms than conduction).
- The temperature field on a given domain is solved approximately by using a discrete model.

The chapter 2 of IEC TR 62095 then gives a general outline of the procedure of establishing an FEM solution. The steps listed for solving for temperature on a given domain/region are:

1. Defining the nodes in the domain.
2. The (yet unknown) temperature at each node is denoted as a variable – it is these nodal temperatures that are to be determined.

3. The domain is split into a finite number of sub-domains called elements, with the nodes located at the vertices (and sometimes also edges) of the elements. The elements are connected in such a manner that they collectively approximate the shape of the domain.
4. The temperature on each element is approximated as a polynomial called an element function, with the polynomial expressed by the nodal temperatures. Matching conditions between adjacent elements are imposed to ensure continuity of temperature between elements. Temperature is solved numerically through a matrix-vector equation, where the solution vector provides the temperatures at each node - i.e., the approximate solution over the entire domain is found.

An example of temperature in a 1D-rod of length L placed along the x -axis is worked through qualitatively. The domain (i.e., the rod) is the region $x \in [0, L]$. Five equally spaced nodes are defined. Two possible discretization schemes are discussed, one leading to linear element functions on each sub-domain and one leading to quadratic polynomial element functions. For the exact procedure for solving this 1D example, the reader is referred to Annex A of IEC TR 62095.

Then follows a description of 2D-regions and elements, after the technical report notes that 2D is more applicable for cable rating calculations. It is described briefly that 2D-surfaces are typically discretized by either triangular or quadrilateral elements, and that the element functions can be linear (planes) or higher-order polynomial (curved) surfaces.

2.3.3 Practical considerations

In IEC TR 62095 considerations are made exclusively for buried land cable systems. The parameters that influence the accuracy of the calculations are reviewed which are:

2.3.3.1 Size of the region to be discretized

The earth surface is an obvious boundary for buried underground cables. The side and bottom boundaries must be selected in such way that the temperature gradient across the boundary equals zero. For a single circuit installation, the TR recommends a rectangular field 10 m wide and 5 m deep, with the cables located in the centre horizontally and at their burial depth below the top boundary vertically, so that satisfactory results can be achieved in the majority of practical cases.

In transient analysis, the boundary region can be selected as an envelope around the circle of the isothermal line around each cable, with a distance r from the cable axis to ensure that the temperature rise over the environmental temperature is sensibly captured (e.g., $\theta_r=0.1$ K). This radius r will change along the time and can be estimated from analytical equation as stated in [6]:

$$\theta_{r,t} = \frac{W_I \rho_s}{4\pi} \left[-E_i \left(\frac{-r^2}{4\delta t} \right) \right]$$

Equation 2.1

where

$\theta_{r,t}$ is the threshold temperature value at distance r from cable axis

W_I is the total cable losses

ρ_s is the thermal resistivity of soil

r is the radial distance from cable axis

t is the time from start of application of heating

2.3.3.2 Size of the elements constructed by mesh generator

The normal way of controlling the mesh sizes and some basic guidance on the placement of different element sizes in buried cable simulation is discussed. The basic messages here are, the smaller the element sizes are, the more accurate the simulation results will be, but the computing time will rise exponentially with the number of the elements and the smallest elements should be placed in the cable domain and in its direct vicinity. Even though this is fundamentally correct in the scope of the TR, the number of elements is not linear to the gain in accuracy. This is shown on the example of 6 cables located in a concrete bank with two different finite element sizes representations. The simulation results show a temperature difference of 3.2 K but an increase of computing time by a factor of 25. The WG notes that this applies to modelling thermal conduction, which is used for buried

cables, whereas for fluid dynamics, which are not covered by the TR, the meshing requirements are more complex and may differ.

2.3.3.3 Type and locations of region boundaries

FEM allows representations of different boundary conditions and arbitrary boundary locations. In particular, circular boundaries around cable, water, or steam pipes can be handled with ease. This section introduces three different boundary conditions that frequently occur in cable current rating tasks:

1. Isothermal boundary condition exists if the temperature is known along a portion of the boundary.
2. Convection boundary exists where heat is gained or lost at the boundary due to convection.
3. Constant heat flux boundary, which will mostly be required when there are other heat sources in the vicinity of the cable and their heat generation is known.

2.3.3.4 Representation of cable losses

In the IEC TR 62095, cable component's losses are represented in the FEM studies as heat sources where the values of these losses are re-calculated at each time step, using methods given in IEC 60287. Since the losses are temperature dependent, an iterative procedure is required. Furthermore, the application of dielectric losses in transient calculations in FEM are addressed.

2.3.3.5 Selection of the time step in transient analysis

IEC TR 62095 states that the selection of a time step is a trade off between the reduction of the amount of computation and the required accuracy, which is correct. Some guidance provided by Electra 87 [11] and Libondi [12] on the selection of suitable values of the time step is introduced. The workgroup notes that since the computing power even of basic desktops and laptops as well as the sophistication of modern FEM software has increased significantly in the last two decades, these considerations are, at least to some extent, out-dated.

2.3.4 Examples of application of the finite element method for cable rating calculations

The chapter contains three cable rating examples.

- Example 1: A 30 kV 3-core XLPE insulated cable with 300 mm² copper conductor and a lead sheath laid directly in soil in a depth of 1 meter. This simple example is used to show the good agreement between IEC 60287 and the finite element results.
- Example 2: This example contains different cable types with several soil layers and a convective heat boundary condition and represents a case, which cannot be solved analytically but can be easily solved with the finite element method. However, apart from the geometric dimensions and the thermal resistivity of the soil, no information, e.g., about the cable designs, the losses, the conductivities of the different concrete materials, the flow rate of the drain contained in the example or the convection coefficient of the convective Boundary Condition, is given in the example. The example merely presents a case that cannot be solved using the analytical method, but does not give the reader enough information to recalculate the temperatures or to fully understand the methodology used.
- Example 3: Example number 3 begins by briefly addressing the issue of soil dry-out, although this is not implemented in the example. Instead, it deals with a three-phase cable system in flat formation within a thermally stabilised backfill material and a convective heat transfer boundary condition, varying the value of the convection coefficient and comparing the results with those of the analytical method of IEC 60287. Although the thermal and geometric parameters are listed here, there is no information on the cable used, its losses or how the convective boundary condition should be selected. For the latter point, reference is made to a specialist publication, but without discussing its content or results.

2.3.5 Annex A of IEC TR 62095

The Annex A of IEC TR 62095 deals with the mathematics of the finite element method using the example of the basic two-dimensional heat equation as well as the mathematical formulation of the corresponding Dirichlet (Equation A.2) and Neumann (Equation A.3) boundary conditions, even though they are not referenced as such. Moreover, these two types of Boundary Conditions are

generally applicable to FEM problems of various physical types (e.g., electromagnetic calculations where the magnetic vector potential at the boundary is zero). Joule heating is given as the heat source, which is derived from the simplified version of the Ohms' law. However, in FEM, this equation is not necessarily used.

The text indicates that the application of the principle of superposition in the IEC approach is not accurate, while numerical methods allow a more accurate representation not only of the physics but also of the media surrounding the cables.

In the following subsections, the basic principles of the approximating polynomials and the definition of the general finite element equations for the time-dependent and steady state analysis are discussed. The appendix ends with the five following examples:

- single triangular element: calculation of the temperature in one point of the element considering the nodal temperatures.
- single triangular element: calculation of the position of an isotherm within the element considering the nodal temperatures.
- single triangular element: calculation of the heat generation vector given heat convection on one surface, heat flow on another surface of the element, an internal heat source and a given ambient temperature.
- a domain with three triangular elements: calculation of nodal temperatures for the element from example 3 extended by two additional adjacent triangular elements with different specific thermal conductivities and a zero temperature gradient boundary.
- a domain with several triangular elements: illustration of the application of the finite element method for modelling cables in extended backfills.

Although Appendix A gives a good basic insight into the finite element method using the example of the 2D heat conduction equation, it has limited added value for the average user of FEM software. As the scope of this TB does not include discussions of the mathematical background of FEM, no further discussion is included here.

2.4 What information is currently missing in IEC TR 62095?

The working group regards IEC TR 62095 as being ahead of its time and as a solid ground work for the application of the finite element method in cable rating calculations, especially for engineers who are only familiar with the analytical approach. However, in the two decades since its publication, not only has hardware and software performance increased exponentially (e.g., computing power has increased more than a thousandfold), but above all, the user field and the applications have changed as well. Both cable designs and installation arrangements have become more complex and the calculation requirements increased significantly due to the massive increase in cable installations and changes in the cable load (e.g. interconnectors or offshore connections). This can be confirmed when looking at Figure 2 - 1, which shows the percentage of votes from the WG members on the different areas of interest.

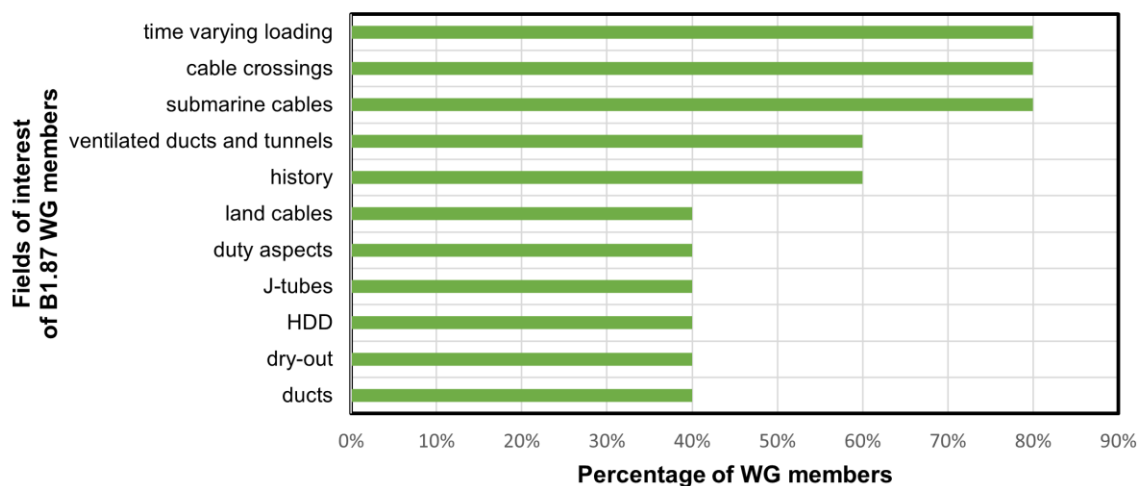


Figure 2 - 1 Topics in cable rating calculations of greatest interest to the group members

For example, the IEC only deals with the heat conduction problems, but not electromagnetic ones, such as the determination of screen currents and losses. Additionally, it mostly concerns land and simple air installations. When considering the necessary development areas for this TB, the working group members gave the ranking shown in Figure 2 - 2 for the required changes to the IEC TR 62095 document, with the maximum score of 10.

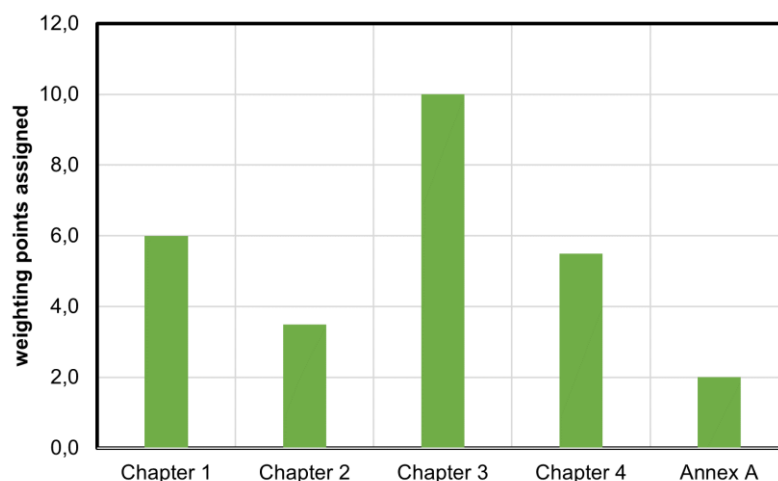


Figure 2 - 2 Assessment of the need to revise the individual chapter

In addition, the working group members have compiled a list of the missing content. An extract from this list with some of the key areas without particular order is as follows:

- Include a list with the main points of attention/pitfalls when modelling FEM
- A more in-depth introduction (qualitative & quantitative) on convection, conduction and radiation, as many complex cable rating scenarios where FEM is used will require more than just conduction
- Include the main/important conclusions and observations from the referenced literature
- A must-have list of requirements before actually using FEM
- 3D FEM-modelling → Maybe list the situations in the IEC standard where 3D modelling is recommended
- Detailed treatment of boundary conditions
- Detailed information on representation of cable losses (for different conductor and screen/armour designs) → see also CIGRE TB 908 (Losses in Armoured Three Core Power Cables) by WG B1.64
- Treatment of Multiphysics
- Submarine cables (Depth of Burial, J-tubes, risers, etc.)
- Inclusion of a step-by-step flowchart on the fundamental steps and settings when performing rating calculations using FEM
- Higher degree of detail in given examples
- Verification and validation methods/recommendations (sensitivity analysis, step-by-step model, intermediate results, etc.)

This resulted in four categories of content that is missing and/or could be added to the IEC TR 62095 if it is revised in the future. These were also subjected to a weighting process, with the same criteria as for the IEC chapters. The results of are shown in Figure 2 - 3.

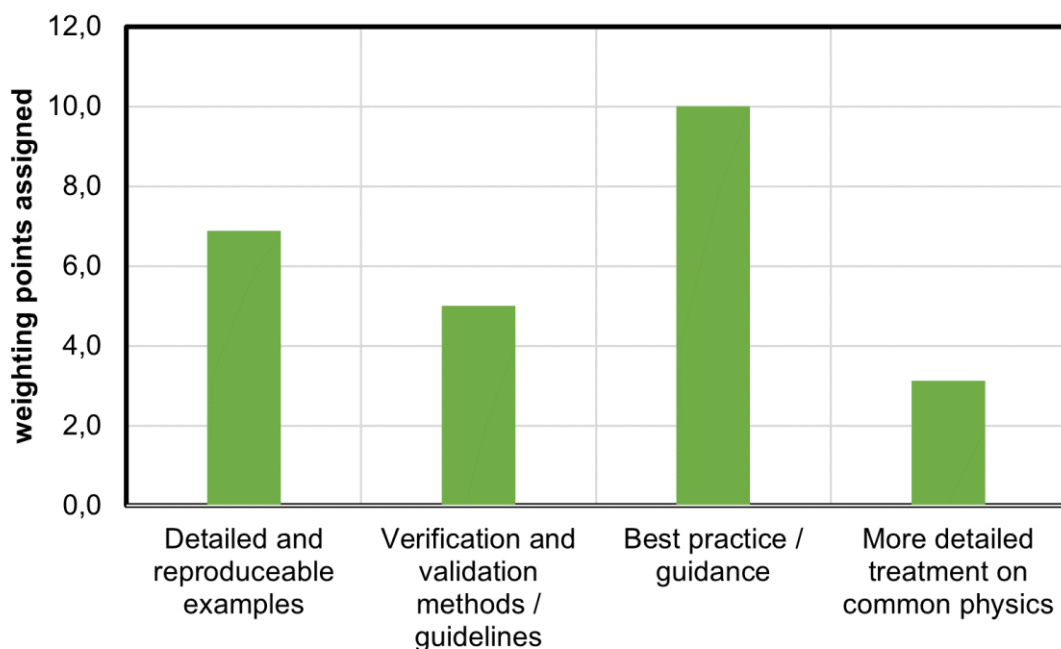


Figure 2 - 3 Assessment of the type of content most desired to be included in the standard

2.5 Recent developments in FEM for cable rating calculations

More than 20 years have passed since the publication of the IEC TR 62095. During this time, intensive research has been carried out on the use of FEM for current rating calculations of power cables. Figure 2 - 4 and Figure 2 - 5, which show the number of the annual IEEE and Jicable publications per conference year dealing with the use of the finite element method in power cable calculations, clearly illustrate this fact.

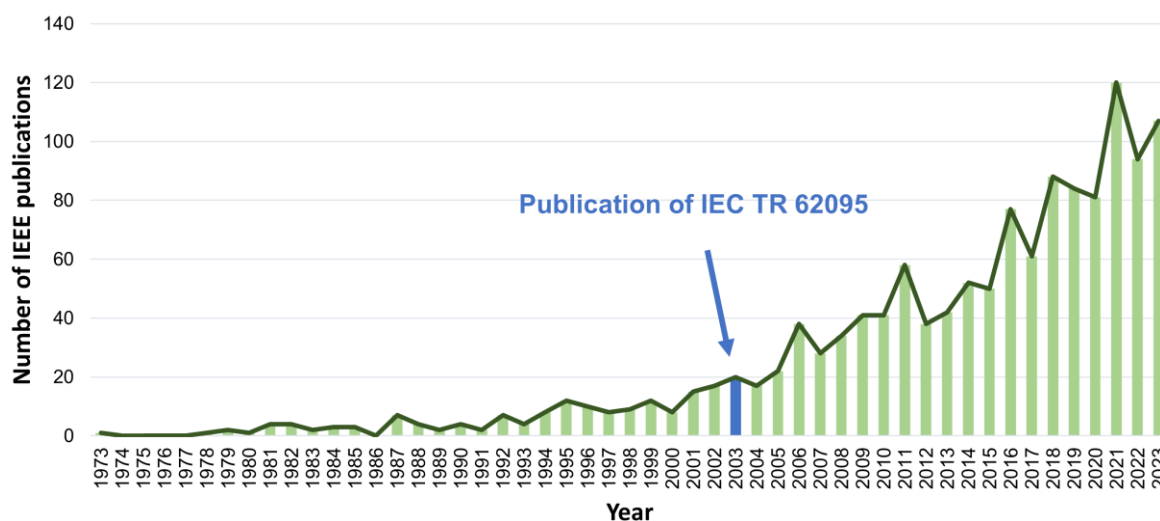


Figure 2 - 4 Development of the annual IEEE publications on FEM in power cables

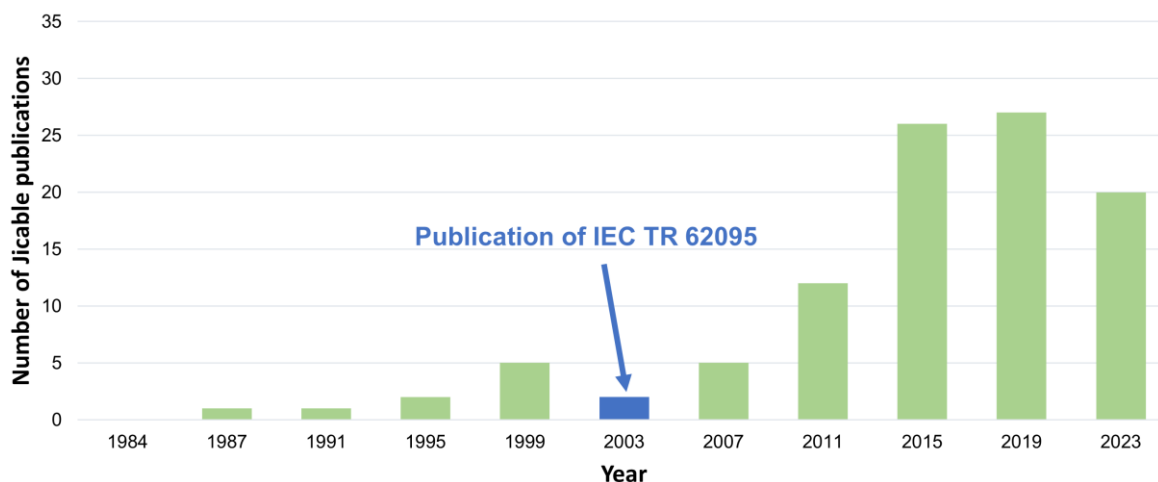


Figure 2 - 5 Development of Jicable publications on FEM in power cables per conference year

These diagrams indicate that overall more than 85% of research papers and publications dealing with finite element analysis of power cables were published after IEC TR 62095. A brief summary of the developments in the recent publications dealing with different domains of cable rating calculation is given below. For this purpose, the recent literature has been divided into several categories as illustrated in the following sections. The intention of including material here is to give the reader a head start when looking for alternative approaches, or an example that FEM has been used to successfully solve a given problem. The content of any referenced paper should be considered secondary to the modelling guidance given in this TB in the event of any conflict.

2.6 3D-Modelling

Three dimensional effects are notoriously challenging to include in analytical calculations, meaning that they have been considered quite extensively by FEM.

In this section, many papers are presented as examples of prior work and are for reference-only purposes.

2.6.1 Cable Crossings

In 2011, M. Chaaban and J. Leduc published a paper on crossing analysis in 2- and 3-dimensional domains [13]. The publication lists briefly the adopted boundary conditions and describes in some detail the modelling approach.

In the 2015 paper of Z Huang et al., a 3D FEM analysis for HVDC subsea cable crossings is described [14]. The modelling approach is described in good detail. Furthermore, symmetry considerations and clever boundary conditions are used to reduce model size and complexity.

The paper by G. J. Anders et al. [15] addresses the thermal challenges of a 230 kV Self-Contained Fluid-Filled (SCFF) cable system crossing three distribution duct banks (DDBs) in downtown Vancouver. The study evaluates different heat mitigation techniques, including alternate backfill, embedded heat pipes, and an underground ventilation vault with both natural and forced convection systems. The paper showcases that for such complex geometries, finite element method needs to be applied. 3D FEM models for the cable crossing and a ventilated manhole are shown.

2.6.2 Joint Bays

J. A. Pilgrim et al. [16] investigated the current rating of cable joint locations using FEM. The paper compares a 2D axisymmetric model of a cable joint with a 3D model, and investigates which simplifications can be made in the 3D-model while retaining a satisfactory accuracy. The paper is quite detailed with respect to assumptions, parameter values etc. Still relevant today, as the 3D models are often in need of simplifications if they are to be solved without being too time and/or resource demanding.

2.6.3 Losses in 3-core armoured cables

M. Høyer-Hansen et al. compare 3D and 2D FEA for three twisted cores with metallic screens of various designs [17]. They show that for the tubular screens there is a good agreement between 2D and 3D, whilst for metallic tape screens in open-helix there is a large difference in the obtained screen losses for the modelling technique used in the paper. However, limited details of the modelling technique or meshing are included.

In [18], by M. Hatlo et al., the analytical method is used for taking into account 3D effects for loss calculation of 3-core cables. A complex permeability is used to take into account both eddy current and hysteresis losses. Comparison with measurements, 3D FEM and the IEC models are made. Analytical models have evolved further today, e.g., taking into account that the magnetic field varies along the wires, something which this paper does not consider. However, the FEM model is not described in detail and the analytical methods have meanwhile evolved further. Nevertheless, the publication shows that an application of the IEC 60287 can lead to conservative armour loss values for 3-core submarine cables.

The paper by S. Sturm et al., also published in 2015, uses 3D FEM to analyse losses in 3-core submarine cables and compares them to the 2D FEM and IEC 60287 [19]. A non-linear type of armour steel $\mu=\mu(B)$ is used in 2D FEM in addition to steel with linearly changing permeability values. Furthermore, a model length equal to the crossing pitch of armour and cores is used. The 2D method allows circulating current in each armour wire. However, there is also no detailed description on the FEM models.

Y. Matsumoto et al., use 3D FEM to analyse losses in 3-core submarine cables and compare the results to 2D FEM and the IEC 60287 [20]. The studies are limited to unilay cable designs with a small difference in pitch between armour wires and cores resulting in 2D and 3D FEM in almost perfect agreement. The publication investigates sensitivity of sheath and armour loss factors with respect to the core conductor size and lead sheath thickness. Little detail is given for the modelling techniques used. Although unilay cables with equal pitch of armours and cores are not that often used, the paper does show that pure 2D FEM is applicable for such cable constructions.

D. Willen et al., investigate the importance of mesh resolution in- and around armour wires in the paper published in 2019 [21]. They show that armour wire loss is not sensitive to mesh resolution (with wide wire spacing). Details are given on modelling methods and meshing, e.g., the use of periodicity vs magnetic insulation for the ends of the cable model, length of the model, and so on. Furthermore, a comparison is made between 3D FEM and measurements.

In the paper by J. Del-Pino-Lopez et al. [22], a comparison is made between 3D and 2.5D FEM, to determine when 2.5D is a sufficiently good approximation and to show the important limitations of 2.5D FEM. Furthermore, a brief discussion on the use of the periodic boundary conditions and crossing pitch to reduce model size is offered. A 2.5D model is a computational approach used to simulate physical phenomena that are essentially two-dimensional but include some three-dimensional effects. This method is particularly useful in scenarios where full 3D modelling would be computationally expensive and 2D modelling would be insufficiently accurate.

R. Benato et al. in [23] use 2D and 3D FEM to model a 3-core cable. First, it presents 2D analysis results performed in several available FEM programs, and finds they are in agreement. The model length is equal to the core pitch of 3 m, with the armours having a pitch of 2 m. It does not give a detailed description of the modelling techniques and choices taken for the 3D models.

The paper by R. Benato et al. [24] presents a method named multiconductor-cell analysis and applies it to a 3-core submarine cable. The method has previously been used for 2D applications but was extended to 3D in this paper. The method includes the pitch of both cores and armour wires. The method is based on an approach where each conductor in the cable is represented by several smaller so-called sub-conductors. The size of these sub-conductors is finite but small enough that their current density can be assumed uniform. Lengthwise the cable is split into cells. Analytical formulas for self- and mutual-impedance are then applied for each sub-conductor and pair, and arranged in a matrix-vector formalism for solving for current and voltages. It uses 3D FEM as verification of the proposed method. The paper by R. Benato et al. [25] covers the same multiconductor cell analysis aspect in more detail.

E. Plumed et al. [26] uses 3D FEM to model twisted Litz wires. However, this is focused on electronics applications with very different design compared to power cables. Nevertheless, some modelling aspects discussed are relevant to WG B1.87; the paper discusses how steep twisting is accounted for

in the FEA model as the circularly shaped cross-section of the individual wires will appear as ellipses in the bundle cross-section.

An algorithm for determining loss allocation from impedance measurements of 3-core cables is presented in the paper by C. Del-Pino-Lopez et al. [27]. The method is based on determining the amount of losses in cores, screens and armour based on the values available measurements when working on a real cable. A “virtual laboratory” is presented, i.e., a parametrized 3D FEM model of 3-core cables on which the algorithm is tested. Results are also compared to the IEC 60287.

The paper by R. Giussani et al. [28] proposes an alternative 3D method for calculating losses occurring in the armour of 3-core cables. The method handles the twisting of both cores and armour wires. It is a combination of analytical and numerical methods. The armour wires are modelled as non-conducting with anisotropic permeability, and the expressions for the relative permeability in the parallel-to-wire and normal-to-wire directions are given by analytical formulas. The wires are discretized in the length-direction, following helical paths. The three cores are represented by three twisted elementary conductors, and for the magnetic field generated by the cores analytical formulas for the magnetic field surrounding three elementary conductors are used. It is further assumed that the magnetization, being a dependent variable, in each small segment of wire is uniform. A linear system of equations is derived, to which the solution is the magnetization in each wire. This linear system of equations can be solved numerically and much faster than a 3D FEM model of comparable geometric size. The proposed model is compared to a FEM 3D model, and generally shows good accuracy. The FEM modelling is not described in detail as it is not the area of focus for this paper.

The paper by C. Del-Pino-Lopez et al. [29] presents measurements on several 3-core cables, ranging from short samples to full length cables (~ 100 km). The measurements are used as experimental validation of the so-called Ultra-Shortened 3D FEM Models (USM). A small novel development of USMs is presented in the paper, called Simplified USMs (SUSMs). Measurements and FEA results are in good agreement. The USM modelling approach is an important step forward in FEA modelling of 3-core cables, drastically reducing computation time. The paper is well written and detailed.

The paper by M. Bau et al. [30] describes the main 2D FEM simulation mechanism in 3-core submarine cables: the validity of a 2D model, boundary conditions, and dominating physics in the simulation of cable losses is discussed. Only resistive losses are considered, while dielectric losses in insulation and hysteresis losses in armour wires are neglected. The exciting mechanisms and net voltage cancellation implementation methods in the FEM model were discussed. A large difference in the armour loss factor and cable ampacity between IEC approach and FEM simulation results was observed.

T. Kim et al. [31] discusses the development of a low-loss cable by optimizing steel armour. The study highlights that the IEC60287 standards overestimate cable armour losses, prompting the design of a loss measuring system and a 3D FEM analysis model to ensure consistency between testing and analysis. The research identifies the main factors affecting cable loss and proposes an optimal cable design, emphasizing the impact of armour material and the direction of twisting (unilay vs. contralay) on cable losses. Relevant topics for the modelling such as periodic boundary conditions, least common multiplier or cross-pitch are described. The results show that the used 3D FEM model is in very good agreement with the measurements that were performed.

2.7 Submarine Cables

2.7.1 J-tubes / Offshore Platforms

In their 2015 publication, R. Chippendale et al. introduce a 3D FEM simulation for steady state Ampacity of submarine cables in J-tubes [32]. This paper does not contain any detailed description of FEM model of submarine cables or J-tubes. Assumptions in internal and external (boundary) heat transfer mechanisms used in the FEM simulation are described. The sensitivity of the cable ampacity regarding the seasonal effects (solar radiation), air section length, and the gap size between the cable outer surface and J-tube inner surface is studied.

Reference [33] by L. Gang et al. presents a 2D FEM calculation of current rating of a single core submarine cable in air subjected to various wind speeds. There is no detailed description of the model nor the simulation process.

The paper by O. Thyrvin et al. [34] discusses the thermal rating of dynamic three-core AC cables, presented at the 11th International Conference on Insulated Power Cables (Jicable'23). It evaluates methods for calculating thermal resistances, screen resistance, and screen current, comparing results from FEM modelling with real installation measurements. The paper highlights the unique design and

installation challenges of dynamic cables, such as the lack of a lead sheath and the use of copper wires or metal sheaths for different voltage levels. It also addresses the impact of factors like corrugation, bend stiffeners, and environmental conditions (e.g., solar radiation, wind) on the thermal performance of these cables. The study concludes that FEM modelling provides accurate results for dynamic cable installations, emphasizing the need for careful analysis of specific cable designs and installation conditions.

2.7.2 Seabed

Publication by T. Hughes et al., from 2015 contains a description of the heat transfer equations in FEM simulation for submarine cable buried in sediments [35]. The size of fine meshing and coarse meshing was mentioned. Steady-state cable ampacity was simulated in a 2D FEM model, under various sediment permeability, porosity, and thermal conductivity values to see their influence on the submarine cable rating.

The FEM model developed and presented in the paper by C. Emeana et al. [36] was used to verify the experimental measurement of the thermal distribution around buried submarine cable in an experimental tank. The transition zone between conductive and convective heat transfer modes in submarine cable current ratings considering the permeability of the sediment was plotted.

J. Lux et al., utilize a 2D FEM thermal simulation of submarine cables to verify the analytical model of cable depth of burial estimation [37]. However, there is no detailed description of the FEM model or the simulation processes.

W. Wang et al., describe cable thermal and losses simulation using 2D FEM simulation in a Multiphysics tool [38]. Thermal conduction inside the cable and between the cable and environmental soil is the dominating mechanism, while the seabed surface and sea water are simulated by heat convection according to the Newton cooling formula. The influencing factors on the thermal distribution inside the cables (current ampacity, soil conductivity, and burial depth) are analysed.

Reference [39] by H. Brakelmann et al. compares the main principles of an analytical model (LSM) and FEM approach for ampacity calculation for submarine cables buried in seabed. The authors concluded that there was good agreement between both approaches. There is no detailed description of the model building process of the FEM simulation in this paper.

The paper by T. J. Hughes et al. [40] presents a 2D FEM model for current rating of submarine cables in seabed taking into account thermal conduction and convection. Dominant equations in the heat transfer mechanism of buried submarine cables in saturated seabed sediment are explained in detail in this paper. It also gives explanations about the geometry, mesh setting and the boundary conditions implemented in the FEM model. The influence of sediment parameters and burial depth on the cable current ratings is studied in sensitivity analysis. Comparison of the FEM and the IEC 60287 results was carried out and possible explanation of the correlation between the agreements was given.

K. Bitsi et al. [41] presents modelling of an offshore rigid sea joint. Comparison is made between 2D and 3D simulations for both joint and bend restrictors showing large conservatism in 2D simulations for this installation.

2.7.3 Mechanical Considerations

In their 2019 work, W. Wang et al., describe a 3D model of mechanical simulation in a single core 500kV submarine cable with single layer flat armour wires [42]. The length of model, the number of nodes and meshes, and the contact types between different cable layers in the model were described in the paper. Cable stress and strain distribution on subcomponents under tensile forces were simulated and discussed. FEM simulation results were compared with those obtained from the experimental testing and good agreement was found.

In [43], by D. Tjahjanto et al., a 3D FEM model for mechanical performance simulation in an HVAC 3-core submarine cable is described. The simulation length and boundary conditions, as well as meshing techniques of the FEM model were briefly introduced. The simulated cable elongation and torsion performance under tensile loading were compared with analytical and experimental results.

The paper by D. Dharmawan et al. [44] discusses the mechanical behaviour of a trefoil cable bundle during offshore laying, focusing on a configuration consisting of two main HVDC cables and one metallic return (MR) cable. FE models were developed to analyse the mechanical response of the cables and bundling yarns/ropes, providing insights into the structural stability of the trefoil cable bundle under different bundling tightness scenarios. The study concludes that a semi-tight bundling configuration

offers the best balance, maintaining the trefoil shape while allowing for necessary slips to ease excessive axial deformations during installation.

2.7.4 HVDC

In the paper by M. Hu et al. [45] a 2D FEM simulation for the electrical field distribution in the HVDC cable insulation at rated load and no load is introduced. It is not mentioned which simulation tool was used.

K. Carvalho et al. provide a 2D FEM model simulating the frequency response of a DC submarine cable [46]. Frequency response from 0 to 150 kHz was extracted using the FEM model, and the simulation results were verified using the cable model in other tools, which proves that the FEM model is adequate for transient simulations of the DC power subsea systems.

In the paper by C. Y. Hao et al. [47], a 3D FEM model of the submarine HVDC cable is presented in detail along with a discussion on how to simplify the cable model to reduce the simulation time. The analysis results in the distribution of the temperature field within the cable layers at its nominal ampacity. The results obtained by the full and simplified cable models were compared. The results show that the maximum temperature difference of the insulation layer is only 0.13 K between these two models. Whereas the simulation time is reduced from 70 min to 4 min, respectively.

2.8 Tunnels / CFD

In 2011, J. A. Pilgrim et al. presented a paper in which they treat 2D-FEM modelling of air-cooled cable joints installed in tunnels [48]. The paper discusses how to overcome the model limitations set by the CIGRE Electra 143. The authors explain how to set up the model and how to represent cable losses and the joint boundary conditions. An example calculation for a tunnel installation is shown.

Traditional IEC 60287 method using heat transfer model cannot calculate cable ampacity concerning various structural details such as complex trench geometry. In the paper by S. Jiang et al. [49], an FEM model to calculate the temperature field outside the cables in combination with the analytical method is discussed. The analysis is limited to directly buried cables and cables laid in tunnels. Comparison is made between FEM and the IEC 60287. The cable surface temperature is calculated. There are, however, some concerns regarding modelling that could lead to questions on the quality and reliability of this source.

2.9 Radiation / Convection

FEM is particularly useful when considering installations which experience heat transfer by convection and radiation.

2.9.1 Cables in Ducts

In the paper by C. Balzer et al. [50], 3D FEM is used for MV cables installed in air filled protection pipes. The paper changed to 2D FEM in case of short cable lengths (here: 3 m). The paper contains the details of the investigation involving a simultaneous presence of 3 heat transfer mechanisms in and around the pipes, thus, considering conduction, convection and radiation. The proposed approach avoided the often used simplification, which describes the heat transfer mechanisms for isothermal, concentric cylinders. FEM is proposed to study the dynamic behaviour and heat capacitance with any type of dynamically evolving load. The paper gives clear highlights on the physical backgrounds of each heat transfer mechanism. Also some practical FEM guidance was included such as how to model a helix shaped wire screen of cables. For future studies, it is suggested to also take into account the thermal expansion of the cable components and the semiconductor layers and swelling tapes.

The paper by T. Saito et al. [51] stresses the importance of those elements that could serve as a basis for increasing precision of FEM calculations for high voltage cables in directly buried ducts in a thermally favourable or unfavourable soil environment. For a very specific location and test setup involving long observation times, precision was enhanced by considering daily and yearly (seasonal) load fluctuations, weather data measurements (air temperature, wind speed, sunlight amount, radiation, rainfall and humidity), as well as soil temperature measurements and its historical data. Also, the results of the measurements with integrated optical fibres in the cable conductor and wire shield were presented. The paper includes a brief description of different heat exchange mechanisms between the ground surface and the atmosphere, as well as a discussion of the time and temperature dependence of soil thermal resistance. There are, however, some shortcomings in this paper: some unclear/blurry figures, overall lack of analytical & technical details on the new conductor temperature

estimation program and what role FEM plays in the newly developed program. A reader seeking to gain more in-depth knowledge on specific FEM topics will not be satisfied.

The paper by Y. Liang [52] lists some interesting references for rather early methods to calculate thermal fields around cables: boundary element, finite difference and FEM. It deals with 3-phase AC cables installed in buried PVC ducts that are internally filled with air. The mass conservation equation, momentum conservation equation and energy conservation equation of the air between external cable surface and an internal duct surface are described. Streamline-upwind/Petrov-Galerkin (SUPG) stabilized 2D FEM is applied to solve the air governing equations for the flow field and an iterative procedure is adopted to solve the nonlinear-coupled electromagnetic-thermal model of underground cables. The basic principles on the mathematical model are explained, while not ignoring details on assumptions made and boundary conditions. For all calculation methods, it is however not mentioned which mathematical procedure or recommendations could apply to obtain a convergence number δ that is acceptable. The paper presents a Multiphysics approach where thermal models of heat governing equations combined with electromagnetic phenomena. The paper also describes the setup and results of an experiment to verify the accuracy of the analytic SUPG model with temperature measurements in the field for a situation without (1) and with convection (2) in the duct inner air layer between cable and duct. In case (1): ampacity results of FEM and measurements in very close agreement, the results from IEC 60287 are too conservative. In case (2): SUPG FEM results match closely to experimental results, convection is more predominant than heat conduction. During the experiment an interesting observation was made: natural air convection in adjacent PVC ducts without cables is also rather strong. Unfortunately, the possible effect of this observation, an improved heat spread/evacuation, is not further analysed in detail. The last part of the paper describes interesting observations on the relationship between ampacity and inner diameter of ducts. The main conclusion is that as the inner diameter of the ducts increases, the natural convection is enhanced and the ampacity is increased. Note that multiple options are available for duct filling materials, with differing impacts on the rating. Refer to TB 640 Section 7.2 for further discussion.

Reference [53] by S. M. Hellesø et al. describes modelling of medium and low voltage cables in duct banks with the ducts filled with air. The paper proposes a more computationally efficient FEM approach in rating of electric power cables installed in ducts. High level experiences, advantages and disadvantages of a commercially available FEM software are discussed. The paper makes a suggestion towards the integration of the proposed computationally simplified approach in the form of a pre-built graphical user interface within an FEM software. A thorough introduction on the classic theories of conduction and convection between cable and duct (or pipe) is given. A formula is presented to take into account the eccentric position of the cable inside the pipe. The latter is also related to an explanation why the assumption of the isothermal surfaces breaks down. Within the convective paragraph, interesting literature references are given on how to deal with eccentricity of laying of the cables in the pipe and with other geometries except a circular shaped pipe.

The paper by K. S. Thinn et al. [54] evaluates simplified heat transport models for power cables in air- or water-filled pipes. It highlights that the industry often uses simplified numerical models combining conduction, convection, and radiation into an effective thermal conductivity based on IEC 60287 formulas, which can be inaccurate for air-filled pipes. The study compares full FEA models with simplified models and finds that the simplified approach underestimates ampacity for air-filled pipes by 10-30%, while it is more accurate for water-filled pipes with an error margin of 0-3%. The paper concludes that radiation plays a significant role in heat transfer for air-filled pipes, and the simplified models should consider conduction and radiation but not convection for better accuracy. The authors recommend further laboratory work to verify numerical models, especially for water-filled pipes.

2.9.2 Groups in Air

In their 2015 paper, G. J. Anders and B. Bochenski propose FEA for the cable rating calculation for multiple 10's of LV circuits laid close to each other [55]. Normal LV cable laying standards are no longer applicable due to non-standard conditions as described in the paper. The impact on ampacity (by use of derating factors) is shown as a function of two variables: Cable size and number of cables and conduits. The paper refers to the IEC 62095 for the basics behind FEM modelling. It is mentioned that the IEC 60287 method can be used for rating calculations for a few circuits, but not for a large number of them. Where exactly the distinction lies in the number of circuits, is not investigated. The paper mainly presents the results of FEM calculations. Although very practical in nature, the paper does not fully cover aspects such as how the model on each duct was constructed, which boundaries were used and how the mutual magnetic and thermal dependencies between the circuits were considered.

Paper by F. Sedaghat et al. [56] presents a history about the improvements of the method presented in IEC 60287-2-1 on the calculation of the heat dissipation coefficient “h” for cables in free air (heat transfer between cable and supporting system is negligible and cable is isolated from solar radiation). The improvement takes into account the fact that emissivity, cable surface temperature and ambient temperature may diverge from what was initially assumed in the background of IEC 60287-2-1. The most recent improvements included in the IEC 60287-2-1 were proposed by V. T. Morgan’s approximations. A major conclusion of this paper is that these are not very accurate. The paper investigates the effect of all critical parameters and proposes an equivalent analytical thermal-electrical circuit for the accurate steady-state rating for the most common installation configurations for cables in air. Additionally, more accurate equations for the calculation of the total heat dissipation coefficient for cables touching and separated for horizontal and vertical grouping are proposed. All contributions in the paper are verified by FEM simulations (non-isotherm modules) and FEM parametric studies, analytical investigations and experiments. Interesting features of the paper are that it describes in detail how exactly to set up the FEM model and its boundaries and meshing, gives a reference paper to transient heat-transfer FEM and explains the limitations and capabilities of a commercially available FEM software.

2.9.3 Troughs

Paper by M. Terracciano et al. [57] mentions that the calculation of the temperature rise inside unfilled troughs applying the formula in IEC 60287-1-1 could not be done accurately in case the trough is flushed with the surface and the cover is exposed to the sun. The following aspects are ignored in the IEC standard: trough aspect ratio, position of cables, ambient air (external), soil and trough thermal conductivities, cable surface emissivity, and intensity of solar radiation. The paper presents a new analytical approach for the calculation of the temperature rise in unfilled troughs. One of the aims while developing this new method was to find its validity range by applying FEM. The parameters varied include the size (perimeter) and the aspect ratio of the trough as well as the position of the cables in the trough. The proposed method is based on successive relaxation. The justification for using FEM is related to the need of properly representing the unpredictability of local vortexes of the air convection process in the trough. The FEM is also used to find the relation between 3 types of adiabatic assumptions and their influence on the transient heat-transfer response. For the readers who are unexperienced in FEM, there is an interesting high level description on how the presented, commercially available, Multiphysics FEM software calculates the combined equations and mutually dependent effects on all types of heat exchange for the complete computational fluid dynamics (CFD) problems. The paper also describes in detail how to treat and calculate the thermal resistance of air. The latter is interesting for the beginners in FEM.

2.10 Physics / Multiphysics

2.10.1 Thermal / Mechanical

S. Eckroad et al., [58], deal with the thermo-mechanical FE analysis of an XLPE insulated power cable with a large cross section (2000 mm²). The key findings are that the magnitude of axial force is lower than extrapolated from the smaller conductor cables and thermo-mechanical patterns form but are less pronounced in the magnitude of lateral deflection.

In 2015, O. Sekula et al., [59], published a paper that deals with FE analysis of mechanical forces/stresses in the GIS/transformer terminations. Further, a case study of FE analysis of the development of 300 kV termination was presented. The FE analysis results show that the cable insulation diameter in the termination’s stress cone has not deformed even at the maximum temperature limit.

2.10.2 Thermal / Electromagnetics

In their 2007 paper, D. Schackleton et al. discuss the use of the finite element models to simulate the electromagnetic and thermal conditions of umbilical cables [60]. They state that two types of electromagnetic analyses can be carried out using the FE analysis:

1. Transient magnetic (variable currents and systems with different frequencies) and
2. Steady State AC magnetic (circuits with the same frequency).

The transient magnetic FE analysis is helpful to get induced currents and dissipated power by the Joule effect. The Steady State AC magnetic FE analysis is helpful in finding currents induced in the conducting regions (eddy currents). It also considers the skin and proximity effects. Analogous to the electromagnetic FE analysis, transient and steady-state are the two types of thermal FE analyses.

Both FE analyses can obtain the temperature distribution in the cable system and thermal flux at the boundary surface. Further, an FEM case study was presented in this paper.

In their paper, T. Henriksen et al., [61], deal with an accurate electrical analysis of frequency-dependent parameters in a complex cable construction. FEM analyses are recommended in the cases where thermal and electromagnetic simulations can be combined to get a real and accurate behaviour of the cables. The paper deals with the instrument and the umbilical cables, for which the ampacity calculations are not normally performed. The paper models the inductive coupling between the cables. Further, an FEA case study was presented in this paper.

The paper of T. Boucher et al., [62], deals with power cable installation in steel tubes to mitigate the magnetic field effects on neighbouring cable installations. The Joule losses in the steel pipe should be considered in the ampacity calculations. A case study is presented, and results are compared between analytical and FEM approaches.

The paper by I. Chrysochos et al., [63], discusses the electromagnetic interference (EMI) issues that occurred during the power cable design. The special focus is on capacitive and inductive coupling between cores and screens. The EMI results in screen losses affecting the cable ampacity. Another consequence of EMI is a possible overvoltage on the cable screen. A case study is presented, and calculations are made using three different methods: 1) Complex Impedance Matrix (CIM), 2) EMTP-like software, and 3) FEM.

In [64] by A. Prša et al., the effect of skin and proximity effect caused by the electromagnetic field is discussed. These effects increase the resistance of the metallic component of the cables and consequently the losses. The increased losses reduce the ampacities of the cables. Further, a real double-circuit 3-phase HV cable system is analysed using FEM software. The presented results illustrate a successful attempt to solve a concrete, real problem of applied power electromagnetics.

N. Korovkin et al., [65], discusses the need for a Multiphysics solution for cable ampacity calculations. The Multiphysics problem considers simultaneously applying the electromagnetic and heat transfer modules for cable ampacity calculations. The coupling of these modules is needed for the solution of the electromagnetic field and Kirchhoff's equations for grounding circuitry. A 3-phase buried cable system's Multiphysics case is analysed using the result obtained by FEM. The discussed approach is also applicable to transient ampacity analysis with symmetrical and asymmetrical short-circuit.

Paper [66], by A. Ariyasinghe et al., discusses the circulating current losses in the sheath and armour of a solid bonded single-core MV underground cable system. Comparative studies are made between the result obtained by analytical calculations according to the standard IEC 60287-1-1 and numerical simulation run by FEM software. The study investigates the effect of various parameters, including cable laying/configuration, spacing, conductor size, armour wire resistivity, and their numbers, and so on, on sheath and armour circulating current losses.

The paper by D. Willen et al. [67] discusses the design and performance of low-loss conductors for high-voltage and extra-high-voltage cable applications. It focuses on the AC resistance relative to DC resistance, the impact of the number of segments, wire size, centre wire/rod, and wire pitch lengths and directions. The study uses FEM modelling and practical measurements to evaluate the performance of large-size Milliken conductors, highlighting the importance of transposition, non-conductive centre elements, and optimized pitch directions to minimize AC losses. The results show that optimized conductors can achieve a K_s value well below 0.25, significantly reducing AC resistance compared to traditional designs.

2.11 Environmental factors

During Jicable 2007, M. Coates et al. published a paper in which the design of a duct bank with 45 circuits is discussed [68]. For the magnetic field calculation and shielding, finite element method is used. However, no details are given how the FEM was set up.

In 2011, J. A. Pilgrim et al., [69], published the paper on evaluating the temperature of power cables. In the paper, FEM is used to model the thermal environment around the cable where the thermal conductivity is represented as a function of moisture content. Therefore, the two coupled differential equations of the Philip and De Vries model are implemented. The paper gives valuable insight in the model setup and also into the different transient calculations that have been performed.

Also, during Jicable 2011, S. Lloyd and G. Stapleton published a paper in which FEM was used to calculate the thermal resistance T_4 for a non-standard trench arrangement [70]. The equations and the resulting matrix approach are discussed.

M. Cabau et al., [71], published work on EMF mitigation techniques for power cables in which they investigated different passive loop layouts for joint bays as well as steel pipes for the power cables by means of FEA. For the latter, the model is described including the non-linear material characteristics of the steel pipe.

The publication by Y. Liu et al., [72], complements work by M. Cabau et al., [71], as it examines the interaction between power cables and pipelines. Here, a 2D FEM model is used to perform a parametric study.

In their 2011 paper, H. Brakelmann et al., [73], consider a cable design with integrated electromagnetic shielding using FEM. The mathematical formulation of the problem of ferromagnetic materials in magnetic fields is shown and a comparison is made between cables with and without electromagnetic shielding.

Another publication dealing with the magnetic field shielding was published at Jicable 2015, [74]. In their paper, G. Sun et al., used 2D FEM to simulate the effect of shielding plates for a 275 kV power cable.

In the paper by F. Ainhirn et al., [75], published at Jicable 2019, the authors discuss the sensitivity of different FEM model parameters such as size of region to be discretized, degree of detail in the modelling of the conductor and metallic screen and variation of the type of boundary condition on the computed cable temperatures. The variations of the boundary conditions include isothermal boundaries on all edges, isothermal boundary on all edges with convection on the surface, isothermal boundaries with different temperatures at the surface and at the bottom and isothermal boundaries with different temperatures at the surface and at the bottom with convection on the surface. The results are compared to measurements of a 400 kV cable system loaded with 1400 A.

3. Value Proposition

3.1 Introduction to FEM

Historically, cable engineers have been performing hand calculations or using tables and basic computer spreadsheets for cable ampacity calculations using analytical methods, i.e., the Neher-McGrath method, which later became the basis for the IEC standards [9], [76], [10], [77]. Thanks to the advancement in computing technology, it has been possible to solve more complex engineering problems using both analytical and numerical methods, such as the finite element method (FEM). In recent times, more sophisticated computer programs have been developed based on both analytical and numerical methods to solve the current ratings of power cables. The aim of this chapter is to compare the analytical method (e.g., IEC method) and the finite element method (FEM), to show the possibilities and limitations of the methods, and to give recommendations for cable engineers as to when FEM is recommended over analytical method and vice versa. The FEM involves solving partial differential equations arising in engineering and mathematical problems, and therefore the user is expected to have a basic understanding of the laws of physics and solving engineering and mathematical equations. FEM should be used where it is not possible to solve problems using analytical methods or where the analytical computation is time consuming, difficult to model or is known to yield inaccurate or imprecise results.

The FEM is used to obtain an approximate solution to differential equations. In the specific case of cable ampacity calculation, it is used to solve the heat transfer equation that links heat dissipated by the cable with the distribution of the temperature in the cable and the surrounding soil.

3.1.1 Analytical Method

The analytical methods are based on the procedures developed in the 1930's and summarized in the 1957 paper by Neher and McGrath [78], which estimates the steady-state temperature of power cables for some commonly encountered cable installation configurations. The method uses a lumped parameter representation of the cable thermal circuit and considers either directly buried installations or cables laid in a thermal backfill or in a duct bank. The Neher-McGrath model has been widely accepted and is being used worldwide. The original model was further developed and enhanced by the International Electrotechnical Commission (IEC) and published in the IEC 60287 series of standards.

3.1.1.1 Methodology

The analytical (IEC) method for the calculation of current ratings broadly contains the following steps:

- Gathering of cable dimensions and material properties (generally via the manufacturer's published cable datasheet).
- Information about installation and operating / environmental conditions (obtained from the client's scope of work document).
- Solving the equations (formulas from IEC standards):
 - Calculation of losses [9]
 - Calculation of thermal resistances [9]
 - Calculation of permissible current rating (also denoted Ampacity) ref [9]
 - Calculation of cyclic and emergency current ratings [76], [10], [77]

3.1.1.2 Thermal Equivalent Network

The analytical methodology for calculating cable ampacity can be represented using a thermo-electric equivalent (TEE) circuit, where thermal resistances and capacitances can be calculated using analytical IEC formulas for the computation of steady-state and cyclic ampacities. An example of a TEE circuit for a single-core armoured cable for a steady-state ampacity calculation is depicted in Figure 3 - 1. Here, W_c , W_d , W_s , and W_a represent conductor, dielectric, metallic sheath, and armour losses, respectively; T_1 , T_2 , T_3 , and T_4 are the thermal resistances per unit length of the insulation (between conductor and metallic sheath), of the armour bedding (between metallic sheath and armour), of the external serving, and of the external environment (between surface and surrounding medium), respectively; θ_c , θ_a , and $\Delta\theta$ are the conductor temperature, ambient temperature, and the difference between θ_c and θ_a , respectively.

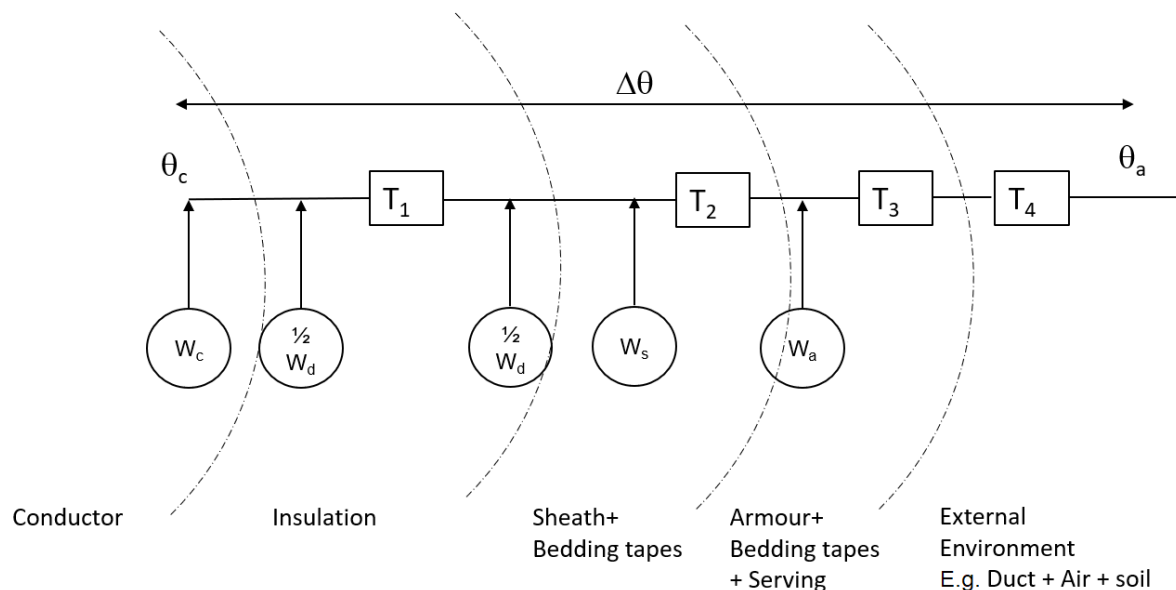


Figure 3 - 1 Thermal Equivalent Network - Example of a single-core armoured cable

3.1.2 Finite Element Method

FEM involves solving partial differential equations (PDE's) describing the physics related to the problem (e.g., electromagnetic or heat transfer phenomena) in power cables and the environment in which it is installed, using numerical approximations.

In the ampacity calculation of power cables, FEM is used to calculate the temperature field in the modelled domain. The temperature field is a result of the losses generated in various components of the cables and possibly other heat sources in their vicinity.

3.1.2.1 Methodology

The geometry of the problem is discretised and solved by generating a mesh of polygons (for 2D problems) or polyhedrons (for 3D problems), called elements. In each element, the heat transfer equation is linearized. The combination of equations for all elements results in a linear differential equation system, which can then be solved numerically. The solution is only calculated at the summit of all elements (nodes), and the values are then interpolated to the whole geometry.

FEM is a highly versatile methodology, in which the computational speed depends on the complexity of the problem i.e., geometry, simple or Multiphysics, 2D or 3D etc.

Figure 3 - 2 below shows an example of 2D mesh system for conventional 3-phase AC single core cables installed in ducts buried in soil.

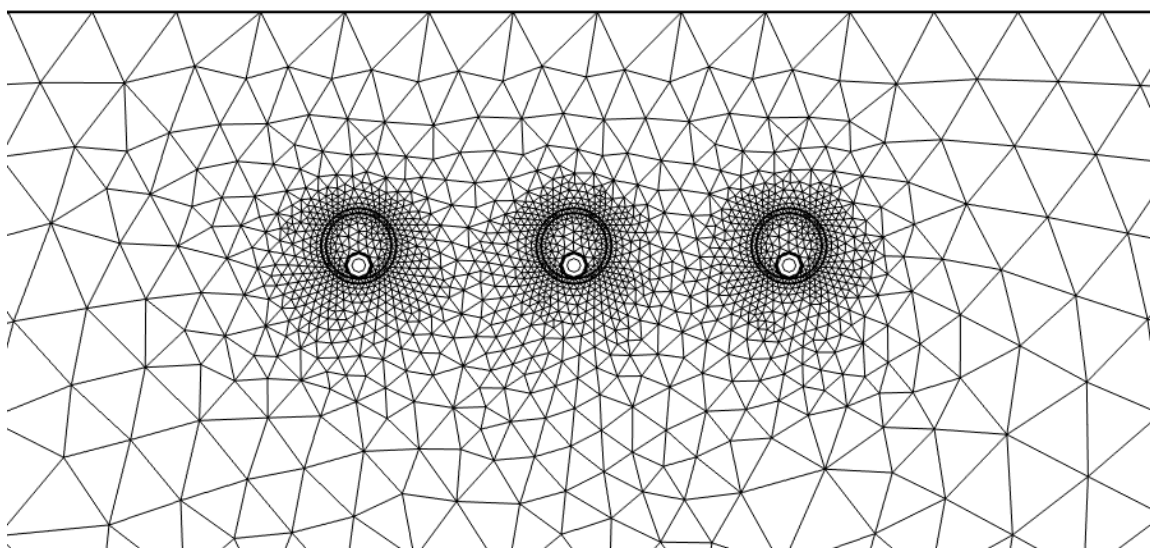


Figure 3 - 2 Example of 2D mesh around AC cables installed in buried ducts

3.1.2.2 General Principles

In FEM, the basic modes of heat transfer (governed by the laws of thermodynamics) described below are applied to solve cable current rating problems.

- **Conduction:** conduction heat transfer occurs both within solid materials and also between objects through direct contact. This is applicable for most underground cable installations. The physical mechanism of conduction occurs as the atoms making up the material are able to vibrate; as an atom absorbs energy, the amplitude of vibrations increase, and allows thermal energy to pass from atom to atom.
- **Convection:** in liquids and gases, heat can be transferred by convection. Convection occurs when particles absorb thermal energy, for example from a heated surface. The movement of the heated particles away from the heated surface allows thermal energy to be transported, for example away from the surface of a cable installed in direct contact with free air. Convection may be natural or free (buoyant) convection where the movement is induced by density changes in the liquid or gas, or forced where the liquid or gas is subject to an external force which drives the movement. For cables installed in air, convection typically occurs in conjunction with radiation heat transfer.
- **Radiation:** in radiation, heat transfer occurs when radiation is emitted from a heated surface towards a cooler surface or ambient environment. In cable rating applications when dealing with radiative heat transfer, infrared spectrum is the main spectrum to consider, but it is not the only significant spectrum for radiative heat transfer in general. The radiation typically occurs in tandem with convection, for example in the case of cables installed either in free air or in a tunnel.

An overview of the general approach to FEM for calculating cable ampacity can be represented generically as shown below:

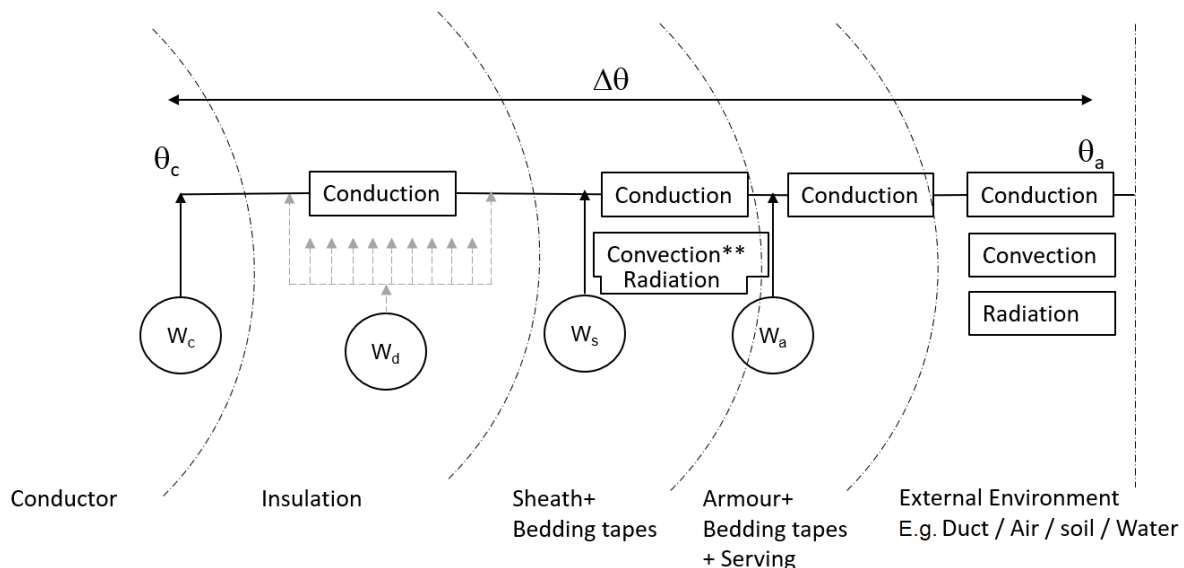


Figure 3-3 Thermal principles - FE Method

** Convection heat transfer is applicable for a small number of cable designs containing fluids, for example pipe type cables.

3.1.2.3 Procedure

Typical steps to be taken in time-static FEM consist of:

1. Deciding on the physics modules needed for simulations
2. Choosing the proper geometrical simulation space, i.e., 2D, axisymmetric, or 3D
3. Building the geometry of the cable (generally via the manufacturer's provided cable datasheet)
4. Defining and verifying material properties of the components in the model for the intended simulation type (e.g., time-static)

5. Obtaining information about installation, operating, and environmental conditions (obtained from the client's scope of work document)
6. Applying boundary and (initial) conditions
7. If needed, setting up equations governing couplings, constraints, and/or numerical stop conditions that are not inherently provided within the physics modules of the simulation tool
8. Meshing the geometry
9. Solving the installation system
10. Postprocessing, i.e., plotting and visualizing the results

3.2 Limitations of Analytical Methods

3.2.1 Isothermal Earth Surface

The challenge of solving partial differential equations analytically stems mostly from the difficulty of computing the temperature distribution and heat flow in the environment surrounding the cable. An analytical solution for heat transfer equations can be obtained when a cable is represented as a line source placed in an infinitely homogenous environment. Since this is not a practical assumption for cable installations, another assumption is made, namely that the earth surface is an isotherm. In practical cases, the depth of burial of the cables is of the order of ten times the overall diameter of the cable, and for the usual temperature range reached by such cables, the assumption of an isothermal earth surface is a reasonable one.

For this approximation to be valid, the cables must be sufficiently far away that the heat from the cable does not locally increase the ground surface temperature or wind speeds are sufficiently high that convective heat transfer would hold the ground surface to the isothermal condition. However, in general, seabed installations should allow shallower burial depth than land based installations since the water increases the heat dissipation from the seabed surface compared to a ground surface in air. Moreover, the IEC does not take into account the effect of solar radiation and its effect on soil ambient temperature, which can affect shallow buried installations more than deep ones [9]. For shallow installation, S. Y. King and N. A. Halfter proposed a calculation method to solve thermal problems involving the creation of an artificial soil layer at the earth surface [79] and [80].

Refer to section 3.4.1 for details on how FEM can help to solve this problem.

3.2.2 Ducts and Casings

In the case of cables installed in non-magnetic ducts, the analytical formulas in IEC have limitations, which limits the applicability of the IEC standard to selected installation arrangements. Therefore, the IEC method is not applicable to all situations of power cables installations in ducts or tunnels. The formulae in the IEC standards for the thermal resistance of the air or water in the duct do not take into consideration the size of the duct or casing, respectively. The formula was developed experimentally in the 1930s for small ducts with a diameter not greater than 50 mm. When the cables are placed in non-magnetic ducts or casings, e.g., HDPE or concrete, these ducts can be larger than 50 mm. Furthermore, IEC assumes horizontal ducts only and there is no standardized approach for incline ducts such as is the case for HDDs.

There is no recent study that has investigated the diameter of a duct for which the IEC approach is valid or at least where the results are conservative. However, work has been done comparing the results of IEC with numerical methods incl. FEM for cables in free air and cable installations in unventilated tunnels. Especially the comparative calculations for cables in free air show that the IEC approach is underestimating the resulting hottest cable temperature in touching trefoil formation and for cables in flat formation with no distance of more than 45 mm distance between cables.

3.2.3 Tunnels

A popular method for cables in ventilated tunnels was the numerical method based on a lumped parameter thermal – electrical analogue approach (1D thermal network) provided by Electra 143 and 144. Extension to the Electra model was developed by Pilgrim et al. [81] which still utilizes an equivalent thermal network but is able to calculate ratings for multiple independent cable circuits in one tunnel and is relatively easy in its application.

The situation becomes even more complicated when the casing, duct or pipe is made of steel or iron. The hysteresis and eddy current losses in the magnetic material very much depend on how the AC power cables are placed inside the casing. Such casings are often used for road, railway, or river

crossings, and it is common practice to place all cables at the bottom of the pipe. IEC 60287-1-1 provides empirically-derived equations for the calculation of Joule losses in steel pipes. The equations given are for two common configurations, cradled at the bottom of the pipe when the ratio of pipe inner-diameter to cable skid wire diameter is 1.6 or greater or triangular for when this ratio is smaller than 1.6, of the three cable phases in a pipe-type cable system. More complex arrangements can be calculated analytically, which can be cumbersome as it involves the evaluation of Bessel functions and their derivatives. Here, the use of the finite element method is advised.

3.2.4 Time-varying

In the transient analysis discussed in IEC 60853, separate computations are performed for the internal and external parts of the cable. Coupling between internal and external circuits is achieved by assuming that the heat flow into the soil is proportional to the attainment factor of the transient between the conductor and the outer surface of the cable. The validity of the methods did not rest on an analytical proof, but on an empirical agreement of the responses given by the recommended circuits and the temperature transients calculated by more lengthy, but more accurate, computer-based methods.

The time-dependent analysis in IEC 60853 deals with the response to a load step, cyclic and emergency rating. In case of cyclic ratings the load variations are described by a 24-hour load curve. Modern generation stations based on renewable resources, such as wind or sun, do not exhibit such regularity of power production. The complex dynamic loading of power cables for such installations is not covered in the given equations.

3.2.5 Groups of Cables

In the analytical methods described within the IEC 60287 series and the IEC 60853 series of standards, the case of a group of cables is dealt with on the basis of the restricted application of superposition. To apply this principle, it must be assumed that the presence of another cable, even if it is not loaded, does not disturb the heat flux path from the first cable. This allows separate computations to be performed for each cable, with the final temperature rise being an algebraic sum of the temperature rises due to the cable itself and the rise caused by the other cables. Besides, IEC 60853 cannot model groups of cables with different cyclic/emergency loads. For better accuracy the temperature rise caused by the simultaneous operation of all cables should be considered.

3.2.6 3-Core Cable Construction

Another limitation of the IEC's simplified approach to the calculation of the thermal resistance between the sheath and armour of a 3-core cable is that the jacket around each core, fillers, and armour bedding are treated as materials with the same thermal properties, but in reality, these materials have much different thermal properties. Submarine AC cable constructions for example commonly have 3-cores in a laid-up formation with armour wires helically applied over laid-up cores with varying pitches. The analytical (IEC) method does not consider this geometry, resulting in an overestimation of armour losses. The analytical method as specified in the IEC 60287 series leads to conservative sheath and armour losses at higher frequencies. Cigre TB 908 provides more information on the subject and methods for calculating losses more accurate.

However, the IEC method also requires the use of the geometric correction factor to approximate the complex geometry of a 3-core cable. This can lead to overestimation of cable ampacity [82], [83].

3.2.7 Heterogeneous soils

Cables can be buried in conditions where the thermal resistivity of the soil varies due to the natural variability of its composition or due to the application of selected sand and backfill for protective or thermal purposes. Several methods are available to solve multiple thermal resistivity problems. IEC 60287 provides a method based on [78] that has been extended by [84] for the inclusion of a single backfill or duct bank. An interesting formulation is proposed in [85], examining a defined number of cable configurations. The use of conformal mapping method is also presented in [86] to address the heterogeneous soil problem. [86] recommends the conformal mapping method can solve most cases requiring calculation of effective thermal resistance of non-uniform soil layers, but recognises that for certain cases where the cables are closely spaced ($\leq 20\%$ of cable diameter), an error was introduced from the application of superposition at values of cable spacing much smaller than those that are permitted under the restrictions implied by its use. The errors could be larger when using metallic sheath designs.

It must also be noted that the analytical methods developed thus far, are suitable for ampacity calculations in steady state conditions but not for transient or dynamic rating applications in heterogeneous soils. In some cases, especially when the limitations of the analytical methods and conformal mapping solutions do not represent the actual installation conditions, it may be appropriate to use FEM.

3.2.8 Multiple Cables in a Duct

Ampacity calculations of a group of cables inside a duct are not described by the IEC standards, and analytical methods were not found in the literature. Whilst the modern day HV cables are most commonly installed in individual ducts i.e., one cable per duct, it is not uncommon to find groups of MV/LV cables installed in an unfilled duct. IEC 60287 treats the group of cables, but only in directly buried installations and not in ducts. Ducts filled with bentonite or other cementitious material can be treated similar to direct burial and hence the IEC method stands valid.

In the specific installation condition of multiple cables inside a duct, both in the case of equally and unequally loaded circuits, analytical (IEC) method exist to calculate the electromagnetic coupling between parallel installed circuits but for complex cases accurate analytical calculation may not be possible.

3.2.9 Multiple Ducts Inside Casing (e.g., multiple ducts inside a single large HDD, micro tunnel or a concrete duct bank)

As per the previous installation condition in 3.2.8, IEC standards do not provide a comprehensive approach for ampacity calculations of multiple ducts inside casings. As described in [87] chapter 7.2.2, this installation can present several variables: duct/duct bank materials, duct/duct bank inclinations, duct filling material (fluid or solid), and duct in an environment with non-homogenous properties. Approximate ampacity calculations in some of these situations can be performed analytically using the IEC method or formulas found in the literature with correction factors, but there are some cases (such as inclined ducts) where accurate calculation results are not achievable. In case the specific condition is not described by standards and literature or when the implementation of analytical and empirical methods is complex, FEM may be more appropriate to obtain results closer to reality.

3.2.10 Region with boundaries and Heterogeneous Soils

Heterogeneous soils, such as multiple soil layers or layers with changing thermal properties cannot be conveniently calculated analytically. Only limited solutions for simplified drying-out of soil and simple duct banks or backfill areas exist. If the medium surrounding the cables is not uniform and the earth's surface boundary is not isothermal, the IEC's assumption of isothermal earth does not hold well for heterogeneous soils.

3.2.11 Other Environmental Factors

Additionally, complex installation conditions involving multiple duct banks and multiple cable crossings cannot be solved using analytical approaches without severe simplifying assumptions. Furthermore, analytical methods include an assumption of no wind for cables exposed to air, which is a very conservative assumption for windfarm installations, which are most loaded in windy conditions.

3.3 Hybrid Models

The previous section listed the limitations of the analytical methods as commonly used and defined by the IEC standards [9], [76], [10]. Examples of the limitations are summarized as below:

- Isothermal earth
- Non-magnetic ducts and casings
- Time-varying effects
- Groups of cables
- 3-core cable construction
- Heterogeneous soils
- Other environmental factors

Meanwhile, within FEM, there can be some difficulties. As an example:

- 3D modelling of stranded and segmented conductors, concentric neutrals, corrugated sheaths, small (medium voltage) 3-core cables, and armours can be completed in FEM, but with a high cost of computation time and effort.

To overcome the challenges of both methods, some hybrid methodologies are consequently proposed and applied in [88], [89]. In the proposed hybrid methodologies, the following approaches are reported:

1. Losses of cable conductors or external heat sources are calculated using analytical methods while considering the skin and proximity effects of the cable conductors. The results are then used as inputs to the FEM (mostly 2D).
2. Calculations of time-varying, cyclic, or emergency loads are completed using the analytical procedures to simplify the time varying load curves to a step function, or to cyclic rating factors and loss load factors, which can then be used as input in the FEM methods (2D or 3D).
3. Cable crossing calculations using analytical procedures to calculate the derating factors [9], which can be used as input in the FEM in a 2D plane.
4. Inclusion of field temperature measurements, e.g., from distributed temperature sensing or individual thermocouple measurements, as input to the FEM.
5. Calculation of the equivalent thermal resistance of heterogeneous soil using FEM and using this as input into analytical calculations. The reverse is also possible.

3.4 Modelling of Complex Installations and Cable Designs

Finite element method can deal effectively with current rating calculations of complex installations and cable designs, those where analytical solutions are not available, where simplified assumptions would lead to imprecise results, or where the analytical solution would render the results unrepresentative of the actual installation.

In this subchapter, we include examples already discussed as limitations of analytical methods in Chapter 3.2, to describe how FEM could address and resolve these limitations.

3.4.1 Shallow Burial Installations

FEM permits the convective and radiative heat transfer from shallow buried cables into the ambient environment. The study of ratings presented in [90] compared between analytical and numerical methods, concludes that there is good agreement between both methods for depths in excess of 0.9 m and isothermal ground surface boundary condition. However, as mentioned above in 3.2.1 for shallow burial installations, the assumption of isothermal ground does not hold well, especially for highly loaded cable circuits. FE method can eliminate this potential overrating error in the IEC method by allowing the user to model the heat flux boundaries for ground/air interface simulating ground temperature increase due to still air and solar gain conditions. Care must be taken with adding solar radiation as solar radiation varies over a 24 hour period. Even at shallow depths the thermal capacitance of the ground will limit peak temperatures. One method to take account of solar gain is to calculate via FEM (or measure on site) the temperature rise due to solar radiation and adjust the ground ambient temperature suitably.

3.4.2 Multiple Cables in a Duct and Multiple Ducts Inside Casing

The limitation from IEC method discussed in 3.2.9 can be resolved using FEM where the electromagnetic coupling between multiple circuits can be modelled to consider the thermal interaction more accurately.

The paper in [91] presents a systematic comparison of IEC calculations versus FEM simulations for the case of multiple ducts in a casing. It can be derived that both analytical (IEC) and numerical (FE) methods give identical results for simple problems, i.e., fewer cables and equal loading. However, FEM leads to more representative conductor temperature for the complex installations involving a large number of cables that are unequally loaded.

3.4.3 Multimode Heat Transfer Installations (e.g., ventilated troughs, J-tubes, CPS)

Cables installed in ventilated troughs, J-tubes, submarine cable protection systems (CPS), and vertical risers on a tower or pole at a transition between underground and overhead line connections are subjected to multimode heat transfer mechanisms. In order to perform ampacity calculations in such conditions, it is necessary to represent the various components of heat transfer, all of which are essential for the successful calculation of the rating.

For ventilated tunnels, the CIGRE method (Electra 143 & 144) is available, but a number of other numerical methods exist and can be more or less exact depending on the level of detail required from the calculation.

In relation to ampacity calculations of cables installed in J-tubes or vertical risers, published analytical methods do exist in the literature, e.g. [92], [93] and [94], but in some circumstances, it may be desirable to conduct FE analysis, particularly if the air-filled section of the tube or casing is short or has perforations, meaning that some longitudinal cooling can be expected. Refer to case study 3 in section 7.3 for an example of multimode heat transfer solved using FEM.

Concerning cable protection systems such as bend stiffeners or concrete mattresses, these devices can represent local hot spots for submarine cables. No guidelines are available on how to represent such multimode heat transfer installations or if they do represent a thermal limit on the cable route. FEM can represent a viable solution for ampacity calculations in such cases because each single item can discretely be defined with its properties and each heat transfer mode applied where appropriate.

3.4.4 Force Cooled Systems

Force cooled cable systems are used to increase the ampacity of cables, especially at hot spots. The cooling system works by absorbing and transporting away heat generated by the loaded cables, lowering the ambient temperature in the cable environment. Generally speaking, when dealing with forced-cooled extruded dielectric cables, two options of coolant flow are possible, namely closed circuit and open circuit. Many different arrangements are possible; the two main options for a closed circuit are lateral and integral cooling as shown in Figure 3 - 4 below. The example of open circuit would be a ventilated tunnel or a trough.

A forced cooling system can be modelled using an equivalent thermal circuit proposed in [95]. The equivalent thermal circuit implementation is based on analytical equations that are difficult to solve, so it may be convenient to perform ampacity calculations through numerical methods.

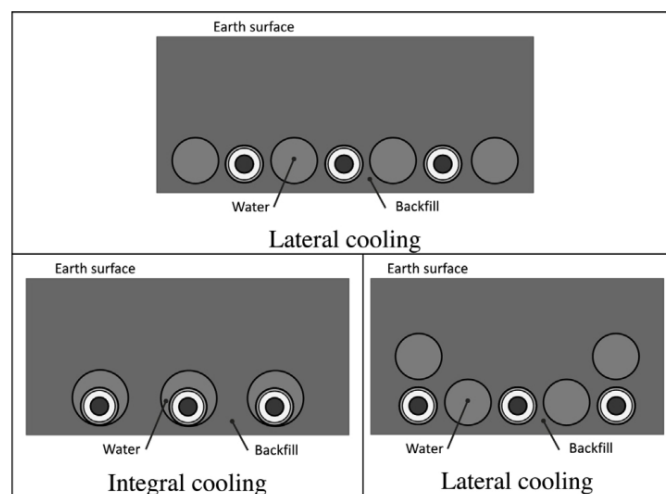


Figure 3 - 4 Typical water cooled cable systems

3.4.5 Joints and Terminations

FEM allows accurate modelling of:

- Complex geometry between cables where the cables are installed with variable depth of burial and phase separation within the joint bay.
- Complex geometry of joint body, containing several different insulating materials and creating a local increased temperature as well as longitudinal heat transfer in metallic layers of the cable.
- Thermal interaction between multiple cable circuits installed in a joint bay with unequal spacings and heat losses. The analytical method is too complex and time-consuming.
- Thermal interaction of a water-cooled cable system(s).

Refer to Case Study 4 in Chapter 7.4 for an FEM example of 3 single-core cables and joints installed in a joint bay.

A typical arrangement of cable formation at a joint bay is depicted below, highlighting the key differences in installation and laying parameters.

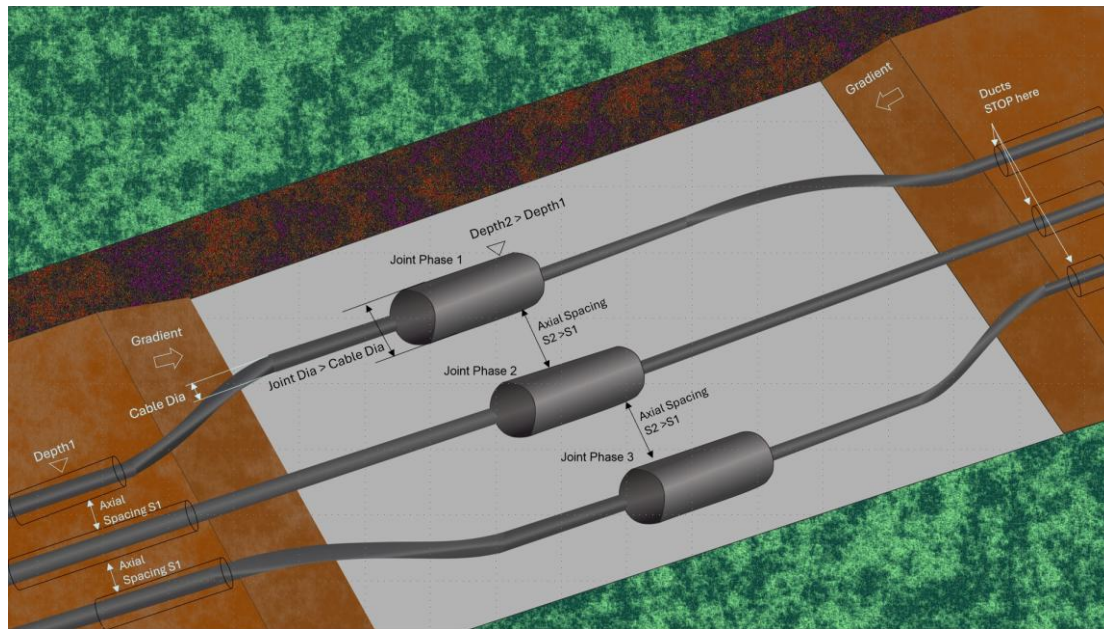


Figure 3 - 5 Typical Joint Bay Layout and temperature distribution

3.5 Skills, Time and Effort

3.5.1 Skills Required

With the advancement of computer technology, where tools with the ability to model and solve complex engineering problems are increasingly available in the hands of engineers, FE programs are one type of tool that is commonly used by cable engineers. As FEM will attempt to solve anything fed into it, proper training, competence, and understanding of the underlying concepts are essential to using these tools. The essence of any simulation or modelling should be to interrogate the solver with well-formed and appropriate questions.

The most qualified person to undertake FEM is someone who understands the physics of the problem and could potentially perform the analysis (at least approximately) without the computational support of an FEM. The computer simulations and models are only as effective as the knowledge and competence of the person developing them. If, however, computer-based simulations are set up and used correctly, highly complicated mathematical models can be solved to an extent that is sufficient to provide cable engineers with accurate information about how the products will perform in real life. The simulation models can be changed, modified, and adapted to suit the various known or anticipated operating conditions, and solutions can be optimised. Thus, cable engineers can be confident that the real products should perform efficiently, reliably, and safely. Like other engineering calculation methods, the results generated using FE tools should always be thoroughly reviewed and interpreted by someone competent in FEM to help determine relative accuracy and applicability. A verification process as described in section 6, should also be established.

3.5.2 Effort Needed

Detailed models of complex cable installation or problems involving Multiphysics may require significant time and effort, even for experienced users of FEM. Therefore, for any calculation, it is always worth considering whether an FEM calculation is necessary or whether more simple analytical or hybrid methods can provide acceptable results. This is especially relevant when the input data for the calculation is preliminary or based on assumptions.

FEM can, however, save time for future runs of a problem of the same or similar nature that requires solving. FEM can be efficient in the analysis of the model results, for example, by editing material properties or analysing electrical stresses at a point of interest.

3.6 General Characteristics

A high level comparison of characteristics between Finite Element Analysis and analytical methods is described below:

3.6.1 2.5D / 3D Capability

Most FEM based software applications are 3D / 2.5D capable. However, non-computer based analytical methods (e.g., spreadsheet-based applications) are either 1D or 2D only or too complex and demand high user skill set and effort to solve 3D problems. Some commercially available analytical software tools are capable of performing these 3D calculations such as cable crossings or electro-magnetic coupling calculations.

3.6.2 Computational Burden

FEM requires relatively high computation burden compared to a similar problem when solved using analytical methods. However, when solving a complex engineering problem e.g., time varying effects, both FEM and analytical methods demand high computational burden.

3.6.3 Model Creation Effort and Accuracy

Analytical methods do not generally require a model geometry to be created to solve a problem. Some commercially available analytical tools have built in cable and duct block models. Furthermore, analytical tools do not rely on dimensional accuracy of the model. FEM on the other hand, requires a scaled model geometry to be created for meshing purposes. The accuracy of FEM output relies on the precision of model geometry. Most commercial FEM applications and some commercial analytical softwares are equipped with a library / database from where cable models and their material properties can be imported. For complex engineering problems or bespoke applications, a new model may need to be created from scratch.

3.6.4 Multiphysics Capability

Multiphysics capability is a key attribute of FEM and a benefit over analytical methods. Analytical methods can solve multimode heat transfer for limited applications e.g., cable ratings in tunnels.

3.6.5 Versatility / Adaptability

Both analytical and FEM are versatile and adaptable in their scope of applicability. Analytical method is more easily adaptable to changing dimensions and cable parameters. FEM is more versatile in terms of re-running & post processing of a model and analysing the output for different physical parameters.

3.6.6 User Skill / Competence

Whilst both FE and analytical methods demand user to have reasonable understanding of concepts of physics and mathematics, FEM requires the user to have know-how in solving the partial differential equations, model creation, meshing and defining boundary conditions etc. FEM, thus requires higher user skill and competence levels in physics, mathematics, computer and material science than analytical method.

3.6.7 Visual Presentation

FEM when compared with basic analytical methods (e.g., spreadsheets or traditional Fortran based applications), has better visual presentation capability i.e., temperature & field plots, 3D visualisation etc. However, it has to be recognised that some commercially available analytical programs now have the equivalent visual presentation capabilities.

3.7 Communication of Results

The ability of FE tools to present the data in a visually pleasing format is often appealing to clients. Following are some of the perceived benefits of using FEM tools for current rating calculations.

1. Heat plots to spot locations with limited thermal ratings.
2. Sensitivity analysis and presentation of results based on varying geometry and boundary conditions.
3. Visually pleasing presentation (aesthetics) of a report.

3.8 Benchmarking of Analytical Calculations vs FEM

The technical papers referenced in Table 3 - 1 below highlight similarities and differences in calculation results performed using analytical and FE methods for some of the well-known installation scenarios.

Table 3 - 1 Benchmarking of Analytical vs FEM

Case ID	Example Case	Analysis	WG B1.87 Recommendation
1	Simple direct buried cable	[96] Both IEC and FEM relatively agree with each other when the cables are buried deeper i.e., depth to cable diameter ratio >10. However, FEM is more accurate for shallow buried cables when the effect of convection at the ground surface is modelled unless the IEC method is modified as described in Section 3.2.1. [97] IEC method is accurate for simple configurations but starts losing accuracy when larger metallic screens are present and when there are groups of cables. [91] Almost identical results for simple configurations but start differing a lot when cables are installed in duct banks with many cables. [66] Metallic sheath and armour losses of solidly bonded cables with various formations and effects of harmonics on the cables rating are studied.	User may choose either Analytical or FEM. The standardized analytical method should be the first choice for the cases where it can provide accurate results, to minimize the calculation time and effort. With larger metallic screens and groups of cables FEM is preferred especially where IEC 60287-1-2 and IEC 60287-1-3 cannot easily be applied.
2	single core and 3-core cables in ducts in heterogeneous soil	[98] Results are comparable when modelled in a commercially available FEM software and IEC method when the soil is considered to be homogeneous. However, significant difference in computed ampacity is observed under heterogeneous soil condition. FEM results in 10% derating for directly buried cable and 15 - 20% for ducted systems. [53] A simplified FE model using heat source and heat sink to replace convection has been created. The result shows little deviation between more precise and simplified models. The temperature deviations are 1.5 K for each core in individual ducts, 0.6 K for all three phases in a single duct in cradle formation and 4.2 – 4.7 K for all three phases in a single duct in trefoil formation. [99] IEC method matches FEM accurately on directly buried cables. IEC method has slightly worse results when there is backfill but the difference in results diminishes as the installation depth increases. FEM allows controlled backfill to be used on top of the main backfill while IEC does not.	FEM may be necessary in cases with heterogeneous soils or thermally complex scenarios. For homogeneous soil cases the IEC methods remain applicable.

Case ID	Example Case	Analysis	WG B1.87 Recommendation
		[52] Convective heat transfer is much higher than conductive heat transfer in the air gap between the cable surface and the inner surface of the duct. Convective heat transfer increases as the inner diameter of the cable duct increases. FEM allows modelling the convective and conductive heat transfer in the air gap. [100] FEM is used to calculate cable ratings in heterogeneous soil and soil with drying out.	
3	Armour Losses of 3-core Submarine Cable	[101] The paper compares sheath and armour loss factors for four different cable designs and finds 3D FEM results align with experimental measurements closely. The values calculated using IEC method results in over estimation of the armour loss factor. [102] FEM proves that armour losses are overestimated by the IEC method due to wire armour stranding. The error increases with the armour diameter. [103] Fractional Frequency Transmission Operation has been studied using both IEC method and FEM. It shows that this non-conventional approach using significantly lower frequency reduces losses significantly due to reduced skin effect. Both IEC method and FEM are used in this study. IEC method is showing more conservative results when compared to FEM.	As discussed in detail in CIGRE TB 908 there are analytical methods, but FEM can also be used, the choice of method will be case specific.
4	Cables in Steel Pipe	[104] An analytical solution for calculating losses in steel pipe is presented, showing errors smaller than 3% for all tested cases when compared with FEM. [105] Test, simulations and mathematical derivations show that the steel pipe losses are negligible for a balanced system in trefoil located at the centre of the pipe. FEM agrees with their results. However, this may not hold true for unsymmetrical geometries where the cables are not located in the centre of the pipe i.e., cradle arrangement.	User may choose either Analytical or FEM. The standardised analytical method should be the first choice.
5	Multiple Cables in a duct bank	[91] FEM takes greater account of the high degree of electromagnetic coupling between cables than the IEC. Almost exact results for simple configurations but start differing a lot when cables are installed in duct banks with many cables.	For simple configurations, user may choose either Analytical or FEM. The standardised analytical method should be the first choice. WG recommends use of FEM for complex arrangements.

Case ID	Example Case	Analysis	WG B1.87 Recommendation
6	Cables in unfilled surface troughs	[106] IEC is conservative. The paper demonstrates that up to 28% increase in rating can be achieved when modelled via CFD technique using FEM. [107] Analytical methods studied in this paper generally neglect two factors: conductive heat transfer from the cable to the base of the trough and the restriction in natural convection due to the trough geometry. A new analytical equation that matches the measured values has been presented in this paper.	User may choose either Analytical or FEM (as presented in [107]). The standardised analytical method should be the first choice for simple cases.
7	Cable Crossings	[108] A technique using SPS (Single Point Source) method has been presented in this paper as an alternative to FEM for multiple cable crossings.	User may choose either Analytical or FEM. The standardised analytical method should be the first choice for simple cases.
8	Dynamic and Emergency Ratings	[109] A method using TEEs (thermo-electric equivalent) has been presented to determine the conductor temperature for dynamic and emergency loads. The results match with FEM. For any time-dependent condition or input, FEM should be used. IEC methods do not factor in changes in time, which is why the user is forced to run multiple iterations.	User may choose either Analytical or FEM. The standardised analytical method should be the first choice.

4. Guidance for Clients

This chapter is intended as a short, stand-alone overview of the specificities of the Finite Element Method (FEM) and is intended for Clients contemplating an FEM ampacity study.

After a basic overview of the principle of the finite element method, the cases where using it over traditional analytical methods adds value are discussed. The dialogue between Client and Contractor is then addressed, first by discussing what a study request should include, then with considerations on what should be included in the report for this study.

Following terms and their meanings are applicable when referenced in this TB.

"Client" - a person or organisation who instructs / specifies an FEM ampacity study.

"Contractor" - a person or organisation who performs / presents / reports on the FEM study & calculated results to the Client.

"Contract" - refers to the agreement between the Client and the Contractor in relation to performing FEM ampacity study.

Chapter 3 describes the overview of FEM including the general principles, calculation methodology and procedure, where value proposition of FEM for current ratings in comparison with analytical calculations is discussed in some detail.

4.1 When is an FEM study useful?

As detailed in Chapter 3 (Value Proposition), the WG recommends to use analytical methods when a problem to be solved is defined under the scope of relevant industry standards (e.g., IEC or CIGRE) or other proven analytical methods, and where there is no significant difference in the expected result when compared to FEM. Also if the analytical methods provide results which are known to be conservative they can be used to screen which cases that require detailed FEM evaluation or not (if approved in design criteria by a more conservative technique). Numerical analysis using FEM is a complex process that requires an accurate understanding of physical phenomena and the ability to translate them into mathematical equations, often requiring considerable time and effort.

Chapter 3 provides more detail on the occasions where FEM adds the most value, but some examples include scenarios with:

- Non-isothermal Earth Surface
- Magnetic ducts and casings
- Time-varying effects
- Groups of cables, especially if of differing construction/loading
- Multicore cable construction
- Heterogeneous soils, or cases where the thermal environment around the cable is affected by other structures or assets.
- Joints and Terminations
- Most cases where full 3D modelling is required.

4.2 How should an FEM study be requested?

4.2.1 Information provided by the client

As discussed in the previous sections, a Finite Element study allows the engineer to take into account phenomena that cannot be covered by the above-mentioned analytical methods. Any request for FEM studies should therefore come with the information needed to build the model. It is thus only useful to perform the FEM study when the input parameters are well defined for the problem being solved, since the accuracy of the model will have a significant influence on the usefulness of the results. Starting an FEM study without sufficient knowledge of the relevant input parameters would have an adverse impact on the usefulness of the final results, often to the extent that the solution obtained would be no more representative of the real installation than would a simplified hand calculation.

Given the open-ended nature of the problem to be modelled in FEM, drawing an exhaustive checklist of the required information is often difficult, time consuming, and expensive. It is also recognized that

the client may not have experience of using FEM, making it difficult to know what should be provided. The working group therefore recommends that FEM study specifies the following as a minimum.

- Purpose of the study: the amount of effort and refinement necessary for a model is different depending on whether it is part of a feasibility study, an optimization problem, or a verification of existing conditions. The objectives of the study and its desired outcomes should be clearly laid out in the initial stages.
- General parameters, common for all cable thermal calculations: as a general rule, all the input parameters needed for an analytical study (e.g., those according to IEC 60287) are also needed for an FEM calculation. This includes the thermal resistivity and temperature dependence of all the relevant material properties and dimensions of layers, the cable designs involved in the model, the exact geometry of the installation environment, the current flowing in each metallic conductor (including their load profile, if relevant). For a time-varying FEM simulation additional minimum material properties will need to be provided such as density, specific heat, and diffusivity. A detailed guide laying out the main considerations to be addressed in calculations is available in Chapter 3 of CIGRE TB 640 "A Guide for Rating Calculations of Insulated Cables".
- Information specific to FEM studies: when FEM studies are performed to model physical phenomena that are not covered by existing standards, indications should be given on how to model them, i.e., using numerical or analytical or hybrid methods and a reference to a recognised best practice guide or a technical paper. There is, for instance, not a single universal way to model soil dry-out or boundary conditions. The manner it should be accounted for is to be clearly specified, and any aspect where the Contractor is expected to use their own judgement to select an appropriate condition should be highlighted.

It should be stressed that the client is responsible for the quality of the relied-upon data, which ultimately determines the quality of the study. Some iterations and information exchange are to be expected with the Contractor to reach a common agreement on how to interpret the data and what exact approach is to be considered.

4.2.2 Considerations for the modelling phase

During the contractual phase for the FEM study, after an agreement has been reached between the Contractor and Client on the work to be performed, discussions should still be expected (e.g., the Contractor may require more information about the thermal influence of 3rd party objects, the Client would then help inform the Contractor of the changes in the system design that affect the model and extension of the range of parameters). As mentioned above, there is often no single way to model some phenomena and some assumptions may need to be validated before further work can be done. The Client should make themselves available and ready to make decisions regarding how the model should be built, based on any inputs the Contractor may provide. If the Client requests handing over of the model files after the project is complete, it must be known from the beginning as part of the contract, so that the Contractor can prepare properly for the request.

4.2.3 Estimation of study complexity

When considering FEM approach in thermal rating calculations, it should be noted that more complex models require not only more time and resources to solve a model, but much more time to define and build it.

In general, a concentric cable can be modelled in 1-dimensional space and this model can probably be solved with a pen and paper, thus it is quite simple to define and solve. The main disadvantage of this kind of model is the simplification of cable surroundings and lack of graphical representations. The first reason is that the main constraint of this approach limits it only to a single cable (as opposed to 3-phase AC systems) in a uniform concentric environment, which is why it is not used.

2-dimensional (2D) approach seems to be the most common and most versatile approach to modelling heat transfer in power cables. Some phenomena that are seemingly out-of-plane can also be modelled (e.g., current flow, forced cooling) but they increase the complexity of the model. In particular, adding natural convection, or longitudinal air/fluid flow, or radiation to a complex geometry, may greatly increase modelling time and computing hardware requirements. In the majority of cases a modern workstation should be able to cope with steady state calculations in 2D models in less than an hour. Modelling time dependent problems (transients) depends greatly on the duration of the event modelled, complexity of the model as well as the experience and skills of the modeler. Complex

models of transient events lasting many days may require days to solve while less demanding models may require only hours to obtain solutions.

Last but not least, is the 3D approach, which is mostly used for cases that would not provide representative solution in 2D. This usually pertains to cable crossings or complex geometries where bends or longitudinal heat transfer cannot be ignored. The main advantage of the 3D models is that the most complex geometries can be accurately modelled. However, there are few constraints of 3D models, namely the size of the model and its mesh, the number of physics being modelled and whether two-way interactions are included. Building 3D models, definition and meshing may require multiple days to complete and some details of the model may need to be simplified. For this reason, the cost of the study should be expected to be notably higher than for a simpler 2D model.

4.3 A simple example

This section provides a short introduction to the fundamental concepts behind the finite element method. The example that will be considered is that of heat transfer within an annulus, with an inner circle with a diameter D_e of 120 mm, and an outer circle with a diameter of 40 m. The outer circle of the annulus is set to a fixed temperature of 20 °C, and the inner circle of the annulus has a heat flux q of 106.1 W/m². The thermal resistivity, ρ , of the material between the inner and outer circles is 0.8 Km/W. This example is displayed visually in Figure 4 - 1.

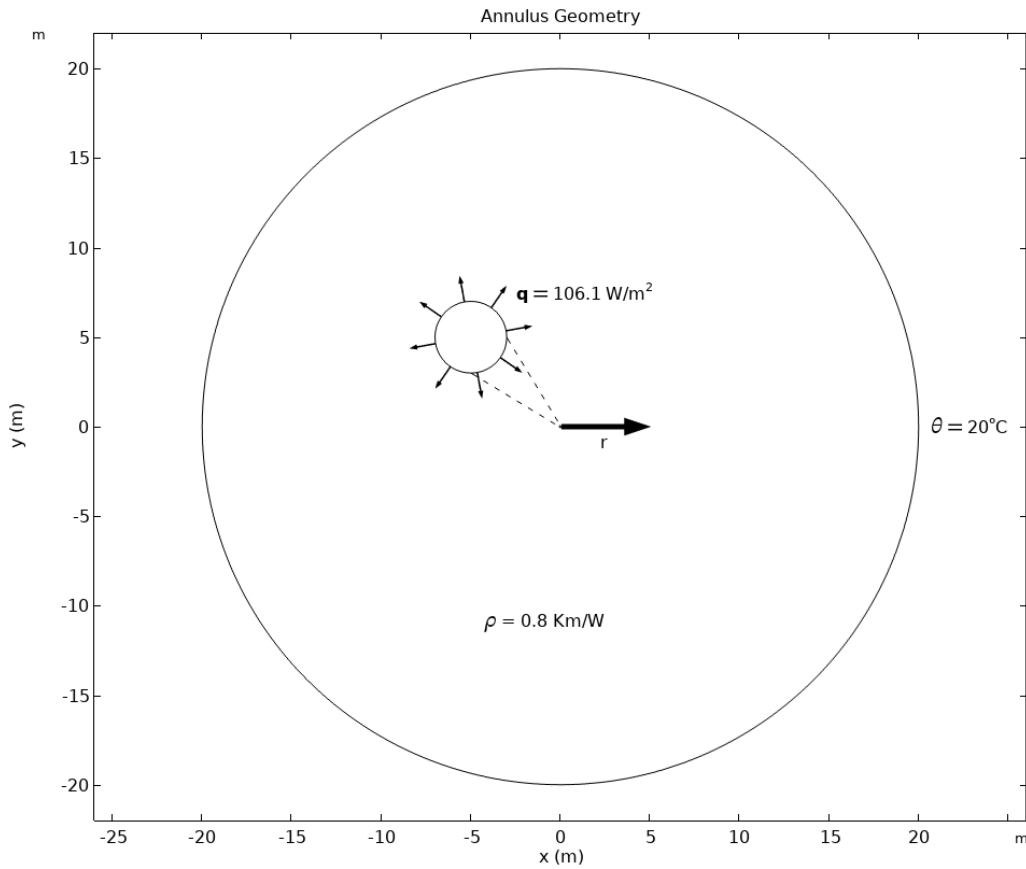


Figure 4 - 1 Annular geometry example considered in this section. A zoom of the inner circle is shown for clarity. The radial coordinate r is indicated.

While this example may seem esoteric, thermally it is a similar scenario to a cable buried at half the diameter of the outer circle ($L = 20 \text{ m}$) with the same diameter as the inner circle (120 mm). The total heat emitted by the cable is 40 W/m, i.e. the product of q and its circumference (πD_e). The maximum temperature calculated at the inner circle is therefore analogous to a cable surface temperature.

We can calculate the temperature, θ , in this simple geometry analytically as a function of the radial coordinate r

$$\theta(r) = 20 - W \frac{\rho}{2\pi} \ln\left(\frac{r}{20 [\text{m}]}\right) ^\circ\text{C}$$

Equation 4.1

Where W is the generated loss (40 W/m in this example) and ρ is the ground resistivity (0.8 Km/W).

In general, for complex geometries and physics an analytical solution is not available. Here we will use this analytical solution to compare the accuracy of different numerical solutions for temperature using a commercial FEM software.

It is first worth noting that the computational domain in which we will calculate temperature is one dimensional, as we only need to consider dependency on the radial coordinate r , which has a minimum value of 60 mm and a maximum value of 20 m, as shown in Figure 4 - 2.

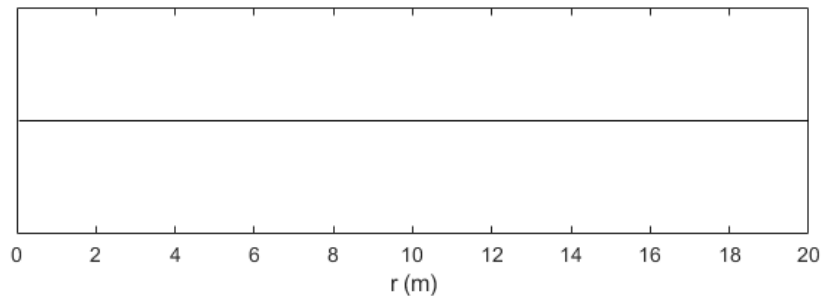


Figure 4 - 2 One dimensional computational domain in which temperature is to be calculated numerically. Note that r does not extend to 0, its minimum value is 60 mm.

The FEM solves for temperature numerically on a mesh over this computational domain. The numerical mesh consists discrete non-overlapping sections, called elements. In one dimensional space, these elements are all line sections, but in higher dimensional spaces different element types are available, such as triangular and rectangular in two dimensions, or cuboids and tetrahedra in three dimensions.

It is then assumed that the dependent variable, in our example θ , can be expressed in each element as the sum of shape functions, ϕ . The simplest shape functions are linear polynomials, straight lines. Higher order shape functions, e.g., quadratic and cubic, can also be used, but at increased computational cost. Often the default setting for heat transfer simulations is quadratic because the heat equation is a second order differential equation, and quadratic elements allow the second derivative of temperature to be calculated. Examples of one-dimensional shape functions are provided in Figure 4 - 3.

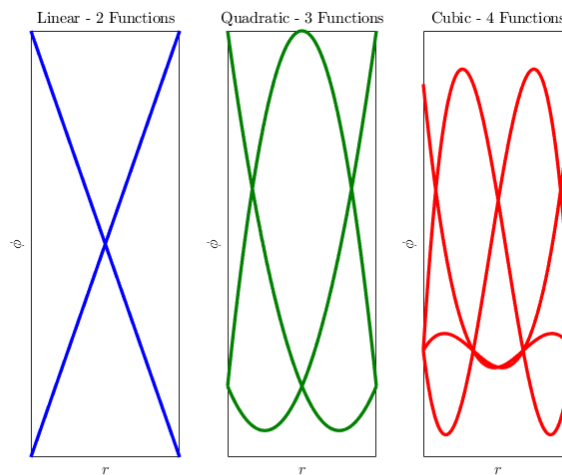


Figure 4 - 3 Shape functions of different orders in a one-dimensional element. These shape functions specifically are Lagrange polynomials.

The scaling factors to apply to the shape functions are calculated by substituting the approximation of the dependent variable, in this case θ , into the weak form of the governing differential equation, in this case the heat equation. The concept of the weak form will not be discussed here, further information can be found in several standard textbooks on FEM.

Returning to our example, if we only use a single mesh element for the entirety of the computational domain and use a linear order shape function within that single mesh element, the numerical solution is shown in Figure 4 - 4.

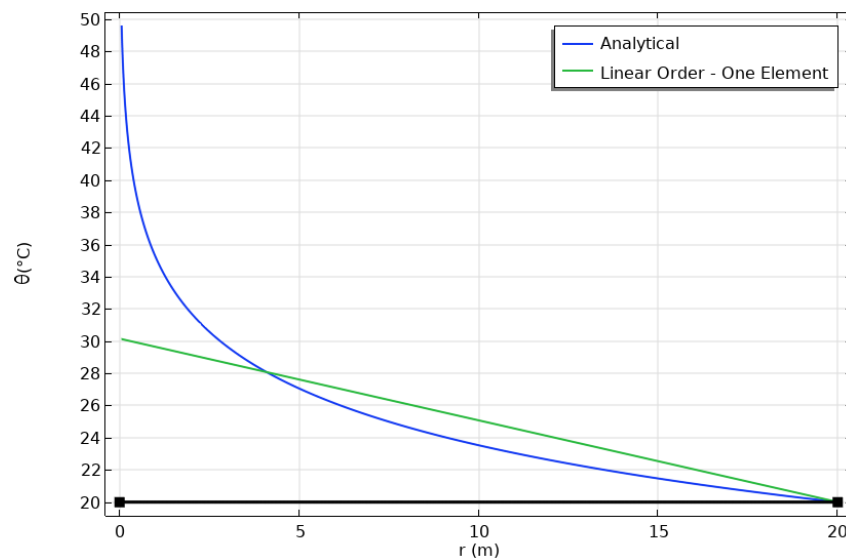


Figure 4 - 4 Analytical calculation of temperature compared to a numerical solution with a linear order shape function using a single mesh element. Spatial discretisation indicated in black.

If we increase the number of elements we improve the accuracy of our result, see Figure 4 - 5. Note that within each element the dependent variable, temperature, must be a linear function.

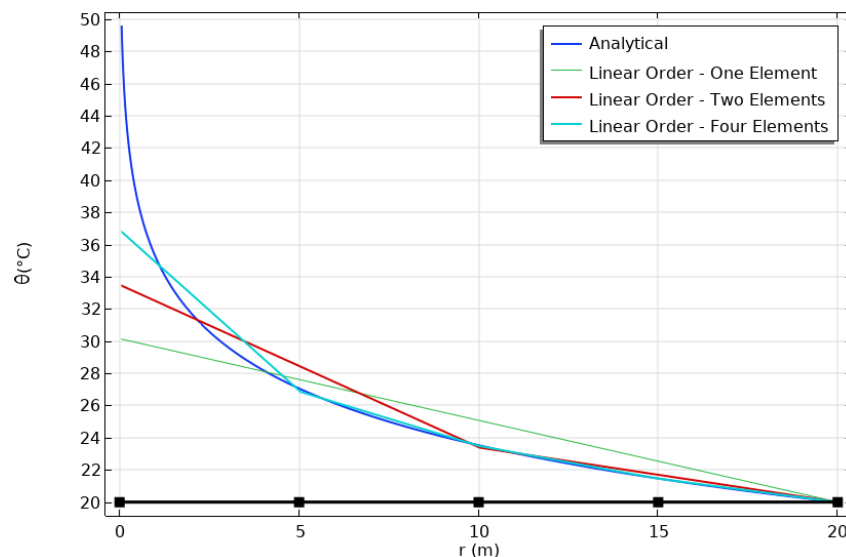


Figure 4 - 5 Analytical calculation of temperature compared to numerical solutions with linear order shape functions and different numbers of mesh elements (1, 2 and 4). Spatial discretisation indicated in black for four element case.

An alternative way to improve the accuracy of our result is to increase the order of our shape functions. Returning to using a single mesh element, a comparison of different element orders is provided in Figure 4 - 6. Even when using a single element, a greater range of spatial variation in temperature can be determined when using higher order shape functions, which makes intuitive sense given that θ is expressed as a weighted sum of the functions in Figure 4 - 3.

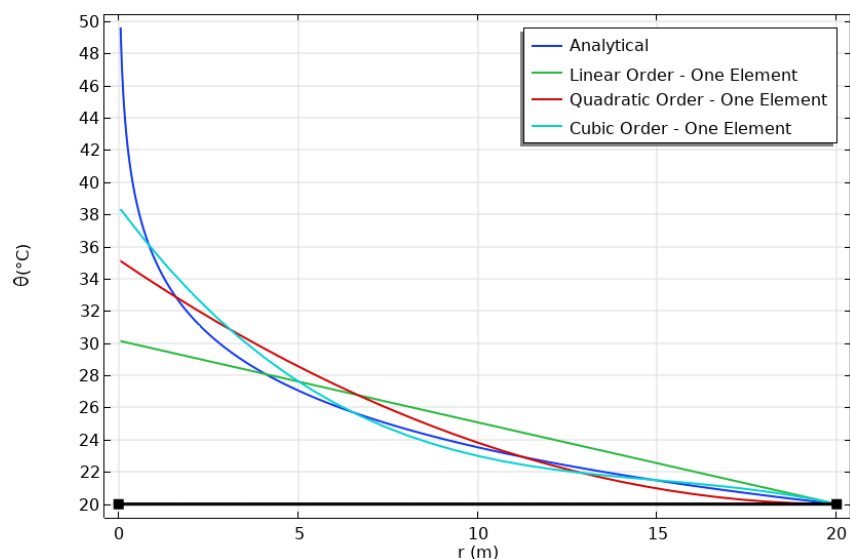


Figure 4 - 6 Analytical calculation of temperature compared to a numerical solution with different order shape functions using a single mesh element. Spatial discretisation indicated in black.

Clearly, we improve the accuracy of our numerical solutions by increasing the number of mesh elements, or by increasing the order of shape functions. As mentioned previously, the common default is for quadratic order elements for heat transfer problems, and it is unlikely that this will need to be changed. The number of mesh elements can be determined automatically, but it is suggested that mesh refinement studies are undertaken to quantify the accuracy of the numerical solution.

As a final demonstration that higher numbers of mesh elements improve the solution, the analytical solution is plotted against the numerical solution with 20 mesh elements and quadratic order shape functions in Figure 4 - 7, excellent agreement is observed.

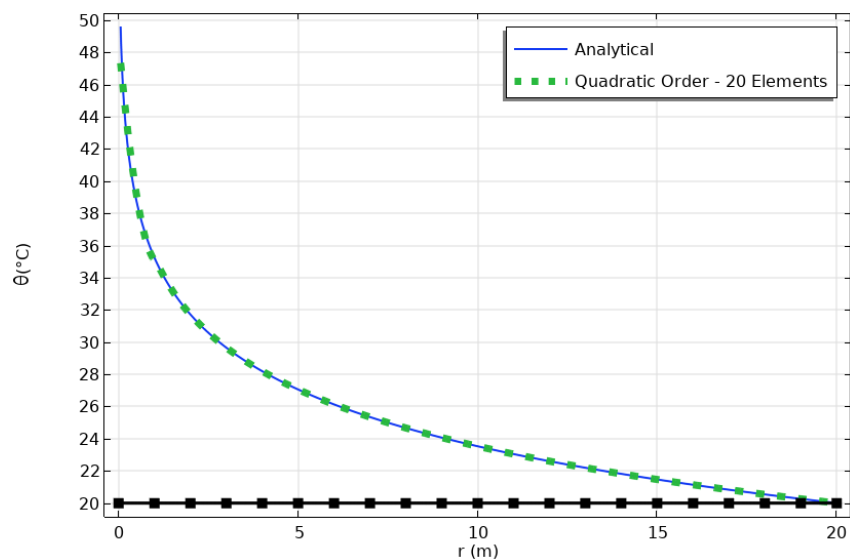


Figure 4 - 7 Comparison of analytical and numerical solution for 20 mesh elements (indicated in black) with quadratic order shape functions.

4.4 FEM study reports

4.4.1 Content to be expected in the report

When receiving the report of an FEM study, the Client can expect some key elements to be present.

First of all, the results of the requested calculation cases are presented. Typically, the final written report will contain one or more plots (images) of the model to get a quick overview of the modelled area and items. This will be accompanied by required results of the calculations. Depending on the task, results could be one or more numbers for, e.g., resulting temperatures or ampacities, or it could

be detailed plots or graphs showing, e.g., a temperature field or gradients along a line. The provided detail is usually limited to a meaningful representation of the requested items and does not include other possible variations of the model or intermediate results.

Second, information on data used to create the model is included in the report. That enables the Client to compare and check the data against the inquired case and possibly perform a second study with the same data to verify the results, if so desired. This information includes the cable and system data of each cable system used in the model with relevant geometry, material properties, electric parameters, bonding arrangement, and minor section lengths (if applicable). Likewise, the modelled surroundings of the cables are described, including material properties and geometry. Where available, a reference to common standards is sufficient (e.g., parameters given in IEC 60287).

The Contractor cannot be liable in the case that the Client has not provided all necessary information. Any assumptions made by the Contractor due to missing information from the Client must be clearly stated in the report.

As FEM models require the definition of certain boundary conditions, they also should be disclosed in the report. This means conditions of the lines, areas or volumes (e.g., isotherm, insulating, heat flow), heat or power sources and sinks, respectively, and how the related numbers are obtained (e.g., by fixed precondition or by secondary calculation).

The exact amount of information varies depending on the case. Irrelevant data (e.g., intermediate results during iteration, temperatures at remote locations in the model, heat flow through boundaries, specific settings of the FE software) is at best omitted to prevent cluttering of the report. In case parameters are varied during the study, they are referred to the appropriate result.

It is a good practice to have the name and version of the used software in the report and references to applied guidelines (e.g., this brochure). While not strictly necessary for reproduction of the results, it may help to sort out possible deviations later. On the other hand, details on the mathematical representation of the model (mesh, solver, node properties) are not typically part of the report, neither are equations or step-by-step calculations of the numerical solution (due to very large matrices). Contents not listed here as standard that are required by the Client should be requested from the beginning during negotiation of the scope.

4.4.2 How to check if the results are valid?

Before going into how to judge FEM results it is crucial to define two terms related to that task:

Definition of verification and validation: Verification focuses primarily on the mathematics and software implementation of the finite element analysis, while validation concerns the accuracy of the model.

In other words:

Verification should evaluate/clarify --> Has the problem been approached and solved correctly.

Validation should evaluate/clarify --> Has the correct problem been solved.

One has to keep in mind that FEM is approximate and it is not robust, meaning that small errors in modelling, boundary conditions or input data can lead to significant errors in the results.

This inevitably leads to the fundamental question: How can one know how large the error of an analysis is if one does not know the correct result, which was the original reason for carrying out finite element analysis in the first place?

The possibilities for assessing FEM results are diverse but have varying degrees of complexity and demand on the assessor, especially regarding knowledge on finite element method. In addition, these possibilities vary significantly in terms of cost and time.

Some of these methods, ranked to their suitability for assessing FEA results, are:

1. Comparative measurements
2. Comparative analytic calculations
3. Comparative numerical calculations
4. Sensitivity studies / error estimation

Comparative measurements

Although measured data is the most suitable method for evaluating FEM results, it is also the least applicable in practice, as such data is usually not available at the time of modelling work being completed. For this reason, comparative calculations, analytical or numerical, are much more common and applicable in practice. Here, the procedure for analytical and numerical comparative calculations differs.

Comparative analytical calculations

The comparability to analytical calculations depends on the type and scope of the numerical cable model.

In practice, numerical cable models are usually too complex to be calculated in a uniform and closed manner using analytical methods such as those given by IEC 60287 and IEC 60853. In such cases, the numerical model should be simplified for verification purposes to such an extent that it can also be calculated using analytical methods. For example, for multiphysical numerical cable models, one possibility would be to divide the multiphysical model into the individual physical domains, e.g., electromagnetic, thermal and fluid dynamic, and compare the individual domains with the results of analytical methods.

This is usually, to some extent, also feasible for the creator of the numerical model with little additional effort, as in FEM, ideally the entire cable rating problem is not modelled at once, but individual components and subsystems are modelled and validated before the final cable rating problem is assembled. Each of these individual components should be validated with simplified loads and boundary conditions.

Furthermore, it is a common practice that numerical cable models are not created from scratch each time, but existing, similar models are used and adapted for the new cable rating problem. Therefore, it is sufficient if the proof of the correctness of the finite element model is provided by means of comparable cable arrangements and not necessarily with the exact same arrangement that is represented in the individual complete model. This also keeps the effort and thus the additional costs for the model builder low, since a limited number of benchmark models only must be created once, which can then be used again and again for verifications.

If a Multiphysics model is used, another good practice is to check the primary line constants, which depend on material and geometry. Line constants describe the capacitance, the inductance, the series resistance in the line direction and the transverse conductance across the line direction of a line in relation to the line length. These values can easily be compared with the data sheets of the cables, although it should be noted that the values in the data sheets are also subject to certain tolerances and to a certain extent to the individual interpretation of the calculation engineer.

Comparative numerical calculations

In numerical comparative calculations for the evaluation of FEM results, there are roughly two variants.

- Comparative numerical calculations of the same complete problem using an alternative simulation software and/or by a third party (consultant, university, etc.).
- Comparative calculations on parts or even the entire numerical model using similar models where the reason for any difference can be well understood.

The first variant requires that the model input data are handed over to a third party, which should be able to reproduce the results based on this. In addition to reproducing the results, this also allows to question and validate the assumptions used to build the model, presented in the report. Therefore, this method might be the most suitable for evaluating FEM results with the help of numerical comparative calculations, especially when only limited FEM experience is available to the customer. However, this is also the most time-consuming and cost-intensive variant, since the entire numerical modelling, validation and calculation effort is effectively doubled.

The second variant can be achieved through techniques such as utilising a 2D model where it might be more conservative than a 3D model, or by removing time or temperature dependent properties to verify that the differences in results with and without these features can be readily explained.

Sensitivity analysis

As mentioned earlier, numerical models are ideally not created in full at once, but in several intermediate steps, ideally also checked by simplified loads and boundary conditions. Furthermore, it is crucial in this process that sensitivity analyses are also carried out. Some forms of sensitivity analysis are:

- Variation of meshing densities
- Variation of boundary conditions and loads
- Variation of domain sizes
- Variation of degree of detail in model
- Variation of material properties

These sensitivity analyses should be performed by the Contractor and referenced in the report, but not normally included in full. The degree of verification should be included in the scope of work and agreed upon prior to commencement of the project, as it may significantly affect the cost of the project. While a certain minimum level is expected in accordance with Good Industry Practice (see discussion in Chapter 6), if a client has very specific requirements or expectations these should be communicated in advance and priced in to the study.

5. Guidance for Modelling

The purpose of this Chapter is to provide guidance to engineers who will be conducting FEM studies. It is assumed throughout that the engineer is familiar with the concept of FEM, and is proficient in the use of their chosen software. It is outside of the scope of this TB to provide an introduction to the methodology of finite element in a general sense; many textbooks, each much longer than this TB, have been written on the subject.

The following subsections take the engineer through the modelling process and the key steps associated with it, starting from the pre-requisites for a study and moving through the setup and solution of the model before ending with the postprocessing of the results. The Guidance Points are indexed and referred to within the Case Studies in Chapter 7. Guidance is given on topics where there may not be an obvious solution, or where multiple options could exist. Reference is also made to other CIGRE publications, such as CIGRE TB 640, as much of the guidance there remains applicable to FEM. The reader will perhaps not need to read this entire section word by word in a linear fashion, but instead scan through to locate the Guidance Points most relevant to their specific application.

5.1 Before Modelling

Before building the model in the FEM software, the engineer should make some checks and decisions upfront to prevent unnecessary iterations and build the model to purpose. Clarification on intention, available data and possible outcomes should be discussed with the client. It is assumed here, that the requirement or decision to use FEM is given already. In general, everything that can be sorted out in the beginning makes the tasks easier and the results more fit for purpose. Furthermore, it is always recommended to read documentation provided with the software package and follow their guidelines, as they may differ for various software packages.

The points listed below in guidance point 1 are not exhaustive and included here as an example that provides some useful advice on what aspects to consider while modelling in FEM.

Guidance Point 1 What is the purpose of the model?

Issue	Purpose of the model
Guidance	Get clarity on: - What are we trying to calculate? - Current rating or resulting temperatures and losses for given load - Effects of surrounding environment, e.g., isotherm, heterogeneous soil conditions - Time dependent calculations, e.g., how long does the soil take to dry-out under time varying load conditions - How do the results need to be presented? - Sensitivity analysis - Optimize a case - Verify a case - Why is FEM used/needed?
Notes	A written specification document from client describing the intent/purpose and expected outcomes of the calculations is essential.
Relations	None

Guidance Point 2 When to use 2D or 3D model

Issue	Decide on using 2D or 3D model
Guidance	Use 2D if: All cables or other services part of the problem such as heat sources/sinks etc, are in parallel and heat flow is in a single plane, or If it can be shown via analytical calculations that any 3D effects are negligible in comparison to the desired accuracy of the study, or The problem can be solved by multiple 2D calculations e.g., helically laid cores of a multicore cable. Use 3D if: Thermal conditions such as heat generated, thermal properties etc, along the axis of cable and other services in proximity change and use of multiple 2D calculations is not feasible. Always prefer 2D if feasible.
Notes	Example 2D: steam pipe in parallel to cables Example 3D: steam pipe crossing cables
Relations	None

Guidance Point 3 Required information

Issue	Is all required information to build the model available?
Guidance	• Ask client for data they can provide • Make assumptions where necessary ◦ Confirm those assumptions with the client (or recommend that they are followed) and ensure that assumptions are documented clearly • Confirm with client that data is sensible and reflective of actual site conditions • Agree with client on all relevant objects within the zone of thermal influence for the model, e.g., other heat sources or heat sinks
Notes	Example: Soil thermal resistivity, ambient temperatures and ground conditions
Relations	IEC 60287

Although FEM is typically used where standard calculations are not suited to the task, it remains important to make use of the information available from standards. A good example of this is that the cable data will often make reference to thermal and electrical properties detailed in IEC 60287 series.

Guidance Point 4 Standards and recommendations

Issue	Applied standards
Guidance	Clarify what standards and recommendations to apply for e.g., (not limited to) • Material properties • Ambient behaviour • Calculation methods and use them accordingly. • Keep track of used sources, noting that standards are occasionally revised and that some content will change between editions. • The selection of properties, values and methods etc. should be in the following order of priority:

Notes	a. Measured or other values provided by client b. IEC standards and recommendations c. This Technical Brochure d. Accepted publications on the subject by recognized sources (CIGRE, IEEE, CIREN etc.) e. Other publicly available sources (books, publications) f. Suggestions by FE software provider g. Own derivation/interpolation/extrapolation for methods/values • Make note of the publication date of the sources and prefer newer ones. Treat CIGRE TBs equal to IEC standards in this regard.
	Other sources, e.g., 3rd party measurements, believed to provide more accurate data can be used when agreed with the client. Avoid using material properties suggested by the FEA software if no source is given or other sources are available. When considering material properties, attention must be paid for the requirements between a time-static or time-varying models. For time-varying thermal models, the material density, specific heat, and diffusivity parameters must be considered. Use own derivations/interpolations/extrapolations where no other source is to be found and disclose to client. CIGRE TBs often supplements IEC standards; this TB supplements IEC TR 62095
Relations	IEC 60287, IEC 60853, IEC 60228

It is rare that an FE model can be built with all minor details fully represented as they would be in the real world, and indeed to do so would often be wasteful in terms of engineering hours and computational time. The complexity of the model must be considered very carefully to ensure that it is appropriate for the maturity of the information available at the relevant project stage.

Guidance Point 5 Complexity of model

Issue	Complexity for FE model could be very high if all details are included
Guidance	Check if parts of the model can be calculated analytically. Assess impact on the model in terms of: • Accuracy • Modelling effort • Calculation time Feed analytically calculated values into the FE model where appropriate. Iteration over multiple steps may be required where analytical results are depending on FE solution, e.g., conductor temperatures.
Notes	
Relations	IEC 60228, IEC 60287

5.2 Cable geometry

FEM comprises several steps, from modelling of the geometry, through material definition, meshing, solving and post-processing. Geometry is the integral part of the FEM and, since it may greatly affect both the solving process and the accuracy of the solution, extreme care must be exercised during the process of modelling. High voltage power cables consist of several components and layers that play various roles and have various properties. Some of them can be simplified to make the model solvable. Some elements can be simplified, or they can be modelled depending on the purpose of the

model and its size, e.g., 3D models may require more simplifications than 2D, or an electromagnetic model may not need all the layers that the thermal model would require. Several elements of the cables and the common modelling tips are provided below.

5.2.1 Conductor

The conductor of HV cables is usually made of individual stranded wires, which either form a round conductor or segments that after stranding form a round conductor. Inclusion of all details, however, would result in a very large number of mesh elements and, especially in 3D models, may render the model difficult/impossible to solve. Therefore, the following can be considered in modelling:

1. Most of the time, there is no need to model the individual strands in the geometry and instead, the stranded conductor may be modelled as a solid one.
2. The outer diameter of the model conductor should match the one from the datasheet as this affects the heat transfer.
3. DC resistance of the solid conductor in the model must correspond to the one of the stranded conductor, as this affects the conductor losses.
4. There is a difference between the model of solid and stranded conductor, thus it is essential to correct the thermal properties to ensure that the equivalent amount of metal is represented (fill factor). This affects the conductor temperature.
5. Milliken conductors may be modelled in two ways, namely:
 - a. if only heat transfer mode is considered, corresponding DC resistance, proximity effect factor, skin effect factor, and eddy current losses can be taken into account and Milliken conductors can still be represented by a simple circle, or
 - b. by application of different modelling techniques than regular stranded conductors when the full electromagnetic model is considered. In this case, the model should include the insulation between the individual strands and sectors.

Guidance Point 6 Thermal modelling of stranded conductor

Issue	Thermal modelling of stranded conductor
Guidance	There is no need to model the individual strands in the geometry. The outer diameter in FE should match that on the datasheet. For conductors without water blocking material the thermal conductivity and specific heat of the conductor material should be multiplied by the fill factor (nominal cross section area/cross section area in FE). For conductors with water blocking material, a weighted specific heat should be used to obtain the correct total heat capacity of the conductor.
Notes	Inclusion of full geometric detail would result in a very large number of mesh elements, which is not necessary. It is essential to correct the thermal properties to ensure that the equivalent amount of metal is represented. Conductor losses are described in Guidance Point 33. The actual size of the conductor will in most cases differ from the IEC mm² label, as the requirement is on the resistance and not on the area of the conductor.
Relations	IEC 60228, IEC 60287, CIGRE TB 880

5.2.2 Helical factor of metallic screen, armour and cables

Screen and armour have their individual strands twisted in a way that they form a helix. In a 2D model of the cable, the helix cannot be reproduced and the strands are represented as straight wires. The difference between these two cases is twofold, namely:

- the resistance of the straight wire is lower than the helix due to the difference in length, thus the actual resistance should be used in the model, and

- the electromagnetic coupling between the helix and the other layers is different than between parallel strands. In the latter, the coupling reaches maximum, while the helix changes the angle between conductors of different layers, making the coupling weaker.

Therefore, a thermal model of the cable can neglect the helical arrangement of the wires, but has to take into account the actual resistance. Depending on the desired complexity of the model, the individual strands can be modelled or the equivalent thermal resistance of the layer can be used. If the latter option is chosen, the thermal and electrical properties of the wire must be adjusted to ensure that the overall value is correct.

On the other hand, an electromagnetic model may require the full helix as described in CIGRE TB 908, unless it is proven that the simplified model does not introduce significant error, or increased losses are acceptable and acknowledged. The helix can be represented properly in 3D space only, where the meshing process may be very challenging, thus simplification is greatly beneficial.

Guidance Point 7 Helical structures

Issue	Inclusions of helical Factor for Screen and Armour Wires
Guidance	The screen/armour wires could be either neglected (by modelling an equivalent thermal region) or modelled as per the datasheet. If the latter option is chosen, then the thermal and electrical properties of the wires shall be adjusted to ensure that the overall value is correct. Regardless of the approach, the heat source cannot be neglected - see CIGRE TB 880 (sections: 2.5.2, 2.7.1, 2.8.2, 4.4.4, 7.5, 7.6). For an electromagnetic model, it is recommended to model the full helix as described by CIGRE TB 908.
Notes	It is not feasible to represent the helix in 2D and challenging in 3D from a meshing perspective. Also the thermal resistivity of the layers in the core should be considered due to the layup as per CIGRE TB 880.
Relations	IEC 60287, CIGRE TB 880, TB 908

Guidance Point 8 Twisting of 3-core cables

Issue	Helical twisting of cores in 3-core cable
Guidance	For a thermal only model where the losses are represented by equations, the core can be modelled as running parallel to the cable axis. The user should ensure that the losses applied represent that of a unit length of cable, which will be slightly larger than a unit length of core. The specific heat of the materials does not need to be amended to account for the lay factor, as the difference will be sufficiently small and the result slightly conservative. For an electromagnetic model, it is recommended to build an entire model of the twist.
Notes	It is not feasible to represent the helix in 2D and challenging in 3D from the meshing perspective. For further discussion of the impact of layup factors on losses, refer to CIGRE TB 880 section 2.5.2.
Relations	IEC 60287, CIGRE TB 880, TB 908

5.2.3 Thin layers

Metallic and semi-conductive screens, tapes, armour, armour bedding and serving may be very thin when compared to the cable size. This disproportion becomes more pronounced in the scale of the entire model where the aspect ratio may be well over 1000, which can cause meshing issues and slow down computations. Therefore, this situation is usually unwanted and can be mitigated by implementing one of the following:

- Simplify thin layers by combining several components where this is thermally reasonable, and does not affect the radial thermal resistance, or

- Model thin layer by introducing additional boundary condition (thermal resistance or heat source), or
- Design the mesh accordingly.

As an example, the semiconducting screens can be merged with the main insulation, or swelling tapes can be merged with adjacent polymer layer. Another example is that the thin metallic foils could be merged with an adjacent layer and the losses in the foil can be modelled via a boundary heat source. The total thermal resistance and thermal capacitance of the equivalent layer should correspond to the sum of individual layers.

Guidance Point 9 Thin layers

Issue	Thin layers, resulting in high aspect ratio of component dimensions
Guidance	If other means are not feasible, simplify by combining several components where this is thermally reasonable, and does not affect the radial thermal resistance or the longitudinal thermal resistance in a 3D model. A great care has to be exercised and try to avoid combining layers if there is a heat source between them. If such layers are combined then great care is needed to ensure that the correct losses flow through each thermally resistive layer. Such components could include thin extruded layers, tapes, wires or foils.
Notes	Modelling all layers could cause meshing issues and slow down computation speed. Examples could be merging the semiconducting screens into the main insulation layer, or merging a swellable tape into an adjacent polymer layer. Thin metallic foils could be thermally merged into an adjacent layer, with the losses in the foil modelled via a boundary heat source. Alternatively, some FEM softwares allow for thin layer approximations as boundary conditions to model layers that are good thermal conductors compared to the adjacent layers. If such a boundary condition is applied to e.g., a thin aluminium foil (defined as a boundary layer), the heat flux across this layer is neglected (temperature drop neglected), while for 3D the tangential heat flux is computed. For such a case, note must be taken to if induced currents can be neglected or should be included, e.g., in the form of a heat source. If this technique is used, the user should verify that the total thermal resistance and thermal capacitance is correct. This can be achieved by comparison to IEC calculations.
Relations	IEC 60287, CIGRE TB 880

5.2.4 Corrugated sheath

Corrugated sheaths may be difficult to model and solve. Accurate modelling of the corrugated sheath would require a 3D model as the sheath's dimension is axially non-uniform. Sometimes, axisymmetric 2D model along the cable length direction can also be utilized. The difficulty is that there may be a small air gap (with water swelling tapes) between the sheath and the cable, therefore the sheath is a combination of metallic sheath, air and thin layers (discussed earlier). The actual sheath thickness should be as per the datasheet and the sheath outer diameter should be the mean of the outer diameter. The resulting gap beneath the sheath should be filled with a material of thermal properties equal to the layer below. Filling the gap above the sheath should be aligned with what is included in IEC 60287 and CIGRE TB 880. This will also ensure that the outer diameter of the complete cable matches the average outer diameter of the physical object. The electrical resistance of the corrugated sheath must be considered carefully, refer to discussion in CIGRE TB 880. Including surface to surface radiation, although justifiable from the perspective of the heat transfer, it would introduce unnecessary computational burden.

Guidance Point 10 Corrugated sheath

Issue	How to model corrugated sheath?
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Guidance	The sheath thickness should be as per the datasheet. The sheath outer diameter can be modelled as the mean of the outer diameter seen at the trough and peak of the corrugation. The resulting gap beneath the sheath should be filled with a material of thermal properties equal to the layer below. The resulting gap above the sheath should be filled with the jacketing material.
Notes	The above approach should be aligned with what is included in IEC 60287 and CIGRE TB 880. This will also ensure that the outer diameter of the complete cable matches the average outer diameter seen in the physical object. The electrical resistance of the corrugated sheath must be considered carefully, refer to discussion in CIGRE TB 880 section 2.7.2. Additional physics such as surface to surface radiation if not simplified may introduce unnecessary computational burden. Reference is made here to [34].
Relations	IEC 60287, CIGRE TB 880

5.2.5 Small contact interface between cables and other objects (including other cables)

When modelling 3-core armoured cables, trefoil formation with touching cables, or simply cables on a flat surface, there is an area of contact between the round surface of the cable and another surface (round or flat), which needs special attention. When modelled as is, i.e., minimal contact surface, the resulting geometry is very challenging for mesh generators as they may create long, narrow regions. It is therefore suggested to separate the components or overlap them to prevent meshing issues.

Guidance Point 11 Small contact surfaces

Issue	Small contact area between components leading to long and narrow regions
Guidance	Separate components with significant gaps or overlap them to prevent regions with sharp points. Either way should not significantly affect the geometry, but modify it to eliminate the narrow regions.
Notes	Meshing long narrow regions may cause singularities. One example of this is the contact between the power cores in a 3-core armoured cable and the bedding. One solution here could be to merge parts of the filler and the bedding. Extra care must be taken if the thermal conductivity of the components are different.
Relations	IEC TR 62095 Section 3.2

5.2.6 Fillers and voids

Space between cores in a 3-core cable may be filled with fillers and/or with voids. The presence of both may result in a complex geometry and necessity to include radiation, which can significantly impact the computational burden, thus one possible solution could be a combination of both into single solid matter with an equivalent thermal resistivity for time invariant or time-static models. For time-varying models additional equivalent material properties such as density, specific heat, and diffusivity will also be needed. Some example values for these additional properties can be found in TB 640. Another solution could be to model the 3-core cable as an equivalent single core cable using IEC T_2 value.

Guidance Point 12 Fillers and voids

Issue	How to treat fillers and voids in 3-core cables
Guidance	For 3-core cables with fillers with voids it may be computationally expensive to model all the details in the fillers and consider all physics with conduction, convection and radiation. Recommendation is to simplify the geometry and physics in the following way.

Notes	If the geometry of fillers are modelled in detail, it is sufficient to assume stagnant water and only model heat conduction in the filler cavities for any sections where the filler is assumed to be flooded. For sections with filler that are not flooded it may be acceptable to treat the entire filler region as a solid with the same thermal conductivity as the filler itself [110].
	If air voids are modelled in detail, additional physics such as surface to surface radiation may be required.
	Another approach could be to use an equivalent single core cable with losses from all three cores and thermal resistivities of layers based on IEC T ₁ , T ₂ and T ₃ .
Relations	CIGRE TB 880, IEC 60287

5.2.7 Embedded fibre optic (FO) cables

FO can be installed in various ways in power cables, either within cable metal screen or armour in single core cables, or between the cores in the 3-core cables. Multiple fibres are usually enclosed in a single tube, which may be conductive (stainless steel or copper) or non-conductive (polymeric). In general, when the FO is in a non-conductive tube, it can be omitted from the model as it does not contribute to the losses or heat transfer. Also, the FO in single-core cables can be neglected as the tube does not significantly contribute to the losses. In a 3-core cable, however, the metallic tube may need to be considered in a full electromagnetic model as it may generate losses sufficient to locally increase the temperature. This usually takes place when the resistance of the tube is sufficiently low to allow for the flow of the induced current.

Guidance Point 13 Fibre optic cables in power cables

Issue	FO cables embedded inside the overall power cable
Guidance	Small FO cables in the metal screen or armour layers of a single core can be neglected.
	In a 3-core cable, FO cables with non metallic layers can also be neglected.
	In a 3-core cable, FO cables with metallic water barrier only need to be considered in a full electromagnetic model if the resistance is low enough that the current flowing along it could be non-negligible.
Notes	Could be in steel (usually stainless), copper, or polymeric jacket.
	Modelling all layers could cause meshing issues and slow down computation speed.
	The temperature at the location of the FO can be extracted from the model directly if it is needed even if the FO cable is not modelled.
Relations	CIGRE TB 899

5.3 Environment

FEM is frequently used to determine operating temperatures and current ratings of cables installed in a complex thermal environment. Determining exactly which aspects of the environment to include in the model is a critical step, and can have a significant impact on the amount of time needed to obtain solutions. In this section, guidance is given on some common topics. Users must note that this list can never be exhaustive, and it may be necessary to use the underlying principles shown below to define a way forward for a new topic which is not yet covered in this TB.

5.3.1 Buried cable systems

When modelling buried cables, it is essential to determine the correct size of the model such that they do not adversely impact upon the results obtained.

Guidance Point 14 Modelling of soil

Issue	Size and shape of the area to be modelled (cables in soil)
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Guidance	While the modelled area can have an arbitrary form, a rectangular form or semi-circular form are generally recommended (3D: cuboid or semi-sphere). The distance of the sides and ground of the modelled area should be at least 10 m from the cables. However, the actual size of the solid domain should be increased until further enlargement does not change the resulting cable temperatures significantly (for example by no more than 0.1 K). For selection of boundary conditions refer to Guidance Point 44 and Guidance Point 45.
Notes	The soil behaves as an immense thermal capacity compared to the cable. However, the soil itself only heats up in the vicinity of the cable. If a constant temperature or a thermal isolation at the boundaries are chosen, they will influence the heat flow from the cable if they are too close to the cable system. Increasing the soil domain increases the number of elements in the model only slightly, as the elements near the soil boundaries are significantly larger than, for example, the elements used to discretize the cable with its relatively thin layers. Therefore, increasing the size of the soil domain has only a minor effect on the total calculation time.
Relations	IEC TR 62095

Guidance Point 15 Multiple layers of soil

Issue	Modelling of multi-layered soil
Guidance	As suggested by CIGRE TB 640, the most common approach for handling non-homogeneous thermal properties is the use of the finite element technique especially when transients or multiple circuits are concerned. The soil region in the model could be divided in accordance with a soil survey. This can also aid in meshing.
Notes	Quote from CIGRE TB 640: <i>“Note the situation regarding inclined ducts crossing multiple layers of soil especially if there is a fluid inside these ducts (air, water). In this case, the possibility of axial heat transfer along the duct must be considered. In such a case, a deeper layer of soil with bad thermal properties may have an effect at another (higher) location along the drilling. Simply considering the radial heat transfer may be too optimistic.”</i> In these cases a 3D-model is required. Reference is made to Electra 98 article on this subject [111].
Relations	IEC TR 62095, CIGRE TB 640;

Guidance Point 16 Inclusion of other materials in ground

Issue	Modelling of other cables/services/structures (including metallic) in ground
Guidance	All structures with different thermal properties in the ground within the modelled area need to be included if they are considered to have a thermal impact upon the cable. This could include different layers of soil, blocks of concrete, parts of buildings, other services (ducts, pipes etc.), voids and cavities (e.g. underground water stream, cavern) etc. Structures/layers with the same thermal properties can be joined in the model. The consideration of metalwork, parallel metallic paths, parallel circuits (equally and unequally loaded) as well as circuit crossings should be in accordance with CIGRE TB 640. As a general rule, any thermal source, such as steam pipes, other cables or magnetic material, as well as any thermal sink, such as water or cooling pipes could be implemented if considered thermally significant.
Notes	For guidance on the definition of the thermal properties within those structures see Appendix D. IEC 60853 and TB 640 provide some general guidance on the thermal properties such density, specific heat, and diffusivity for some materials common to cable system environments.
Relations	IEC 60287-3-3, IEC 60853 and TB 640

Often the ground surface must be represented in an idealized way within an IEC 60287 compliant model. The use of FEM allows different surface conditions and structures to be taken into account, as opposed to analytical calculations which mostly assume an isothermal ground surface.

Guidance Point 17 Modelling of ground surface

Issue	Modelling of surface and structures above ground
Guidance	Usually the surface can be modelled as a straight line, or plane surface. However, in case of sloped, very uneven terrain or structures or buildings above the cable, residing on the surface, the additional change in geometry should be reflected in the model. Both uneven surface and structures above the ground should be modelled only if they change the heat dissipation from the ground surface.
Notes	Examples: road embankment, bridge pillar, etc.
Relations	None

Ground water conditions, both free water and moisture trapped locally within the soil structure, can have a significant thermal impact upon the cable. When rating a circuit, it is essential to consider the variation in conditions that might be seen seasonally and during the life of the cable in determining an appropriate set of assumptions for the thermal design of the cable system.

Guidance Point 18 Ground water migration

Issue	Influence of ground water migration
Guidance	There are two primary influences of the ground water. The first is the influence on thermal resistivity under normal conditions, the second is on the dry out phenomena. Thermal resistivity: the impact of the ground water level should be captured by the thermal resistivity values specified for different soil layers. This means that it is not necessary to directly model the concentration of water within the soil in the majority of cases. Consideration must be given to any annual variations in ground water level that might adversely impact the cable rating. Dry out: it is possible to implement the simplified "two-zone" dry out model described in IEC 60287-2-1 based on a critical temperature. This is best achieved by setting a temperature dependent thermal resistivity, with a smoothing function so that the change between the wet and dry value does not introduce numerical instability. Provided the cable system is installed below or close to the ground water table, soil drying out can be neglected. For installation in the unsaturated soil zone, moisture migration can be implemented using the Philip and De Vries model [69]. However, thermal properties of the soil and hydrological data on the ground water table, e.g., seasonal variation, are required for this approach.
Notes	The implementation of the Philip and De Vries model is demonstrated in [69]. Implementation of such models requires very detailed geophysical property information which is rarely available to cable engineers, and it is not recommended for anything other than the most detailed / critical studies.
Relations	None

The development of a rigorous equation for the thermal resistance of the filling medium for cables in ducts is quite involved and the expression depends on the cable surface temperature. This equation, even though amenable to computer implementation, is not suitable for standardization. In FEM analysis, one could build a full heat transfer model for the water or air-filled ducts, including conductive, convective and radiative components. However, the required complexity is not justified in the majority of applications since the thermal resistance of the air or water in the duct has a small effect on the cable current rating. An example of both IEC based method and full convection and radiation modelling is available in case study 1. Therefore, the simplifications given by Neher and McGrath (1957) and in IEC 60287-2-1 are proposed for this purpose.

Guidance Point 19 Modelling of duct filling

Issue	Modelling of duct filling
Guidance	If it is filled with grout, then model the filling material directly using its thermal conductivity. If it is filled with air or water, then use the IEC approach (IEC 60287-2-1) to calculate the thermal resistivity of the medium in duct and implement that in the numerical model as a material with customized thermal resistivity given by: $\rho_{eq} = \frac{2\pi U}{[1 + 0.1(V + Y(\theta_m - 273))D_e] \ln\left(\frac{D_d}{D_e}\right)}$ where D_e and D_d, are the cable diameter and the inner diameter of the duct, respectively. In this equation, the values of the constants U, V, and Y depend on the installation and are given in the IEC 60287-2-1 and the temperature θ_m is the temperature of the medium filling the duct in Kelvin. In the development of this equation, it is assumed that the inner surface of the duct (diameter D_d) or pipe is isothermal. This assumption is usually valid for metallic conduits. For ducts made from materials having poor heat transfer properties, the average temperature inside the duct should be assumed. Additionally, since the above equation assumes the cable to be in the centre of the duct, it is recommended that this assumption be followed in the FEM studies. For ducts filled with air, one could use the equations presented in “Simplified model for heat transport for cables in pipes” [53] to account for the eccentricity of the cable(s) in the pipe. A similar procedure could be developed for water filled ducts, but this would require further research. In case of multiple cables in a duct, an equivalent diameter representing the area of all the cables inside the duct should be used.
Notes	The IEC method is proposed. The constants in the above equation were established empirically. An important consideration is that the constants were developed for the cable external diameter between 25 - 100 mm for cables in ducts and to 75 - 125 mm for three cores in pipe. However, due to lack of more information, the same constants are currently used for larger cables and larger ducts, see also TB 880 GP 49.
Relations	IEC 60287-2-1, TB 880

Metallic casings represent a special case of a ducted system, as unlike plastic ducts there can also be an additional heat source due for example to hysteresis losses or eddy currents within the metal casing.

Guidance Point 20 Metallic casings

Issue	Modelling steel or other metallic casings containing power cables
Guidance	Two cases can be considered, depending on whether the casing is magnetic or not. Non-magnetic metals: The magnetic field generated by the cable can lead to eddy currents being induced in the metal components, which will generate heat. Typically the heating effect will be small for 3-core cables, or 3 single cores inside a common pipe and then the thermal impact is likely to be negligible. Single core AC cables in individual pipes may see greater losses and need to be considered more carefully, but this situation is rare. This would require a coupled thermal and electromagnetic model, where the electrical conductivity of the pipe can be represented. Magnetic metals: Cables can sometimes be installed in steel pipes, or surrounded by steel structures to act as magnetic shielding. The losses in magnetic pipes, casings or containments can be significantly higher than for non-magnetic cases. A full electromagnetic model will be required which represents the currents in metallic

Notes	components of the cable and also considers the complex permeability of the steel to ensure that both hysteresis losses and eddy currents are captured.
	Reference to CIGRE TB 559 is encouraged prior to commencing any modelling.
	CIGRE TB 559

5.3.2 Cable systems in air

J-tubes and similarly constructed risers are not covered by IEC 60287 at the present time, although the WG is informed that a method is under development within IEC TC20 WG19. Given the prevalence of such thermal sections when connecting cables to offshore platforms, FE is commonly used to determine expected cable temperatures. Guidance is provided here on how to create a suitably accurate model, including the convective and radiative heat transfer, without excessive computational time. A Case Study exhibiting this is shown in Chapter 7 of the TB.

Guidance Point 21 Risers, J-tubes and bend stiffeners

Issue	Modelling of cable installations in riser poles, J-tubes, and bend stiffeners
Guidance	Cables in J-tubes or risers may contain sections where the tube or riser is water filled, and parts which are air filled. The following guidance applies: - Heat transfer from cable surface to tube/riser inner wall: while it is possible to model the fluid flow directly using CFD techniques, a semi-analytical approach can be perfectly adequate. In the latter case, a heat flux boundary is defined on outer surface of the cable and an empirical correlation is used to define the Nusselt number. As the annulus is typically thin and has a large aspect ratio, correlations designed for cylindrical annuli can be used. This requires the following steps to be taken: - The heat flux boundary is defined in the cable surface and an integral function is used to find the total heat transferred across the boundary. The heat flux from the cable is specified as negative. This flux may have contribution from both convection and radiation. - The air or water inside the annulus is modelled as a solid material for the purpose of capturing conductive heat transfer only. - A second heat flux boundary is defined on the inner wall of the tube. The positive heat flux entering this boundary is equal to the total heat leaving the cable surface, but distributed uniformly over the tube inner wall. - Heat transfer from tube outer wall (in water): a heat flux boundary is defined on the outer wall, and the heat source leaving this boundary is calculated using an empirical Nusselt number correlation. The heat is assumed to leave the model, i.e., to dissipate immediately into free water where the local water temperature cannot be increased by the heat dissipated into it. - Heat transfer from tube outer wall (in air): as per the above case, a heat flux boundary is specified to account for the convective heat transfer. In addition, a second boundary condition is applied to account for surface to ambient radiation; this can either be using a direct radiation boundary condition in a software, or an additional term can be added to the convective heat flux boundary to account for the radiation. Solar radiation and possibly wind (forced convection) on the outer wall should be considered; while neglecting wind will only lead to conservative results, neglecting solar radiation may result in too optimistic results. Wherever empirical coefficients are used, it is essential to ensure that they are only used within their range of validity.

	For sufficiently long J-tubes in air with valid analytical heat transfer coefficients, a 2D model can accurately describe the location with the highest temperature. For J-tubes with air-filled section below 5 m in length, the longitudinal heat flow becomes significant, which requires a 3D model to obtain accurate results. Note that a 2D model will in this case give conservative results, and may therefore be used depending on the situation (for example if the 2D calculation shows that the section is not thermally limiting for the circuit). For cables in bend stiffeners the longitudinal heat flow is typically of importance, due to the varying thickness of the bend stiffener. Therefore a realistic model of cable installations in bend stiffener requires a axial symmetric 2D model or a full 3D model. As an alternative a conservative analysis on bend stiffeners can be conducted by assessing the sections with the lowest and biggest bend stiffener thickness, in a 2D model which will yield conservative results and can be used in certain conditions, especially when they are not the limiting of the circuit.
Notes	See case study 3 in chapter 7.3
Relations	None

Guidance Point 22 Troughs and tunnels

Issue	Modelling unventilated, naturally ventilated, or forced air cooled troughs or tunnels
Guidance	2D vs 3D: The first decision to make is whether a 2D or 3D model is necessary. A 2D model is valid for unventilated and naturally ventilated troughs and tunnels where longitudinal air movement is negligible and no other geometry changes along the cable axis need to be accounted for. Where longitudinal air movement is not negligible, a 3D model is required. However, a 3D model could be computationally very large and consideration should be given to the use of Electra 143 style numerical calculations as a first preference. Heat transfer out of trough/tunnel walls: for the purpose of modelling heat transfer through the surrounding soil, any guidance given for modelling the environment around a buried cable remains equally valid, e.g., considerations for boundaries, domain size etc. The exception is for troughs where the lid is laid in level with the ground surface and directly exposed to air. In this case, it is inappropriate to force an isotherm at the interface to the ambient air and a convective condition should be considered instead. Refer also to Guidance Point 44. Convective Heat Transfer from the cable surface to the air: determination of appropriate heat transfer coefficients for natural convection inside the enclosure is difficult, especially if cables are installed vertically such that air warmed by one cable reaches another. It is recommended to solve the air velocity profile directly using a full CFD model, with the heat transfer coefficients at the cable surface being determined by the CFD calculation rather than input by the user. Special attention must be paid to the mesh around the cables. Consult software specific guidance on the use of wall functions. Convective Heat Transfer from the air to the trough/tunnel wall: as per heat transfer from the cable to the air. Special care must be given to the interface between solid and fluid domains when simulating heat transfer performing CFD analysis. A boundary layer meshing, such as that in Figure 7 - 65, is always to be developed in order to impose no-slip (zero velocity) wall as a boundary condition. Further guidance on this is available in Guidance Point 48 and Guidance Point 51. Radiative Heat Transfer from the cable to the trough/tunnel wall: in an unventilated trough or tunnel, heat transfer by radiation may be the dominant mode. Due to the complexity of multiple cables existing inside the same enclosure, not all of the radiation exiting a cable might reach the wall. It is therefore recommended to utilise full surface to surface radiation calculations within the FEM calculations, rather than

Notes	simplified heat flux boundary conditions as are used for the J-tube in Guidance Point 21.
	Exchange of air with ambient: Naturally ventilated troughs with (partly) open covers allow (arbitrary) air to flow in and out of the trough. A simple solution to model the exchange of air is not currently available. A model incorporating air flow calculations is required. For a more comprehensive discussion on the use of inflow and outflow boundaries, refer to Case Study 2 in Section 7.2.
	In addition to the guidance given here, please also refer to: Guidance Point 2 When to use 2D or 3D model Guidance Point 48 Boundaries for CFD Guidance Point 51 Boundary layer mesh
Relations	None

Guidance Point 23 Cable on enclosed trays

Issue	Modelling air in enclosed cable trays
Guidance	If neither the IEC nor any relevant analytical formulas encompass the ampacity calculations for the particular enclosed tray configuration in question, CFD simulations may be employed. For enclosed air cavities in general, including enclosed trays with heat sources (cables), it is advisable to setup the fluid flow solver with a compressible formulation and a time-dependent (also for stationary ampacity simulations) solver in order to capture any pressure-determined density changes correctly - the evolution of heating air and pressure increase over time, which otherwise would be lost using a stationary solver, is captured using the time-dependent solver. The air fluid within the usually small air enclosures of cable trays can be expected to be of laminar type, and thus, a laminar solver coupled with the above-mentioned settings is recommended.
Notes	For further explanation of the modelling of fluid flow within enclosed trays, please refer to the closed tray variation of Case Study 2 in chapter 7.2.
Relations	None

Guidance Point 24 Cable in open air

Issue	Modelling of cable installations in open air
Guidance	Ampacity calculations for cable installations that are fully exposed to free air can easily be performed using IEC 60287 formulas. For example, this applies to single cables or groups of cables in free air, where it is assumed that the heat from the cables dissipates rapidly into the surroundings without affecting the air temperature increase. In situations where heat dissipation is somewhat limited, such as cables located near obstacles like trays, and where IEC formulas do not apply, CFD modelling within FEM simulations can be utilised. CFD modelling of cable exposed to open air is different than the modelling of cable in air enclosures as described in Guidance Point 23 above. For most open air cable installation scenarios it is not needed to account for the density dependency of pressure, as the pressure difference is insignificant as opposed to the closed tray scenario, where the pressure/volume interaction with the enclosure should be included in the simulations. Thus, it is usually sufficient to setup the open air CFD model with incompressible solver settings; the density dependence on temperature should, however, still be accounted for - the most widely used approach is here to include the so-called Boussinesq approximation, which is readily available in commercial CFD software, and which accounts for the density variations in non-isothermal simulations in a simplified manner for low computational efforts. For large air domains, it is advised to implement a turbulent flow model, such as not to resolve the complete flow fields (as opposed to the enclosed simulation described in Guidance Point 23 above). Also, as the pressure dependence is neglected here, a stationary solver can be used.

Notes	For further explanation of the modelling of fluid flow in open air domains, please refer to the open tray variation of Case Study 2 in chapter 7.2.
Relations	IEC 60287-2-1

5.3.3 Cable systems in water

Cables installed directly in contact with water boundary conditions can be handled with a heat flux boundary using appropriate heat transfer coefficients. If the cable is covered with porous material, such as rock, extra care must be taken to consider future sediment ingress stopping water flow.

Guidance Point 25 Modelling of cables in free water

Issue	Modelling of open water around power cables (e.g., dynamic cables)
Guidance	When modelling a power cable directly exposed to a large body of water, it is recommended that the heat transfer from the cable surface to the surrounding water is modelled by the use of a heat flux boundary with appropriate heat transfer coefficients. Modelling the outer surface as an isotherm with water temperature could be used as a non-conservative approximation.
Notes	Formulas for heat transfer coefficients for forced and natural convection around horizontal cylinders, thin vertical cylinders and vertical walls can be found in standard heat transfer textbooks. Whether formulas for a thin vertical cylinder or vertical wall should be used depends on the ratio between the cable diameter and the height of the vertical cable section. Derivation of the heat transfer coefficient for submarine cables may require consideration of the depth of laying as it affects the water pressure and temperature, the salinity of the water affecting the heat expansion coefficient in the Grashof number and the water temperature as the dynamic viscosity and the Prandtl number depend on this parameter. Range of validity of Nusselt number correlations (containing the previously mentioned Grashof and Prandtl numbers) should be checked. Note that this may require either manual checks or more advanced post-processing within the software.
Relations	None

Guidance Point 26 Rock cover and concrete matts

Issue	Modelling of sea cable installations with concrete matts, crushed rocks, etc.
Guidance	When setting up a model of a subsea cable covered by a water permeable layer (rock bags, mattresses, rock berms), the following factors should be considered: 1) Sediment mobility: the first step is to confirm if the rock layers are expected to fill with smaller particles such as sands/sediments. This information would be known from a seabed mobility assessment. If the rock layer is expected to become filled, then the thermal condition will be much closer to that of a buried cable. It would be appropriate to model the rock covering as an additional solid layer, with the thermal conductivity defined as a volume weighted average of the rock and sediment conductivity. This avoids the need for modelling of fluid flow or convective behaviours. 2) Pore size: if the rock layer is not expected to fill with sediment and the pore size is large, such that free water can flow in and out of the berm, the rock will have a modest impact on the cable temperature. An example of how to treat this case can be found in [112]. The rock layer may still be represented as a solid medium, but with equivalent thermal properties. In any case, the sea water above the rock layer may be treated as an isotherm, meaning that it is not necessary to model the fluid flow itself.

Notes	When modelling, take care to consider the full life cycle of the cable installation when assessing the most limiting assumptions.
Relations	None

5.4 Material Properties

The material properties of the cable components and the environment can have a significant effect on the current rating of the simulated cable system. Material properties at a project specific level need to be clearly identified and agreed between the client and contractor. Prior to the start of the simulation work, an information request sheet should be prepared to inform the design basis. A general recommendation is to avoid the use of material properties embedded in software for cable properties. Where defined in IEC standards, guidance documents and component manufacturing data sheets, material properties shall be used according to the defined values. For environmental material properties, such as soil measurement results, intermediate steps are often required to arrive at the usable material properties as input to the simulation models. It is important to clarify who (split between client and contractor) is responsible for the design basis of the system, i.e., who is responsible for the necessary intermediate data analysis, simplifications and interpretations to arrive at the usable material properties as input to the simulation models.

The following guidance points give some advice on what to consider.

Guidance Point 27 Required material properties

Issue	Identify the material properties required in the simulation model
Guidance	First define the type of problem, for example stationary/transient, thermal analysis / electrodynamic analysis / coupled analysis. Then identify the required material properties in the simulation based on the equation set that must be solved, noting that the software menus may request more input than actually needs to be provided.
Notes	As an example, heat capacity is needed in transient simulation, but not in steady state simulation.
Relations	IEC 60287, IEC 60853-2

Guidance Point 28 Selection of Material properties

Issue	Selection of material properties
Guidance	Choose material properties from IEC standards when available. As a general principle it is advised to start from a "blank" material, and add properties as needed. When needed material properties are not listed in IEC standards, refer to specialized well recognized literatures and treat information provided by the cable supplier as the priority in the event that the values specified deviate from those in standards. When the material properties are not defined in the IEC standards, or in the manufacturers' published data, refer to CIGRE TB 640. Material properties can also be made dependent on other variables such as temperature, electric field or frequency. The most common application of this is modelling of soil dryout. The user should ensure that the temperature dependent function is continuous across the desired temperature range and if required implement smoothing rather than allow discontinuous properties that change significantly for a small change in temperature. Check also the Guidance Point 4.
Notes	Please consider that material properties in FEM program build in libraries often will differ from the IEC values. Note also that for thermal conduction calculations where a software will not accept a volumetric heat capacity, it is normally possible to use the IEC 60853 volumetric heat capacity data (C_v) by setting a density of 1 kg/m^3 and then specifying the specific heat capacity (C_p) equal to the C_v value given by IEC 60853. This approach is acceptable only for domains containing solid cable materials and only for thermal

Relations	problems; applying a similar approach for domains containing fluids, or analysis where mechanical properties are accounted for, is clearly invalid.
	IEC 60287-2-1, IEC 60853-2, CIGRE TB 640, TB 880

Guidance Point 29 Material properties from surveys

Issue	Implementation of environmental material properties from surveys to the FEM model
Guidance	Agree with client about how to perform the preliminary analysis, and what data will be entered in the simulation. Also agree with client who should perform that analysis and who is liable for the results.
Notes	It is important to clarify who (split between Employer and Contractor) is responsible for the Design Basis of the system. As an example, soil sampling does not generally provide usable thermal resistivity information that can be directly used, and some processing is needed for the data to be usable in a calculation. In a similar manner, absorption and emission coefficient of the J-tubes are dependent on multiple factors such as the colour and material of the J-tubes painting and although this is a critical input, it may be unclear who is best placed to provide the data (or make a valid assumption).
Relations	None

Guidance Point 30 Meshing of non-homogeneous properties

Issue	Meshing of domains containing materials with non-homogeneous properties
Guidance	Some FEM software allow different domains in the same meshing element. One should try not to have materials with significantly varying properties within the same meshing element, as the model would then consider an average of the different values as property for that element, i.e., meshing should be created for different layers separately.
Notes	As an example, if a soil has different thermal resistivities depending on the depth, a meshing border should be included between the two different depths.
Relations	None

Guidance Point 31 Soil properties for transient simulations

Issue	Transient simulation and soil thermal diffusivity
Guidance	Transient simulations will also require specific heat, which is interrelated with diffusivity and density. In total, in addition to thermal resistivity, for a transient simulation the user will need density, specific heat and diffusivity. Use direct geotechnical measurement results at location if applicable. In case of no direct soil diffusivity measurement results being available, use IEC 60853-2 defined equations to calculate soil thermal diffusivity from soil density, moisture content by weight, and thermal resistivity.
Notes	The quality of the measured values is of great importance for the accuracy of simulation results. Special attention should be taken regarding the units of provided material properties.
Relations	IEC 60853-2, CIGRE TB 880, TB 640

Guidance Point 32 Time dependent environmental properties

Issue	Variation of the environmental material properties over time under transient simulations
Guidance	Either use time dependent environmental material properties as input in case there is statistical information available, or use constant worst case scenario values throughout the simulated time window.
Notes	As example, ground soil temperature and ground water level varies over the years. Proven statistical annual data from reliable data source could be used. In case of lacking proven data, the highest soil temperature and lowest ground water level should be used as input for the simulation.

Relations	None
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5.5 Heat Sources

When modelling cables in FE under different design and installation conditions, understanding the behaviour of heat sources is crucial for obtaining accurate simulation results. Heat sources in cables can be caused by various factors, e.g., electrical resistance, external environmental conditions or neighbouring heat-emitting elements. When dealing with heat sources, various aspects should be examined, e.g., the identification of these heat sources, the electrical power dissipation, the thermal conductivity of the materials, the boundary conditions, the heat transfer mechanisms, etc. This section presents the guidance points for dealing with heat sources and explains how to overcome some challenges in modelling heat sources.

Guidance Point 33 Conductor losses

Issue	Determining when to allow software to compute conductor losses using full electromagnetic model
Guidance	Where there is a valid IEC approximation for proximity effect, then the IEC calculations should be used to analytically calculate the conductor loss. It is advisable to use the IEC values for the DC resistance and the calculation of the AC resistance. These values can be entered into the finite element software and made temperature-dependent in the model itself, then heat sources defined using this resistance. If this is not possible, e.g., due to special cable design or bundle configurations or groups of cables, it is instead necessary to allow the FEM software to calculate the conductor losses. However, great care must be taken in the definition of the conductor in order to correctly capture the electromagnetic phenomena and at the same time ensure a correct geometrical realisation to correctly represent the thermal properties. In particular the DC resistance in FEM must match the data sheet value. For Milliken conductors it is extremely difficult to model the conductor losses in FEM and the IEC approximation should normally be used.
Notes	The filling factor of any stranded conductor being the ratio of the indicative diameter representing the conductive material only to the diameter of the actual outer conductor (including filling material), can be used in scaling of the electrical conductivity of the conductor material. The implemented geometry may, however, affect the induced losses to the metallic screens, and does not truly represent the genuine layout of the modelled cable. 3D effects, such as in Milliken conductors cannot be truly captured from 2D simulations, either.
Relations	IEC 60228, IEC 60287

Guidance Point 34 Sheath losses

Issue	Determining when to allow software to compute metallic sheath losses
Guidance	Where there is a valid IEC approximation for proximity effect, then the IEC calculations should be used to analytically calculate the sheath loss. It is advisable to use the IEC values for the DC resistance. These values can be entered into the finite element software and made temperature-dependent in the model itself. If this is not possible, e.g., due to special cable design or bundle configurations or groups of cables, it is instead necessary to allow the FEM software to calculate the sheath losses. The induced eddy current and/or circulating current losses may be influenced by the proximity effect and should be modelled using a Multiphysics approach. However, great care must be taken in the definition of the metallic sheath in order to correctly capture the electromagnetic phenomena and at the same time ensure a correct geometrical realisation to correctly represent the thermal properties. It is often overlooked that fixed λ_1 values are used over the entire temperature range, typically at the maximum permissible conductor temperature. However, in scenarios where calculations aim to determine temperatures at specific currents, it is advisable

	to include λ_1 as a function of conductor and sheath temperature, as described in the IEC standard formula, but with a variable temperature. Care must be taken to ensure that the conductor and sheath resistance are evaluated at the correct temperature. It is essential to correctly represent the grounding connections using external circuit models, the way this is done is extremely software specific.
Notes	If electromagnetic model is used for the sheath losses it must also be used for the conductor losses.
Relations	IEC 60228, IEC 60287

Guidance Point 35 Heat sources

Issue	Volume heat source vs boundary heat flux
Guidance	When simulating cables, it is advisable to represent heat sources as volume heat sources. The exception to this would be heat sources arising from very thin foils where it may not be desirable to directly model the foil, in this case a boundary heat source can be used, see Guidance Point 9. Boundaries exposed to radiation i.e., from the sun, can be modelled as a surface/boundary heat source.
Notes	Other heat sources, like parallel pipelines, could be modelled as a surface heat source, particularly if the internal dynamics of the source are unimportant or unknown. In such cases, it is important to ensure the accuracy of the total heat per unit length.
Relations	IEC 60287-2-1 section 4.2.1.2

Guidance Point 36 AC dielectric losses

Issue	Modelling of AC dielectric losses in insulation materials
Guidance	Model the voltage-dependent dielectric losses as a uniform 2D heat source entered using the standard dielectric losses formula from IEC 60287 [9], which is as a function of voltage, permittivity, capacitance and loss tangent.
Notes	Since the distribution of the electric field within the insulating material, between conductor and metallic screen is non-uniform, the heat generation due to dielectric losses is also non-uniform and should theoretically be modelled as such. However, the final contribution to overall heat losses and temperature profile under steady-state and cyclic loadings is estimated to be very small. For cables at 400 kV or above it is possible to represent the dielectric loss as function of radius [90].
Relations	IEC 60287

Guidance Point 37 Leakage current

Issue	Integration of the leakage current effects in the insulation of HVDC cables for thermal simulations
Guidance	In general, the dielectric loss is very low for both mass-impregnated (MI) and XLPE HVDC cables compared to the conductor loss and can be neglected at all operating temperatures. Nevertheless, modelling of the dielectric loss should be taken into account at very high voltages, e.g., during models which represent test conditions.
Notes	The electrical conductivity of insulation materials (MI and XLPE) increases with temperature and applied voltage (electric field strength), and thus the dielectric losses also increase, which, if not taken into account, can lead to thermal instability (runaway) and electric breakdown, as the electric field strength is dependent on the electrical conductivity itself. However, in normal steady-state (U_0) and cyclic operation (note: capacitive phase of dielectric loss), dielectric losses are very small (greater for XLPE than for MI) and can be neglected in thermal and ampacity simulations. The temperature increase due to insulation losses can usually be described by so-called conductivity coefficients of temperature and electric field.

<i>Relations</i>	None
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Guidance Point 38 Induced losses from neighbouring cables

Issue	AC losses caused by adjacent systems
Guidance	Neighbouring cables or other electrically conductive and/or magnetic structures should be included to account for the additional induced losses. If the inductive coupling between the cables or other metallic/magnetic structures leads to additional losses that cannot be adequately captured by the known analytical methods (see Guidance Point 33), a Multiphysics approach should be used.
Notes	Cables and pipes that cross at an angle of around 90 degrees have only a minor electromagnetic influence, but may need to be taken into account from a thermal point of view.
Relations	None

Guidance Point 39 Losses from harmonics

Issue	Consideration of additional losses due to harmonics
Guidance	In some cases, harmonics are taken into account by adding a certain percentage to the current at the fundamental frequency, e.g., specified by the supplier of the variable speed drive/motor system. If the currents at each frequency are known, the most accurate method is to calculate the total losses generated by all cable components for each frequency. The total heat loss is the sum of all heat losses for each frequency. See CIGRE TB 640 section 5.3.5 for more information.
Notes	Harmonics are normally damped by the transmission in the power cable. Thus, the amount of present harmonics changes with increasing length. Harmonics are also present in DC cables, depending on the filtering in the converter stations. Depending on the degree of distortion, the impact of harmonics on rating may be negligible.
Relations	CIGRE TB 640

Guidance Point 40 Active cooling

Issue	Modelling of external active cooling pipes
Guidance	Forced convection using water can typically be treated as a surface of constant temperature. In some cases, with long pipelines with liquid cooling close to or inside of the cable, the heating of the cooling liquid from the cables must be accounted for. This must be calculated by knowing the fluid velocity, fluid heat capacity, and the heat transfer from the cable to the fluid. Computational fluid dynamics can be used if no other methods are applicable.
Notes	Reference is made here to [113] and [114]
Relations	None

Guidance Point 41 Magnetic armour losses

Issue	Magnetic armour losses
Guidance	Armour losses can be estimated using 3D finite element models or numerical/analytical methods. However, the use of a coupled electrothermal model, which takes into account both armour losses and heat transfer, is usually not computationally efficient. Extensive guidance on this topic is given in CIGRE TB 908 and therefore not reproduced here.
Notes	In case of cables with stainless steel armour, the armour losses are negligible.
Relations	CIGRE TB 908

5.6 Heat Transfer Modes

This section aims to describe the modelling of heat transfer modes in cable ampacity calculations.

In cable systems heat is transferred by three main modes, i.e., conduction, convection, radiation. Understanding the significance of each heat transfer mode present in the system enables an accurate representation and efficient calculation of the analysed cables' current capacity.

From conductor to cable surface, heat is transferred primarily by conduction. The heat transfer rate is dictated by the material thermal resistivity. See in Appendix C.1 for the typical thermal resistivities. For cables with corrugated sheath, surface-to-surface radiation could be used to model the air gaps between corrugation sheath and bedding, although it can be simplified with equivalent sheath diameter and thermal properties (see Section 5.2.4 for details).

From cable surface to the surroundings, in case of directly buried cables, conduction represents correctly the heat transfer (IEC 60287 has covered this case extensively); while convection and radiation are appropriate for cables in fluid-filled ducts and fluid. Depending on the complexity of installation, convection and radiation can be modelled using boundary conditions or a CFD module or neglected for more efficient computation, see Guidance Point 42 on neglecting convection effects.

Solar radiation onto objects can be from direct sunlight, but it can also be sky radiation, reflected short-wave radiation and long-wave radiation. For simple cases, solar radiation can be simulated using boundary conditions on cable surfaces that are exposed to such radiation. For more advanced finite element tools, solar radiation may be modelled directly, such that more accurate results can be calculated.

Guidance Point 43 shows the comparison between direct modelling and heat flux boundary modelling of solar radiation.

Guidance Point 42 Neglecting convection effects

Issue	When can convection be ignored?
Guidance	It can be very difficult to predict when the cooling effects from natural convection within a cavity can be ignored, as it depends on the size of the cavity. For example, natural convection may be neglected for a rectangular air cavity of height <10 mm, however it should also be examined for the specific air cavity of interest. For cable installations where small air cavities are present, it is needed to examine if the total heat loss/temperature profile is significantly affected by neglecting natural convection within said cavities. One approach could be to perform a sensitivity analysis of an air cavity of the size and shape of the one(s) in the relevant case study, but in a separate FEM simulation, including expected max temperature gradients across the cavity, to analyse if the inclusion of convection has any effect on the final heat transfer across the cavity itself.
Notes	Depending on where the air cavity is located, i.e., what the surrounding elements are; for example, air cavities formed between touching cables in a cable bundle should not be modelled as the conductivity of air alone, but instead using either the thermal conductivity of the outer jacket material of the cables forming the air cavities, or by including radiation within the cavities. For cables in ducts also see Guidance Point 19 for guidance on how to model air using IEC 60287-2-1.
Relations	None

Guidance Point 43 Solar radiation

Issue	Solar Radiation (direct modelling or heat flux boundary modelling)
Guidance	For large scale models, representation of solar heating via a heat flux boundary is relatively efficient and simple to set up. However, direct modelling of solar radiation could be used if desired.
Notes	Solar radiation can be modelled as a boundary condition applied to the cable surface, see case study 3 as an example, usually in the form of application of a W/m ² heat influx. Provided standard values may be corrected for the solar azimuth angle using analytical formulas. It allows fast computation but may result in lower

	accuracy as it may not take into account the geometry of the cable and its surrounding environment.
	Direct modelling of solar radiation in the form of solar irradiance physics directly implemented within the simulation, allows simulation of radiation distribution on the surface of the system as well as associated physical phenomena for different environments. It is more versatile for various arrangements and it can be set up more easily for some software tools.
Relations	None

5.7 Boundary Conditions

This section aims to provide guidance on how to set boundary conditions used in FE simulations for thermal rating. Using the correct boundary conditions ensures that the model accurately represents the problem and behaves according to the involved physics.

This includes for instance how the outer boundaries of the surrounding soil is defined in case of a buried cable, or the cable surface in case of cable in free air. In addition, some advice is given on how to exploit symmetries to reduce model size and complexity.

The advice is given based on the shared experience of the group based on commonly available FE software. The user should explore the capabilities of the FE software that is being used, and apply the guidance accordingly.

Guidance Point 44 Surface boundary for buried cables

Issue	Buried cables - isothermal surface
Guidance	The isothermal surface approximation can be used for submarine cables, and for land cables if the burial depth is sufficiently large that the heat from the cable would not significantly increase the ground surface temperature (could be determined by a sensitivity analysis), reference made to [90]. The temperature at the surface should be set to the maximum expected ground temperature at the burial depth, as if the cable was not there. For shallow burial where isothermal surface is not possible the ground-air surface should be modelled with a convection heat transfer coefficient or convection and surface to ambient radiation.
Notes	If land cables are located close to the surface, with a depth of burial less than ten times the overall diameter of the cable, the isothermal assumption could be optimistic.
Relations	IEC 60287 and IEC 62095

Guidance Point 45 Side boundaries for buried cables

Issue	Buried cables - thermal insulation (no heat flux)
Guidance	If the simulation model is sufficiently large, the no-flux condition can be applied at the boundaries. A sensitivity analysis should be performed to ensure that increasing the model size further does not affect the results. Some software may allow to simulate an infinite region at the edge of the modelled domain. This could also be used as an alternative to increasing the size of the simulation model.
Notes	The thermal insulation boundary condition means no heat flux is going through it.
Relations	None

Guidance Point 46 Symmetries

Issue	How and when to include symmetries
Guidance	Symmetries can be used to reduce model complexity - when it is applicable. For instance, for a single core cable including an accessory such as a CPS or joint, the axial symmetry could be exploited. In a 90° crossing between two cable systems, it is sometimes sufficient to model one quarter of the installation due to symmetries (see Case study 5 in chapter 7.4 of this brochure). Some software allow periodic boundary conditions ("short twisted 3D models"), which is helpful for e.g. reduced the model size when there are twisted geometries. Geometric symmetry must never be mistaken for full physical symmetry; a classic example is for cables in flat formation with solid bonding, where the outer leading and outer lagging phases have different sheath currents and therefore different losses.
Notes	When implementing symmetries, boundary conditions need to be carefully considered depending on the physics used in the model. As an example, see Guidance Point 45 on thermal insulation, which is equivalent to having the exact same model (although mirrored) on the other side of that boundary. Since both sides of the symmetry lines would be identical, there would be no heat flux through that boundary. Thus, thermal insulation is equivalent to a mirror symmetry, and can be used to reduce model size by exploiting symmetries.
Relations	None

Guidance Point 47 No magnetic flux outer boundary

Issue	Electromagnetic - no flux outer boundary
Guidance	When using a coupled electrothermal model the outer boundary can be modelled with no magnetic flux. The simulation model should be sufficiently large to not affect results. This could be determined e.g., by a sensitivity analysis. The size of the domain for an electromagnetic solution does not need to be the same as for a thermal solution.
Notes	The required size of the simulation model depends on the application. For example a short-circuit with current returning in ground requires a significantly larger model than a balanced 3-core cable. For electrical simulations involving frequency content, if the model does not include an infinite edge or domain, the model has to be sufficiently large so multiple electrical wavelength can be simultaneously present to allow certain transverse electric and magnetic modes to occur.
Relations	None

Guidance Point 48 Boundaries for CFD

Issue	Choosing the right specific boundaries for CFD simulations
Guidance	For CFD simulations, boundaries are of great importance, and it is crucial to model the boundary conditions correctly as these can have a huge impact on the adjacent or surrounding fluid domain. Several types of CFD specific boundaries exist, and depending on the type of fluid dynamic simulation, e.g., open air vs closed air, and natural vs forced convection, etc., proper boundary conditions must be defined. For CFD simulations in both 2D and 3D, it should always be checked if the geometry exhibit symmetry, and hence can be split up, i.e., only simulate some part of the domain, for example one half of the domain - if so, the symmetry type of boundary condition should be defined on the axis of symmetry. E.g., for a symmetrically laid cable formation with cables exhibiting identical sheath losses. Some care must, however, be taken with regards to symmetry boundaries for CFD simulations, especially ones that may include vortex shredding where small perturbations can

	lead to large scale impacts on the fluid flow field. Here, despite geometry obeying symmetry, the symmetry condition can break down and not be valid. One of the most important boundary types is the no-slip boundary condition which enforces zero bulk velocity of a fluid at a solid boundary. The no-slip condition at solid boundaries is generally used for the majority of fluid flows relevant to fluid engineering, and should by default be defined at solid boundaries of cable ampacity CFD simulations as well. However, in some scenarios it can be beneficial to deploy the slip boundary condition, for example in situations where the open air domain is defined by some external boundaries mimicking an infinite domain. Here it can improve robustness of the converging of the solver by using slip conditions to prevent additional friction to take place. Another used kind of boundary condition type is the inlet/outlet definitions. The inlet boundary condition specifies conditions for fluid entering the simulation, such as velocity, temperature, and pressure, similar to the outlet type, which defines conditions for fluid exiting the simulation, often involving exit pressure or flow rate. The inlet and outlet boundary conditions are normally both used at the same time, and they can be utilised for stable convergence of several types of fluid flow simulations, including natural convection types, e.g., cable in open air where the inlet/outlets may be set to constrain the pressure at specific boundaries.
Notes	All of the above boundary types have been used in Case Study 2 to highlight each of their usability in CFD modelling for cable ampacity simulations, and how the unique features of each type can be exploited to setup a fast computing model for an engineering problem involving cables surrounded by air.
Relations	None

Guidance Point 49 Heat transfer coefficients

Issue	How to select appropriate heat transfer coefficients
Guidance	Depending on the geometry, for example a long horizontal cylinder could represent a cable in air, there are formulas available in literature to calculate a heat transfer coefficient. Some extra guidance is given in the following bullets. • Characteristic length selection of the geometry which is employed in characteristic numbers, such as Nusselt or Rayleigh numbers. For instance, the cable outer diameter, D, is the characteristic length in the long horizontal cylinder case. In the same fashion, annulus gap, δ, is the characteristic length in the J-Tube analytical formulation and the cylinder height, H, is the characteristic length in the case of vertical thin cylinder. • Geometry which the respective convective heat transfer coefficient is applied on and the possible convection effects which may not be compatible. For instance, long horizontal cylinder formulae shall not be applied on a vertical thin cylinder, because the flow is expected to be significantly different due to buoyancy effects. • Consider the mutual convective effects. Although long horizontal cylinder formulae may be applied in case of a horizontal flat formation with sufficiently spaced cables, it may be debatable in cables laid in a closer horizontal flat formation or a vertical flat configuration where the velocity field of one cable clearly affects that of an adjacent one. • Under natural convection conditions, surface-to-surface radiation is often the dominant heat transfer mechanism.
Notes	Note that empirical formulas do have uncertainties even within the range of applicability. Care should be made to include safety margins for this.

Relations	Some more information on empirical heat transfer coefficients are given in Appendix C.3.
	Case study 3 and 6 shows examples on how heat transfer coefficients can be used.
	None

5.8 Meshing

This section describes the meshing procedure and its pitfalls. Most FEM software provide some default mesh to the geometry based on physics settings. In 2D simulations this is in most cases good enough, possibly with adjustment of mesh size.

In 3D simulations the geometry usually is too complex for the default mesh to be both accurate and small enough. The meshing scheme depends on the geometry that is being studied and needs to be carefully considered to represent the geometry and the physics in a correct way. As cables are modelled as concentric cylinders in 3D a swept mesh is often a good choice both in good geometric representation and in limiting the mesh elements.

A FEM simulation computes results for each mesh node, and in-between each node the result will be interpolated. The interpolation method is described as element order and selection of this can have significant impact on the result. It is recommended to validate by ensuring that the result is not affected by increasing the order, or consulting recommendations from the software provider.

It is recommended to check that the mesh is fine enough by performing a mesh refinement and check that the results are not affected by this.

Guidance Point 50 Default mesh

Issue	Mesh driven by the physics
Guidance	FEM programs often provide a default mesh for the geometry with element size based on the physics to be solved. It is usually possible to adjust the mesh element size to coarser or finer level. A mesh refinement study should be undertaken to ensure that the mesh is suitable. If meshing of certain geometric elements results in mesh elements of poor quality, one solution could be to update the geometry. See chapter 6.
Notes	In 2D simulation, this is often sufficient while in larger 3D models it might result in either too many elements or a not representative geometry. See also Guidance Point 53.
Relations	None

Guidance Point 51 Boundary layer mesh

Issue	Boundary layer meshes for air flow
Guidance	When conducting flow simulations, to completely resolve the boundary layer the mesh must be very fine close to the wall. Wall functions are a possible option to reduce the need for a fine boundary mesh. For further discussion see case study 2.
Notes	
Relations	None

Guidance Point 52 Thin conductive layers

Issue	Meshing thin conductive layers
Guidance	Thin layers in geometry will require a very fine mesh to resolve, resulting in many mesh elements and by that long simulation times. Avoid thin conductive layers if possible, for example by combining it with the nearest conductive layer. If the layers cannot be avoided, an alternative would be not to define them within the geometry, but to set a thin boundary layer with the appropriate physics.

Notes	See also Guidance Point 9.
Relations	None

Guidance Point 53 Mesh in 3D models

Issue	3D meshing strategies for cable specific applications
Guidance	Cable geometry is not changed along a cable which makes it suitable for swept meshes. The recommendation is therefore to create a swept mesh along the cable(s). The remaining objects are meshed with tetrahedral mesh.
Notes	Adding a larger cylinder or block around the cable just for meshing purposes can in some cases help to solve issues with matching tetrahedral elements to the swept surface. In some cases, also the ground can be swept. When using the swept mesh in a domain, it is important to coordinate the meshing of the adjacent domains. Significant difference in the mesh size of the adjacent domains can result in a skewed and low-quality swept mesh.
Relations	None

Guidance Point 54 Mesh quality

Issue	Mesh quality
Guidance	The FEM Software provides information on mesh quality based on size ratio of elements. If element quality is poor this could affect the result. However if the poor quality element is in a region with a low gradient, the effect might be neglectable. To validate the mesh a sensitivity study should be performed. The results should not significantly change when the size of mesh elements is reduced.
Notes	See also chapter 6.3.7, Suitability of mesh.
Relations	None

Guidance Point 55 Conducting layers

Issue	Meshing of conducting layers
Guidance	When modelling current and induction it is important to consider the skin depth of the metallic layers and ensure that the mesh size is sufficient to resolve the skin effect.
Notes	Validation could be achieved by hand calculations of the skin depth of the specific layer or by mesh refinement until the result is stable.
Relations	None

Guidance Point 56 Element order

Issue	Selecting of element order
Guidance	The element order describes how the results are interpolated in-between each mesh node. The interpolation could be of different orders, for example linear or quadratic, and this could affect the result. The choice is a trade-off between accuracy and computational resources. A sensitivity analysis could be performed to ensure that it does not influence the result.
Notes	Some software provide settings of this based of selected physics. This is a general setting and depending on your modelling this might still need to be changed just as mesh size. See also chapter 6.3.7, Suitability of mesh and the discussion in Section 4.3.
Relations	None

5.9 Solution Process

The final step in modelling is the process of solving the matrices of equations that define the model. The solvers are simply computer programs that execute numerical algorithms that solve matrix equations. The user often has no control of the algorithm, but can choose the solver used and adjust various parameters of the solvers. It should be mentioned, that every solver is different and there is no single solver that would allow for solving all cases, but rather each of them is tuned for certain cases while not for others (some are better for 2D while others are better for 3D). For a beginner, it is recommended to use the default solvers recommended by the software package used. The choice of solver depends on the size of the model which, in turn, depends on the dimensions (1D, 2D, 3D) of the model, the mesh size and the equations defining the physics and the boundary conditions. In general, the bigger the model, the more difficult it is to solve, but the type of physics may significantly affect the feasibility of finding a proper solution of the model.

It is possible that a model converges to a solution, but the results are incorrect (e.g., temperatures are too high or too low, or lower than ambient at the presence of heat sources, etc.). This means that there is a convergent mathematical solution for an incorrectly defined model, but the solution is also incorrect. The verification process is described in chapter 6.

One of the most common issues during the solving process is lack of convergence. There may be various reasons for this, and while some models may take long time to converge, the others may benefit from the following:

- Verify that the model is properly defined. Conflicting boundaries, areas of extremely high flux, or singularities may not allow the model to converge.
- Split Multiphysics problems into separate models.
- Decrease the number of non-linear dependencies or material parameters (approximate to linear or constants if possible).
- Improve mesh (see guidance points above).
- Use transient to get the initial conditions.
- Change the solver (to more stable one if possible).
- Users familiar with numerical solvers may be able to adjust parameters of the solver to increase stability.
- Gradually ramp up the load by running several stationary computations, each time increasing the load and using the results of the prior study as a starting point for the next study.

Monitoring the solution process during transient calculations (where the variables change in time) may give an indication of the convergence process and the quality of the model. Some software packages may provide other metrics describing the solution error with the steps taken by the solver, which can also be beneficial to the experienced user.

Another issue of the solution process may be the time required to solve the model. This is mainly applicable to two cases:

1. 3D models
2. Transient calculations

Both can benefit from the simplification of the model described in guidance points above. In addition, the user may consider increasing the time step in transient calculations, but this requires extra caution so the results follow the actual physics. For example, the temperature profile (solution) should follow the load profile (input) and if it does not, smaller time steps may need to be enforced.

Guidance Point 57 Large models

Issue	Solving large models
Guidance	Simplify the model, search for symmetry, combine thin layers into a bigger one, modify mesh
Notes	Simplification techniques are provided in CIGRE TB 640, TB 880, TB 908
Relations	CIGRE TB 640, TB 880, TB 908

Guidance Point 58 Model fails to converge

Issue	Convergence of the simulation
Guidance	Is the model properly defined? Are there conflicting boundaries, areas of extremely high flux, or singularities? Improve mesh. Use transient solver to get the initial conditions, or instead of steady state solvers. Simplify model. Change the solver to more stable one if possible. Adjust parameters of the solver to increase stability. Check for non-linearities in material properties and if feasible, substitute with linear equations, e.g., the order of the equation representing thermal conductivity of air can be reduced for the expected temperature range (0-100 °C).
Notes	Convergence issues may also relate to wrongly defined values of material properties, such as a temperature dependent material with a steep change.
Relations	None

Guidance Point 59 Convergence criteria

Issue	Defining when the solution has converged
Guidance	Check default convergence criteria in the software. A smaller target value will improve accuracy with cost of simulation time. Where the convergence criteria is relative, not absolute, and temperatures are calculated in Kelvin verify how large a temperature error could be permitted by the convergence criteria and consider if this is appropriate in the context of other sources of error in the model.
Notes	Consult user guide for software package as convergence criteria depends on software. A refinement of the convergence criterion can be made to assess the change in resulting temperature.
Relation	None

Guidance Point 60 Accuracy of calculation

Issue	Accuracy of calculation according to provided data
Guidance	Depending on accuracy of the input data, the tolerances of the model should be adjust. With more accurate input data the tolerances should be reduced and with less accurate data the tolerances should be increased. The least accurate data governs the overall accuracy. Adjust the expectations to the significance of the results. With more accurate data increase confidence in results, and vice versa. Adjust the accuracy of the reported data according to the provided data by limiting the number of decimals based on the input data.
Notes	Be cautious of apparent accuracy in results when provided data is lacking thereof. Intermediate results of calculation shall not be rounded unnecessarily, e.g., let calculation software store values in its standard or maximum precision.
Relations	None

5.10 Transients

Where there is a variation over time in the cables' current or ambient conditions, the model requires some additional attention. Most obvious, the variable conditions need to be known and fed into the model. Further, in addition to steady-state problems, some extra properties are now relevant, like the thermal capacities of the materials. The timesteps for the changes and for the solver need to be carefully chosen to be meaningful in the solution process. On the other hand, the computational burden on the calculation increases significantly. This usually requires a balancing of details in

geometry and timesteps against the required computation time without losing significant information in the results.

As the basis for any transient studies usually is an existing steady-state model, its validity must be checked to begin with. The guidance points below then provide useful hints at what to look for during creation of the model. In any case, a sensitivity study of the transient model is advised to check if the model reacts to the time-variable parameters as expected.

Guidance Point 61 Volumetric heat capacity

Issue	Use of volumetric heat capacity term
Guidance	Many FEM software will expect to receive the heat capacity in units of $\frac{\text{J}}{\text{kg}\cdot\text{K}}$ as input. Make a conversion from volumetric heat capacity C_V (in $\frac{\text{J}}{\text{m}^3\cdot\text{K}}$) to specific heat C_p (in $\frac{\text{J}}{\text{kg}\cdot\text{K}}$) by using the relation $C_p = C_V/\rho$, where ρ is the mass density in $\frac{\text{kg}}{\text{m}^3}$. Internally in some software only the product ρC_p is used. In such software, if using IEC 60853 specific heat capacity values, the density can be set to $1 \frac{\text{kg}}{\text{m}^3}$ and the volumetric term with units $\frac{\text{J}}{\text{m}^3\cdot\text{K}}$ can be used directly provided that the region in question has only conductive heat transfer, and that the density is not used for other calculations.
Notes	Care must be taken for the latter approach for any convective heat transfer calculations where the model will also use the density in defining the buoyancy force.
Relations	IEC 60853-2

Guidance Point 62 Optimized meshing for reduced computation time

Issue	Reducing computation time by optimized meshing
Guidance	Follow the guidance in Guidance Point 9 concerning the handling of thin layers. In addition: • For combined layers, apply a volume weighted heat capacity. • Consider the use of structured mesh, especially in the axial direction if the temperature change is gradual.
Notes	
Relations	None

Guidance Point 63 Loss-load factor

Issue	Use of the loss load factor
Guidance	Where the loss load factor for a circuit is known, it can be used to: • Initialise a transient model to the expected long term average steady state condition. • Represent the average losses from an adjacent or crossed circuit, which is not the circuit being rated by the model • In case no load profile is known, to create a load profile matching the loss load factor, e.g., using the flat top load cycle mentioned in IEC 60853-2.
Notes	
Relations	Definition of loss load factor according to IEC 60853-2

Guidance Point 64 Initial conditions for circuits in operation

Issue	Definition of initial conditions for a circuit that has been in operation for a long period of time
Guidance	Loading: If a transient response must be calculated on the assumption that a circuit has been in operation for a long period of time, the best option is normally to initialise the transient model using a steady state calculation under a load which represents the average losses over a long period of time. Due to the heating effects from reactive currents, it is essential to use the long term average apparent power (unit MVA), not the long term average active power (unit MW). Environmental conditions: If following the above approach, then the user must carefully consider the correct effective ambient temperatures to use. Depending on the thermal time constant of the model, the choice is either to use the ambient conditions applicable at the start of the transient, or to use the long term averaged values for the ambient temperature.
Notes	If following the above guidance, the user must always consider carefully the risk of discontinuity of ambient temperature at the start of the transient. For example, if a transient calculation is intended to represent a 24 hour period during high summer, the ambient temperature seen by the cable may be above the long term average. If this is a concern, then it may be necessary to initialise a short term transient using a long term transient model.
Relations	None

Guidance Point 65 Solver step size in transient models

Issue	Solver step size and total simulation time in transient models
Guidance	In order to choose the correct time step to use, the user is recommended to follow the steps below: 1. Review the factors driving the transient (loading or environmental conditions) and assess the size of the important features. 2. Consider the time constant of the model itself (e.g., surrounding of a cable in air has a very short time constant compared to a deeply buried cable). 3. Fix the maximum step size such that all important features from step 1 are captured (e.g., if a peak is 6 hours long, time steps not bigger than one hour might be appropriate). 4. Consider forcing the solver to calculate at certain time steps, or manually refine the maximum time step in periods of time with big changes in load. 5. During verification, compare the transient conductor temperature profile with those parameters which are varying in time, and ensure that the expected features are being captured by the model. If in doubt, solve the model with a smaller time step and compare the solutions to confirm. 6. For continuous and cyclic loads, the total simulation time shall be set considering the time constant of the model, to ensure that steady state is reached.
Notes	The above guidance is general, and running sensitivity studies to verify the performance of the model under different time stepping constraints is highly recommended. Be aware that the output times of the model is not the same as the model step size, and some software can provide output data for time steps which were never solved for.
Relations	None

Guidance Point 66 Time varying ambient conditions

Issue	Handling time varying ambient conditions
Guidance	The impact of variable ambient conditions will vary significantly depending upon the thermal time constant of the model. The user should consider the following factors: 1. For a buried cable at depths greater than 0.5 m, daily variations in ambient conditions will not impact the temperatures at burial depth. Therefore it would be appropriate to use a longer term moving average to capture the seasonal changes, while disregarding the daily fluctuations. This can be achieved by specifying a time varying isothermal ground surface boundary. 2. For shallow buried cables such as those in filled surface troughs, the time constant will be closer to that of a cable in free air. If the user seeks to model short term transient responses (i.e., less than one day) then the applicable thermal conditions must be chosen with great care, and in full consideration of what assumptions may be suitably conservative. 3. For cables in free air, especially those exposed to solar radiation, conditions may vary rapidly and the time step needed in the solver may be dictated by the ambient conditions rather than the load.
Notes	The above guidance is general, and running sensitivity studies to verify the performance of the model under different time stepping constraints is highly recommended.
Relations	None

Guidance Point 67 Simulation of losses in transient models

Issue	Transient models featuring electromagnetic calculations of losses
Guidance	Where the cable losses are represented by analytical equations, the time step must only consider the requirements of the thermal part of the model. Where the cable losses are directly solved for in a coupled electromagnetic and thermal model, the time constants of the two different sets of equations are vastly different. Correctly representing the electromagnetic problem will require time steps in the order of microseconds to correctly resolve the sinusoidal variations in voltage and current. Clearly it would be impractical to apply the same approach to the thermal model, as even a 6 hour transient thermal response would then feature many millions of time steps. This would result in a huge computational burden. There are two possible strategies to overcome this: 1. Some software packages allow the use of combined frequency dependent and time dependent solvers. This allows the electromagnetic problem to be solved in the frequency domain, the losses extracted and then passed to the thermal solver. The software then automatically iterates the electromagnetic problem as the temperature change begins to affect the conductivity of the metals. 2. Where the above option is not available, it may be necessary to manually couple the two models. This can be achieved by a segregated solution process, where the electromagnetic solver is run to obtain the losses under the initial thermal conditions, and then the losses are passed into the thermal model. The user will then need to define when to update the loss calculation based on changes in the temperatures of the conductor, screen and armour. 3. A third possibility is to use the electromagnetic model to derive the losses in each component as a function of temperature and current in that component. The user would then derive a set of polynomial functions to represent the losses. While this can appear to be an elegant solution, it is important to note that the presence of proximity effects could make these polynomial functions very complex. This also tends towards the base case of

	modelling only the thermal response, and using the IEC equations to capture the losses.
Notes	In all of the above cases, the user must take great care to avoid accumulated error in the losses. More frequently updating the loss calculation will ensure that the right amount of heat is dissipated into the medium surrounding the cable, but at the expense of slower solution times and larger memory requirements.
Relations	None

6. Verification Process

This chapter presents a summary of the verification process that should be undertaken before results from a finite element model are used in a cable project. It is primarily aimed at engineers who are undertaking the modelling work. Note that verification is not considered the same as validation; the purpose is not to “prove” that the FEM results are correct, rather to verify that appropriate assumptions have been made and that no mistakes have affected the solutions.

6.1 Why is verification needed?

The mere fact that an FE model solves problems and produces results that can be visualized does not guarantee the correctness of the FE model or the accuracy of its output. It is possible to obtain solutions which are mathematically valid, but are complete nonsense in physical terms (for example, inverted gravity, negative electrical resistance etc). When conducting FEM, users are faced with numerous decision points and are required to input various parameters manually. Errors can manifest in various facets of the modelling process, including:

- **Physics:** For example, applying incorrect boundary conditions, relying on wrong simplifying assumptions or utilising non-representative heat transfer models.
- **Numerical Solution Procedure:** Errors may arise from aspects like choice of mesh, solver options, time-stepping methods, and more. Convergence may be achieved even though these errors exist.
- **User-Input Errors:** Mistakes in typing parameter values or auxiliary formulas can lead to inaccuracies.
- **Post-Processing:** Post-processing may often require a steady knowledge of calculus and vector calculus, and mistakes may occur when extracting resulting values or plots.

These errors, regardless of their nature, can significantly influence the simulation results to varying degrees.

The more complex the model, the more prone for error it may be. Firstly, a more complex model requires a greater number of decisions and manual inputs. Secondly, complex problems may be harder to solve numerically and require changes to the default solver setup, which can introduce additional risks. Lastly, the complexity often makes it more challenging to compare results with analytical formulas, limiting the applicability of certain verification methods.

As the probability is not zero for such aforementioned errors to occur, verification should be performed to reduce the risk that errors in FEM go unnoticed. If such errors do go unnoticed, it could be detrimental, as decisions and actions may be based on the erroneous results.

6.2 What is out of scope?

The verification process should focus on the model, hence the following are considered out of scope:

- The way the software works - this should be verified by the vendor through benchmark cases, where the analytical solutions exist,
- Validity of physics models in software or their implementation,
- Performance of solvers, as they are simply an implementation of numerical methods, thus are physics agnostic.

6.3 General principles for all models

The sections below identify some general principles which should be applicable to almost all cable thermal calculations which are conducted by finite element analysis. For a relatively simple 2D model containing a small number of cable circuits, this may be sufficient by itself. Section 6.4 which follows goes into more details for models which have a range of different features.

6.3.1 Verification of geometrical parameters

Verifying the geometry of the system “as calculated” is the first step of a verification process. Geometry should be checked visually to identify major errors related to main item dimensions and position in space. Special attention must be paid to identifying unintentional overlapping areas, which must be avoided. In order to avoid output errors due to inconsistent units, numeric data associated to geometry should be also verified in detail.

Some of the common mistakes listed in CIGRE TB 880 (2.4) also apply here:

- Diameters and thickness are usually expressed in millimetres (mm), while the engineer usually relies on depths and axial distances expressed in metres (m) given in a dimensional drawing.
- Confusion is possible for developers or users concerning the angles provided in degrees and provided in radians for trigonometric calculations.
- Cable depth is sometimes expressed as the depth of the cable centre, sometimes the depth to cable top, duct top etc.
Similarly, the distance between cables/circuits may either refer to the distance between axes or the free space separating them; see Figure 6 - 1 below as an example.

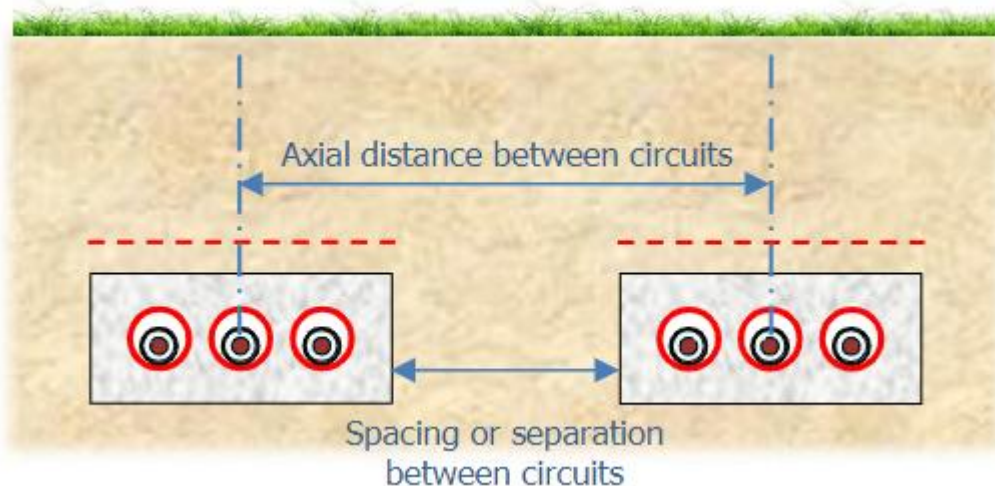


Figure 6 - 1 Example of two different distance definitions of underground circuits in backfills

Contradictions may appear between the cable diameters and its stated thicknesses in the datasheet, in such case it is recommended to check that the diameter of the model match the one in the datasheets (CIGRE TB 880, guidance point 11). Note that some thin layers (swelling tapes, glue above laminated foils) or complicated ones (wire screen) may be simplified for modelling purposes provided that their thermal influence is captured correctly.

This is especially important when models are generated by changing parameters: a consistent geometry may not remain so after changing parameters, either due to unexpected situations (e.g., elements being placed at a location where they become crossed by a separation) or simple inputting errors (hard-coded values not reflecting the change in parameters).

6.3.2 Material properties

The material settings in the model shall be verified to ensure that the expected material properties are used.

For this purpose, when building or modifying a model it is recommended to check that:

- The materials are correctly assigned to the corresponding elements of the model, and that each material has the required set of properties to solve the equations.
- The values and units of the material properties are as intended. When using default materials from the software libraries, it should be checked that the material properties are as intended for the specific model.
- For material properties defined as an analytical function of a variable in the model (typically temperature dependent properties), it should be checked that the functions are correctly implemented. In case of step functions, for example soil dry-out, there should be a smooth transition both to reproduce reality more accurately and to avoid convergence issues when solving the model (ideally the function should be continuously differentiable).
- For materials defined by an external set of data, it should be checked that the set of data has sufficient resolution, and that it has been correctly interpolated by the software.

After executing the FE simulation, it is advised to confirm that the behaviour of the materials was as expected. This can be achieved by:

- Sensitivity analysis of relevant material properties (i.e., the properties with the largest impact on the simulation results).
- In particular, one should mind that the primary heat transfer mode is not always direct conduction. In some conditions, radiation becomes the most important factor, and the parameters governing it need to be checked.
- Plots of the material properties defined as a function of another variable in the model.

6.3.3 Validity of intermediate equations

The validity of user-defined equations in an FE model shall be verified to ensure that the equations are implemented as intended.

As a first step, it should be checked that:

- The units of the equations are correct (especially K vs °C). For example, if using the heat transfer coefficients for cables in air from IEC 60287, these are in units $W/m^2 \times K^{5/4}$ and great care should be taken when applying them.
- The definitions of the variables are consistent across the model.
- The user-defined equations are not overridden by the software's internal equations.

Dimensional analysis checks can be performed to confirm the dimensional consistency of the equations.

Furthermore, plotting the intermediate equations as graphs can help to verify that the equations are correctly defined.

After executing the simulation, further verification may include:

- Plotting the dependent variables of the equation solved by the FEA software to confirm that the results are as expected.
- Sensitivity analysis of the dependent variables for different model inputs.

6.3.4 Initial Conditions

Software will often provide a default initial condition if nothing else is specified. Users should verify that the initial conditions used are consistent with the assumptions and do not give a risk of mathematical errors in the initial calculations. This is even more important in transient calculations, which are addressed in Section 6.4.1.

6.3.5 Do not break the laws of physics

Respect the laws of physics while solving a model in FEM. This may sound like an obvious statement, but it is essential to remember that a solution can be physically invalid while also being mathematically valid. For example:

- Conservation of Energy: Energy can neither be created nor destroyed; it can only change forms. Is heat generated in steady state equal to the heat leaving the model? In transient cases, the cable and its environment (i.e., soil) have thermal capacities which are meant to retain the heat and the sum of heat generated should match the sum of heat leaving the model boundaries plus that stored in the thermal capacitances in the model.
- No machine is 100% efficient. This usually means that energy is lost to heat in any system and you can never get back all the energy you put in.
- Temperature cannot reach absolute zero and normally cannot go below the ambient boundary temperature. Errors related to negative temperatures in Celsius becoming negative temperatures in Kelvin (impossible!) can easily occur if care is not taken.
- Check for physically invalid results, e.g., negative electrical or thermal resistances
- Variation Principle: Solve with very low loads to confirm that small load currents result in only a small change in temperature.

6.3.6 Cross check of heat sources

A useful parameter to check (both for troubleshooting and verification) is the consistency of the heat flux.

Checking that heat sources input the same amount of heat as the sinks receive is a first step. The total amount of heat (in W/m) should be in the right order of magnitude considering the conductors/applied current. A quick check on the heat sources against a hand calculation or IEC 60287 compliant method is also useful.

Sensitivity analysis (check input/output behaviour): as a general rule of thumb, small input values, e.g., small load currents on the cable, should lead to small changes in the model, e.g., low cable temperatures, while a large load current should lead to a high cable temperature. Another example could be to vary the bonding to see if it affects the screen losses as expected while the charging currents stay more or less the same. Furthermore, in order to consider and fundamentally evaluate an entire system, it makes sense to apply conservation laws such as the conservation of heat flux, which can be evaluated by checking that the heat generated in the power cables also leaves through the boundaries under stationary conditions.

6.3.7 Suitability of mesh

When verifying the suitability of the mesh for the specific problem, a couple of straightforward methods can be employed. After solving the model and generating temperature field plots, manually inspecting these plots in areas of interest can help identify anomalies. For instance, if the mesh structure becomes visible in the temperature field plot, it may indicate that the mesh is too coarse to accurately resolve the solution.

Additionally, conducting a manual inspection of the mesh itself can reveal potential weaknesses, such as poor mesh quality characterized by, for example, triangular elements with small areas relative to their circumference. Such issues can lead to convergence problems. Manual inspection can also assess whether thin layers have been meshed appropriately. In addition to manual checks, many FEA software packages offer built-in tools for analysing mesh quality.

Another simple yet effective technique, suitable for many models, involves performing a sensitivity analysis on mesh refinement. Starting with the base case, the mesh can be locally or globally refined and the model re-solved. Comparing the base case solution, such as conductor temperature, with the sensitivity results can determine if the original mesh was adequate for the problem.

In flow models, ensuring sufficient meshing of boundary layers is crucial. In large and complex models, the objective is not only an accurate solution but also computational efficiency, requiring a careful balance between the two goals.

For electromagnetic simulations, e.g., where conductor and metallic sheath losses are computed and two-way coupled to heat transfer, correct meshing of the metallic parts is crucial, as in to properly account for electromagnetic phenomena such as skin and proximity effects. For example, very thin metallic layers should be resolved correctly in the mesh sequence to account for the skin depth. Please refer to Guidance Point 55 for further explanation on this topic.

6.3.8 Review convergence plots

If available in software, the convergence plots show the error estimate for each iteration of the FE computation and are useful to verify that the calculation has converged for the given tolerance, resulting in an acceptable error in the outcome of the simulation. If not available, try to find other means to review and evaluate convergence behaviour of the model or software.

To verify that the solution of the simulation is reliable and the computation time is optimal, it is recommended to:

- Check that the convergence tolerance is realistic. A high tolerance may lead to inaccurate results, while a too low tolerance may result in a slow or no convergence.
- Use the convergence plots and the time step size convergence plots to see if they are showing reliable convergence. Note that the time step size convergence plots are particularly useful when using a segregated solver, as they can provide more straightforward information of solver instability.
- Plot temperature gradients and confirm that there are no high gradients in unexpected areas.

- Examine the temperature profiles in the vicinity of boundaries to ensure that they are not artificially constraining the solution.

6.3.9 Post-processing

If temperature values or plots are the only output needed, the risk of errors in the post-processing step is low. However, for more complex analyses where detailed results are needed, a few steps can help to verify the post-processing:

- Double-check that the extracted results have the correct dimensions.
- Verify that built-in quantities provided by the software actually correspond to the specific parameters you are interested in.
- Pay attention to whether the quantities used or obtained in post-processing calculations represent root-mean-square (RMS) or peak values.
- Have in mind the spatial location of where the results are extracted from, i.e., perhaps it is not the hottest spot in the model that is of interest in a particular simulation.

6.4 Recommendations for advanced models

In this section, models that involves more complexity are considered. This includes time dependent calculations, three dimensional models and the modelling of fluids and electromagnetics.

6.4.1 Time dependent calculations

As a first step, verify that the chosen time step is suitable for the simulation, such that variations in input parameters are correctly reflected as variations in the output. This could be done by a sensitivity analysis, following the general guidance in Section 5.10.

Some FEM engines may use adaptive time step in time-dependent calculations. This can be a source of error, as it can cause the time step to become longer than, for instance, a short load variation, causing the software to essentially ignore it.

Additionally, one should be mindful of the initial conditions when performing time dependent calculations that lead to a steady state result: the time constants of buried cables can be high, and a thermal equilibrium may only be reached after years, with temperature variations difficult to see if the simulated time is too short. This applies for example to continuous loads in models with a high time constant, or cyclic loads where the resulting steady state temperature is reached after a number of cycles (see Figure 6 - 2). Please refer to Guidance Point 65 for recommendations on the selection of the total simulation time and time step.

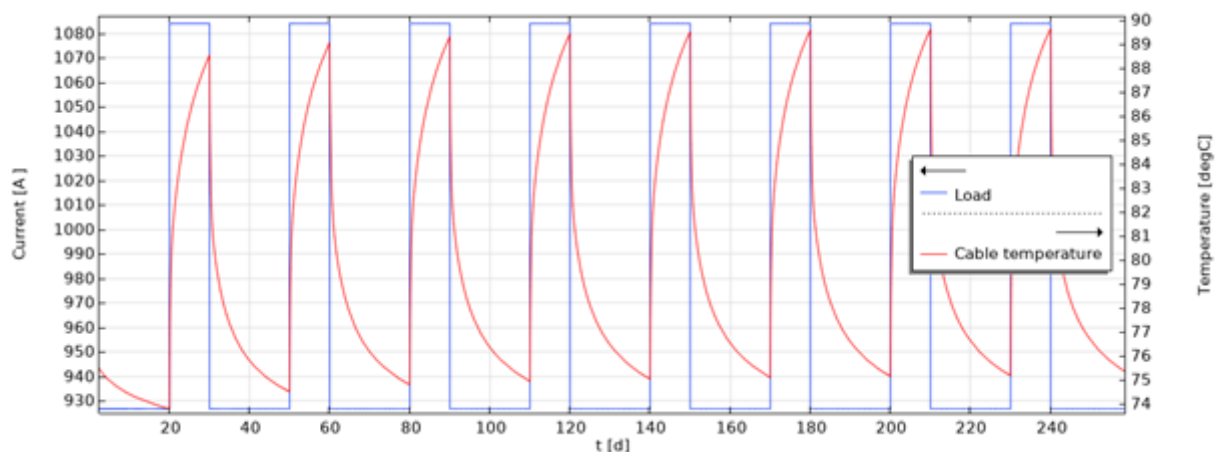


Figure 6 - 2 Simulation of cable temperature for a cyclic load

Having in mind the concerns related to time dependent calculations in FEM, hybrid methods can be considered a valid alternative for such calculations. As detailed in chapter 3.3, hybrid methods consist of applying analytical methodologies to calculate time-varying, cyclic, or emergency loads then included in the FE methods (2D or 3D).

6.4.2 Three dimensional models

The main challenge of the FE 3D models is typically the increased computing time, in particular for detailed geometries. Common strategies to reduce the complexity of a 3D model and optimizing the computing time are reducing the size of the model, removing minor details in the geometry and coarsening the mesh.

The following approaches may help to verify that a simplified 3D model produces reliable results:

- Making 2D models of the ends of the cable system and confirm that the end of the 3D model matches the 2D expectation.
- Making 2D models comparing the detailed and simplified geometry, to verify that there is no significant difference in the simulation results at the areas of interest.

For the 3D models especially the sensitivity analysis of the model size is relevant, to ensure that the boundaries of the model are far enough to not have an undesirable impact in the simulation result, and the sensitivity analysis of the mesh to verify that it has a suitable size and sufficient resolution at the areas of interest.

6.4.3 Modelling of fluids

For small cavities where heat transfer is dominated by conduction and modelled as such, e.g., for water-filled cavities in polyethylene filler elements in horizontally laid 3-core cables, the reduced complexity of the modelling limits the risk for errors. If necessary, a sensitivity analysis can be performed to verify this assumption.

Instead, if heat transfer coefficients are used at interfaces between solids and fluids to model heat transfer in fluids, quite complex expressions may need to be entered into the FEA software, often featuring temperature dependency. Ensuring that these expressions are correctly entered is important. Many of the parameters these expressions consist of are temperature-dependent. If fluid parameters are calculated only once for a single temperature, this should be checked to be a valid assumption.

In the case that CFD is used for modelling the heat transfer in fluids, it is important to verify that an appropriate fluid flow model has been used. General flow models can be more accurate over a broader range of flow types, but they will likely be less precise in predicting heat losses at a specific flow condition, like true laminar flows, which would otherwise be computationally more demanding to solve as the standard Navier-Stokes equations are usually directly computed. Sometimes turbulence models can be used on coarse meshes for rough estimates, where full velocity fields are not solved, and where coarse mesh is exploited for high numerical diffusion for fast computation. Such an implementation is utilised in Case Study 2 of Chapter 7 for cables on an open tray.

In addition to the flow model itself, check if the boundary conditions imposed are realistic. E.g., for a power cable in a vertical riser or J-tube, open vs closed boundaries may have a significant impact on the final temperature solution.

Manual inspection of the solution may have some value. Inspect if the velocity field looks realistic, if there are any anomalies, if mass and energy are conserved, and if the velocity magnitudes and directions are realistic. This underlines the importance of understanding the actual physical response that should be expected.

Benchmarking with a simplified case is an option, e.g., by using FEM with heat transfer coefficients on interfaces instead of CFD.

Compare the FEM solution with analytical or semi-analytical methods, if possible. If not, check if it is possible to simplify the problem slightly to compare with other calculation methods.

Another check involves disabling fluid flow physics to solely model conduction, assessing whether the difference in achieved core temperatures is realistic. This can also be benchmarked with analytical methods when only conduction is considered.

6.4.4 Full electromagnetic modelling of losses

The electromagnetic losses in the cables occur at conductors, insulation, screens, and armour wires. There could also be losses outside the cable if there are magnetic materials present, or other components with induced currents. Below are the steps that can be taken to verify the FEM model used for evaluating cable electromagnetic losses.

Visualization: Visual verification using graphical tools of the FEM software can be done two times, before and after running the simulation.

Visualization before simulation: This ensures that whether the geometry reflects the defined problem and that the relative positioning and orientation of conductors, screens and armour wires are correct. Further, it can be checked whether the assigned meshing is appropriate to address the complexity of the model. The mesh size should be smaller than the respective skin depth inside the conductors, wire screens, and armour wires to ensure accurate estimation of the skin and proximity effects.

Visualizing the mesh can also help detect geometry errors, especially when checking the mesh element density around material boundaries and domain interfaces. An extremely high cell density can indicate a geometry error or other area of concern.

Visualization after simulation: This check can be done by generating contour plots that directly show the resulting distribution of the magnetic and/or electric field around the cable and compare the model behaviour with mathematical and scientific expectations.

Dimensional analysis: Verify if the cable model has consistent units and dimensions for every component/layer.

Material property scrutiny: Recheck whether all the cable components are correctly inputted with the electrical and thermal properties for a given temperature and if required for the defined frequency.

Sensitivity analysis: This can be done by changing the input values and comparing the changes in the FEA simulation results and these should be in the expected range. e.g.: permeability, radius, temperature, and resistance etc.

Bench-marking: A two-dimensional FEA model is commonly used for electromagnetic loss evaluation. Since power cores, screens, and armour wires are helically laid up, a three-dimensional FEA model represents all the physical aspects responsible for the losses in the cable.

Below are the examples which can be used for bench-marking, noting that an exact match on results would not be expected.

- Conductor diameter and conductivity should be modelled such that the results of AC and DC resistances obtained from the FEM simulation match with the results of the mathematical model (for AC resistance using Bessel functions) and datasheet values.
- Losses in the armours can be analytically calculated using the method defined in CIGRE TB 908 and compared with the FEM simulation results etc.

6.5 Comparison against analytical calculations

Comparing finite element analysis results against analytical calculations for simpler cases is a valuable way of verifying the outputs from the more complex model. It is recommended to do so whenever feasible and, in well defined models, it is often possible to reach good agreement. More complex FEM cases may have to be simplified for the purpose of comparison with relevant analytical solutions.

However, the comparison is never as straightforward as it may seem, and it is easy to draw the wrong conclusions. It is essential that the user considers carefully the simplifications which are inherent in both: analytical and FEM calculations. For example, these may include:

- **Metallic layers being isothermal;** this allows infinite circumferential heat conduction. The use of a realistic thermal conductivity for the metal in an FE model, particularly in the case of lead or steel, will lead to some temperature difference around the circumference of the metal sheath.
- **Geometrical simplifications;** analytical calculations may neglect the impact of very thin layers, or may use empirical or theoretical approximations to account for their thermal impact (e.g., thermal resistance T_2 in 3-core cables).
- **Uniform dissipation of heat within regions;** most analytical calculations will take the assumption that heat is dissipated uniformly throughout a region. This could be eddy current losses within a sheath for example. In reality, there may be a proximity effect present. When combined with the finite thermal conductivity of metal in an FE model, this may affect the maximum sheath temperature.

- **Geometrical simplifications of the conductor within the FE model;** certain simplifications of the cable conductor cross-section usually must be taken within FEM, especially for Milliken conductor designs. In IEC the AC resistance of said conductor designs is calculated using empirical factors, which can be used within an FE model as well, however, if the conductor resistance of the FE model is based on geometry and material properties only, great care must be taken.
- **Metallic screen induced currents;** most analytical calculations and IEC is based on screen loss factors that are a function of the screen bonding configuration and the screen design itself. When using an FE model, the metallic screen losses can be defined using the analytical values or using geometry and material properties, e.g., in an electromagnetic coupled simulation.
- **Submarine cable crossings;** analytical formulas from the IEC 60287-3-3 are based on the mirrored image theory where an isothermal ground surface is assumed; temperature profiles of a cable crossing involving, e.g., a post-laid rock berm can readily be simulated using e.g., an equivalent thermal resistivity of the rock berm in an FE model, whereas the analytical approach fails to properly take into account the difference in thermal resistivity of the ground plane and the rock berm in the superposition calculation, and the calculated derating factor cannot directly be compared to results from the FE model.

6.6 How can the verification process be streamlined?

Depending on the strategy taken by the user building and solving the models, it is sometimes possible to streamline the verification process. One of the common ways of streamlining the process of verification, would be to utilize published and previously verified numerical models (scripts). Another method would utilize structuring the model in a way where elements may be checked once, and then reused repeatedly.

6.6.1 Analytical equations for losses

A common approach to implementing the electrical losses in finite element software is to use the analytical equations from IEC 60287 to determine the heat sources, rather than a full electromagnetic solution. This can be advantageous in many ways, particularly if the electromagnetic aspects of the problem are relatively simple. Some software packages will allow sub-routines to be imported in various script languages, or run through an API. If this approach is used, then the individual equation sets within the loss calculations may be verified once, and then the checked version can be used repeatedly without further need for additional verification. This can be achieved by comparing the outputs of the script to known answers, such as those in CIGRE TB 880, which contains examples of all of the major clauses in IEC 60287 (keeping in mind that differences will exist between a purely FEM calculation and one performed analytically).

Where this approach is taken, the extent of the verification will be reduced, but there is still a need for checking that the correct inputs are assigned to the calculation of losses, and that the outputs from that calculation are being used in the correct way. For example, make sure they are being applied to the correct cable, and have they been evaluated at the correct operating temperature.

6.6.2 Use of simplified models for preliminary analysis

Simplified models, such as assuming constant electrical resistance at maximum operating temperature, can be used for initial analyses. This approach reduces computational complexity and time, allowing for rapid assessment of the design's viability. By identifying potential issues early on, detailed simulations can be focused more effectively, optimizing the overall verification process.

6.6.3 Use of template cable models

Where a user expects to model the same cable many times (for example, those working for a cable supplier with a known product range), it may be useful to create a template model of the cable in question. Future models can then be built around a cable geometry, material property data etc, which has been fully verified.

6.6.4 Sensitivity analysis

Sensitivity analysis involves determining how different variables influence the simulation outcomes. This is crucial for understanding the robustness of the cable design under various conditions. By identifying which variables have the most significant impact on ampacity, engineers can prioritize

these factors in the design and optimization process, leading to more effective and reliable cable systems.

6.6.5 Automation of routine tasks, e.g., scripts for geometry creation, mesh, etc.

Automating repetitive tasks, such as geometry creation and mesh generation, through scripting can significantly streamline the verification process. This not only saves time but also minimizes human error. Automation ensures consistency in the setup and execution of simulations, leading to more reliable results and freeing up engineers to concentrate on more critical aspects of the analysis. Some values can also be verified automatically, e.g., by checking if they are in an expected or physically possible range.

6.6.6 Early verification

Verification should be integrated in the process of creating the model early on. When some verification is already done on basic parameters and the basic geometry for example, potential errors can be resolved early and not only after solving the model. Later verification steps should also be considered during the modelling phase e.g., by outputting intermediate values or adding defined probe points in the model. Possible items to check early are input parameters and their units, results of external calculation or customized equations, geometry of first items etc.

6.7 How should the verification process be reported?

There are two types of records that should be kept for the verification process. Firstly, the engineer working with the model should ensure that a detailed (but perhaps informal) record is kept internally to their organisation. This should be sufficient to give a colleague confidence that the model has been verified. An example of why this is important would be another employee working on a project, where the model needs to be opened and updated but the original author of the model is unavailable. The second use case is where a contractor needs to summarise the verification work that has been done and provide the summary to the Employer.

6.7.1 Internal reporting

The internal verification report should include a more detailed, but potentially informal, summary of the work done. The verification process described in this section is mainly addressed to the engineer undertaking the modelling, to confirm the model is acceptable for its intended use. Therefore, not all the steps described below need to be reported to the client. This should include as a minimum:

- List of assumptions, properly justified and reasoned.
- List of the limitations of the model and the software.
- Verification of the cable model in a simple configuration against the IEC analytical calculations, or against already validated calculations, simulations or test results. The extent of this will depend on the type of model, as described in Section 6.5.
- Summary of the checks carried out in the model building and simulation process. This may not include all of the results, just a confirmation that the checks were made and that the results were acceptable.
- Documentation of any relevant functions implemented by the user; this mainly applies to larger pieces of code which are interfaced with the model and not to individual equations. The documentation should be sufficient to identify which code has been used, whether that code is itself verified and any major limitations to its use.
- Reasons for the use of specific boundary conditions and major model inputs confirmation that any relevant sensitivity checks were completed.
- Summary of the sensitivity analysis of the mesh, with a note on the recommended settings.
- Version number of the software.

6.7.2 External Reporting

The reporting of the FEM calculation should include a dedicated section or separate report covering the verification of the model. A proposed level of detail of the external report on model verification is shown below. Any additional details will typically be agreed in advance with the client.

Recommended content for the external report is listed below:

- i. List of assumptions, properly justified and reasoned.
- ii. List of the limitations of the model which could affect the final outcome.
- iii. Software name and version number.
- iv. Confirmation that the recommended checks in this TB have been performed, including a high-level summary of what those checks were.

7. Case Studies

7.1 Case 1: Cables buried in single ducts

7.1.1 Case definition

Three power cables are contained in three air-filled single HDPE ducts, respectively, buried in a flat formation. The ducts were surrounded by soil and backfill layers with varying materials of different thermal properties, including sand, backfilling material, and asphalt. In the base case, a multi-point special sheath bonded configuration is chosen, while one of the model variants explores a single-point sheath bonding arrangement. Key input parameters for the system are outlined as follows:

- Cable conductor cross-section: 630 mm² Cu
- XLPE insulation
- Aluminium laminated sheath
- PE oversheath
- Voltage level: 132 kV
- Solid (multi-point) cable sheath bonding
- Air-filled HDPE ducts

7.1.2 Input data

7.1.2.1 Electrical parameters

Table 7 - 1 Electrical parameters case study 1

Electrical parameters	Value	Unit
System voltage level	132	kV
Frequency	50	Hz

7.1.2.2 Cable parameters

The cable type employed in this case study is based on CIGRE TB 880 Case Study 0, specifically a 132 kV land cable featuring a stranded copper (Cu) conductor with a cross-section of 630 mm², and a single laminated (Al) foil screen with a cross-section of 170 mm². The detailed dimensions, encompassing outer diameters and thicknesses of the cable layers, are provided below.

Table 7 - 2 Cable parameters case study 1

Cable parameter	Thickness (mm)	Outer Diameter (mm)
Conductor	-	30.30
Conductor semi-conductive screen	1.5	33.30
Insulation	15.5	64.30
Insulation semi-conductive screen	1.3	66.90
Metallic sheath	0.80	68.50
Oversheath	3.50	75.50

7.1.2.2.1 Conductor

The DC resistance at 20°C for a single-core cable's round stranded conductor is obtained from the IEC 60228 standard. The coefficients for skin and proximity effects are chosen with consideration for a round stranded conductor and an extruded insulation system.

Table 7 - 3 Conductor parameters case study 1

Parameter	Value	Unit
DC resistance at 20 °C (IEC 60228, Table 2)	0.0283	Ω/km
Skin effect coefficient (k_s)	1	-
Proximity effect coefficient (k_p)	1	-
Temperature coefficient	3.93×10^{-3}	1/K

7.1.2.2.2 Semiconductive screens

Cable suppliers typically select common semi-conducting layers to enhance the smoothing of the electric field, thereby preventing breakdown over both the conductor and the insulation.

Table 7 - 4 Semiconductive screen parameters case study 1

Parameter	Value	Unit
Thermal resistivity	2.5	Km/W

7.1.2.2.3 Insulation

A standard XLPE insulation layer is employed, with the selection of relative permittivity and loss factor values tailored for voltages exceeding 18/30 (36) kV (unfilled). The manufacturer designs the insulation thickness to withstand the specified maximum electrical field when the voltage is applied between the conductor and earth.

Table 7 - 5 Insulation parameters case study 1

Parameter	Value	Unit
Relative permittivity	2.5	-
Loss factor	0.001	-
Thermal resistivity	3.5	Km/W

7.1.2.2.4 Metallic sheath

An aluminium sheath is selected to serve as a watertight metal sheath. The manufacturer designs the sheath thickness to withstand the specified short-circuit current.

Table 7 - 6 Metallic sheath parameters case study 1

Parameter	Value	Unit
Electrical resistivity	2.84×10^{-8}	Ωm
Temperature coefficient	4.03×10^{-3}	1/K

7.1.2.2.5 Oversheath

A standard PE (Polyethylene) oversheath is selected. The thermal resistivity value is chosen from the "protective coverings" section of the IEC Table. The manufacturer designs the oversheath thickness to provide adequate mechanical protection and withstand the specified induced voltages in the metal screen.

Table 7 - 7 Oversheath parameters case study 1

Parameter	Value	Unit
Thermal resistivity	3.5	Km/W

7.1.2.3 Installation

In this straightforward case study, the cables are installed in HDPE (High Density Polyethylene) ducts. The spaced ducts are arranged in a flat formation, embedded in a backfill with low thermal resistivity. The backfill area is surrounded by native soil, and an additional layer of asphalt on top.

Table 7 - 8 Installation parameters

Parameter	Value	Unit
Depth (from ground surface to centre of ducts)	1.0	m
Duct outer diameter (Do)	140	mm
Duct thickness	10.3	mm
Standard Dimension Ratio (SDR)	13.6	-
Duct bank width	1.0	m
Duct bank height	0.5	m
Axial distance between ducts	0.3	m
Asphalt layer height	0.1	m

The ratio between the internal duct diameter and external cable diameter is 1.58. The width of the asphalt layer corresponds to the width of the model. The modelling dimensions are evaluated in variation 0 in the 'Results' chapter.

7.1.2.4 Operating conditions

The maximum temperature of the conductor is 90 °C. The soil comprises layers with diverse materials and thermal properties, such as backfilling material and asphalt. The ambient ground temperature and the thermal resistivity of the soil are chosen to reflect the conditions of a temperature climate in summer, characterized by regular rainfall, in accordance with IEC 60287-3-1.

Table 7 - 9 Operating conditions case study 1

Operating conditions	Value	Unit
Maximum operating conductor temperature	90	°C
Ambient ground temperature	20	°C
Soil thermal resistivity	1.0	Km/W
Back filling thermal resistivity	0.5	Km/W
Asphalt thermal resistivity	1.33	Km/W

7.1.3 Model

7.1.3.1 Geometry

The modelling of air in ducts has been simplified in accordance with the IEC standard. The calculation assumes that the cable is situated at the centre of the duct. Consequently, the cable is also considered in the FE (Finite Element) model to be positioned in the middle of the duct. This simplification aligns with the IEC standard approach for the analysis.

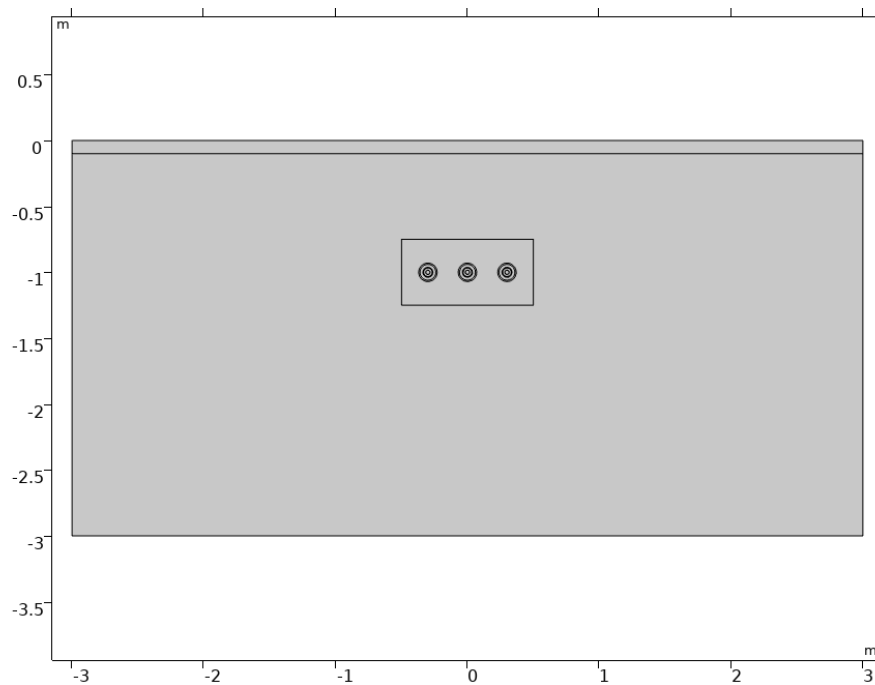


Figure 7 - 1 Installation and cable geometries

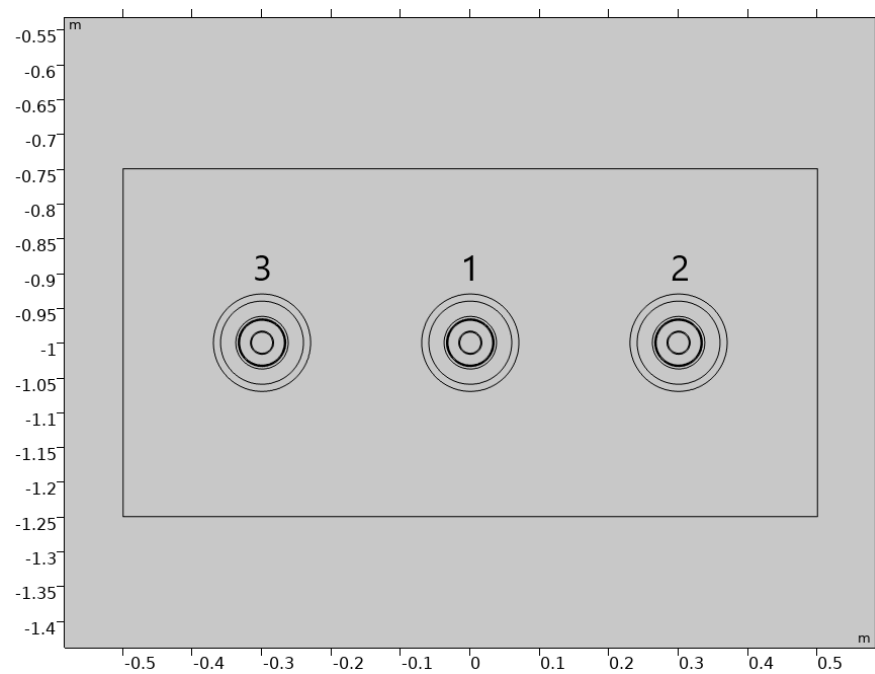


Figure 7 - 2 Overview of duct bank installation. System numbered with 1 for centre, 2 for right and 3 for left in the following analyses

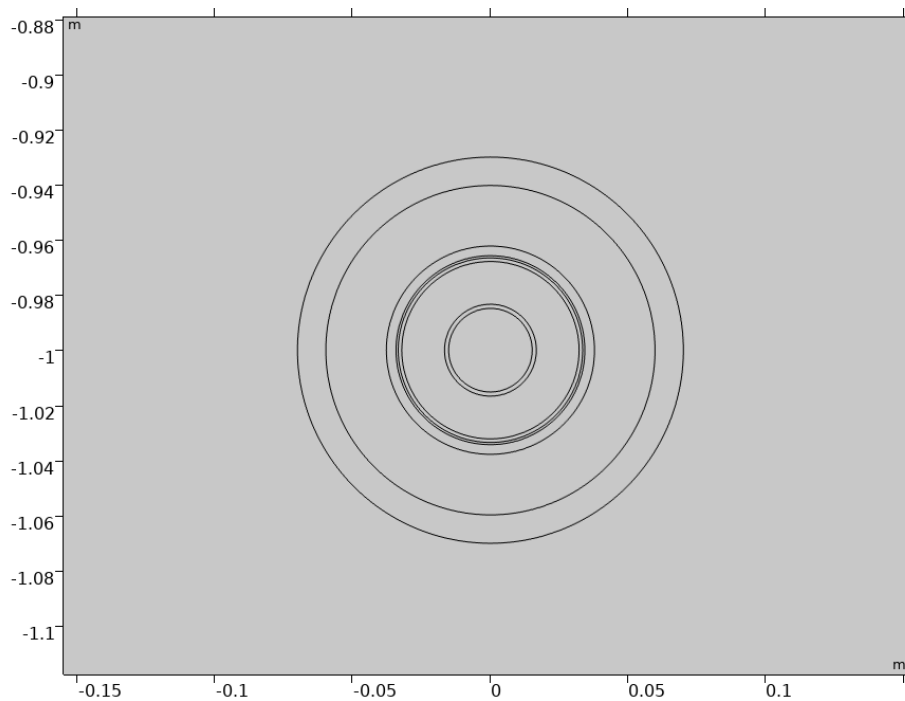


Figure 7 - 3 Cable and duct geometry

The chapter on the results contains tables and plots arranged according to the phase sequence shown in Figure 7 - 2, (Phase 3 - Phase 1 - Phase 2).

7.1.3.2 Material

For material properties not explicitly detailed above, please refer to APPENDIX C. The modelling of air within the ducts follows the guidelines outlined in IEC 60287, as further elucidated in the same sub-chapter.

7.1.3.3 Physics

7.1.3.3.1 Multiphysics approach

The base case and some of the variations employ a Multiphysics approach where the model is set up so that each physic is calculated with its respective physics package instead of simply defining heat sources within the heat transfer physics package.

The magnitude of the current is adjusted or varied to maintain the maximum conductor temperature at 90 °C.

7.1.3.3.1.1 Magnetic fields

The phases are characterised by the relationship between the current magnitude (RMS) and the corresponding phase in a three-phase system:

- Phase 1 (centre): The current in this phase is denoted as I_0 , serving as a reference for the three-phase system.
- Phase 2 (leading): The current in this phase is $I_0 \cdot e^{j \cdot 2\pi/3}$, indicating a leading phase with a phase angle of $2\pi/3$ or 120 degrees.
- Phase 3 (lagging): The current in this phase is $I_0 \cdot e^{-j \cdot 2\pi/3}$, representing a lagging phase with a phase angle of $-2\pi/3$ or -120 degrees.

For the sheaths, the excitation conditions for different bonding systems are specified below:

- Solid bonded systems: The voltage excitation for solid bonded systems is set to 0 V. This condition reflects the scenario where the sheaths are electrically connected.
- Single point bonded systems: For single point bonded systems, the current excitation is set to 0 A. This configuration signifies that the sheaths are bonded at a single point and the current excitation remains neutral.

Both phases and sheaths are modelled as coils with linearised electrical resistivity, each with their respective temperatures and material properties, providing a comprehensive representation of the electrical system.

$$\sigma = \frac{1}{\rho_0(1 + \alpha(\theta - \theta_0))}$$

Equation 7.1

where

σ is the electrical resistivity at temperature θ

ρ_0 is the reference resistivity at 20 °C. For the conductor the reference resistivity is calculated using the maximum resistance of the conductor at 20 °C from IEC 60228: $\rho_0 = R_{con} \cdot A_c$. It is important to emphasize that, when calculating the electrical resistivity of the conductor, the area considered should be determined based on the real diameter of the conductor in line with Guidance Point 6. This is essential for an accurate and representative assessment of the conductor's characteristics.

α is the resistivity temperature coefficient ($\alpha_{cu} = 3.93 \cdot 10^{-3} \text{ 1/K}$, $\alpha_{al} = 4.03 \cdot 10^{-3} \text{ 1/K}$)

θ is the temperature of the conductor/sheath

θ_0 is the reference temperature (20 °C)

7.1.3.3.1.2 Electrostatics

To address dielectric losses, the inner diameter of the insulation is specified as a terminal with voltage U_0 (RMS), while the outer diameter serves as the ground point with a voltage of 0 V. The losses generated within the electrostatics physics node are then incorporated into the heat transfer node as a heat source.

7.1.3.3.1.3 Heat transfer

The burial depth of the cable is assumed to be sufficiently large so that the ambient temperature is set as an isothermal boundary condition on the ground surface following the Guidance Point 44. To assess the impact of boundary conditions on the results, a sensitivity study will be conducted to determine the optimal model dimensions following the Boundary conditions Guidance Point 45. In one of the variations, infinite domains will be employed in the side and bottom boundaries to simulate the semi-infinite nature of the soil. This stretching of the model aims to reduce computation time while maintaining the accuracy of the simulation.

7.1.3.3.1.4 Electromagnetic heating

The magnetic fields and the heat transfer are coupled through the electromagnetic heating interface to account for the resistive heating in the heat equation.

7.1.3.3.2 Heat sources approach acc. to IEC

Certain model variations will embrace the method of defining heat sources in accordance with the principles outlined in IEC 60287. This methodology is easily implementable in FEM software, enabling the specification of AC resistance, the magnitude of the current (RMS) and the temperature relationship. The input formula may adopt units like W/m^3 , representing a heat source, or simply W for a heat rate, depending on the specifics of the FEM software. The equations below assume the former.

7.1.3.3.2.1 Conductor losses

The primary heat sources in the system are the 3 conductors of the three phase system, and the heat source for each of them can be expressed simply as follows:

$$P_c = \frac{I_0^2 R_{AC}(\theta_c)}{A_c}$$

Equation 7.2

where

P_c is the conductor losses per volume unit (W/m^3)

I_0 is the current in one conductor (RMS)

$R_{AC}(\theta_c)$ is the AC resistance in accordance to IEC 60287 calculated at conductor temperature θ_c

A_c is the conductor cross-section area in the model.

7.1.3.3.2.2 Sheath losses

In addition to the conduction losses, there are also sheath losses, which can be expressed as a fraction of the conductor losses:

$$P_{sh} = \frac{I_0^2 R_{AC}(\theta_c) \lambda_1}{A_{sh}}$$

Equation 7.3

where

P_{sh} is the sheath losses per volume unit (W/m³)

I_0 is the current in one conductor (RMS)

$R_{AC}(\theta_c)$ is the AC resistance in accordance to IEC 60287 calculated at conductor temperature θ_c

A_{sh} is the cross section of the metallic sheath

λ_1 is the ratio of total losses in metallic sheath: $\lambda_1 = \lambda'_1 + \lambda''_1$, where λ'_1 circulating currents loss factor and λ''_1 Eddy-current loss factor

For solid bonding: λ' as per IEC 60287-1-1 5.3.4, λ'' following Guidance Point 31 in CIGRE TB 880 and as per IEC 60287-1-1 5.3.6-7.

For single point bonding: $\lambda' = 0$, λ'' following recommendation from CIGRE TB 880 and according to IEC 60287 5.3.7.1 (2.).

7.1.3.3.2.3 Dielectric losses

The dielectric losses are defined by the following formula from the IEC 60287-1:

$$W_d = \omega C_{cab} U_0^2 \tan \delta$$

Equation 7.4

where

W_d is the dielectric losses (W/m)

$\omega = 2\pi f_0$ with f_0 being the operating frequency

C_{cab} is the capacitance per unit length of cable (F/m)

U_0 is the voltage to earth (V)

$\tan \delta$ is the loss factor of the insulation at operating frequency

7.1.3.4 Mesh

The mesh was generated following Guidance Point 50. A coarse mesh was used where the entire geometry was meshed with free triangular elements.

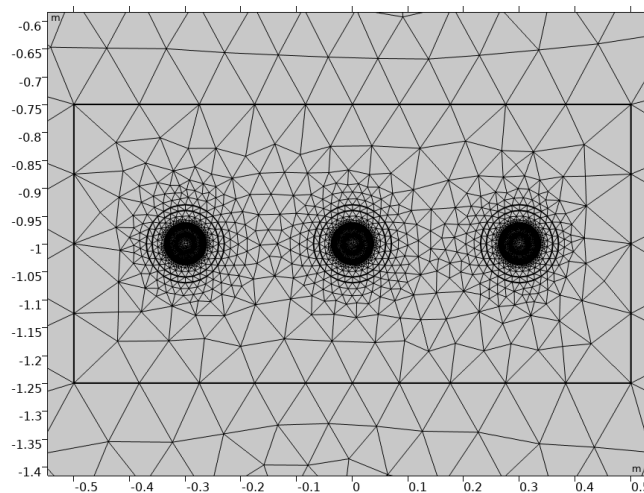


Figure 7 - 4 Mesh

7.1.4 Results

7.1.4.1 Variation 0

As mentioned above, the base model refers to cables buried in single HDPE ducts in flat formation with a solid bonding configuration of the metallic sheaths. A fully coupled model (Multiphysics) is used to calculate the loss on the conductors and sheaths through a temperature dependent thermal resistivity of the metals. For simplicity reasons, the air inside the ducts is considered as defined in the IEC 60287 and described in APPENDIX C.

At first a sensitivity study on the dimensions of the model should be performed following Guidance Point 14. A sweep study is performed in 10 m decrements, starting with a height of 100 m, with the width of the model in each case being two times the height. A further case with the IEC 62095 recommendation has been added at the end for comparison.

Table 7 - 10 Results Case Study 1: Variation 0

Dimensions (W x H)	Ampacity
200 m x 100 m	615.84 A
180 m x 90 m	615.83 A
160 m x 80 m	615.82 A
140 m x 70 m	615.80 A
120 m x 60 m	615.79 A
100 m x 50 m	615.77 A
80 m x 40 m	615.73 A
60 m x 30 m	615.65 A
40 m x 20 m	615.44 A
20 m x 10 m	614.31 A
10 m x 5 m (IEC 62095)	609.85 A

The slight variations observed up to 40 m x 20 m scenario might be attributed to differences in meshing rather than any impact from the boundaries. Hence, for subsequent cases, a dimension of 40 m x 20 m has been chosen.

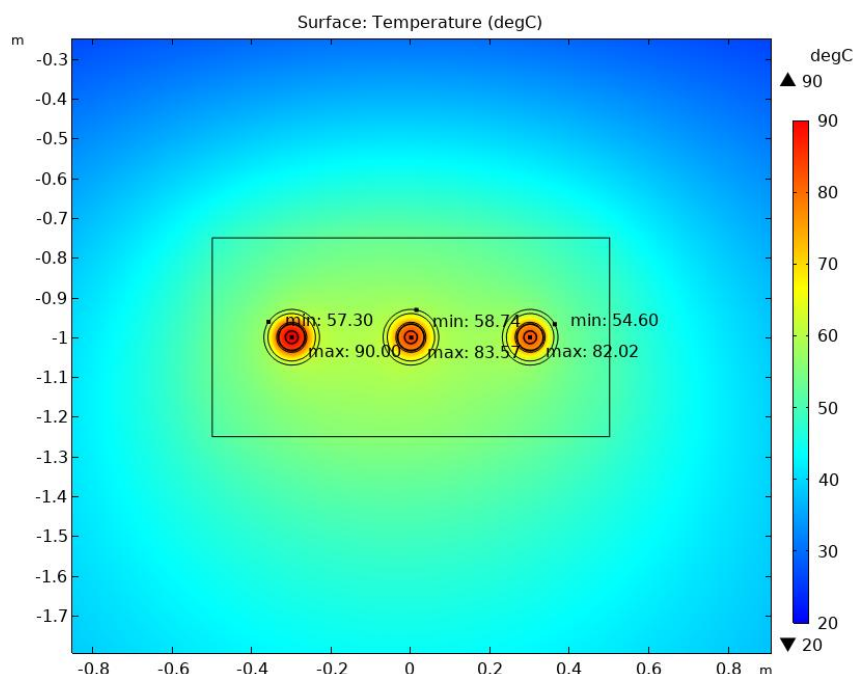


Figure 7 - 5 Results Case study 1: Variation 0

Resistances, losses and temperature of cable components:

Table 7 - 11 Results Case Study 1: Variation 0

Parameter	Unit	Phase 3	Phase 1	Phase 2
I	A	615		
R_{ac}	Ω/m	3.828×10^{-5}	3.766×10^{-5}	3.744×10^{-5}
R_{sh}	Ω/m	2.100×10^{-4}	2.063×10^{-4}	2.047×10^{-4}
W_c	W/m	14.5	14.3	14.1
W_{sh}	W/m	35.4	20.9	24.0
θ_c	$^{\circ}\text{C}$	90.0	83.6	82.0
θ_{sh}	$^{\circ}\text{C}$	83.8	77.5	76.0
$\theta_{duct\ air}$	$^{\circ}\text{C}$	71.2	68.4	66.1

As is to be expected with a solid bonding configuration in flat formation, the middle cable is not the hottest, but the lagging phase due to the strong influence of the circulating currents in the metal sheaths together with the proximity to the other two cables.

7.1.4.2 Variation 1: Modelling air heat transfer in ducts

In this variation, the model is exactly the same as the base model (solid bonding), but focuses on modelling the physics between the cables and the ducts using FEM. In the base model, the air in the ducts is modelled following IEC 60287, which assumes that the cable is positioned in the centre of the duct. However, by explicitly modelling the heat transfer between the duct and the cable, this is no longer necessary, allowing the cables to be placed on the bottom part, in contact with the inner surface of the duct. To model the air in the ducts, the physics of convection and radiation must be incorporated into the model. The convection part is modelled as laminar flow, where gravity is included and non-isothermal flow is assumed. For radiation, surface to surface radiation physics is added,

where the outer surface of the cables and the inner surface of the ducts are set as diffuse surfaces with an emissivity of 0.9. This physics interface uses the radiosity method to calculate the radiation flux, which includes both reflection and re-radiation.

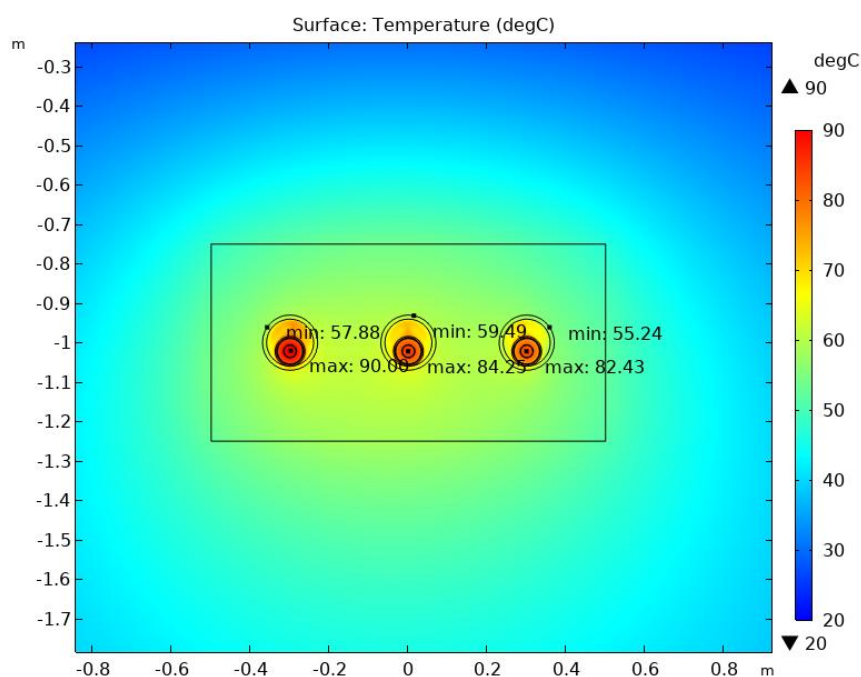


Figure 7 - 6 Results Case study 1: Variation 1

Resistances, losses and temperature of cable components:

Table 7 - 12 Results Case Study 1: Variation 1

Parameter	Unit	Phase 3	Phase 1	Phase 2
I	A	630		
R_{ac}	Ω/m	3.837×10^{-5}	3.782×10^{-5}	3.757×10^{-5}
R_{sh}	Ω/m	2.098×10^{-4}	2.065×10^{-4}	2.048×10^{-4}
W_c	W/m	15.2	15.1	14.9
W_{sh}	W/m	37.2	21.9	25.2
θ_c	$^{\circ}\text{C}$	90.0	84.3	82.4
θ_{sh}	$^{\circ}\text{C}$	83.5	75.9	76.1
$\theta_{duct\ air}$	$^{\circ}\text{C}$	71.5	69.0	66.5

7.1.4.3 Variation 2: Sheath single point bonding

In this variation, the model is exactly the same as the base model, where only the bonding configuration has been changed from solid to single point bonding. Circulating currents are zero and only eddy currents have an effect.

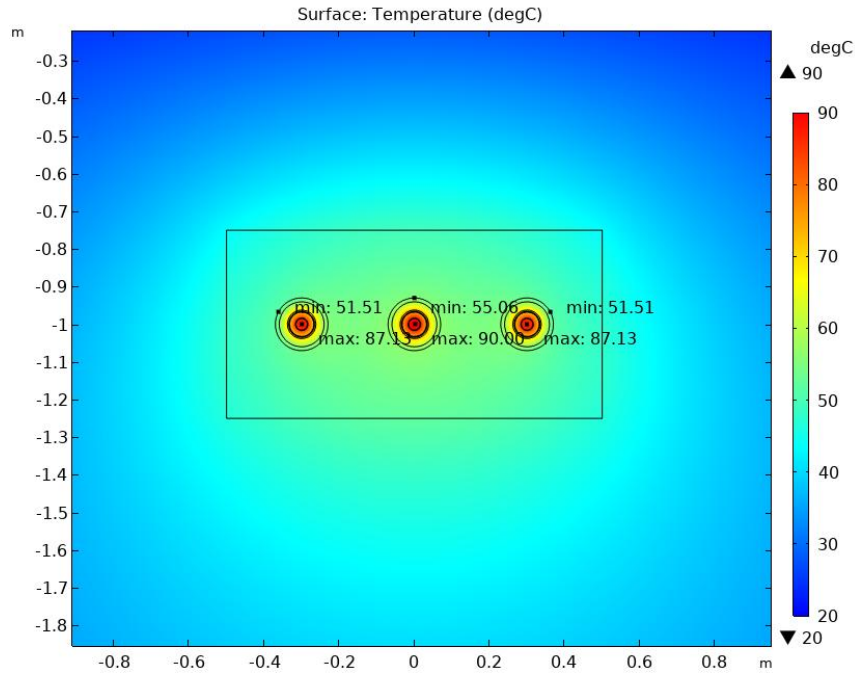


Figure 7 - 7 Results Case study 1: Variation 2

Resistances, losses and temperature of cable components are listed in Table 7 - 13.

Table 7 - 13 Results Case Study 1: Variation 2

Parameter	Unit	Phase 3	Phase 1	Phase 2
I	A	973		
R_{ac}	Ω/m	3.799×10^{-5}	3.840×10^{-5}	3.799×10^{-5}
W_c	W/m	36.0	36.4	36.0
W_{sh}	W/m	0.1	0.3	0.1
θ_c	$^{\circ}\text{C}$	87.1	90.0	87.1
θ_{sh}	$^{\circ}\text{C}$	71.9	74.6	71.9
$\theta_{duct\ air}$	$^{\circ}\text{C}$	62.4	65.0	62.3

7.1.4.4 Variation 3: Bentonite as duct filling material

In this variation, the model is exactly the same as the base model (solid bonding), where only the filling material of the duct has been changed from air to bentonite (0.4 Km/W), which is a common material with low thermal resistivity improving the thermal dissipation in the ducts compared to air. In the base model, the air inside the ducts is modelled following IEC 60287, which assumes that the cable is positioned in the centre of the duct. However, with the introduction of bentonite as a filling material in this variation, it is no longer necessary to refer to the IEC standard. Consequently, it is no longer necessary to assume the cables to be placed in the centre of the duct.

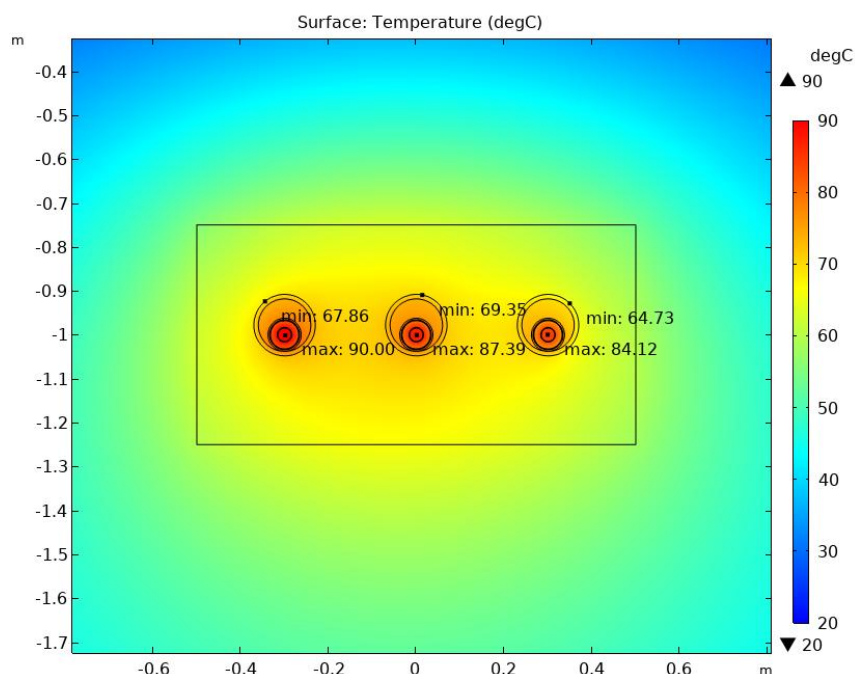


Figure 7 - 8 Results Case study 1: Variation 3

Resistances, losses and temperature of cable components are listed in Table 7 - 14.

Table 7 - 14 Results Case Study 1: Variation 3

Parameter	Unit	Phase 3	Phase 1	Phase 2
I	A	700		
R_{ac}	Ω/m	3.828×10^{-5}	3.806×10^{-5}	3.766×10^{-5}
R_{sh}	Ω/m	2.088×10^{-4}	2.076×10^{-4}	2.049×10^{-4}
W_c	W/m	18.8	18.7	18.5
W_{sh}	W/m	46.0	27.2	31.2
θ_c	$^{\circ}\text{C}$	90.0	87.4	84.1
θ_{sh}	$^{\circ}\text{C}$	82.0	79.5	76.3
$\theta_{duct\ bentonite}$	$^{\circ}\text{C}$	76.3	75.3	71.8

7.1.4.5 Variation 4: Using infinite domains instead of defined model dimensions

Instead of modelling a large soil volume, some softwares have implemented Infinite Element Method [115] which allows to use a smaller volume and at the edges adding a scaling layer of a virtual domain. This method is applied to the sides and bottom boundaries in order to stretch the model and reduce the computational time. The model dimensions have been reduced to 6 m x 3 m with the infinite domains defined with a width of 0.25 m.

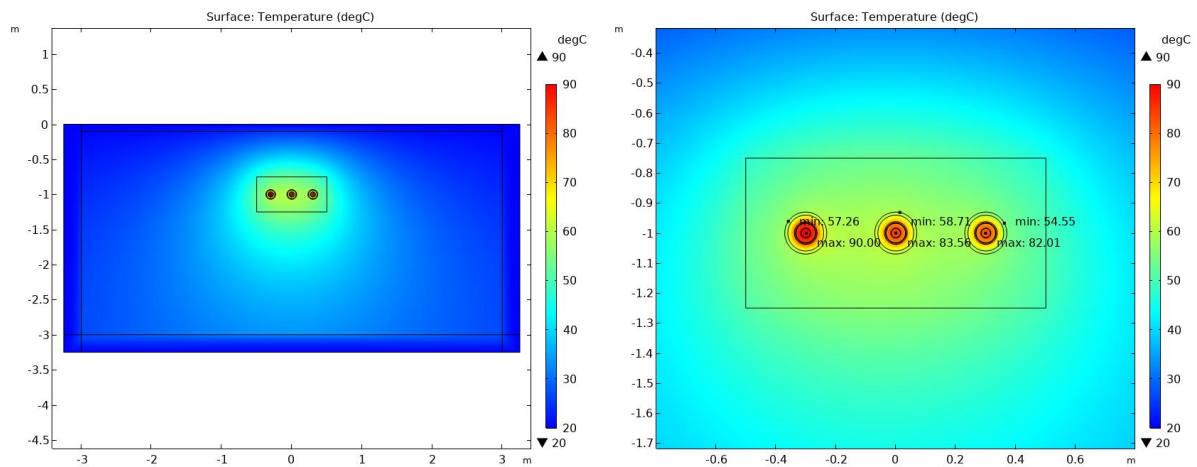


Figure 7 - 9 Results Case study 1: Variation 4

Table 7 - 15 Results Case Study 1: Variation 4

Parameter	Unit	Phase 3	Phase 1	Phase 2
I	A	615		
R_{ac}	Ω/m	3.828×10^{-5}	3.766×10^{-5}	3.744×10^{-5}
R_{sh}	Ω/m	2.100×10^{-4}	2.063×10^{-4}	2.047×10^{-4}
W_c	W/m	14.5	14.3	14.2
W_{sh}	W/m	35.5	21.0	24.1
θ_c	$^{\circ}\text{C}$	90.0	83.6	82.0
θ_{sh}	$^{\circ}\text{C}$	83.8	77.5	76.0
$\theta_{duct\ air}$	$^{\circ}\text{C}$	71.2	68.4	66.0

7.1.4.6 Variation 5: Defined heat sources according to IEC

In the previous and subsequent cases the heat sources were calculated as part of a Multiphysics problem. In this variation the conductors and sheaths are defined as heat sources where the inputs are the current and the resistances as described in the Physics section. The resistances are calculated according to CIGRE TB 880 and all the formulas are implemented within the FE software.

Calculation of temperature independent parameters:

- Capacitance
- Sheath reactance and mutual reactance between sheath of an outer cable and the conductor of the other two
- Electrical resistance of the metallic sheath at 20 $^{\circ}\text{C}$
- Dielectric losses

Calculation of temperature dependent parameters:

- Conductor DC resistance
- Skin and proximity effect factors
- Conductor AC resistance
- Sheath resistance
- Losses caused by circulating and eddy currents

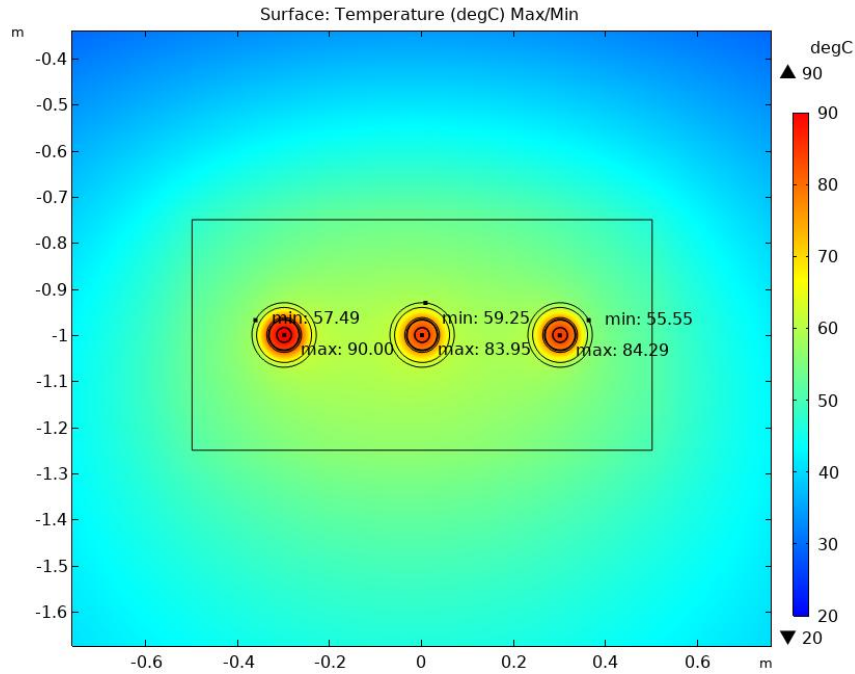


Figure 7 - 10 Results Case study 1: Variation 5

Resistances, losses and temperature of cable components are listed in Table 7 - 16.

Table 7 - 16 Results Case Study 1: Variation 5

Parameter	Unit	Phase 3	Phase 1	Phase 2
I	A	617		
R_{ac}	Ω/m	3.833×10^{-5}	3.770×10^{-5}	3.774×10^{-5}
R_{sh}	Ω/m	2.098×10^{-4}	2.058×10^{-4}	2.061×10^{-4}
λ_1	-	2.378	1.429	1.846
W_c	W/m	14.6	14.4	14.4
W_{sh}	W/m	34.7	20.5	26.5
θ_c	$^{\circ}\text{C}$	90.0	83.9	84.3
θ_{sh}	$^{\circ}\text{C}$	83.8	77.8	78.2
$\theta_{duct\ air}$	$^{\circ}\text{C}$	71.3	68.8	67.6

In terms of computational time, this variation is approximately 65 % faster than the original case in which the entire physics was defined. In addition, the creation of this model requires less effort and is also less error-prone. Due to the nature of this model with a simple cable design and installation arrangement, the heat sources can be defined following IEC standards and entered in the FEM software without the need to define any other physics. This also goes in line with Guidance Point 33.

7.1.4.7 Variation 6: Soil dry out

In this variation, a backfill material resistant to drying out has been selected, with variation 0 serving as the modelling basis. As a result, the thermal properties of the backfill material itself will remain stable, while those for the surrounding native soil will change with temperature. It is assumed that the thermal

resistivity of the native soil stays constant at 1 Km/W until reaching the critical temperature of 45 °C, at which point it increases to 2.5 Km/W. The curve was smoothed to avoid discontinuities in the function and to improve convergence.

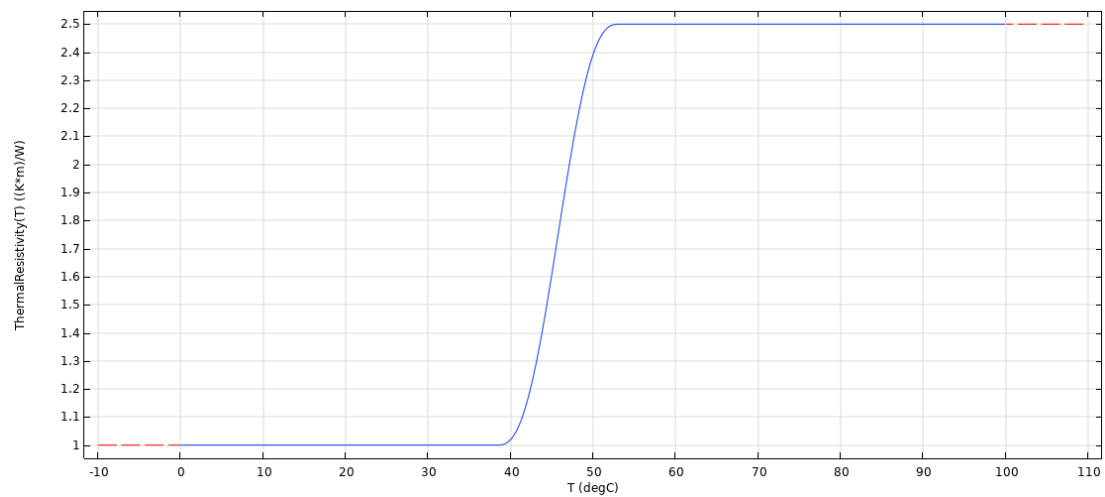


Figure 7 - 11 Native soil dry out curve

Results of soil dry out:

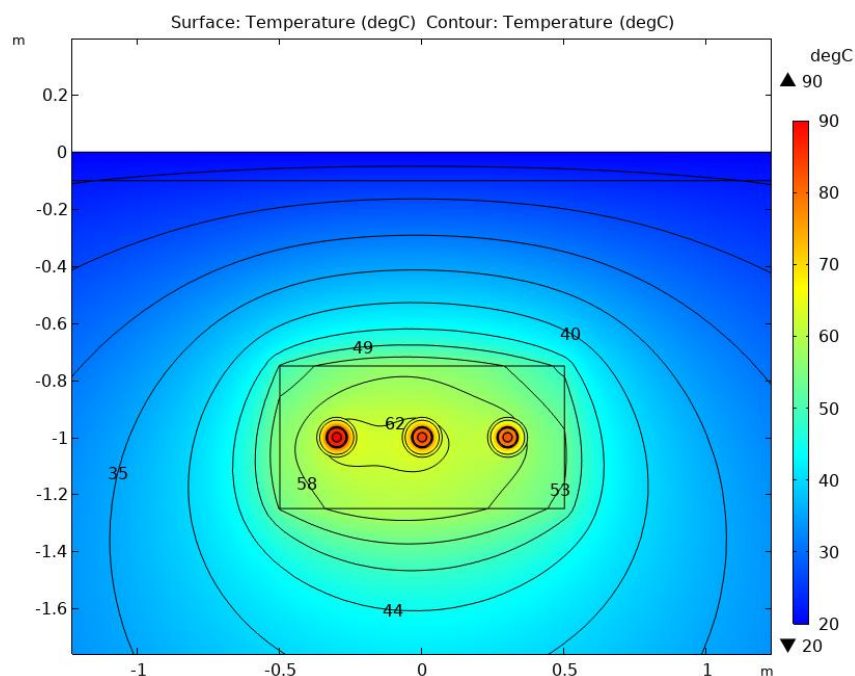


Figure 7 - 12 Results Case study 1: Variation 6

Table 7 - 17 Results Case Study 1: Variation 6

Parameter	Unit	Phase 3	Phase 1	Phase 2
I	A	587		
R_{ac}	Ω/m	3.828×10^{-5}	3.774×10^{-5}	3.746×10^{-5}
R_{sh}	Ω/m	2.103×10^{-4}	2.071×10^{-4}	2.052×10^{-4}
W_c	W/m	13.2	13.0	12.9
W_{sh}	W/m	32.2	19.0	21.8
θ_c	$^{\circ}\text{C}$	90.0	84.3	82.2
θ_{sh}	$^{\circ}\text{C}$	84.4	78.7	76.7
$\theta_{duct\ air}$	$^{\circ}\text{C}$	72.9	70.5	67.7

7.1.4.8 Variation 7: Copper wire screen

The cable design has been modified to include copper wire screen and two swelling tapes, with variation 0 as the modelling basis, including the size of the ducts, which remains unchanged. Considering the mentioned layers the final design of the cable would have the following dimensions:

Table 7 - 18 Modified cable parameters case study 1: Variation 7

Cable parameter	Thickness (mm)	Outer Diameter (mm)	Number (-)
Conductor	-	30.3	-
Conductor semi-conductive screen	1.5	33.3	-
Insulation	15.5	64.3	-
Insulation semi-conductive screen	1.3	66.9	-
Swelling tape	0.7	68.3	-
Earthing screen (concentric wires)	1.77	71.84	57
Swelling tape	0.3	72.44	-
Metallic sheath	0.8	74.04	-
Oversheath	3.5	81.04	-

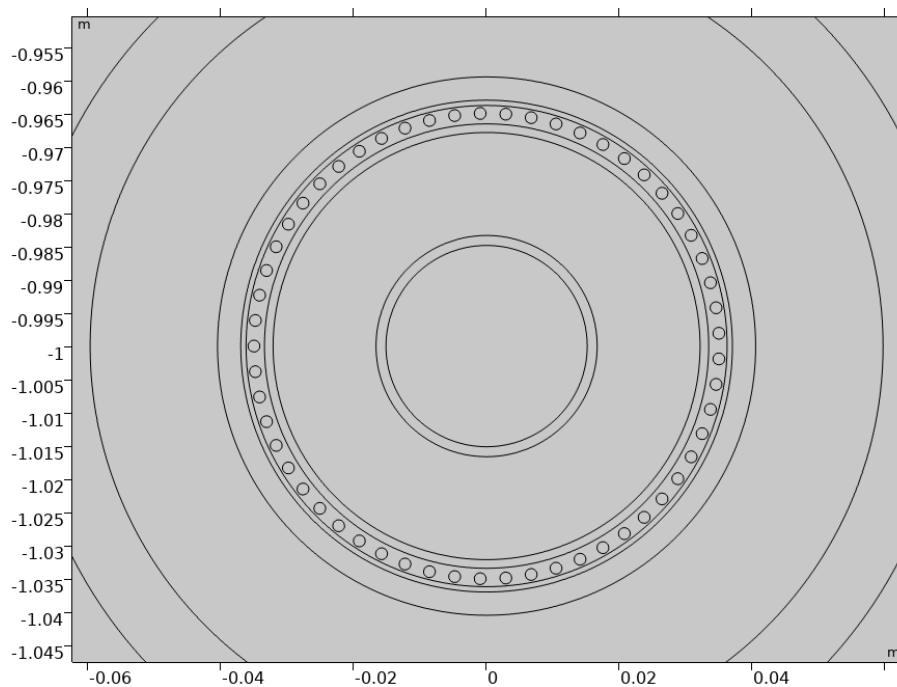


Figure 7 - 13 Modified cable design case study 1: Variation 7

Results are illustrated in Figure 7 - 14 and Table 7 - 19.

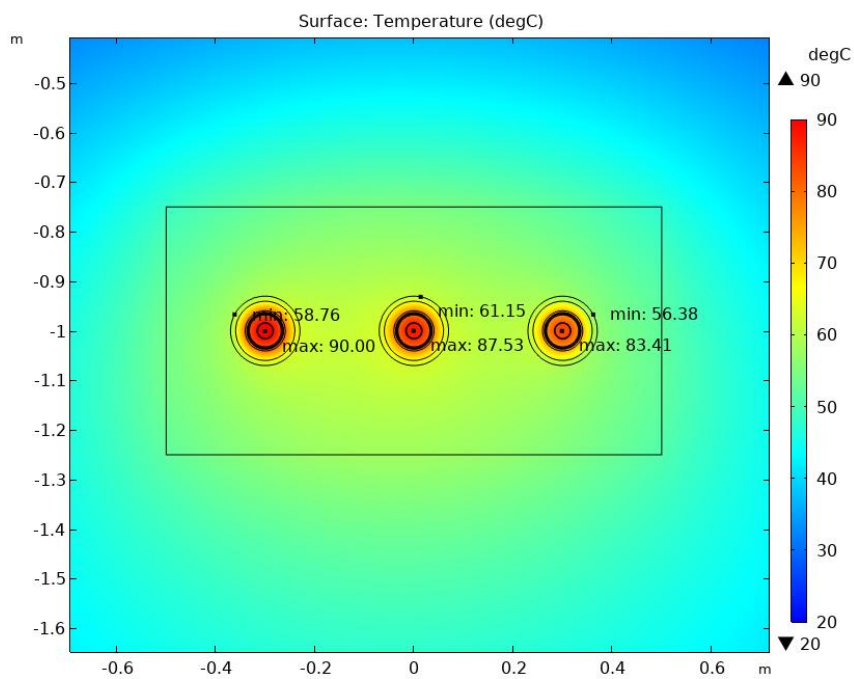


Figure 7 - 14 Results Case study 1: Variation 7

Table 7 - 19 Results Case Study 1: Variation 7

Parameter	Unit	Phase 3	Phase 1	Phase 2
I	A	654		
R_{ac}	Ω/m	3.827×10^{-5}	3.802×10^{-5}	3.757×10^{-5}
R_{sh}	Ω/m	1.551×10^{-4}	1.541×10^{-4}	1.519×10^{-4}
R_{sc}	Ω/m	1.933×10^{-4}	1.922×10^{-4}	1.893×10^{-4}
W_c	W/m	16.4	16.3	16.1
W_{sh}	W/m	18.7	13.8	13.0
W_{sc}	W/m	14.5	10.5	10.1
θ_c	$^{\circ}\text{C}$	90.0	87.5	83.4
θ_{sh}	$^{\circ}\text{C}$	82.8	80.3	76.3
θ_{sc}	$^{\circ}\text{C}$	82.4	80.0	76.0
$\theta_{duct\ air}$	$^{\circ}\text{C}$	71.7	71.2	67.4

7.1.4.9 Variation 8: Time dependent with changing load

The aim of this variation is to determine the temperature of the cable under changing load conditions over a certain period of time, whereby the cable and the environment model remain the same as variation 0. It is important to have suitable initial conditions for the model to represent the load conditions that occurred before the period to be analysed, as suggested in Guidance Point 64. In this variant, the load profile starts at 0 because the cable has been inactive for a long time.

Particular attention must be paid to the selection of the solver step size as described in Guidance Point 65. The solver step size for this variant was defined in such a way that the solver takes an additional time step within the interval between the requested output times, whereby the solver is free to decide when this intermediate step takes place. The intervals are set to 0.1 days.

As recommended in Guidance Point 61, if heat capacities from IEC 60853 are used and the finite element software requires $\text{J}/(\text{kg}^3 \times \text{K})$, a workaround is to enter the volumetric heat capacity as $\text{J}/(\text{m}^3 \times \text{K})$ and set the density to $1 \text{ kg}/\text{m}^3$. In this case, both criteria met, so the recommendation from the guidance point was followed.

The proposed load profile for the time dependent study is described in Table 7 - 20 and the results from simulation are shown in Figure 7 - 15.

Table 7 - 20 Case study 1: Variation 8, load profile

Time (days)	Step (%)
Infinite (preload)	0
50	68
10	34
30	100
10	54

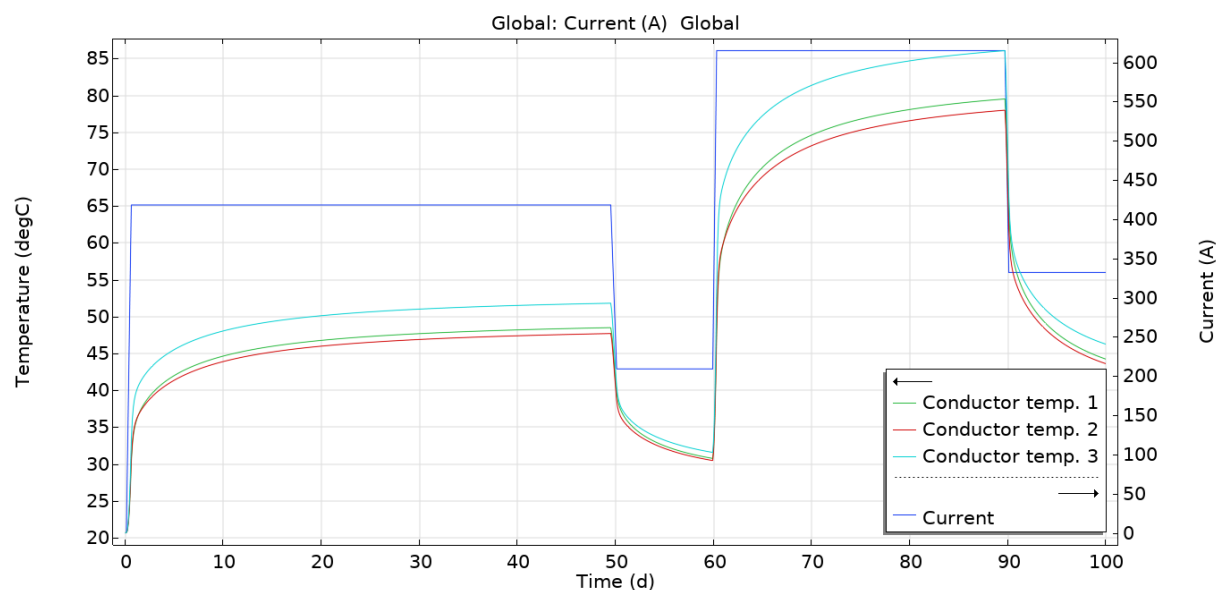


Figure 7 - 15 Results Case study 1: Variation 8

7.1.4.10 Case 1 variations overview

The following table shows the results of the different variations considered in Case Study 1.

Table 7 - 21 – Case study 1: Variations overview

Variation	Description	Ampacity [A]
0	Base model: solid bonding	615
1	Modelling air heat transfer in ducts	630
2	Sheath single point bonding	973
3	Bentonite as duct filling material	700
4	Using infinite domains instead of defined model dimensions	615
5	Defined heat sources according to IEC	617
6	Soil dry out	587
7	Copper wire screen	654
8	Time dependent with changing load (max. temp 85 °C)	615

7.2 Case 2: Single core cables on a non-magnetic tray surrounded by air

7.2.1 Case definition

Single-core distribution cables in metallic (non-ferrous) open and covered trays, placed indoors and surrounded by still air. Two overall simulation variations are studied:

1. Open tray consisting only of a bottom plate on which 12 cables are situated in a two-layer bundle, all cables excited with 300 A (rms) current magnitude.
2. Closed tray consisting of four walls (plates) enclosing 6 cables in a single layer and free air, all cables excited with 400 A (rms) current magnitude.

For both variations, all the cables are touching each other, all cables are equally loaded. Key input parameters are given below:

- **Cross-section:** 240 mm² copper (Cu)
- **Insulation:** XLPE, unfilled
- **Sheath:** Stranded copper wires
- **Oversheath:** PE
- **Voltage level:** 33 kV
- **Bonding:** Single-point bonded
- **Aluminium tray**

Details regarding each variation are given under each relevant section in the following text below.

7.2.1.1 Geometry

7.2.1.1.1 Open tray variation

A single bottom plate defining the open tray itself is assumed (with no tray sides), and two layers of 6 cables, all touching, are examined. The layout of this geometry in Figure 7 - 16 below, where cable numbering is displayed for later referral for tabulated results.

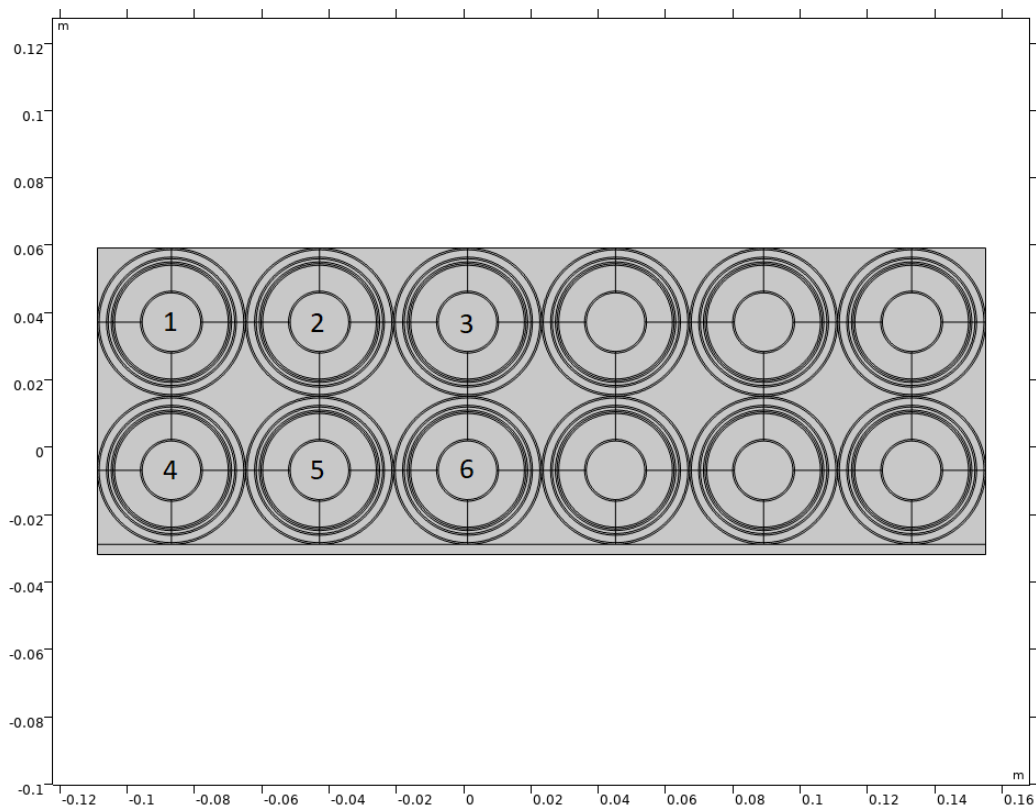


Figure 7 - 16 Layout of the open tray variation with two layers of 6 touching single-core cables.

The thickness of the tray is 3 mm, while the width of the tray is 264 mm, simply corresponding to six times the outer diameter of the cable used; dimensions and layout of the modelled cable can be found in the subsection below. Note the rectangle comprising all the 12 cables, which is for modelling purposes in order to conform with the implementation of the analytical formulas for heat transfer coefficients given in literature for cable bundles on trays - further explanation is given in the Heat Sources Section 7.2.2.3.

7.2.1.1.2 Covered tray variation

An enclosing rectangle with thickness of 3 mm is forming the tray with cover; the (inner) width is similar to above 264 mm, while the height of the sides is 74 mm (inner). For this variation, a single layer of 6 touching cables is considered. The layout of this geometry can be seen in Figure 7 - 17 below.

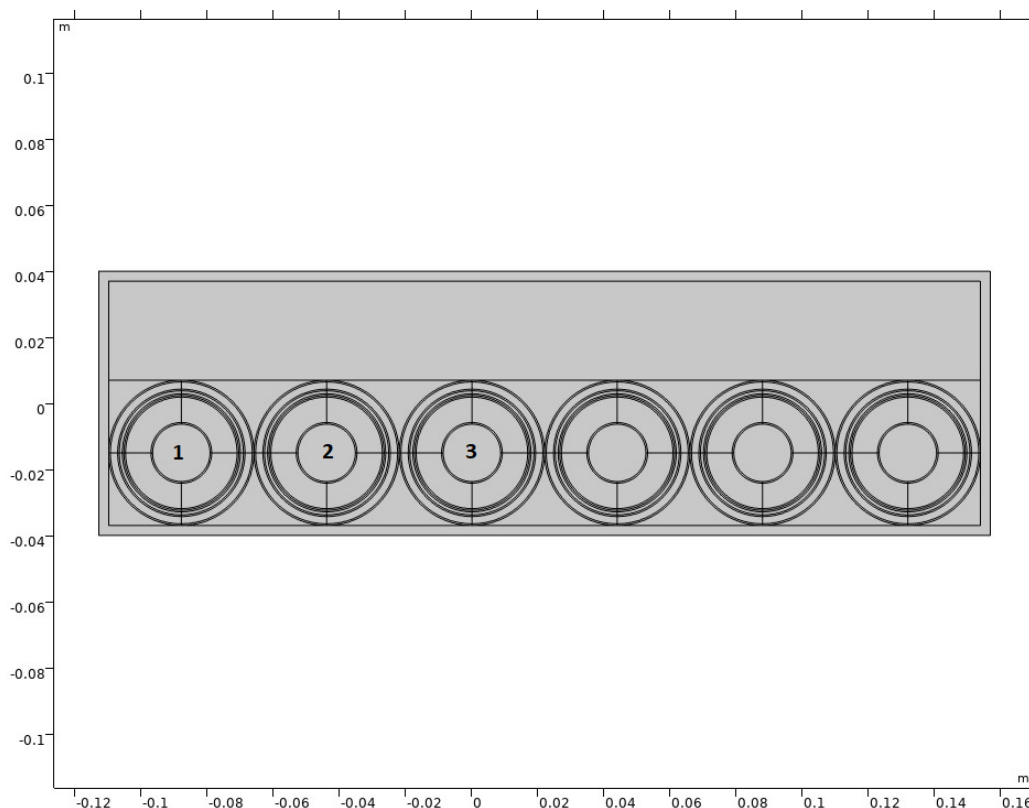


Figure 7 - 17 Layout of the covered tray variation with one layer of 6 touching single-core cables.

Similar to the open tray variation, a rectangle is here also enclosing the cable bundle to conform with the heat flux exchange with the surrounding air, as per formulas from existing literature - more information is provided in Heat Sources Section 7.2.2.3.

7.2.1.2 Cable data

The cable type used in this case study is from CIGRE TB 880 Case Study 4, and is a 33 kV land cable with a 240 mm² stranded copper (Cu) conductor and 36 mm² wire Cu screen. The electrical DC and AC resistances of the conductor at 20 °C are taken from CIGRE TB 880 and equal to 0.0754 x 10⁻³ Ω/km, and 0.0963 x 10⁻³ Ω/km, respectively (bundled cable proximity effects assumed equal to the default value from the TB). The cable dimensions, including cable layer outer diameters and thicknesses are reproduced in the Table 7 - 22 below.

Table 7 - 22 Cable data: Cable layer outer diameter and thickness used for simulations

Layer	Thickness of layer t [mm]	Diameter over (D) [mm]
Conductor (c)	-	18.4
Conductor Screen (cs)	0.5	19.4
Insulation (i)	7.7	34.8
Insulation Screen (is)	0.5	35.8
Tape 1 (Semiconducting bedding) tape1	0.5	36.8
Wire Screen (ws)	0.9	38.6
Tape 2 (tape2)	0.3	39.2
PE sheath (incl. glue from laminate if any)	2.2	43.6
Semiconducting outer layer sheath	0.2	44.0

7.2.1.3 Environmental data

The ambient temperature is equal to 20 °C, which also is used as the external temperature for boundary heat flux conditions.

7.2.2 Model

7.2.2.1 Material

The thermal resistivities of the materials of the cable layers from Table 7 - 22 above are reproduced from CIGRE TB 880 Case Study 4 in Table 7 - 23 below, which corresponds to thermal resistivity values from IEC 60287-2-1.

Table 7 - 23 Cable data: Thermal resistivities used for simulations

Layer	Thermal resistivity [mK/W]
Conductor (c)	-
Conductor Screen (cs)	2.5
Insulation (i)	3.5
Insulation Screen (is)	2.5
Tape 1 (Semiconducting bedding) tape1	6
Wire Screen (ws)	-
Tape 2 (tape2)	6
PE sheath (incl. glue from laminate if any)	3.5
Semiconducting outer layer sheath	2.5
Note 1: Tape 1 and Tape 2 are considered as bedding tapes, not water blocking tapes, therefore the thermal resistivity for both tapes is 6.0 Km/W	

7.2.2.2 Physics

The steady-state temperatures, most importantly, the hot-spot temperature (HST) of the cable bundle, can be computed from formulas given in literature, e.g., [116], for both open and covered trays with. It is within the FEM software advisable to firstly implement a simple simulation, manually inputting the said formulas for the simulation of the final HST. As such, a heat transfer in solids type of physics can readily be used, which only requires the input of material thermal resistivities and well-defined joule heating terms for the heat sources (as function of predefined current magnitudes).

7.2.2.3 Heat sources

7.2.2.3.1 Conductor losses

The primary heat sources are the six cable conductors, which can be easily modelled as heat sources in FEM software. Here, the AC resistance of the specified cable type can be entered along with the current magnitude (rms) and a resistance-temperature relationship.

Depending on the FEM software, the input formula could either have the unit W/m³ as in a heat source or just W as in a heat rate. In the below equation, the former is assumed, and the heat source of each cable conductor is thus simply:

$$P_c = \frac{I_0^2 R_{ac}(\theta_c)}{A_c}$$

Equation 7.5

where A_c is the cross-sectional area of the conductor as it is defined in the geometry of the model, which is needed for the unit adaptation to W/m^3 . The AC resistance, R_{ac} , of the conductor is temperature dependent and calculated in model based on the temperature in the conductor, θ_c . When constructing the conductor cross-section in FEA software, the outer diameter of the conductor is used, resulting in a cross-section that does not match the 240 mm² equivalent copper cross-section, which excludes fillings and other components. Instead, it should match the actual cross-section occupied by the conductor as defined in the software's geometry settings.

Value for α_{cu} can from IEC 60287-1-1 be found to be equal to 0.00393 1/K (copper). The AC resistance can be determined using formulas that account for skin and proximity effect factors from the same standard, applied to the correct cable configuration. For systems with more than one three-phase excitation, like this case, analytical calculations for these factors are not available. It's recommended to use single-phase system approximations or to numerically estimate the equivalent AC resistance, for instance, through magnetic field simulations in FEM or by establishing a Multiphysics link between magnetics and heat transfer. The latter approach is complex, as it must accurately reflect the electrical and thermal behaviours of the cables, meaning the stranded conductor's cross-section must be modelled correctly to represent the proper DC resistance, and the model must conform to the thermal properties. Given the minor expected changes in the proximity factor for a multi-cable bundle of 240 mm² cables versus three single phases at power frequency, and the uncertainties in the simplified heat transfer coefficient terms from the literature, the AC resistance calculated from CIGRE TB 880 Case Study 4 is utilised for simplicity.

Although copper conductors are modelled as solid, circular geometries with an equivalent AC resistance, using the thermal conductivity of copper would not be accurate. This is because the actual equivalent thermal conductivity would be lower due to the presence of filling material (providing longitudinal water tightness) among the strands. To adjust the thermal conductivity value for the conductors' circular cross-sections, a correction factor could be applied to the copper's thermal conductivity. This factor would likely be quite low, as non-metallic filler layers substantially decrease the cross-section's equivalent thermal conductivity. Nonetheless, generally, the thermal resistivity of the heat-generating conductors is considered negligible compared to that of the surrounding thermally insulating layers. Therefore, adjusting the thermal conductivity of the conductors would not significantly impact heat transfer or the final temperature and can often be disregarded. For more information, please see Guidance Point 27.

7.2.2.3.2 Metallic sheath losses

In the metallic screens of the cables, due to single-point bonding, there are no out-of-plane induced circulation currents; only in-plane induced eddy currents are expected. These can be calculated using the screen loss factor formulas in IEC 60287-1-1. Similarly, no analytical formulas exist for multiple three-phase cable bundles, but the heat contribution from induced eddy currents in the stranded, single-point bonded copper screens of the cable bundle is considered insignificant. This aligns with IEC 60287-1-1, which deems eddy-current losses in metallic wire screens, equalizing tape, and foil as negligible. A summary of sheath loss formulation is available in section 2.6 in CIGRE TB 880. Therefore, no heat source should be included in the metallic wire screens for both variations. Instead, they can be modelled as cylinders with a thickness equal to the wire/strand diameter, with a corrected thermal resistivity based on the same principles as the conductor. In this case, the minor air cavities next to the wire screens are considered negligible for the overall thermal heat dissipation from the centre conductors to the ambient environment, and the screen cylinder is assigned a thermal resistivity equivalent to copper.

However, for convenience, a simple formula that can be used for a heat source defined on the cable screen, is provided below, which is based on the above formula for the heat source of the conductor:

$$P_{sh} = \frac{I_0^2 R_{ac}(\theta_c) \lambda_1''}{A_{sh}}$$

Equation 7.6

The above formula is simply a fraction of the conductor heat source where λ_1'' is the eddy-current loss factor, as in accordance with the definition of the screen loss factor definition from IEC 60287-1-1 for a single-point bonded screen. Note, that in the above formula the cross-sectional area of the screen, A_{sc} , should be used instead of the original cross-section of the conductor, A_c , as input to the heat source definition within the FEM software, such that the 2D heat source representation of the screen has its geometric dimensions defined correctly. The heat loss term must still correspond to the joule

loss of the main conductor to comply with the sheath loss factor, λ_1'' , being a fraction of the main conductor joule loss.

7.2.2.3.3 Dielectric losses

Due to the voltage level of the cables being 33 kV, the voltage-dependent dielectric losses, W_d , can be neglected, as also stated in CIGRE TB 880 for this exact cable, where the dielectric losses have been computed to only 0.11 W/m. Such low insulation losses will contribute minimally to the final temperature distribution. The dielectric losses can, however, easily be added to the FEA simulation as heat sources to the insulation layers, if wished (relevant for higher voltage levels).

7.2.2.4 Heat loss fluxes

7.2.2.4.1 Open tray variation

In general, from conservation of energy, all heat generated within the cable bundle must be transferred through the cables, from conductors to cable surfaces, to the surface of the cable bundle, and again from said surface to the air by convection and radiation.

7.2.2.4.1.1 Convection

To model convective heat dissipation, the most straightforward method in 2D FEM simulation tools is to implement convective boundary heat transfer fluxes for heat exchange with ambient (heat sink), using formulas assuming a homogenised cable bundle mass. This can be achieved from heat transfer coefficients for the boundary heat flux on the bottom and top of the cable bundle mass as given in e.g., [116].

In [116], formulas for the convective heat transfer coefficients for the horizontal bottom (subscripted denotation B) and top part (T) of the rectangular cable bundle mass are provided, which below have been reproduced, starting with h_B of the bottom surface of the bundle:

$$h_B = 0.248 kw^{-0.25} \left[\left(\frac{g\beta}{v^2} \right) (\theta_{sB} - \theta_{air}) \right]^{0.25}$$

Equation 7.7

where w is the width of the cable tray in m, k is the thermal conductivity of air in W/(K*m), v is the viscosity of air in m²/s, β is the volumetric thermal expansion coefficient in 1/K, g is the gravitational acceleration in m/s², θ_{sB} is the temperature at the bottom surface of the bundle, and θ_{amb} is the ambient temperature.

And h_T of the top surface of the bundle:

$$\begin{cases} h_T = 0.496kw^{-0.25} \left[\left(\frac{g\beta}{v^2} \right) (\theta_{sT} - \theta_{air}) \right]^{0.25} & \text{for } 10^5 < Ra \leq 10^7 \\ h_T = 0.134k \left[\left(\frac{g\beta}{v^2} \right) (\theta_{sT} - \theta_{air}) \right]^{0.333} & \text{for } 10^7 < Ra \leq 10^{10} \end{cases}$$

Equation 7.8

where θ_{sT} is the temperature at the top surface of the bundle, and Ra, the Rayleigh number, given as:

$$Ra = 0.71w^3 \left(\frac{g\beta}{v^2} \right) (\theta_s - \theta_{air})$$

Equation 7.9

where θ_s is the surface temperature of interest, here the top bundle temperature, θ_{sT} .

Note, that the thermophysical properties, e.g., the density, thermal conductivity and the viscosity, are in general temperature and pressure dependent. For relatively small pressure changes, formulas have in literature been generated by curve fitting, relating said quantities to a specific temperature range. Such terms for the thermal conductivity and viscosity used in Equation 7.7, Equation 7.8 and Equation 7.9 above can be found in the literature, where [116] provides such as well. This is similar to [117] where simpler, non-quadratic fitted terms can be found, the latter reproduced below for the temperature range 0-100 °C (note that θ_f is in Kelvin):

$$\begin{cases} \rho = 352.64/\theta_f & \text{kg/m}^3 \\ k = 2.42 \cdot 10^{-2} + 7.2 \cdot 10^{-5}(\theta_f - 273) & \text{W/K} \cdot \text{m} \\ \nu = 1.32 \cdot 10^{-5} + 9.5 \cdot 10^{-8}(\theta_f - 273) & \text{m}^2/\text{s} \\ \text{Pr} = 0.715 - 0.00025(\theta_f - 273) \end{cases}$$

Equation 7.10

where θ_f is the so-called average film temperature (temperature of fluid inside convection boundary layer), $\theta_f = (\theta_s + \theta_{air})/2$, where θ_s would be the expected temperature of the surface of interest, here the top surface temperature, θ_{sT} , of which convection arises, and θ_{amb} is the constant ambient temperature (far) away from the film surface.

The volumetric thermal expansion coefficient, β , is temperature dependent, and can for ideal gasses be approximated as the reciprocal of the absolute temperature in Kelvin, where the average film temperature should be used, i.e.: $\beta = 1/\theta_f$.

The formulas for the convective heat transfer coefficients h_T and h_B in Equation 7.7 and Equation 7.8, respectively, together with the formulas for the thermophysical properties in Equation 7.10 can within the FEM software be defined as variables. Here the bundle top and bottom temperatures, θ_{sT} and θ_{sB} , respectively, can be defined as evaluation terms on the respective top and bottom surfaces. This corresponds to couplings of measured temperatures (averaged over the surfaces) that are fed into the variable formulas.

The formulas consider a uniform cable bundle mass, focusing solely on heat conduction within, meaning radiation and convection within the air gaps created by adjacent cables are not factored in. For modelling purposes, these cavities can be treated as a solid with thermal resistivity matching the outer PE sheath of the bundled cables. The method to accurately depict heat transfer in these air gaps largely hinges on their size; very small gaps would have minimal convection and, depending on their proportion to the overall bundle, might not significantly impact the final temperature. Small cavities can be highly insulating, but if they are minor relative to the entire bundle, only slight temperature variations are likely, which is often the case in a large cable bundle with numerous low-voltage cables in a tray raceway. Representing these air gaps as merely air with temperature-dependent thermal conductivity could lead to a substantial overestimation of heat loss and, consequently, the final temperature, especially if the gaps are sizable. In such cases, accounting for heat transfer via radiation and convection is also necessary. For simplicity and alignment with the analytical formulas provided the cable bundle is presumed homogeneous without any thermal anomalies from the air gaps, thus adopting a constant (low) thermal resistivity akin to the outer PE cable sheaths. An alternative configuration for the air gaps can be setup to take into account both radiation effects and air thermal conductivity. This configuration has been tested with negligible changes to the outcomes.

7.2.2.4.1.2 Radiation

In [116], the method to determine the final maximum HST involves solving the heat balance equation. This equation equates the heat gained by the cable to the heat lost through convection (which includes the convective heat transfer coefficients mentioned above) and radiation. The latter refers to the radiative heat transfer from the surfaces of the cable bundle to the surrounding environment, and is generally given as:

$$W_{\text{rad}} = \epsilon \sigma_B A_s (\theta_{\text{cable}}^4 - \theta_{\text{air}}^4)$$

Equation 7.11

where ϵ is the emissivity of the radiating surface, and σ_B is the Stefan-Boltzmann constant equal to $5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4$, and A_s is the surface area of the rectangle made up of the cable bundle plus tray bottom in $[\text{m}^2]$, per unit length.

To account for heat loss due to radiation to the ambient surroundings in FEM simulation software, the three sides of the simplified bundle and the bottom part of the tray in Figure 7 - 16 can be assigned a radiative heat transfer coefficient, similar to the convective heat transfer coefficient, with an estimated heat flux in W/m^2 . Alternatively, most FEM tools offer a more straightforward method: ambient radiation boundary fluxes, which require only surface emissivity and ambient temperature as inputs. This method applies the aforementioned expression for radiation to ambient only, excluding reflected surface radiation. Typically, an emissivity value of 0.95 is used for dark, semi-aged surfaces, and the ambient temperature is considered to be 20 °C.

7.2.2.4.2 Closed tray variation

7.2.2.4.2.1 Convection

The thermophysical quantities as given in the formulas in Equation 7.10 are to be used for the closed tray variation simulation as well. Similarly, the formula for the convective heat transfer coefficient of the bottom of the tray, h_B , in Equation 7.7 can be reused for the closed tray case. The coefficient for the tray cover is given in [116] as:

$$\begin{cases} h_o = 2 \cdot Nu \cdot \frac{k}{w} \\ Nu = c(Gr \cdot Pr)^n \\ Gr = g \times \beta(\theta_o - \theta_{air}) \left(\frac{w}{2}\right)^3 / \nu^2 \end{cases}$$

Equation 7.12

where c and n are defined based on the product of the Grashof and Prandtl number, which defines the Rayleigh number, and can be found tabulated in [116]. The film temperature is used to estimate the number, and since it is unknown initially, it can be approximated from a guesstimate, where the product $Gr \cdot Pr$ should lie between 10^4 and 10^7 equating to an n constant equal to 0.250. From [116], the c constant should be set equal to 0.57 for the closed tray simulation.

The heat transfer from convection within the air enclosure formed by the cable bundle top, the tray sides, and the tray cover, can also be taken into consideration using a heat transfer coefficient. From [116], the heat transfer coefficient of the top surface of the cable bundle, h_s , and the inside surface of the cover, h_w , are equal, and can be calculated using the formulas below:

$$h_s = h_w = Nu \cdot k_{air} / \delta$$

Equation 7.13

$$\begin{cases} Nu = 1 & \text{if } Gr < 1708 \\ Nu = 0.195 Gr^{0.25} & \text{if } 10^4 < Gr \cdot Pr < 4 \cdot 10^5 \\ Nu = 0.069(Gr \cdot Pr)^{0.333} & \text{if } 1.5 \cdot 10^5 < Gr \cdot Pr < 10^9 \end{cases}$$

Equation 7.14

$$Gr = g\beta(\theta_{cable} - \theta_w)\delta^3 / \nu^2$$

Equation 7.15

where the Nusselt number Nu depends on the Gr and Pr again, which for this simulation case can be expected to exceed 4×10^5 . The δ is the height of the enclosed air gap.

7.2.2.4.2.2 Radiation

Similar to the open tray case, no radiation heat transfer coefficients are defined in [116], and the contribution to cooling from radiation is instead included as Equation 7.11 in the final energy balance equation also. Thus, again the most straightforward approach is to setup the outer sides of the tray with radiation boundary fluxes to ambient.

The energy balance equations in [116] account for radiation heat transfer within the air gap, akin to ambient radiation but based solely on the temperature differential between the cable bundle's top and the tray cover's interior surface. Since this involves not just radiation to ambient but enclosed radiation with diffuse reflection, the optimal method within the FEM tool is to implement a radiation heat transfer physics interface. This interface employs the radiosity method to calculate the radiative flux, encompassing both reflection and re-radiation.

7.2.2.5 Mesh

7.2.2.5.1 Open tray variation

Typically, it is sufficient to deploy a default triangular mesh type for the entire solid domain in a rather simple 2D FEM simulation like this one. Here, heat transfer physics only involves conductive heat

transfer inside the materials and heat dissipation to ambient via the above heat transfer coefficient formulas. This is illustrated in Figure 7 - 18 below.

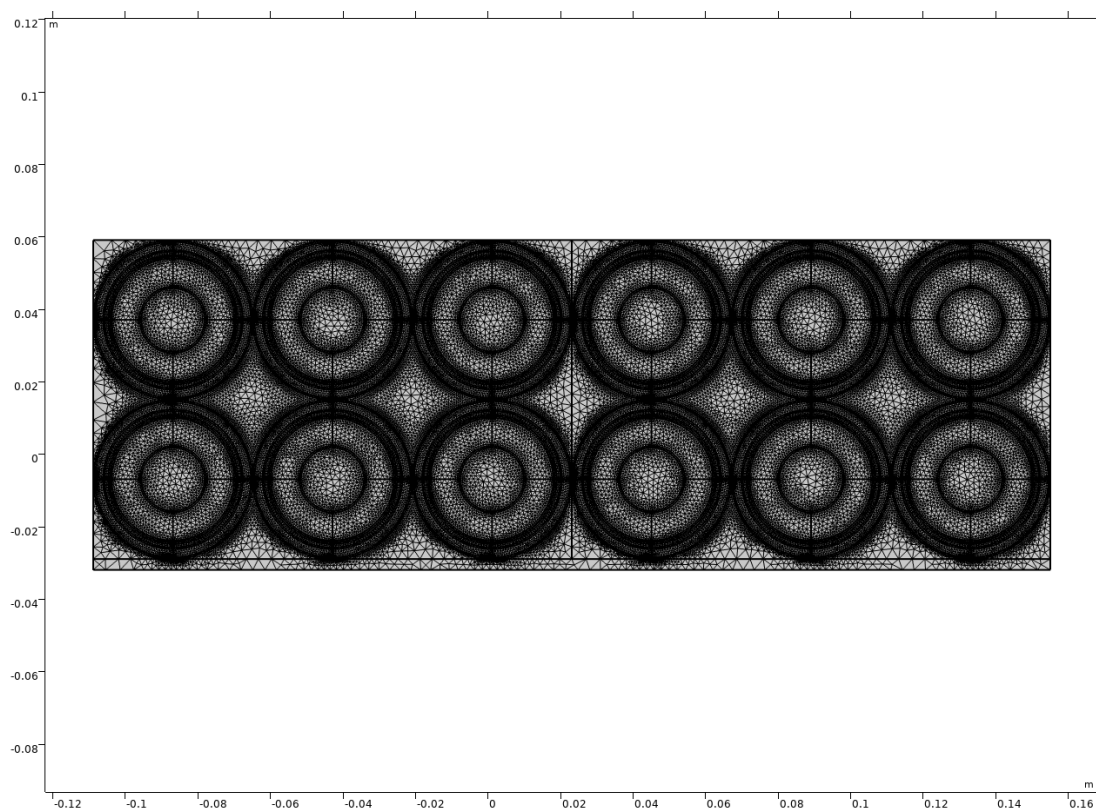


Figure 7 - 18 Mesh plot: Fine, triangular meshing

7.2.2.5.2 Closed tray variation

Similarly, a triangular type of mesh is defined on the domains treated by the software as thermal conduction. Note, that as the internal air domain is treated using the analytical formulas, this domain should not be meshed, but instead the top side of the cable bundle and the bottom side of the tray cover are configured with heat fluxes as given in Equation 7.13, where the outwards and inwards fluxes are equal. The top of the cover is defined with the external heat flux as given in Equation 7.12. Refer to Figure 7 - 19 for the schematic of the final meshing of the closed tray.

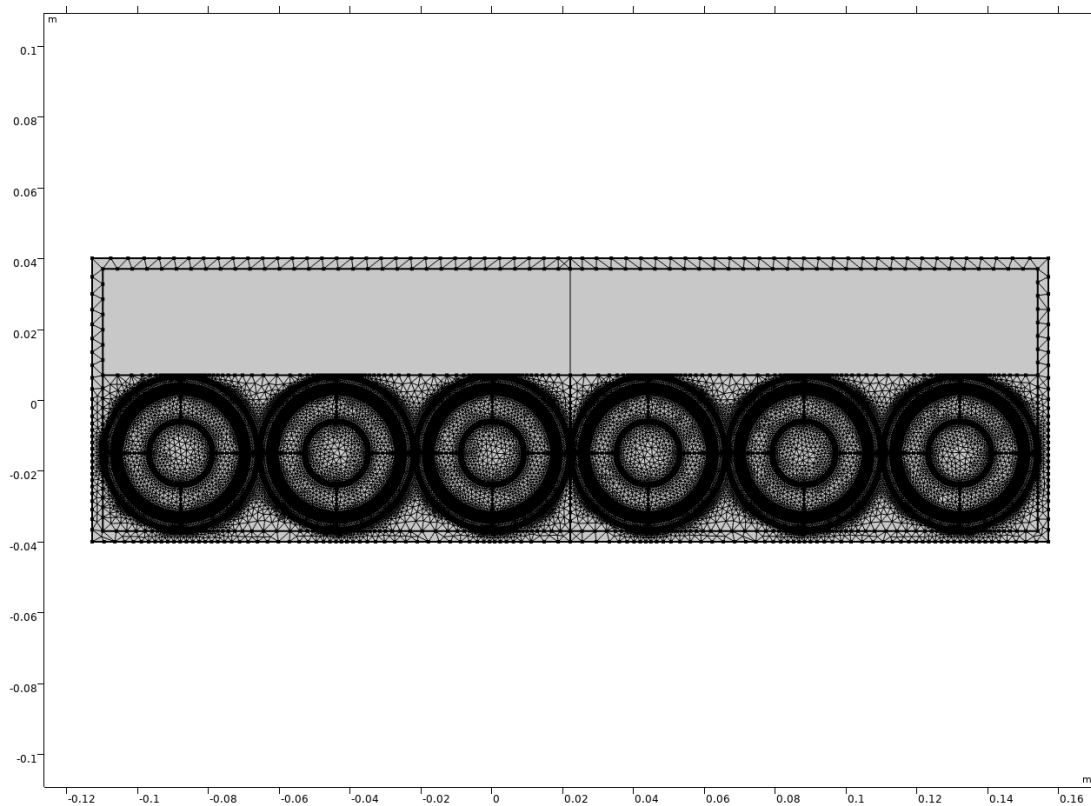


Figure 7 - 19 Mesh plot: Fine, triangular meshing

7.2.3 Results

Below are the temperature results for both variations.

7.2.3.1 Open tray variation

In Figure 7 - 20 below, the simulation results in the form of a 2D temperature plot can be seen, where the maximum temperature (HST) of the complete geometry (bundle) is computed within the software and marked on the same output figure.

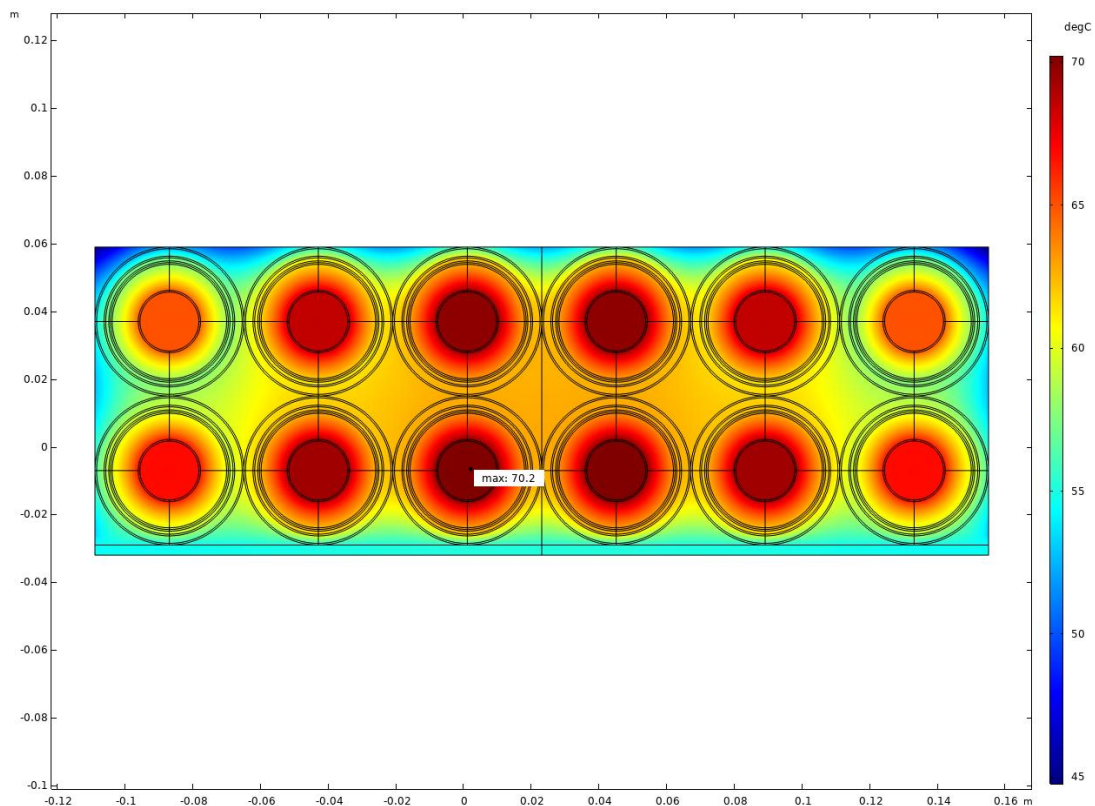


Figure 7 - 20 Surface temperature plot with HST indication

In Table 7 - 24 below, the evaluated temperatures, AC resistances, and heat losses per conductor are listed. Refer to Figure 7 - 16 for the conductor numbering. Note, that only the left part of the geometry is numbered, as the right side is mirrored using a symmetry boundary condition (more information on this type of boundary can be found in the section related to CFD below).

Table 7 - 24 Tabulated results of conductor temperatures, temperature-dependent AC resistances, and heat losses

Parameter	Unit	Conductor 1	Conductor 2	Conductor 3	Conductor 4	Conductor 5	Conductor 6
Temperature	°C	65.0	68.6	69.9	66.8	69.5	70.2
AC resistance	Ω/km	0.1149	0.1162	0.1167	0.1156	0.1165	0.1168
Heat loss	W/m	20.7	20.9	21.0	20.8	21.0	21.0

7.2.3.2 Closed tray variation

The simulation results are displayed in Figure 7 - 21 below in the form of a 2D temperature plot, where the software computes and marks the maximum temperature (HST) of the entire geometry (bundle) on the same output figure. It should be noted that the enclosed air cavity lacks a temperature field computation since the FE model does not explicitly simulate this domain, as the convective heat transfer coefficient expressions were applied to the top and bottom trays of the cable bundle.

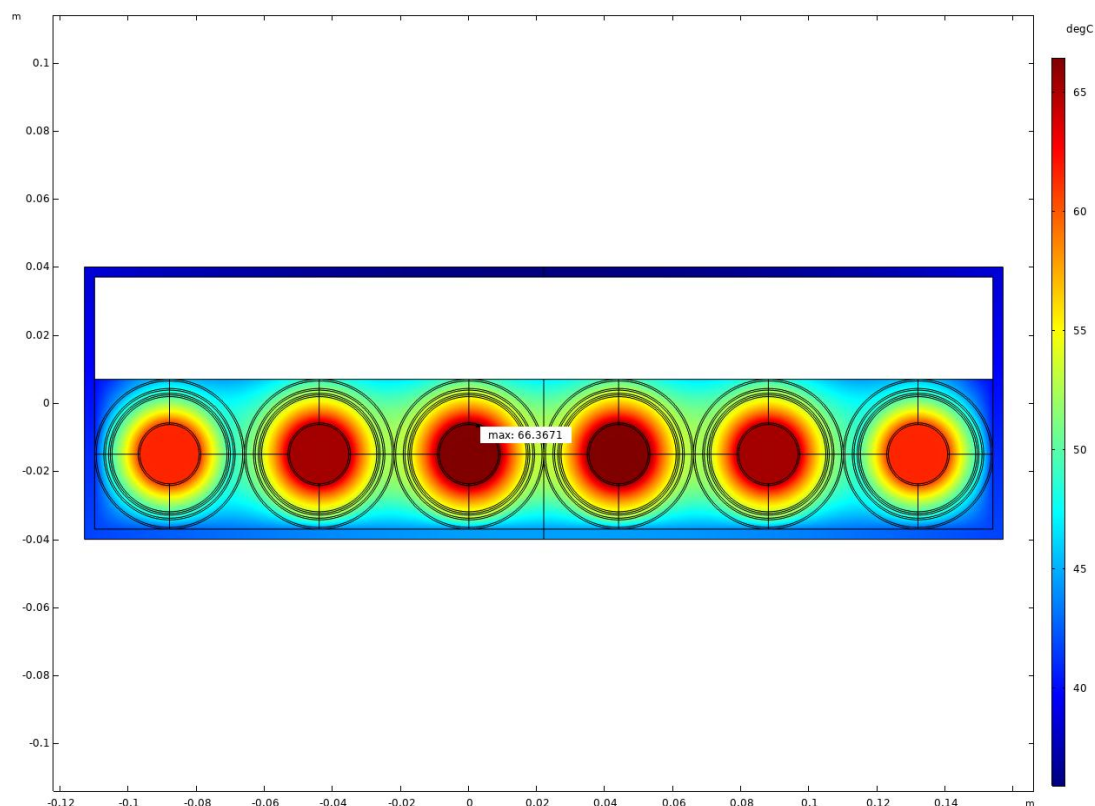


Figure 7 - 21 Surface temperature plot with HST indication

In Table 7 - 25 below, the evaluated temperatures, AC resistances, and heat losses per conductor are listed. Refer to Figure 7 - 17 for the conductor numbering. Note, that only the left part of the geometry is numbered, as the right side is mirrored using a symmetry boundary condition (more information on this type of boundary can be found in the section related to CFD below).

Table 7 - 25 Tabulated results of conductor temperatures, temperature-dependent AC resistances, and heat losses

Parameter	Unit	Conductor 1	Conductor 2	Conductor 3
θ_c	°C	61.6	65.4	66.4
R_{ac}	Ω/km	0.1135	0.1150	0.1154
W_c	W/m	36.3	36.8	36.9

7.2.4 Comparison with CFD

For comparison with the results above that used the analytically implemented convective heat transfer coefficients' approach, the same geometries, i.e., open and closed tray variations, are simulated using CFD within the fluid domains, instead of convective heat transfer coefficients to the ambient, non-simulated domains. For CFD, the air fluid domains have to be explicitly modelled within the FEA software, and fluid dynamics must carefully be considered when setting up such models. The materials and heat sources are the same as above, while the geometry, physics, mesh, and results for the CFD implementation are presented below.

7.2.4.1 Geometry

7.2.4.1.1 Open tray variation

In Figure 7 - 22 below, the geometry used for the CFD simulation approach is shown. Note, that only half of the geometry is shown (and modelled), as a symmetry boundary condition is defined on the "middle", vertical line, hence the whole geometry is mirrored to the right after simulations - more information on this boundary can be found in the section below.

Compared to the previous simulations that deployed analytical terms of convective heat transfer coefficients as the outer boundaries, the CFD approach requires well-defined fluid domains, in this case air, as to model the fluid dynamics surrounding the cable bundle. The layout of the geometry for this study case as seen in the figure is specifically adapted for this exact configuration, and the details necessitating the setup of the air fluid domain are given in the Physics Section below.

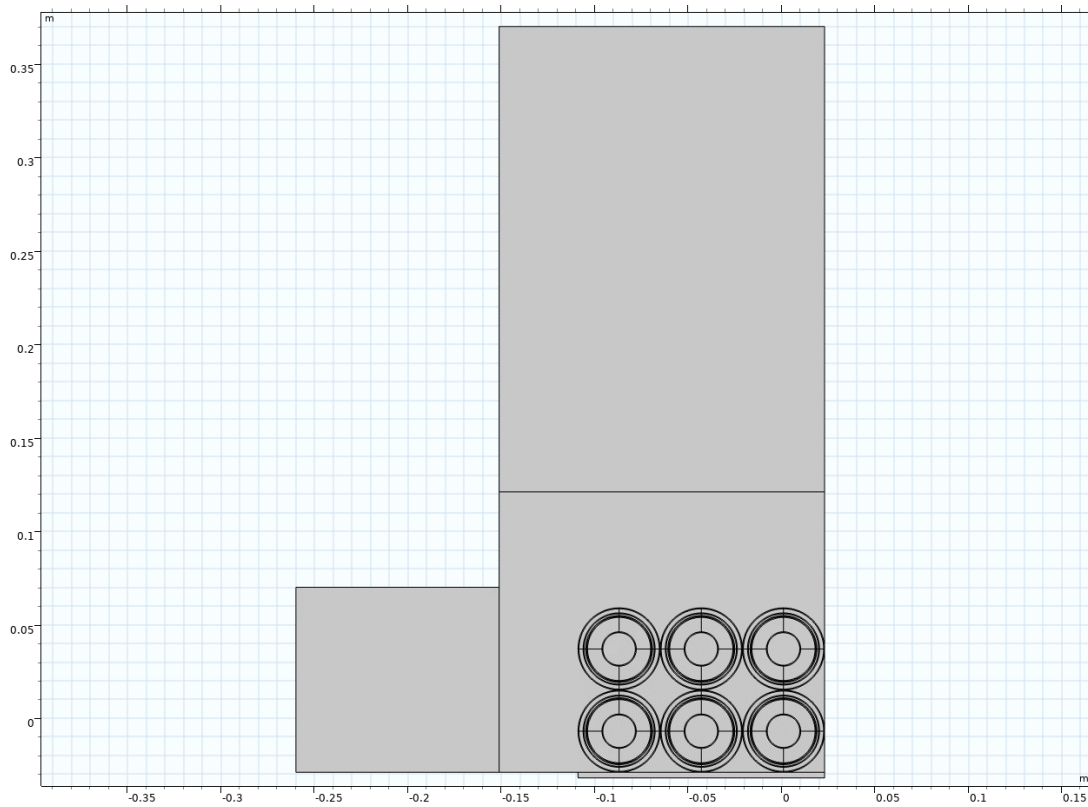


Figure 7 - 22 Inputted geometry to the CFD simulation, where only "half" the model is represented with enforced symmetry boundary conditions

7.2.4.1.2 Closed tray variation

Identical to the geometry of the closed tray geometry for the analytical approach, refer to Figure 7 - 17.

7.2.4.2 Physics

The heat sources are not changed from the previous analytical approach, hence this section focuses solely on the CFD specific physics' part. For simulations including CFD with heat sources, the initial task is to determine the flow type of the fluid surrounding the heat sources of interest, i.e., can the flow be expected to be of laminar or turbulent behaviour, thus is a turbulence model needed or not. Determining the need for a turbulence model based on an estimation of the final max temperature, from which the Reynolds and Grashof numbers can be calculated. The determination of the numbers using expected real-life end temperatures provides a very good indication of the flow type to adapt, and is widely used within the CFD engineering.

7.2.4.2.1 Open tray variation

7.2.4.2.1.1 Convection

Calculating the above-mentioned number would indicate a laminar flow below the threshold, but some turbulent flow may also be expected depending on the max temperature of the cable bundle.

Turbulence model

In open-air cable installations, where airflow can be assumed "infinite", using a turbulence model, particularly the Reynolds-Averaged Navier-Stokes (RANS) model, can provide stable and efficient results. The RANS approach simplifies computations by averaging out small fluctuations in the flow field, focusing instead on large-scale structures like vortex shedding. Beginning with a coarser mesh is effective for such models, as it introduces numerical diffusion, smoothing out minor details while

providing computational efficiency. Subsequent mesh refinements can confirm solution accuracy if mesh sensitivity is suspected. The k-epsilon (k- ϵ) model, a standard in industrial applications, balances performance and accuracy, especially where strong pressure gradients are absent, making it ideal for scenarios like open cable trays where achieving a stable temperature distribution is the main goal.

Compressibility

In this open tray case study, an incompressible flow setup is appropriate given the expected low temperature and pressure variations. Although theoretically compressible, air is treated as incompressible, simplifying computations. For non-isothermal scenarios, where temperature differences induce buoyancy-driven flows, the Boussinesq approximation is used. This captures density variations due to temperature gradients without resorting to fully compressible simulations, maintaining computational efficiency while addressing thermal effects on airflow.

Solution type and solver

For external natural convection in open-air setups, a steady-state solver is suitable, as a stable state can generally be assumed for temperature distribution. However, specific enclosed configurations may require time-dependent solvers, as covered in the Guidance Point 23 and Guidance Point 24.

Boundary conditions

Boundary conditions in CFD simulations must realistically reflect the cable-air interaction while keeping the model computationally manageable. In open tray scenarios, inlet and outlet boundary conditions are effective for establishing and maintaining airflow directionality. Positioning these boundaries at sufficient distances from the region of interest ensures uniform pressure around the cables, reducing the influence of artificial boundaries on the solution. Inlet and outlet conditions also help avoid unwanted recirculation or backflow within the domain, ensuring solution stability. Some FEA software provides alternative "open boundary" conditions for large domains; however, standard inlet/outlet conditions are often more reliable, offering greater control over the simulated airflow, especially in cases requiring precise temperature distribution predictions. Refer to Guidance Point 48 for additional information regarding boundary conditions for CFD simulations.

For the implementation of the boundaries for the open tray variation, refer to Figure 7 - 23 below, where an outlet boundary condition (red colour) is placed as such to extend the fluid domain in the vertical direction, and an inlet boundary condition (blue colour) is defined to the left, also some distance away from the region of interest. Locating the inlet and outlet boundaries in such manner, forming "channels", can help prevent backflow and circulation between (around) inlet/outlet boundaries directly - these channels should not be too fine, as this could impose an artificial restriction on the flow itself. Both the inlet and outlet boundary are defined as pressure constrained boundaries, with static pressures of 0, thus it is equal to the reference pressure of the domain (1 atm) - in practice, the pressure far away from the cables can indeed be assumed to equal 0, while (small) pressure gradients can be expected close to the heat sources, i.e., at the cables. The pressure condition specifies the normal stress, which more or less is equal to the pressure, while the tangential stress component is set to zero.

To further improve the robustness, by imposing further restrictions on the flow, the bottom part of the air domain to the left is modelled as a no-slip wall boundary condition (magenta colour), and the cable bundle surface is also of no-slip type. The no-slip condition sets the tangential velocity at the boundary to zero, blocking the inflow from the left, and the flow can be expected to be more in the vertical direction, as in natural convection.

On the other hand, the left vertical and horizontal wall (green colour) are both set up as slip wall boundary conditions, to prevent these from introducing additional friction, which also helps improve robustness of the converging - no boundary layer develops at slip walls. The slip walls can also be seen as approximations as in preventing fluid from leaving the domain (in practice it could also be added to the magenta coloured part of the air and tray boundary facing the modelled air, not the cable surfaces, which must be modelled as no-slip).

As mentioned in the geometry section above, the right, middle "wall" (cyan colour) is a symmetry boundary condition that should be used when the geometry can be mirrored, to significantly reduce computation time.

In this kind of model some approximations, like the above wall boundary conditions, have to be made, simply because including the "infinite" air domain around the cable bundle would not be possible. However, as the main purpose in this simulation is the temperature fields, such approximations would

be the go-to approach. The effect of varying the distances to the inlet, outlet, and wall boundary conditions should always be checked during simulations, to ensure that the approximated conditions are well established.

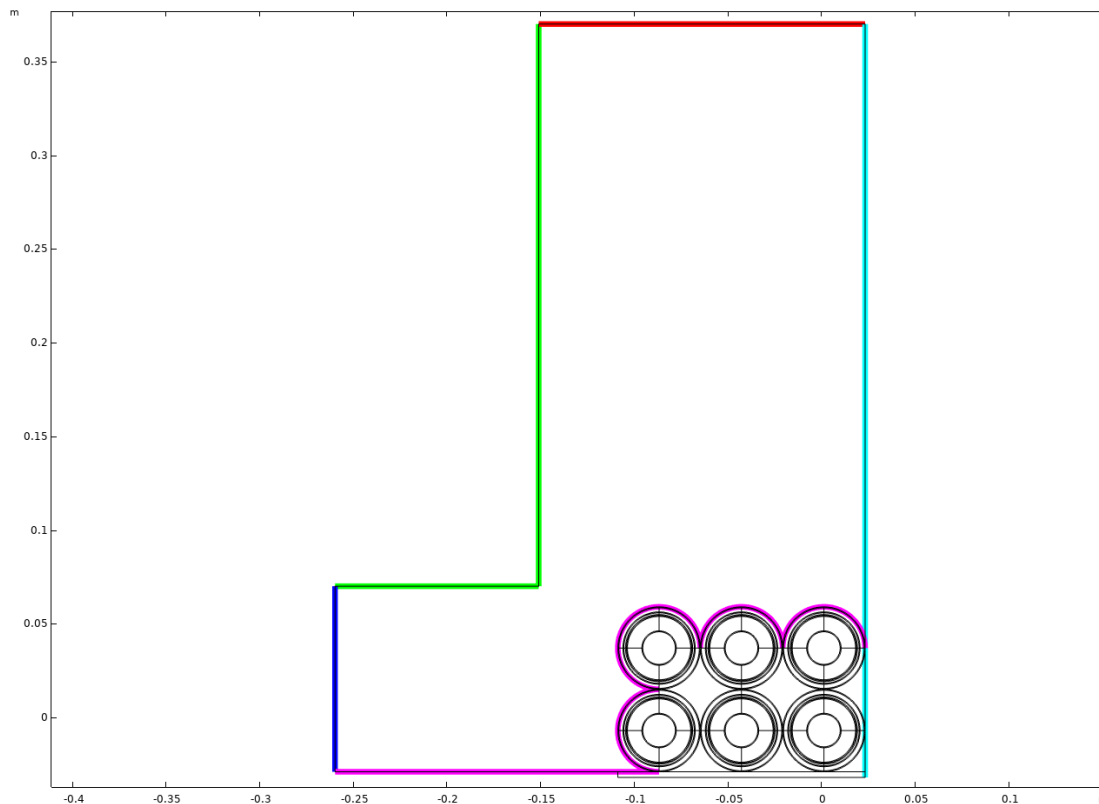


Figure 7 - 23 Boundary conditions related to the fluid domain, magenta: No-slip, blue: Inlet, green: Slip, red: Outlet, cyan: Symmetry

The tray bottom and sides (thickness of tray) that are exposed to the non-modelled fluid domain, simply are fitted with external convective heat transfer coefficients for horizontal and vertical walls from the literature, and likewise an external radiation (to ambient) coefficient.

7.2.4.2.1.2 Radiation

The radiation from the cable bundle surfaces (and small part of the tray cover at each end) within the fluid domain is accounted for using a radiation physics interface, making it a multiphysical simulation. The surface radiation physics includes both surface-surface radiation and surface radiation to the surrounding fluid (ambient). The physics make use of the radiosity method to compute the radiative flux (both reflected and re-radiated), incorporating the hemicube method for the computation of the irradiance, and any obstacles and shadowing effects are accounted for in this method. Only diffuse radiation is accounted for and no wavelength dependence, as no solar radiation (external heating) is included.

7.2.4.2.2 Closed tray variation

7.2.4.2.2.1 Convection

In the domain of CFD modelling, the distinction between closed and open tray variations is profound, primarily due to the differences in fluid domains. With the closed tray, the focus is on an enclosed fluid domain, in contrast to the open tray where the domain is open and seemingly infinite. Enclosed air domain simulations, especially with natural convection, present unique challenges. The behaviour of natural convection in such settings tends to be unsteady, which complicates computations of steady-state solutions. A model's sensitivity can further amplify these challenges, resulting in multiple potential steady-state outcomes. Additionally, factors such as initial conditions and predefined study steps can influence the results. One work-around can be to reduce the solver tolerance. However, for scenarios like the one involving enclosed air cavities with natural convection-driven flow, time-dependent solvers offer the most effective approach. Even when steady-state conditions are desired, these solvers may need to run extensively to capture the desired state. This approach is particularly

beneficial when considering the complexities of heating in an enclosed space. As air heats, pressure rises — a phenomenon often missed by steady-state simulations. This makes the case for adopting a transient, compressible flow simulation. The advantage of the compressible formulation is its consideration for pressure-determined density changes. It recognizes that, as the pressure decreases, the confined air would ideally want to expand. But given the restrictive nature of the surrounding walls, expansion is limited. Consequently, the compressible formulation inherently governs the pressure level during a transient simulation, removing the need for any manual pressure constraints. Thus, a transient approach mirrors real-life flow dynamics in time. Conversely, when treated as a stationary problem, a solver will leap directly to a steady state. While it may meet convergence criteria, the result may not truly represent the natural steady state.

No turbulent conditions are expected for the closed case simulation, and a time-dependent, laminar simulation type is preferable, such that the fluid flow field is correctly resolved (as opposed to many turbulence RANS models) and the pressure/density relationship is carefully accounted for.

Similar to the open tray variation above, heat exchange with the external boundaries are modelled using heat transfer coefficients for both convection and radiation, where the literature provides well-established formulas for horizontal and vertical walls, the first both upwards and downwards facing.

Of course, this 2D problem with enclosed air is in fact a simplification of the physical system, as air flow in here would be allowed to escape at the, normally, open ends of the tray, i.e., not constraining the volume and causing the pressure to increase arbitrarily. However, the limited pressure increase for this 2D natural convection problem is not thought to affect the final steady-state temperatures much, and the 2D model should provide sufficient coverage of the real-world problem.

7.2.4.2.2 Radiation

Same approach as for the open tray case above, where radiation within the fluid domain is accounted for by a separate radiation physics interface.

7.2.4.3 Mesh

7.2.4.3.1 Open tray variation

CFD modelling requires great attention to the meshing of the fluid domain and any boundaries to solid parts, and depending on the turbulence model, the mesh handling of the solid walls. In general, the no-slip walls should be handled by boundary layer meshes, and for turbulence model, extra care is needed, due to the so-called "law of the wall". In turbulence CFD models, the "law of the wall" pertains to the behaviour of velocity in turbulent boundary layers adjacent to solid walls. For a flat plate boundary layer, as one moves away from the wall, the velocity profile transitions from a linear viscous sublayer to a logarithmic "log-law" region, and finally to an outer layer influenced more by external flow conditions than the wall. To practically simulate this in CFD without requiring an extremely fine mesh near the wall—which would be computationally expensive—wall functions are employed. These are semi-empirical relationships that model the near-wall velocity profiles. By using wall functions, simulations can place the first grid point away from the wall in the logarithmic region, and then leverage the wall function to connect wall shear stress with the mean velocity at that point. This approach provides a balance between computational efficiency and the fidelity of capturing turbulent behaviour near walls, and such functionality is found in most FE software tools.

The wall function treatment of the solid surfaces is adapted in this model, which, as indicated above, is very useful for engineering applications, and for this case where the temperature field is the main interest. The mesh must be setup accordingly, where boundary meshing is needed for the no-slip walls. As a rule of thumb, five layers can be resolved over the thermal boundary layer thickness. The thickness of the layer can be estimated using the Prandtl and Rayleigh numbers in combination with the typical length of the geometry.

In Figure 7 - 24 below, the meshing of the domain is shown with zoom in on the boundary layers between two adjacent cable surfaces, where the five layers are evident. Note, that no boundary layers are needed for the slip walls, as no velocity gradient is setup in the tangential direction, thus no boundary layer is formed. The remaining triangular mesh of the fluid is coarsely resolved to increase numerical diffusion as previously explained.

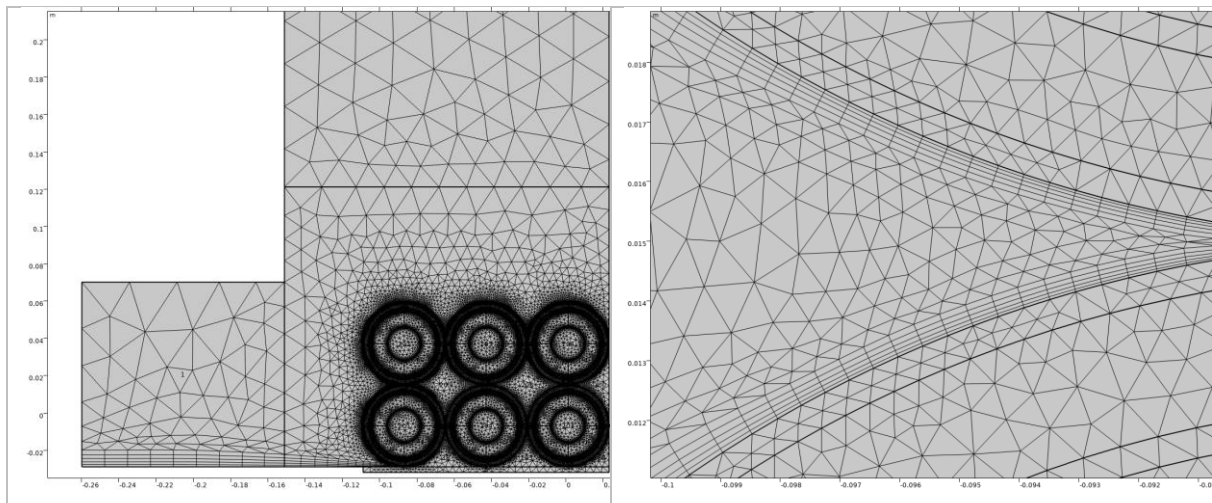


Figure 7 - 24 Left: Meshing of open tray variation with CFD, right: Zoom-in on the boundary layer mesh of the no-slip walls

7.2.4.3.2 Closed tray variation

Similar to the open tray variation above, the closed tray variation is computed (initially) using a coarse mesh. Convection cells can be expected to form in the enclosed air cavity, which by nature are unsteady. To ease the solver, a coarse mesh is here used to compute the physics and run for enough time (200 min) for the unsteady convection cells to form - in order to capture such transient flow field variations, most solvers will typically restrict the time steps - after which a finer mesh is initiated as a step in the solver sequence, and run for very long time (1440 min, or 24 h). Doing so, can greatly reduce the computation time; had the full 24 h been run on the fine mesh, the computation of the fluid field variations in the beginning would take very long. Thus, the mentioned workaround simply computes a "rough" solution first, where convective cells stabilize, after which the solution is continued on a fine mesh and steady-state results reached.

Both the coarse and fine meshes can be found in Figure 7 - 25 below.

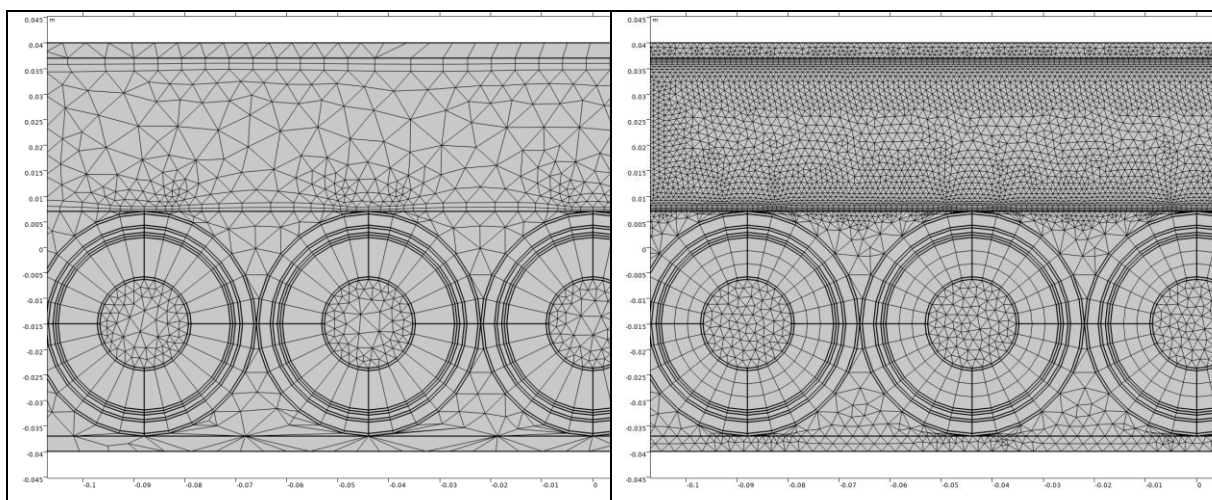


Figure 7 - 25 Left: Coarse meshing of closed tray variation with CFD, right: Fine meshing used in second step of simulation

As evident in Figure 7 - 25, The cables in this design are meshed differently than in earlier versions, where both the cables and their surrounding medium used triangular meshes. In the closed tray with the CFD variation, the cables are now meshed using a mapped mesh type. This change substantially reduces the degrees of freedom, consequently cutting down the computational time required for this fluid flow simulation. The meshing around primary heat sources, where most heat gradients are present, has been notably optimized. This method is a powerful way to reduce computation times across all kinds of FEA simulations, not only one including CFD. Be aware, that when conducting

simulations that involve electromagnetic physics, it's essential to ensure that current-carrying materials are resolved with sufficient layers to account for contributions from skin and proximity effects, particularly at elevated electrical frequencies.

The results acquired from the above meshing has been compared to results from triangular, standard meshing, with insignificant changes in the compared results.

In Figure 7 - 26 below, the solver (reciprocal) step size as function of time steps, where it from the left figure can be seen that the solver restricts (decreases) its step size, due to transient, unsteady fluid flows occurs, after which it increases again (marked by red bars), and the next solver step with finer mesh takes over (right figure), until steady solution in time is reached. The time start of the second step should only be initiated after stabilization of convection cells, i.e., long enough time after for the time stepping to increase again.

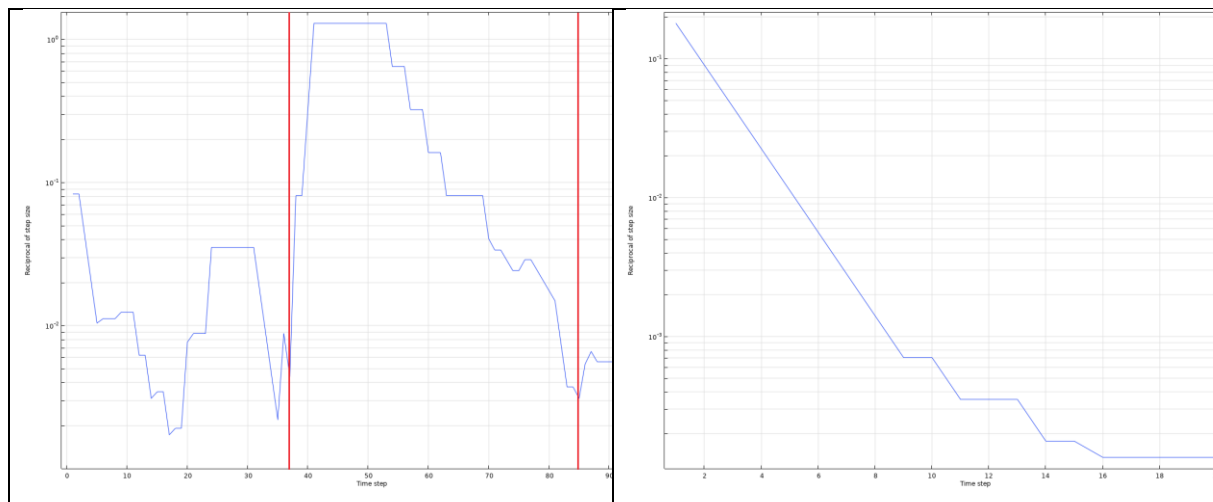


Figure 7 - 26 Solver reciprocal step size as function of solver step, left: Coarse mesh with red indicators of decrease of sizing as a result of unsteady convection cells forming, right: Fine mesh, solver taking over from coarse mesh

From the above solver step development, the setup physics can be concluded to converge well with the approach of the splitting of the transient time simulation in two solver steps. Alternatively, instead of dividing the solving process into two, one could instead reduce the heat capacity (specific heat) of the solid cable bundle to faster reach a steady-state in the time-dependent regime, as the steady-state results should not be affected by this - this method is not valid if transient results are of interest.

7.2.4.4 Results

7.2.4.4.1 Open tray variation

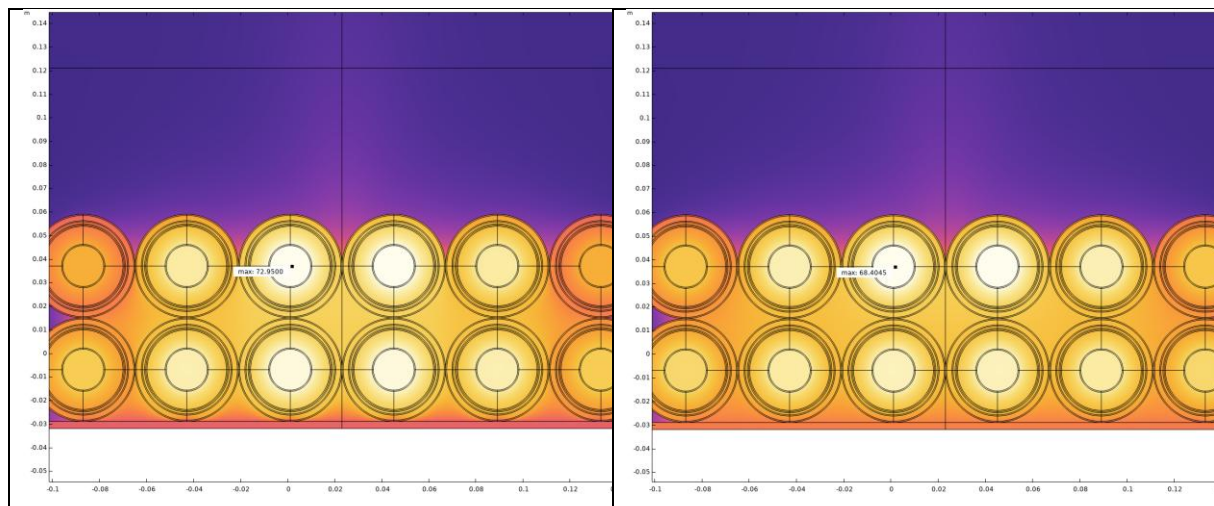


Figure 7 - 27 Temperature results with HST indication of CFD simulations, left: with air conductivity and radiation in air cavities enclosed by cables, right: outer sheath thermal conductivity of air cavities enclosed by cables

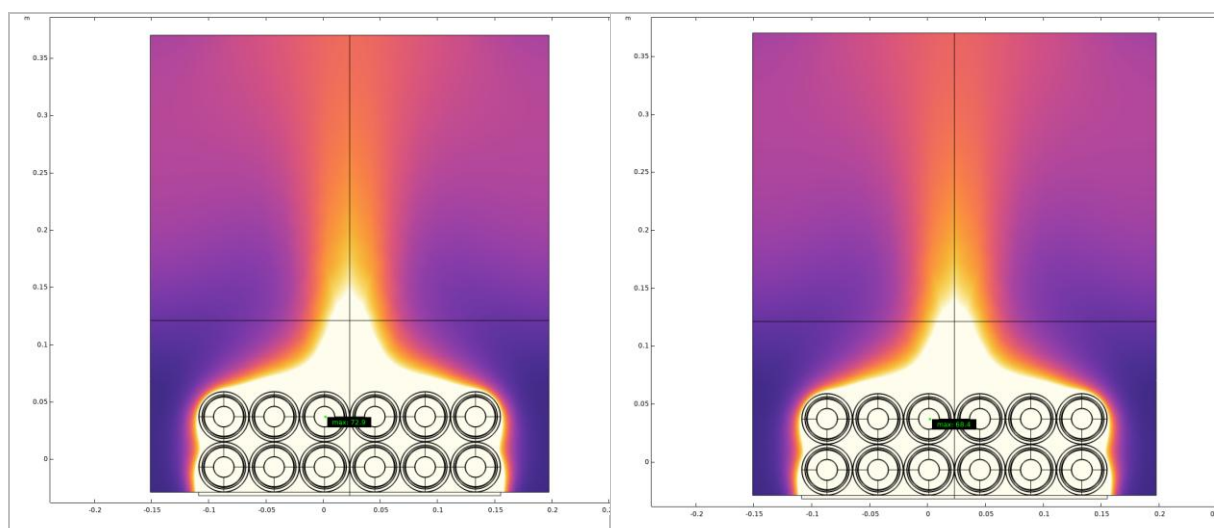


Figure 7 - 28 Same as above figures, but with enhanced contrast to emphasize the fluid temperature field of the surrounding air

7.2.4.4.2 Closed tray variation

In Figure 7 - 29 below, the final temperature field of the closed tray CFD simulation is plotted, i.e., after the transient simulations has run for full simulation time of 24 h.

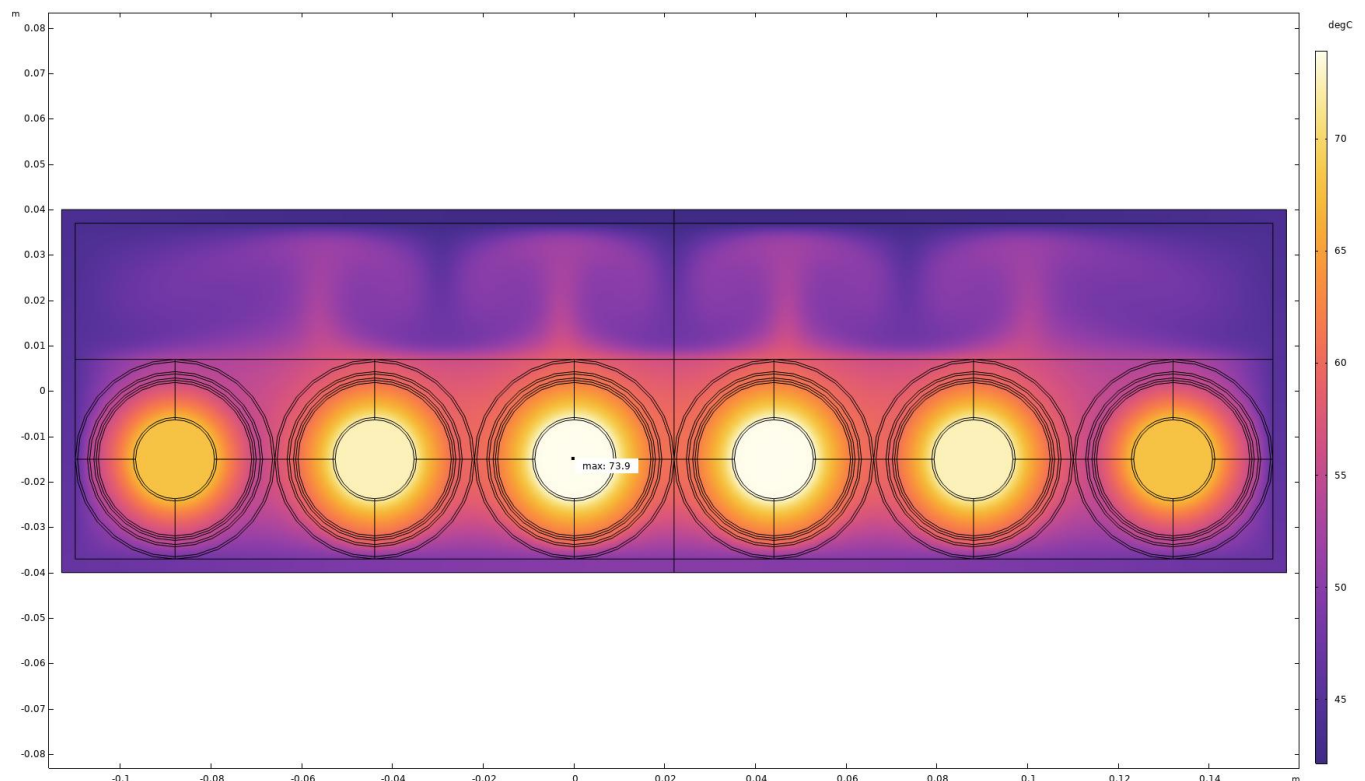


Figure 7 - 29 Temperature results with indication of HST for CFD simulation at $t = 24$ h

In Figure 7 - 30 below, the fluid field velocities have also been plotted at $t = 24$ h. On the figure, the convection cells mentioned previously are clearly evident, and at the 24 h time these are well-developed and steady.

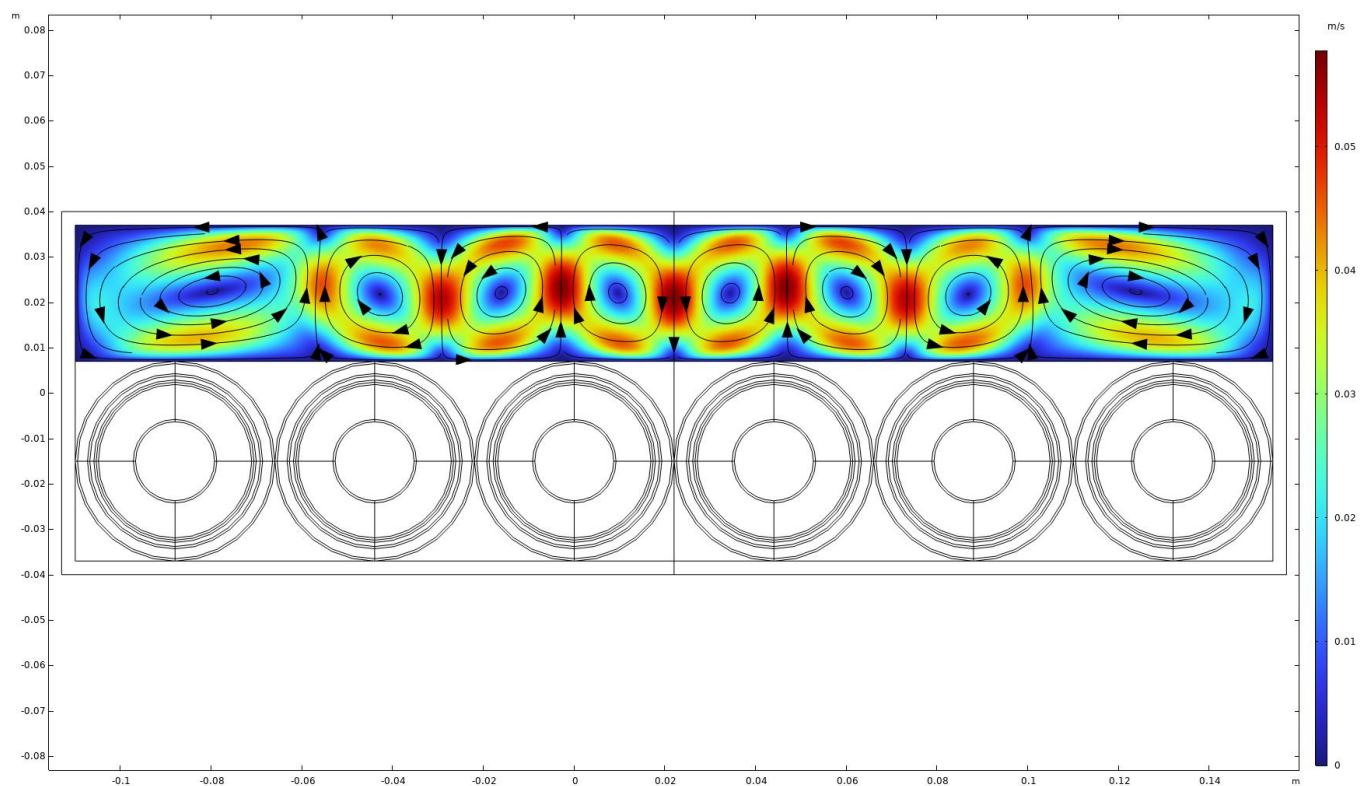


Figure 7 - 30 Velocity field results of CFD simulation at $t = 24$ h

7.2.4.4.3 Comparison of results

A comparison of final temperatures are presented in Table 7 - 26. From the table, it can be noted that the analytical implementation of the closed tray scenario results in a lower computed HST than that of the CFD implementation, hence the analytical model can be seen as non-conservative with risk of underestimation of the temperatures.

Table 7 - 26 Comparison of results for open and closed tray with analytical and CFD method

Parameter	Unit	Open tray, analytical	Open tray, CFD	Closed tray, analytical	Closed tray, CFD
Temperature (HST)	°C	70.2	68.4	66.4	73.9

7.3 Case 3 J-tube

7.3.1 Case definition

We examine a 3-core submarine cable with non-magnetic stainless steel armour installed in a J-tube. The model comprises three distinct sections:

1. **Water-Filled J-Tube Section:** This segment of the cable is submerged in water within the J-tube.
2. **Air-Filled J-Tube Section:** Here, the cable resides in an air-filled J-tube.
3. **Phases Separated in Air After Hang-Off:** In this part, the cable phases are separated and directly exposed to the ambient air.

Key Input Parameters:

- **Cross-Section:** 3x1000 mm² copper (Cu)
- **Insulation:** XLPE (Cross-Linked Polyethylene)
- **Sheath:** Lead
- **Oversheath:** Semiconducting PE (Polyethylene)
- **Armor:** Stainless steel
- **Voltage Level:** 220 kV (Um=245 kV)
- **Bonding:** Solid
- **J-Tube Condition:** Air/water-filled

7.3.2 Geometry

The cable geometry is as described in study case 8 in CIGRE TB 880. The armour is modelled as a tube with thermal properties as steel. This is possible as the armour is made stainless steel and has negligible losses. The power phases have been modelled as straight, however, the losses take into account the twisting of the cores. The lengths of the different sections, and geometry of the J-tube is shown in the table below, and depicted in the figure.

The J-tube is assumed closed at the top in all cases. The difference in rating with an open top is typically small. In most cases assuming the J-tube is closed at the top will give realistic results on the conservative side, and uncertainty in other factors such as emissivity and absorptivity will have a more significant impact on the results than whether the J-tube is open or closed at the top.

Table 7 - 27 Geometric parameters for the J-tube

Parameter	Value	Unit
Outer diameter of J-tube	555	mm
Inner diameter of J-tube	475	mm
Length of J-tube in air	15	m
Length of J-tube in water	10	m
Length of phases in air	10	m

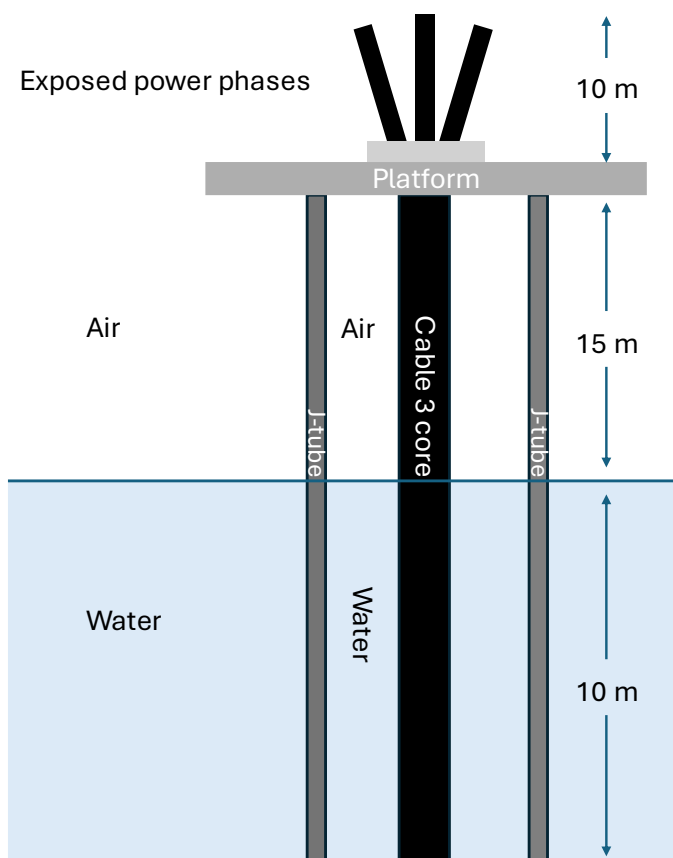


Figure 7 - 31 Overview of the J-tube geometry

7.3.3 Cable data

The cable has been modelled according to the geometry described below. The fillers have been modelled as solid PE, which is likely to be conservative (Guidance Point 12), and gives a higher value for T_2 than IEC 60287. Losses are applied to the different metallic elements, without any temperature adjustment, with a constant AC conductor resistance and sheath loss factor corresponding to the values applicable at maximum temperature. The thermal resistance of the insulation is equal to the value given below ($T_1=0.487$ Km/W).

Table 7 - 28 Cable geometry

Parameter	Value	Unit
Conductor diameter	38.6	mm
External diameter of metal sheath	99.4	mm
Thickness of metal sheath	3.6	mm
Axial separation of conductors	104.6	mm
Armour inner diameter	227	mm
Armour thickness (armour wire diameter)	5	mm
PP-yarn thickness	4	mm
Cable diameter	245	mm

Table 7 - 29 Thermal properties for the cable

Parameter	Value	Unit
Thermal resistance between conductor and metal sheath	0.487	Km/W
Thermal resistivity of XLPE/PE	3.5	Km/W
Thermal resistivity of semiconductive PE	2.5	Km/W
Thermal resistivity of PP-yarn	6	Km/W

Table 7 - 30 Thermal conductivity of metals

Parameter	Value	Unit
Thermal conductivity of Cu	380	W/Km
Thermal conductivity of Pb	35.3	W/Km
Thermal conductivity of Steel	18	W/Km

Table 7 - 31 Electrical parameters

Parameter	Value	Unit
AC resistance of conductor, R_{ac}	2.728×10^{-5}	Ω/m
Screen loss factor, λ_1	0.459	NA
Dielectric losses (per core), W_d	0.963	W/m

7.3.3.1 Operating conditions

The operating conditions are shown in the table below. Note that solar radiation and wind speed has to be evaluated on a case-by-case basis. The values given here are examples used for this case. The correct value for the solar radiation is not assessed using the finite element tool, and is typically calculated with an analytical method appropriate for the specific location. Wind speed should be adjusted based on the application. If the cable is for a wind farm, the wind speed will probably be higher than 5 m/s, but it should be noted that wind speed changes with height above seawater, and the wind speed at the J-tube might not be the same as at the blades of the wind turbines. For other

applications, for example for oil and gas electrification projects, the wind speed may be close to zero at full load/current.

The current is not given in the table, as the current is adjusted such that the conductor temperature is 90 °C.

Table 7 - 32 Operating conditions case study 3

Operating conditions	Value	Unit
Maximum operating conductor temperature	90	°C
Ambient air temperature	30	°C
Ambient water temperature	10	°C
Solar radiation maximum	1000	W/m ²
Wind speed	5	m/s

7.3.4 Mesh

The system has been meshed with normal triangular mesh in the 2D plane, looking into the cable, and a mapped mesh longitudinally along the cable/J-tube.

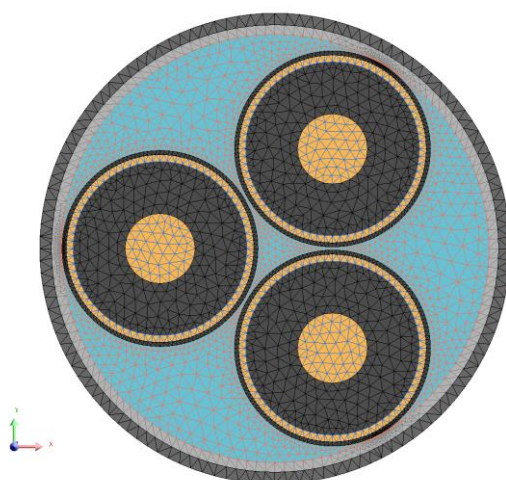


Figure 7 - 32 Mesh in 2D

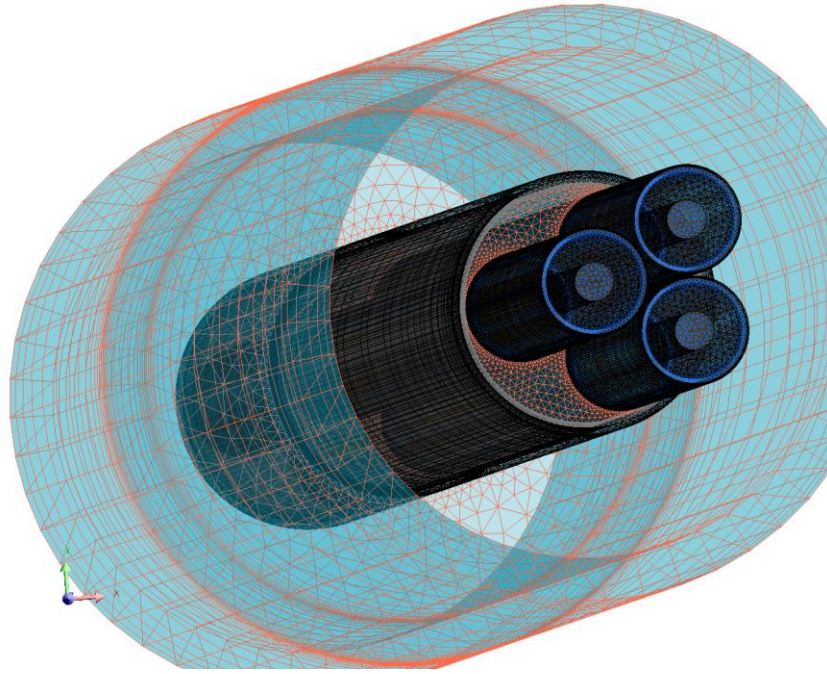


Figure 7 - 33 Mesh in 3D, in software package 2

7.3.5 Losses

The Finite Element model considered is a thermal model, and the losses are input values based on the cable data.

The conductor losses are calculated by the formula (per unit length) below, and assigned to each conductor per m.

$$W_c = R_{AC} I_0^2$$

Equation 7.16

The screen losses are calculated by the formula below (per unit length), and assigned to each screen per m. When selecting λ_1 special attention must be paid to the sheath circulating current and the resulting losses. Standard IEC 60287 equations will not account for the influence of the increased phase spacing above the hang off, which can have the effect of increasing the circulating current.

$$W_{sh} = R_{AC} I_0^2 \lambda_1$$

Equation 7.17

Losses in the armour (stainless steel) and the J-tube itself are assumed to be zero.

Dielectric losses are assigned to the XLPE insulation per m per phase.

7.3.6 Heat transfer coefficients and emissivity

The heat generated by the cable are transferred by several different heat transfer modes:

1. **Conduction Inside the Cable:** Heat is conducted from the inner core to the cable surface.
2. **Radiation and Convection at the Cable Surface:** The cable surface exchanges heat with the inner wall of the J-tube through radiation and convection.
3. **Convection and Radiation from the J-Tube Outer Wall:** Heat is dissipated from the outer wall of the J-tube to the surrounding environment. Convection may occur due to both natural and forced convection.
4. **Solar Radiation Impact:** Sunlight directly affects the J-tube, subsequently influencing the cable temperature.

The schematic representation of these distinct heat transfer modes is depicted in the figure below.

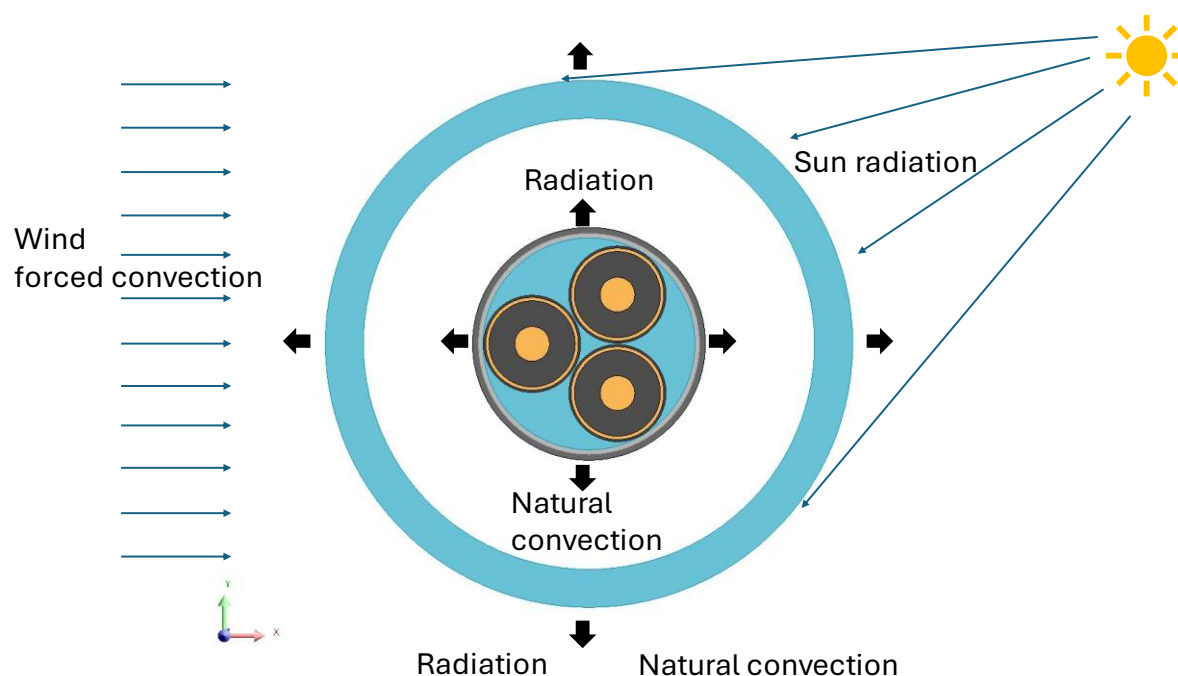


Figure 7 - 34 Overview of the heat transfer by convection and radiation

The input data needed to model the convection and radiation heat transfer modes are given in the table below:

Table 7 - 33 Input data for convection and radiation

Parameter	Symbol	Value	Unit
Gravitational constant	g	9.81	m/s^2
Volumetric expansion of air	β	0.003	$1/\text{K}$
Emissivity of cable and J-tube surfaces	ϵ	0.9	
Solar absorption constant	α	0.4	
Kinematic viscosity	ν	20×10^{-6}	m^2/s
Thermal conductivity of air	k_{air}	0.024	
Stefan Boltzmann constant	σ	5.6704×10^{-8}	$\text{W}/(\text{m}^2 \text{K}^4)$
Prandtl number	Pr	0.71	
Ambient temperature air	θ_{air}	303.15	K
Water temperature	θ_{water}	283.15	K
Outer diameter of J-tube	D_{j-o}	555	mm
Inner diameter of J-tube	D_{j-i}	475	mm
Outer diameter of cable	D_c	245	mm
Length of J-tube in air	L_{J-tube}	15	m

Note that several of the material parameters are temperature dependent. In this study case the parameters have for simplicity been kept constant. If the parameters are modelled as constant it should be checked that they are representative for the temperature in the final model. It is also a possibility to model them with the temperature dependence included in the FE model.

For a given wavelength of radiation, the absorptivity is equal to the emissivity. However, since thermal radiation from the sun is given for a specific range of wavelengths, the solar absorption coefficient is typically not equal to the emissivity. The actual value of the emissivity and absorptivity of the J-tube depends on the surface treatment, coating/paint, and should be checked on a case by case basis.

Natural convection heat transfer coefficient on J-tube surface (and power phases in air) is:

$$h_n = \frac{k_{air} c}{L_{J-tube}} Ra_{J-tube}^n$$

Equation 7.18

where

$$Ra_{J-tube} = \frac{Pr g \beta (\theta_{J,ext} - \theta_{air}) L_{J-tube}^3}{\nu^2}$$

Equation 7.19

With $c=0.021$ and $n=0.4$ for $10^9 < Ra < 10^{13}$ and $c=0.59$ and $n=0.25$ for $10^4 < Ra < 10^9$ [118], [119], where L_{J-tube} is the length of the J-tube in air (alternatively power phases in air).

Forced convection on the J-tube surface can be described by

$$h_f = \frac{Nu_f k_{air}}{D_{j-o}}$$

Equation 7.20

with

$$Nu_f = b Re_f^p Pr^{1/3}$$

Equation 7.21

and

$$Re_f = \frac{V_{air} D_{j-o}}{\nu}$$

Equation 7.22

where V_{air} is the air velocity (m/s), and the constants are dependent on the Reynolds number as shown in the table below [118]:

Table 7 - 34 Constants for calculation of Nusselt number dependent on Reynolds number

Re	b	p
0.4-4	0.989	0.330
4-40	0.911	0.385
40-4000	0.683	0.466
4000-40000	0.193	0.618
40000-400000	0.0266	0.805

The range in the table above covers the typical applicable range for wind strength and J-tube diameter. There might be cases with large wind speeds, where the Reynolds number is outside the range in the table above, however, this is typically for wind speeds above the maximum production of the wind turbine generator, and is most likely not the case of interest for a rating calculation.

The total convection heat transfer coefficient on the outer surface of the J-tube is:

$$h_{conv,ext} = \max(h_n, h_f)$$

Equation 7.23

Heat transfer by natural convection between the cable surface and J-tube is described by:

$$Nu = 0.188Ra^{0.322} \left(\frac{L_{tube}}{\delta_g} \right)^{-0.238} \left(\frac{D_{j,i}}{D_c} \right)^{0.442}$$

Equation 7.24

with:

$$Ra = \frac{Pr g \beta (\theta_{cable} - \theta_{j,inner}) \delta_g^3}{\nu^2}$$

Equation 7.25

and

$$h_{conv,int} = \frac{Nuk_{air}}{\delta_g}$$

Equation 7.26

$$\delta_g = \frac{D_{j,i} - D_c}{2}$$

Equation 7.27

δ_g is the distance between the cable surface and the J-tube inner surface.

Radiation heat flux between cable surface and J-tube inner wall is calculated with:

$$q_{rad,int} = \frac{\pi D_c \sigma (\theta_{cable}^4 - \theta_{j,inner}^4)}{1 + \frac{1-\epsilon_{cable}}{\epsilon_{cable}} + \frac{D_c(1-\epsilon_{j,inner})}{D_{j,i}\epsilon_{j,inner}}}$$

Equation 7.28

Where sigma is the Stefan-Boltzmann constant, epsilon is the emissivity of the surfaces.

Radiation heat flux between J-tube surface and ambient is:

$$q_{rad,ext} = \epsilon_{j,outer} \pi D_{j,o} \sigma (\theta_{j,ext}^4 - \theta_{air}^4)$$

Equation 7.29

In formulae above $\theta_{j,ext}$, $\theta_{j,inner}$, θ_{air} , θ_{cable} are the temperatures of ambient air, external surface of J-tube, internal surface of J-tube and cable surface respectively. All these temperatures are in unit K.

The surface of the power phases in air above the armour hang-off, are described by the same heat transfer coefficients as J-tube surface (with different length and surface temperature). The phases are assumed to be separated, such that the phases do not influence each other thermally (shielding of radiation for example).

The system of equations describing the heat transfer between regions in air are solved consistently within the Finite Element Software, using a transient thermal solver.

7.3.7 Results benchmark case

Here two different methods and software are compared. A simulation is performed with 100 W/m applied to the conductors only for comparison between the two software packages and calculation methods. Only natural convection is considered, and there is no solar radiation.

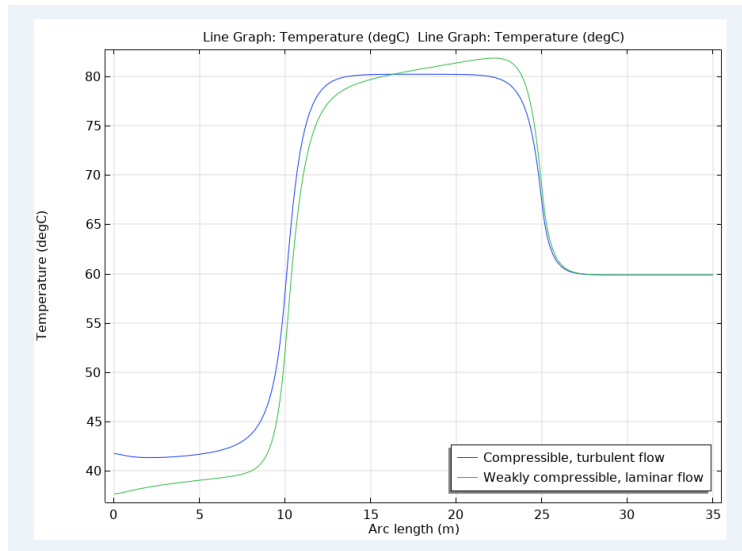


Figure 7 - 35 Result from software number 1. Comparison is made between laminar flow with weakly compressive flow and turbulent flow with compressive flow

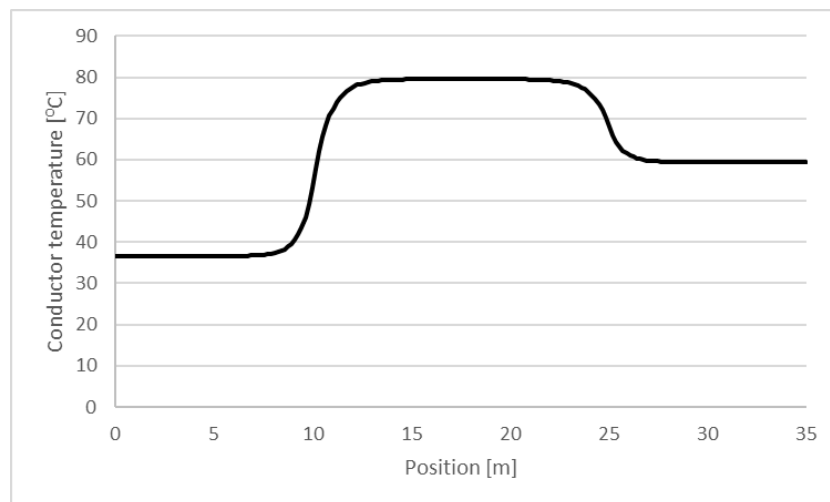


Figure 7 - 36 Results from software number 2 where flow is modelled with empirical formulas

In software package 1, convection is modelled explicitly in air and water between the cable and J-tube, with built-in properties of air and water, including the temperature dependency of the material properties. In software package 2, convection is modelled with empirical formulas, with constant air-properties as given here, and the cable surface in water is set at a fixed temperature, equal to the water temperature. The results are in good agreement, with slightly lower temperature with software package 2 in the region with water in the J-tube. This is as expected, due to the modelling simplification, and the difference is small. In both packages the radiation is modelled between the cable surface and the inner wall of the J-tube, and between the J-tube outer surface and ambient.

In software package 1 a 3D model is utilised. Twisting of the cable cores and armour wires is neglected, and sector symmetry is utilised for all physics (radiation, heat conduction and CFD). As mentioned above, CFD is used in the annular region to model natural convection both in air and water. Heat transfer by convection from the external surfaces (J-tube outer surface and cable surface in breakout region) is modelled by using heat transfer coefficients. Due to the size and complexity of the model, it is necessary to use a highly structured mesh, as shown in the figure below (note that the

figure is scaled). On a desktop computer with 32 GB RAM, the solving time is approximately 2 hours, significantly longer than the approach used in software package 2.

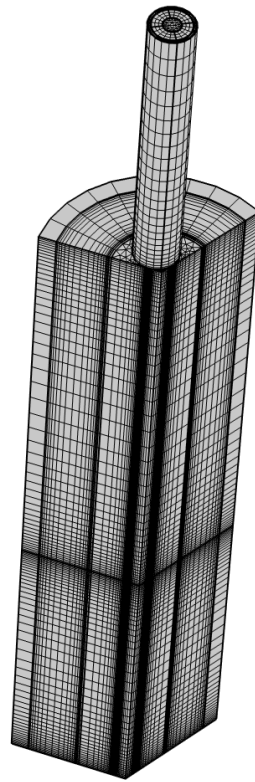


Figure 7 - 37 3D model from software package 1 showing the meshed geometry. Note that the model is scaled for increased visibility. And that sector symmetry is utilized, and this figure only shows one third of the full geometry (one power phase).

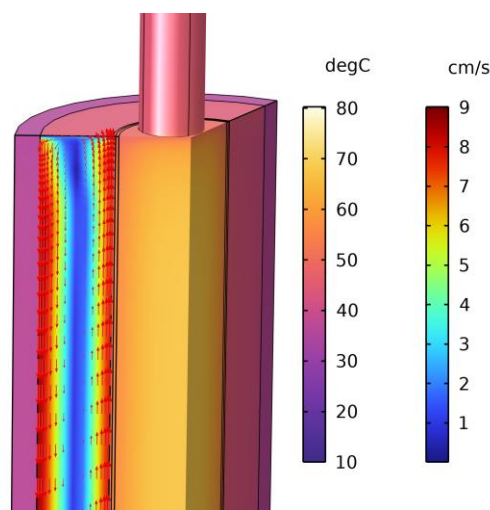


Figure 7 - 38 3D model from software package 1 showing the air flow and temperature

7.3.8 Results: Rating without solar radiation

Given the close correlation between the results of method 1 and 2 in section 7.3.6, the simple approach using analytical formulas to describe the convection is applied when finding the rating. The losses are calculated using the data described above. The rating is found to be 1060 A (see figure below). Only natural convection is considered.

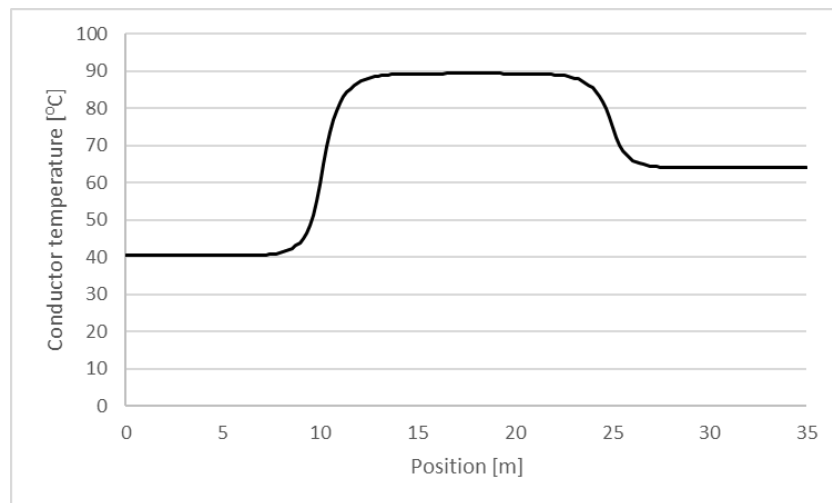


Figure 7 - 39 Result for case without solar radiation using current 1060 A

Table 7 - 35 Temperatures and heat transfer coefficients for case without solar radiation

Parameter	Value	Unit
Max J-tube temperature	39	°C
Max 3-core cable surface temperature	62	°C
Power phase surface temperature (in air)	47	°C
Cable surface heat transfer coefficient	1.7	W/(m ² K)
J-tube surface heat transfer coefficient	2.5	W/(m ² K)
Power phases heat transfer coefficient	3.0	W/(m ² K)

7.3.9 Results rating with solar radiation

1000 W/m² is applied to half the J-tube and surface of cables above hang-off with a solar absorption coefficient of 0.4. The method to calculate the appropriate solar radiation (based on geographic location, angle, time of day) is not covered here, and the solar radiation is simply a power source applied to the J-tube surface, simulating the absorbed solar radiation. To account for the direction of the sun, the solar power absorbed is applied to half the J-tube. Not all the power from the sun is absorbed by the J-tubes surface, which is accounted for by the solar absorption coefficient, which is here assumed to be 0.4. The solar absorption coefficient will vary depending on material and coating/paint. With a solar absorption coefficient of 0.4, 400 W/m² (0.4x1000 W/m²) is applied to half the J-tube surface. Only natural convection is considered in this case.

Rating is found to be 890 A.

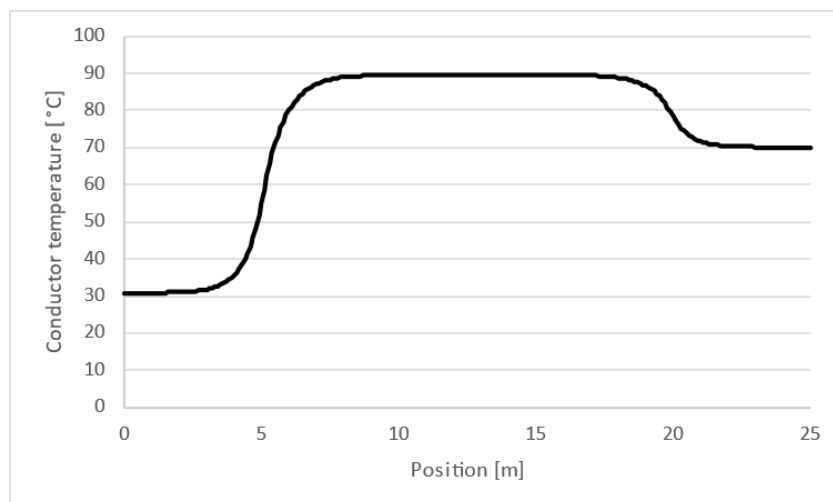


Figure 7 - 40 Result for case with solar radiation using current 890 A

Table 7 - 36 Temperatures and heat transfer coefficients for case with solar radiation

Parameter	Value	Unit
Max J-tube temperature	55	°C
Max 3-core cable surface temperature	70	°C
Power phase surface temperature (in air)	60	°C
Cable surface heat transfer coefficient	1.5	W/(m ² K)
J-tube surface heat transfer coefficient	3.8	W/(m ² K)
Power phases heat transfer coefficient	3.8	W/(m ² K)

7.3.10 Results rating with forced convection and solar radiation

The same case is repeated with solar radiation and forced convection from a wind speed of 5 m/s. Rating is found to be 1020 A. The rating is as expected increased due to forced convection on the surface of the J-tube compared to the case with only natural convection.

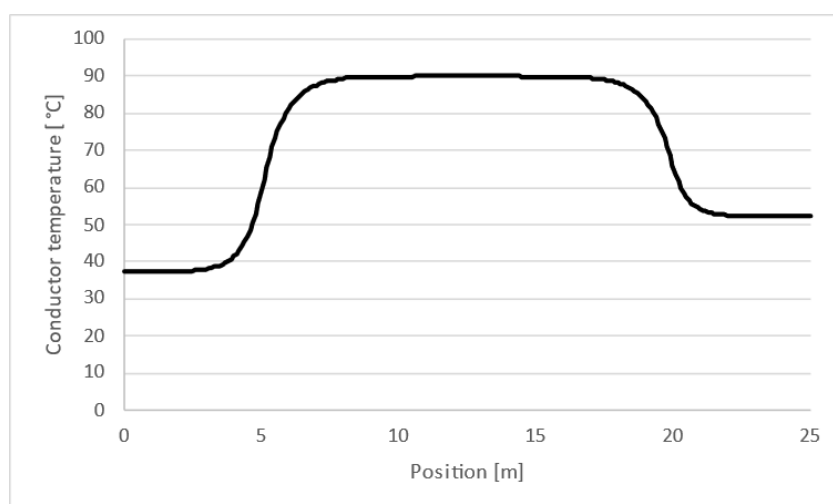


Figure 7 - 41 Result for case with forced convection from wind using current 1020 A

Table 7 - 37 Temperatures and heat transfer coefficients for case including forced convection from wind

Parameter	Value	Unit
Max J-tube temperature	43	°C
Max 3-core cable surface temperature	64	°C
Power phase surface temperature (in air)	38	°C
Cable surface heat transfer coefficient	1.7	W/(m ² K)
J-tube surface heat transfer coefficient	14	W/(m ² K)
Power phases heat transfer coefficient	3.8	W/(m ² K)

7.3.11 Results rating with a shorter air-filled J-tube

The case is repeated with 4 m long air-filled J-tube, the other lengths are kept constant. Only natural convection is considered. In this case the rating is found to be 950 A. Here the cables above hang-off are assumed shielded from the sun. The purpose of this model is to show that when the cable length is sufficiently short, a 2D approximation no longer holds.

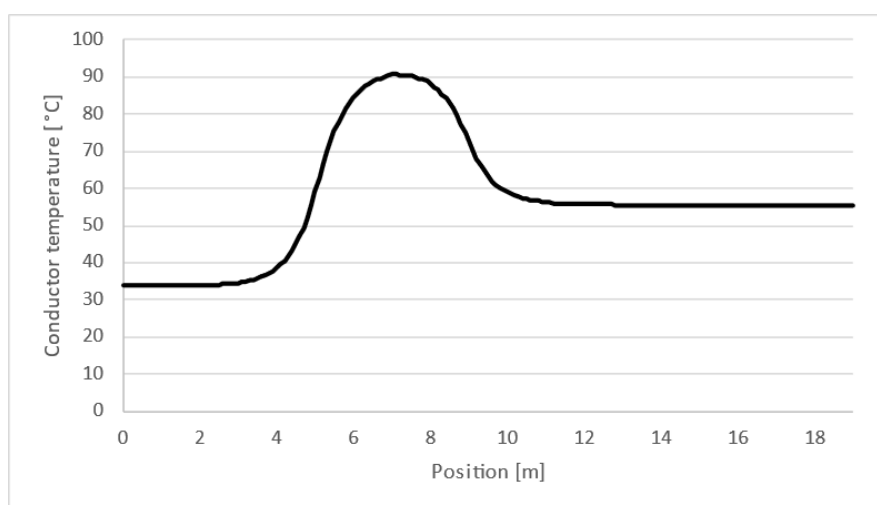


Figure 7 - 42 Result for case with shorter J-tube using current 950 A

Table 7 - 38 Temperatures and heat transfer coefficients for case with shorter J-tube

Parameter	Value	Unit
Max J-tube temperature	56	°C
Max 3-core cable surface temperature	71	°C
Power phase surface temperature (in air)	43	°C
Cable surface heat transfer coefficient	2.0	W/(m ² K)
J-tube surface heat transfer coefficient	3.0	W/(m ² K)
Power phases heat transfer coefficient	2.7	W/(m ² K)

7.3.12 Discussion

The case study shows that it is sufficient to include convection by use of heat transfer coefficients based on analytical formulas. A 3D model is only needed if the air-filled section is fairly short, such

that longitudinal heat flow must be considered. In other cases, it is sufficient to model the J-tube with a 2D FE model, or analytical formulas.

When modelling the J-tube it is important to be aware of the fact that this section of the cable may require a different rated current compared to other sections of the cable route, due to the charging currents. This will also depend on the reactive compensation scheme and the load/generation, and must be evaluated case by case.

As found in the simulations the temperature of the cable in water is lower than in the air-filled section. This means that the lead sheath (or other screen) has a lower temperature and resistance in the water filled section compared to the air-filled section. Thus, the sheath losses in the air-filled area may depend on the temperature in the water-filled part, as the circulating current may be determined mainly by the section in water for large water depths, which might be case for example for floating wind. This will depend on the conducting properties of the outer jacket of each power core, being insulating or semiconducting, as well as the length of cable in water (water depth) compared to length of air-filled. In most cases the section of the cable buried in the seabed will determine the circulating currents, and this section of the cable will in most cases have similar temperature as the J-tube, and the assumption made in this case is valid. However, this has to be considered on a case by case basis.

7.4 Case 4: Joint bay

This case study is an example of a joint bay. The cables are placed in flat formation and the separation is increased when the cables approach the joint bay. Screens are unbalanced cross bonded without transposition.

The cable and the installation are from CIGRE TB 880 case 1.

7.4.1 Case definition

The objective of this simulation is to verify the conductor temperature along the cable in the joint bay to ensure that current rating is not limited by the joint temperature. The problem is solved in a 3D geometry and with an IEC 60287 approach for losses.

7.4.1.1 Geometry

The cables are installed in flat formation at 1 m burial depth and 0.5 m separation. As the cables approach the joint bay the separation is increased to 1 m with maintained burial depth. The backfilling of the joint bay is done with native soil.

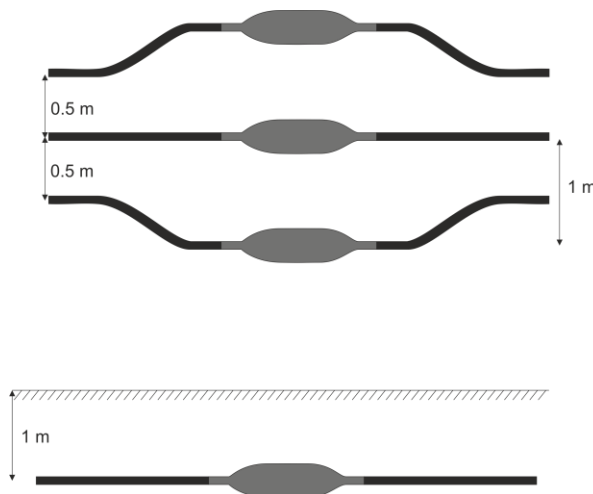


Figure 7 - 43 Overview of the joint bay geometry, from top and from side

7.4.1.2 Cable data

Cable used in this case study is taken from CIGRE TB 880 case 1. It is a 132 kV cable with 1200 mm² segmented copper conductor. The parameters used for the simulations are included in the table below.

Table 7 - 39 Cable data used for the simulations

Parameter	Value	Unit
Conductor diameter	43.1	mm
T ₁ Diameter	84.3	mm
Screen diameter	87.84	mm
T ₂ Diameter	88.4	mm
Sheath diameter	88.8	mm
Cable outer diameter	98	mm
T ₁	0.378908897	Km/W
T ₂	0.01213714	Km/W
T ₃	0.053281319	Km/W
R ₀	0.0151	Ω/km
k _s	0.8	
k _p	0.37	
R _{s0}	0.139752435	Ω/km
R _{sh0}	0.510158057	Ω/km

Rated current for this cable is 1460.45 A.

The cross-bonding minor sections are unbalanced as, 1000 m, 1000 m and 1200 m.

IEC calculations gives the following sheath loss factors, for detailed description on how to calculate these see CIGRE TB 880 case 1.

Table 7 - 40 Screen loss factors for the cable installation

Parameter	Value	Description
λ'_{11}	0.01761440438	leading
λ'_{1m}	0.012451913721	centre
λ'_{12}	0.01341113964	lagging
λ''_{11}	0.00087900366024	leading
λ''_{1m}	0.003462252627312	centre
λ''_{12}	0.0008952797411183	lagging

The screen and conductor current are not expected to change due to the short section with different separation but the temperatures will. To enable temperature dependent calculations the screen current is calculated based on the IEC screen loss factor λ'_1 .

$$I_{s1} = I_c \sqrt{\frac{R_{ac}}{R_{sc+sh}}} \lambda'_1 = 78.67 A$$

Equation 7.30

Based on this the current in each screen/sheath can be calculated

Table 7 - 41 Screen current for each cable

Phase	Screen current (A)
Leading	78.67
Centre	66.03
Lagging	68.65

These currents are then used to calculate temperature dependent losses in the screen and sheath by

$$W_s = I_s^2 R_{sc+sh}$$

Equation 7.31

Where R_{sc+sh} is the parallel combination of screen and sheath.

7.4.1.3 Joint

A joint is a complex geometry containing several small elements that do not significantly affect the result of the thermal modelling. An example of a joint is included in the figure below.

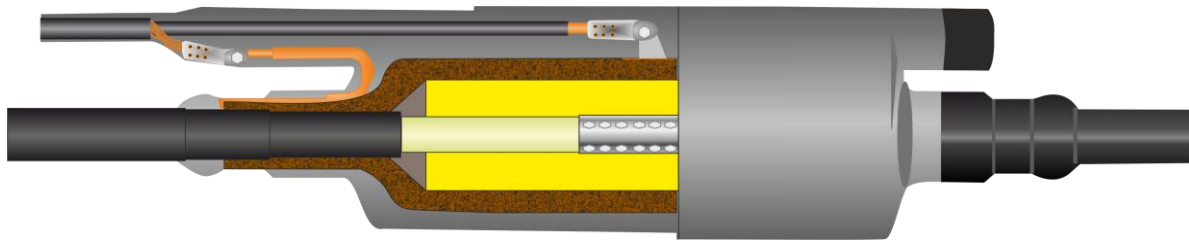


Figure 7 - 44 Example drawing of a joint

The details are reduced to create a joint geometry that is possible to represent as an axisymmetric part which makes it easier to draw in the 3D model. The joint geometry used for the simulation is visualized in the figure below. The outer PE shrink tube is considered to be 5 mm thick and the outer metal casing is 5 mm thick. The rest of the joint layers dimensions are available in the drawing. The specific joint parts are noted by a number and each number is found in the table with description and thermal properties.

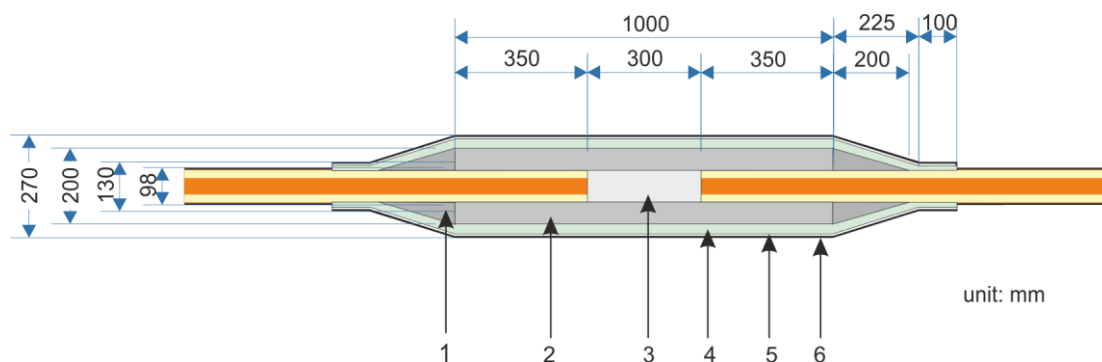


Figure 7 - 45 Joint geometry

Table 7 - 42 Material properties for joint

Id	Description	Material	Thermal resistivity [Km/W]
1	Stress cone	Rubber	5
2	Insulation	Rubber	5
3	Connector	Aluminium	1/240
4	Joint filling material	Resin	4
5	Metallic cover	Steel	1/80
6	Shrink tube	PE	3.5

7.4.1.4 Environmental data

Table 7 - 43 Environmental parameters for installation

Parameter	Value	Unit
Ambient temperature	20	°C
Ground resistivity	1	Km/W

7.4.2 Model

The following sections will describe the different steps required to recreate the model.

7.4.2.1 Geometry

The joint as well as the joint bay is symmetric to the vertical plane through the centre joint perpendicular to its axis (and not along the axis because of the different phases). So only half of the joint bay needs to be modelled. The centre of the joint has been placed in coordinate (X,Y,Z=0,0,-1).

The cable geometry is simplified to avoid thin layers. All materials within the insulation layer such as swellable tapes, semiconducting layers and insulation are combined into one layer, see Guidance Point 9.

Also copper screen and aluminium sheath is combined into one metallic layer. The thin bedding in-between screen and sheath are added to the outer layer to ensure conservative temperatures.

Table 7 - 44 Cable dimensions used in drawing cable inside the joint

Layer	Diameter [mm]
Conductor	43.1
T ₁ layer	84.3
Screen + Al	88.24
T ₂ +T ₃ layer	98

To draw the joint and cable geometry a work plane is created in x-y direction and at Z coordinate -1 m as the burial depth.

The joint geometry based on the description from the previous section is drawn in the work plane as shown in Figure 7 - 46.

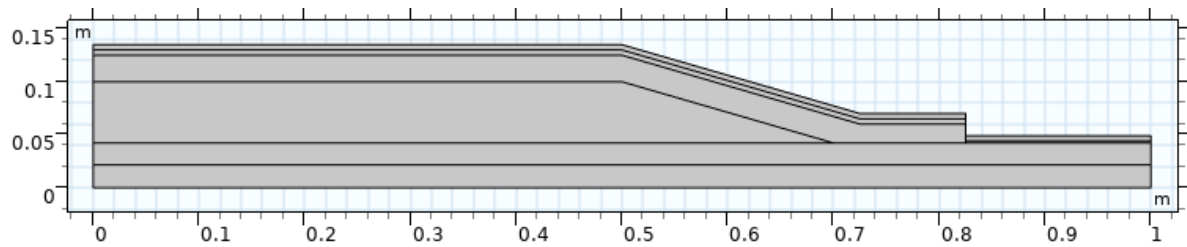


Figure 7 - 46 Joint details drawn in 2D work plane

The work plane is then revolved around the centre to create a 3D model of the joint and cable.

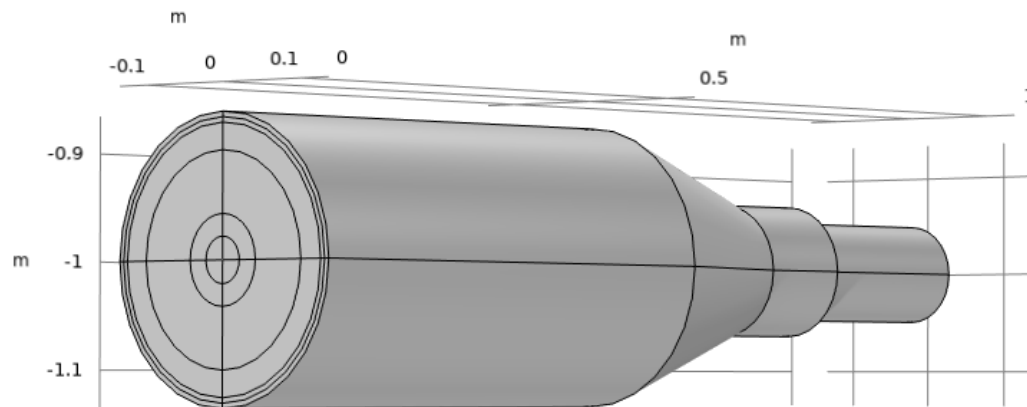


Figure 7 - 47 Joint in 3D after revolving the work plane

When joint and first part of the cable is drawn is the geometry copied such that there are joints and cables with 1 m separation.

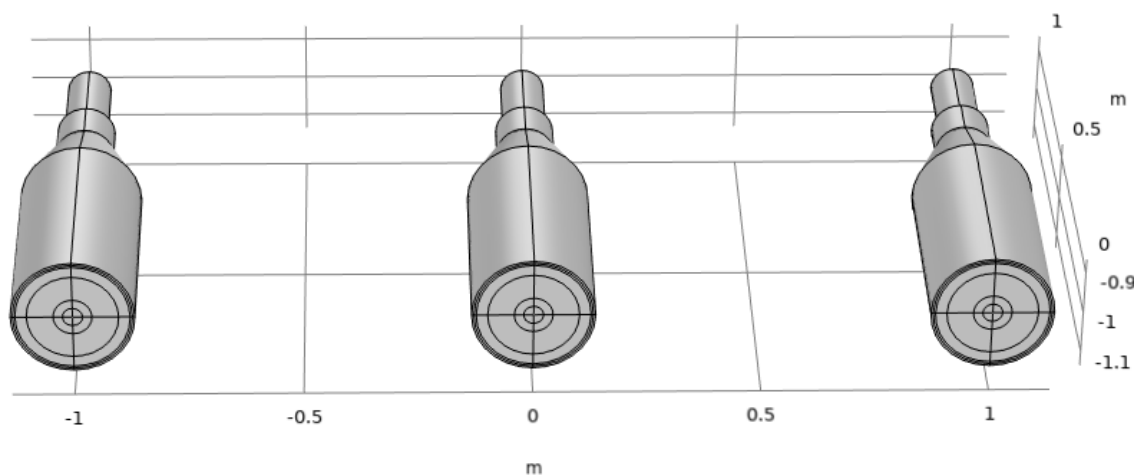


Figure 7 - 48 The joint is copied such that 3 equal joints are placed with 1 m separation

3 lines are created, one from each conductor. The centre one is a straight line for the distance that the model is made for, in this case 20 m.

The two outer lines are started in x coordinate -1 m and 1 m respectively and should at the end be at -0.5 m and 0.5 m.

The lines are therefore drawn as a function of y, example below is for the left cable which starts to separate at y=5 m and has reached final separation at y=10 m.

$$x = \begin{cases} s_{joint} & \text{if } y \leq 5 \text{ m} \\ s_{joint} - \frac{s_{joint} - s_{cable}}{2} \left(1 - \cos \left(\pi \frac{y-5}{5} \right) \right) & \text{if } 5 \text{ m} < y < 10 \text{ m} \\ s_{cable} & \text{if } y \geq 10 \text{ m} \end{cases}$$

where

$s_{joint} = 1 \text{ m}$, separation of joints

$s_{cable} = 0.5 \text{ m}$, separation of cables

Equation 7.32

The cable surface created by the revolution in previous step is used as start point for a sweep along the lines, which creates 3 cables that can be seen in figure below.

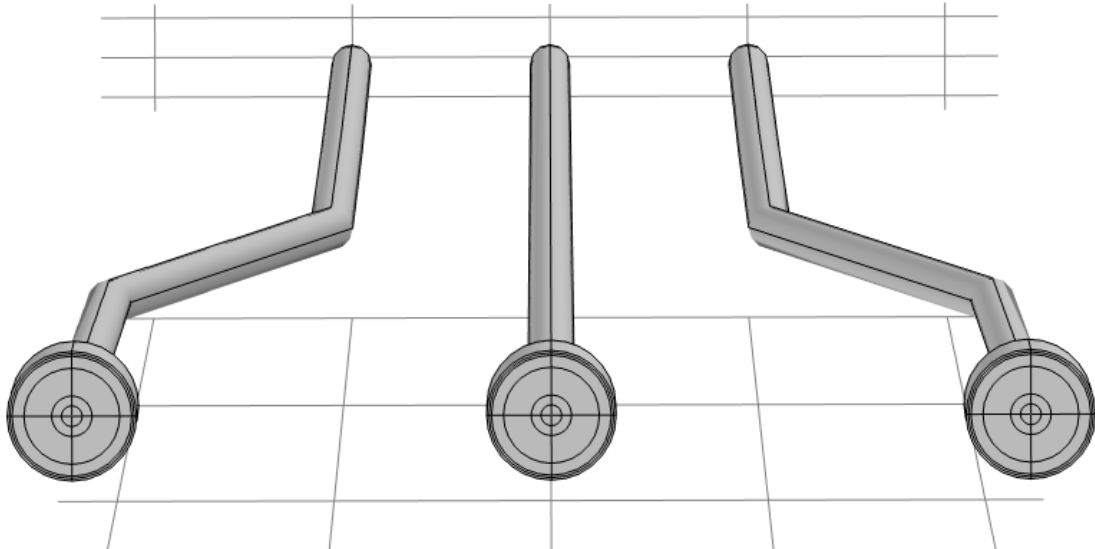


Figure 7 - 49 The cables are extruded along the drawn lines to create the change in cable separation

After this ground is added as a large block. Length of the block is set to 20 m which is equal to the length of the modelled cable and joint. Width and depth of the box should be sufficient such that it does not impact the result. In this example, 100 m width (50 m at each side of the centre cable), and 40 m depth is found to be sufficient.

7.4.2.2 Material

As this case is a stationary thermal problem the only parameter required is the thermal conductivity of the materials.

The cable is built up of several thin layers with different thermal conductivities. Drawing all these layers is not feasible in a large 3D model. Instead, the metal layers are included and in-between are the layers estimated based on the IEC thermal resistances, T_1 , T_2 and T_3 . With this approach, the thermal effect of each layer is included without needing to include every layer. See Guidance Point 9 for more details.

Example shown for T_1 .

$$C_{T_1} = \frac{\ln \left(\frac{D_{T_1}}{D_C} \right)}{2\pi T_1} = \frac{\ln \left(\frac{84.3}{43.1} \right)}{2\pi \cdot 0.378909} = 0.28178 \text{ W/(Km)}$$

Equation 7.33

The copper screen and aluminium sheath are modelled as one combined sheath layer instead of separate layers and individual screen wires. As this cross-section area of this equivalent layer is larger than the area of the wires and aluminium and copper has different thermal conductivity the material properties need to be compensated. This can be done by adjusting the thermal conductivity based on the effective area of the materials compared to the FEM layer.

$$C_{screen} = \frac{C_{Cu} \cdot A_{Cu} + C_{Al} \cdot A_{Al}}{A_{FEM}} = \frac{400 \frac{W}{Km} \cdot 0.0001234 m^2 + 240 \frac{W}{Km} \cdot 0.0000557 m^2}{0.0006118 m^2} = 102.5 W/(Km)$$

Equation 7.34

A summary of the material properties in the model is presented in Table 7 - 45.

Table 7 - 45 Material properties used in FEM simulation

Material	Thermal conductivity [W/Km]
Cu-Conductor	400
Insulation layer	0.28178
Screen sheath combination	102.5
Outer layer (T ₂ +T ₃)	0.23983
Joint rubber	0.20
Joint filling	0.2
Joint metallic cover	80
Shrink tube	1/3.5
Ground	1

7.4.2.3 Physics

The physics used in this case study is heat transfer. The boundary conditions for this model are set as follows.

Ambient temperature at burial depth, 20 °C, is set as isothermal boundary to the top ground layer. The rest of the ground boundaries are set to thermal insulation.

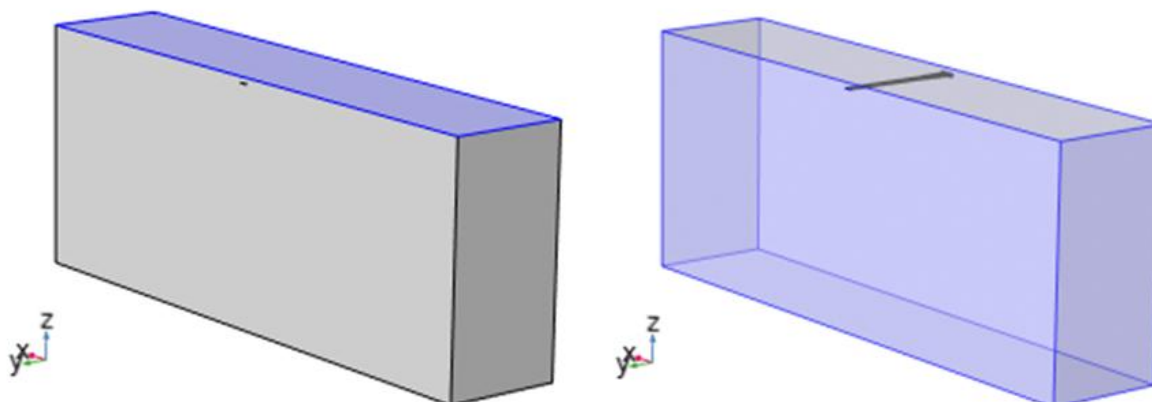


Figure 7 - 50 Boundary conditions for the soil box. Left shows surface where temperature boundary is set and right shows where thermal insulation is set.

Heat sources are placed in the conductor, screen and sheath based on the current and temperature dependent resistances. IEC formulas are used to derive the eddy current losses, Dielectric losses derived from IEC and added as in IEC half to conductor and half to screen layer due to the fact that the semiconducting screens are not directly modelled.

$$W_c = R_{ac}(\theta_c)I^2 + \frac{W_d}{2}$$

Equation 7.35

$$W_s = R_{s+sh}(\theta_s)I_s^2 + W_c\lambda_1''(\theta_s) + \frac{W_d}{2}$$

Equation 7.36

Losses are added to the corresponding layer as per volume (W/m^3) meaning that the calculated losses are divided by the cross section of each layer.

The screen connection in the joint is assumed to have equal resistance as the cable screen and loss is added based on the screen current and the resistance of the cable screen.

Eddy current losses are considered in the joint using IEC formula and thickness of the metallic layer (5 mm).

The total screen losses added to the joint metallic layer are 3.2 W/m in this case.

The connector is considered to not add any extra loss than the normal conductor resistance.

7.4.2.4 Mesh

The mesh is created by adding a 2D mesh to the inner joint surface and cable, choice of quad or triangular mesh can be made depending on how the geometry is shaped.

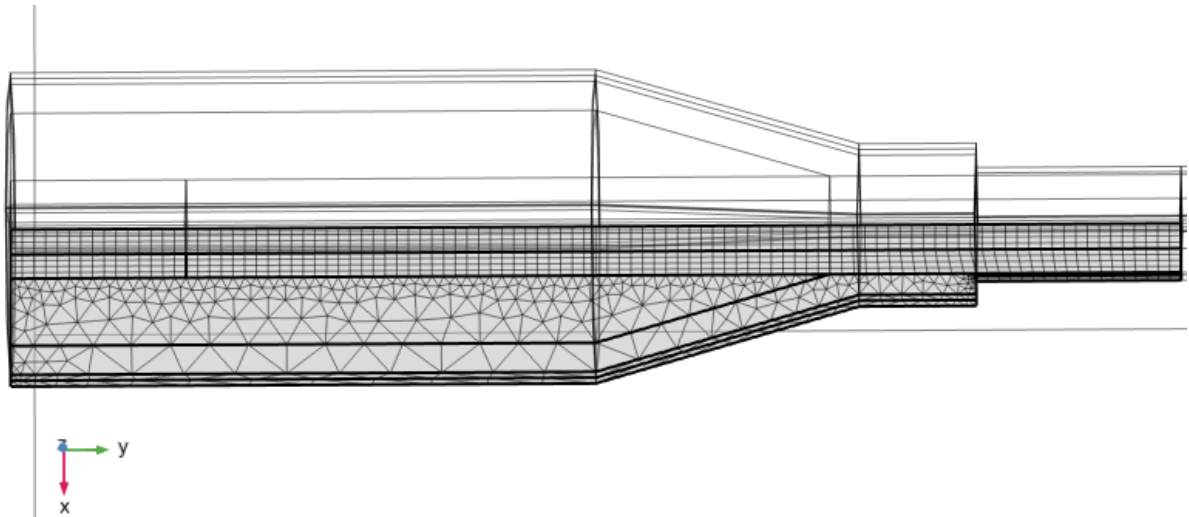


Figure 7 - 51 Mesh is created at the surface layer at the work plane

The mesh is then swept around the axis of the joint in 32 steps to create a round 3D mesh of the joint and cable.

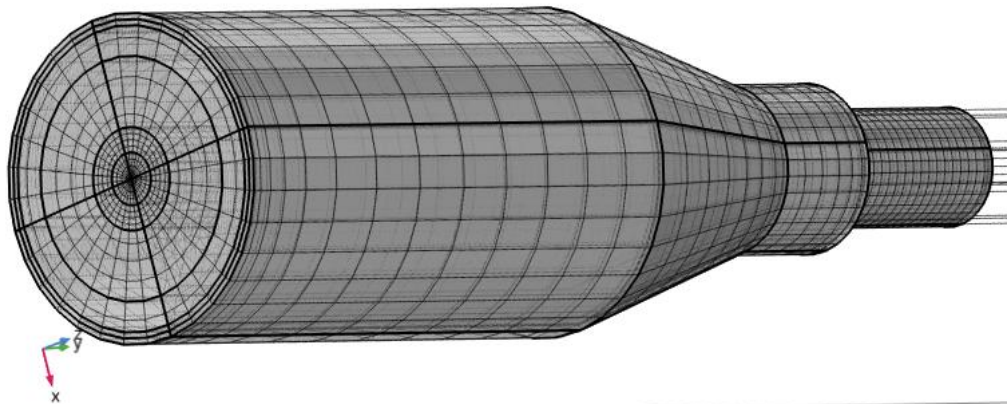


Figure 7 - 52 Mesh of the complete joint after sweeping the 2D mesh around the joint axis

The mesh of the joint can then be copied to the other two joints or recreated in the same manner.

The cable cross section surface can then be swept along the cable away from the joint using a maximum step length of 0.05 m.

Finally, the remaining geometry is meshed using a tetrahedral mesh.

This gives a final mesh of 5.1 million domain elements.

7.4.3 Results

Results from simulations are shown in figures below. It is clearly seen that the temperature decreases in the cable when the separation is increased.

Temperature of the cables far from the joint bay can be compared to the IEC calculation of the cables, which is also case 1 of CIGRE TB 880 and is presented in the following table. It is clear that the FEM simulation gives similar results as the IEC calculation.

Table 7 - 46 Comparison between FEM and IEC results for cable far from joint bay

Cable	IEC	FEM
Leading	85.1 °C	85.1 °C
Centre	90.0 °C	90.1 °C
Lagging	84.9 °C	85.0 °C

Figure 7 - 54 shows that the conductor temperature of the cables are reduced when cable separation increases. The conductor temperature then increases again at the joint where the thermal insulation is larger. The maximum temperature inside the joint is 86.3 °C for the centre cable while being 82.4 °C and 82.3 °C for the outer cables.

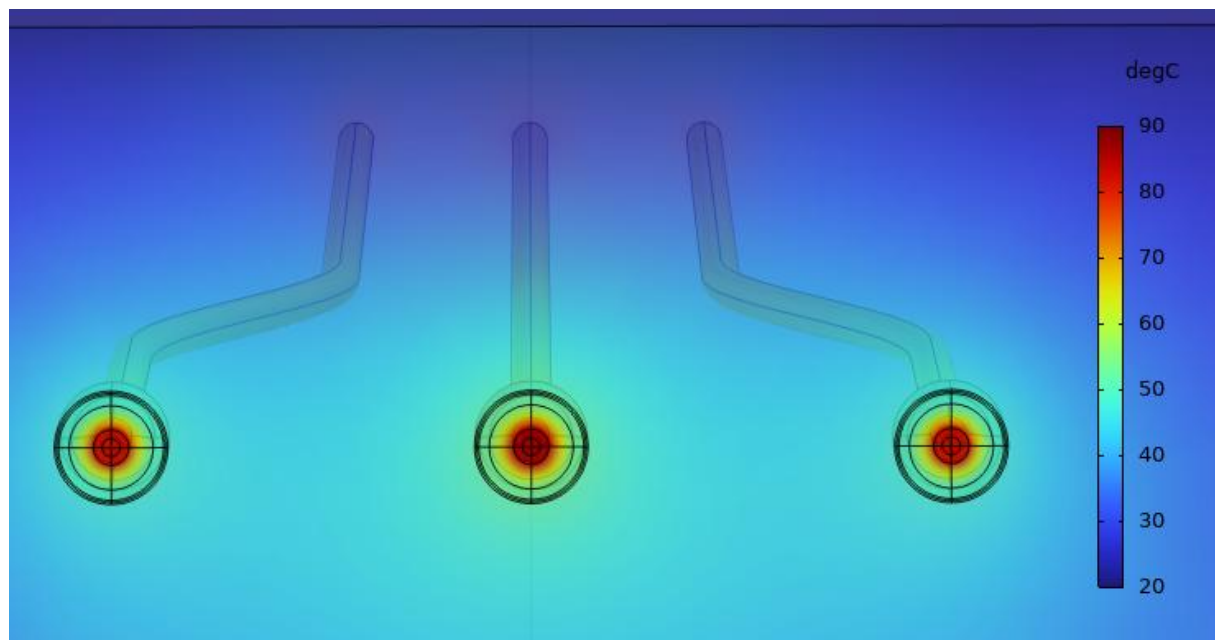
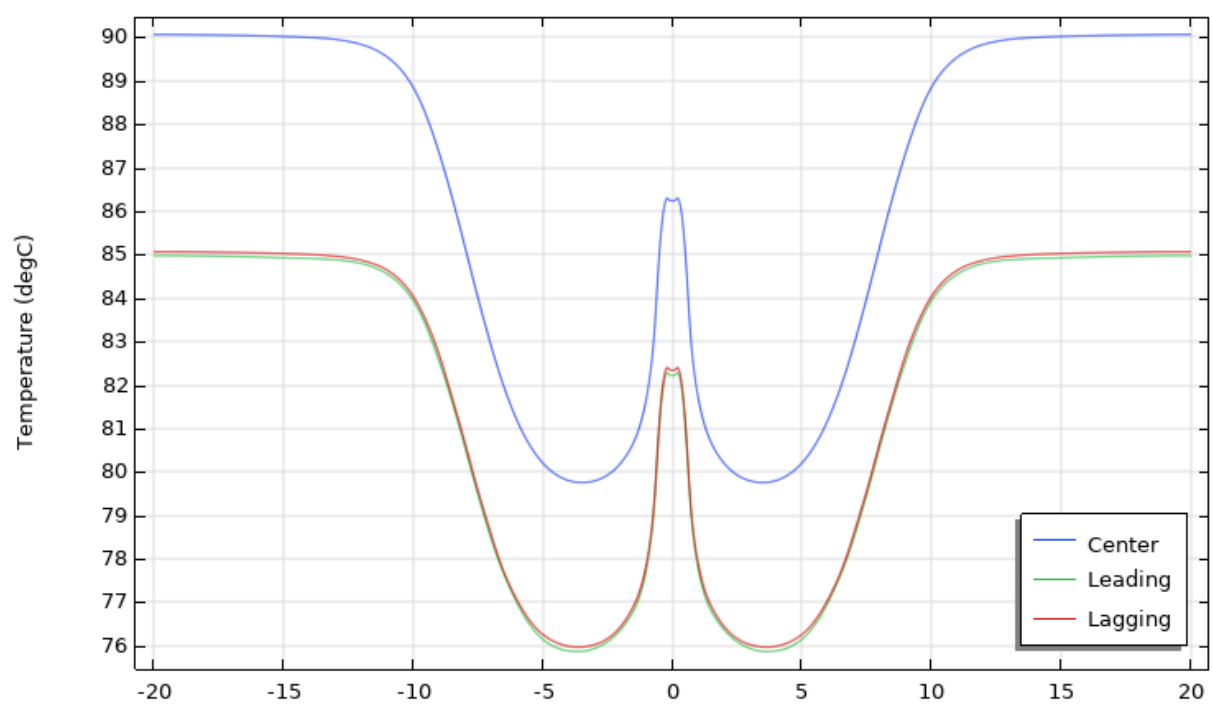


Figure 7 - 53 Temperature overview

Figure 7 - 54 Temperature along the conductors of each cable. Centre of joint is at $x=0$

7.5 Case 5: 3D crossing case including an HDD with multi-layer soil

This case study features a 3D crossing where a pair of HVDC cables in air-filled HDDs cross below an AC cable trefoil. In addition, the HVDC cables cross a horizontal layer of soil with high thermal resistivity.

7.5.1 Case definition

The objective of this simulation is to verify the conductor temperature in the DC cables to ensure that current rating is not limited by the crossing AC cables or the soil layer with higher thermal resistivity.

7.5.1.1 Geometry

In case of a 90° crossing, there are two symmetries, meaning it is sufficient to model one quarter of the geometry. This is illustrated in Figure 7 - 55 where half of the AC trefoil can be seen as the straight, bold line in the top left, and one of the HVDC cables as the bold, curved line. The size of the box is 60x50x50 m. The AC trefoil is located with centre at 1.0 m depth. The horizontal layer is located between 6 and 8 m depth.

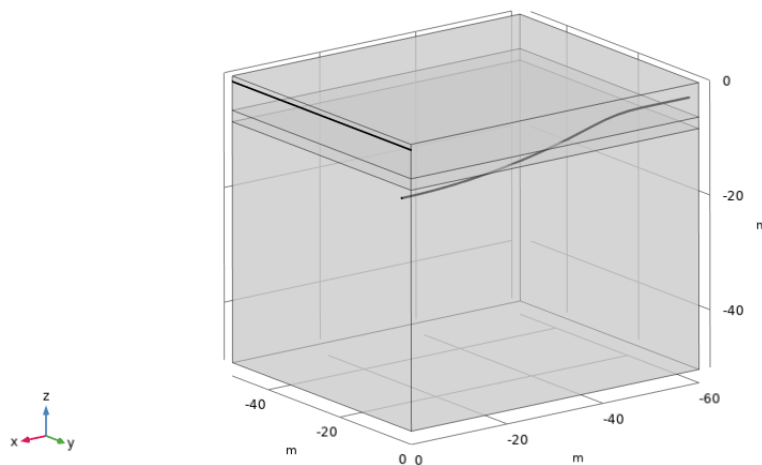


Figure 7 - 55 Crossing modelled in 3D in FE software. Due to symmetries, it is sufficient to model one quarter of the full geometry.

The HVDC HDDs reach a maximum depth of 10 m (centre) and have a minimum depth (at the edges) of approx. 3.2 m (centre). The HDD is modelled as a bi-arc, consisting of two circular arcs. The large arc has a radius of 150 m and meets the symmetry plane at $x=0$ in a right angle. It connects to the smaller arc with a radius of 37.5 m at $x=-40$. The smaller arc meets a straight section of 10 m length at $x=-50$ also at a right angle, and the two circular arcs have the same tangent where they meet. Note that the ratio 4:1 applies both to the radii of the circular segments (150 m and 37.5 m), as well as their projected length on the x-axis (40 m and 10 m). The resulting geometry can be seen in Figure 7 - 56.

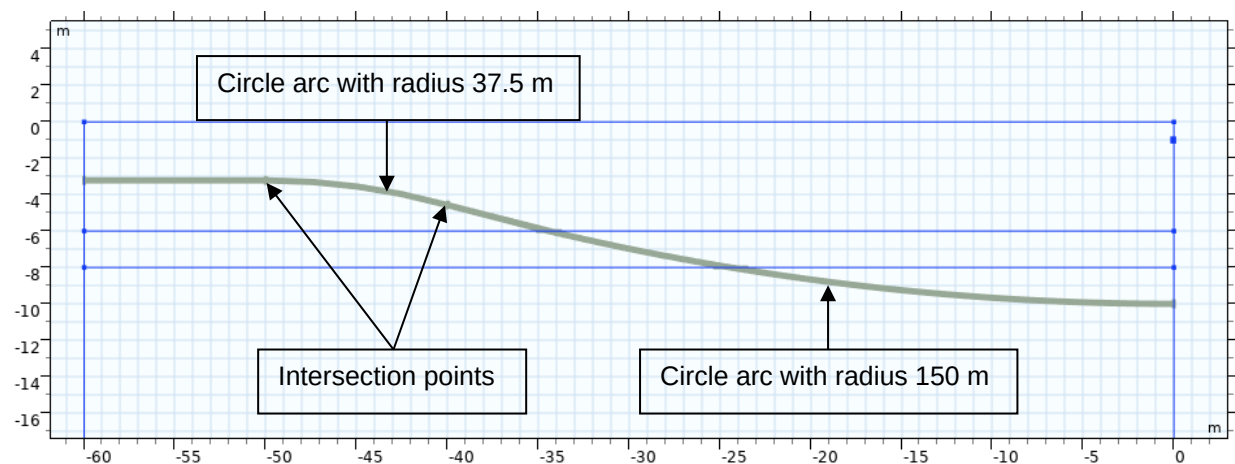


Figure 7 - 56 The bi-arc used to sweep the HVDC cables and HDD (including 10 m of straight section to the left)

7.5.1.2 Cable data, AC cables

Data for the AC cables used in this case study are taken from CIGRE TB 880 case 0. The parameters used for the simulations are included in the table below. Note that fixed losses are used for the crossing AC cables.

Table 7 - 47 Cable data, AC cables

Parameter	Value	Unit
Conductor diameter	30.3	mm
T ₁ Diameter	66.9	mm
Sheath diameter	68.5	mm
Cable outer diameter	75.5	mm
T ₁	0.4199	Km/W
T ₃	0.0867	Km/W
R ₀	0.0283	Ω/km
R _{ac} (90°C)	0.0395	Ω/km
W _c	21.6	W/m
W _d	0.4	W/m
W _{sh}	6.3	W/m

Rated current for this cable section in isolation is 822 A, but the load on the circuit in this case is only 740 A due to the thermally limiting part of the AC circuit being within an adjacent thermal section. The loss per meter values in the table are given for the derated cable, but assuming a conductor temperature of 90 °C.

7.5.1.3 Cable data, DC cables

Data for the DC cables are taken from a fictive cable with 3000 mm² copper conductor. The parameters used for the simulations are included in the table below. In the FE model, the metallic layers will be omitted, and an equivalent thermal resistivity calculated for the combined T₁/T₂/T₃ layer.

Table 7 - 48 Cable data, DC cables

Parameter	Value	Unit
Conductor diameter	63.1	mm
T ₁ diameter	130.5	mm
Copper screen diameter	134.0	mm
T ₂ diameter	135.2	mm
Al laminate diameter	135.6	mm
Cable outer diameter	146.6	mm
T ₁	0.4197	Km/W
T ₂	0.0085	Km/W
T ₃	0.0434	Km/W
T ₁₊₂₊₃	0.4716	Km/W

Parameter	Value	Unit
Conductor thermal resistivity	1/400	Km/W
R_0	0.0060	Ω/km
HDD duct, outer diameter	280	mm
HDD duct, inner diameter	229	mm
Separation, HDD (C-C)	5.0	m

The cable geometry is simplified with several layers put into one common layer. The correct thermal conductivity is then derived from the IEC T_1 , T_2 and T_3 and diameters presented in Table 7 - 48. The HDDs are assumed to be air-filled with equivalent thermal resistivity calculated according to IEC 60287.

Rated current for this cable for air-filled ducts made of PE material and 5.0 m c-c distance can be seen in the table below. These values are taken from IEC calculations assuming uniform soil. The FE model results should be slightly conservative due to the omitted metallic layers.

Table 7 - 49 Rated current, assuming uniform soil thermal resistivity

Burial depth	Thermal resistivity of soil (uniform)	Rated current
3.2 m	1.0 Km/W	2159 A
7.3 m	2.0 Km/W	1626 A
10 m	1.0 Km/W	1957 A

7.5.1.4 Environmental data

The environment parameters are shown in Table 7 - 50.

Table 7 - 50 Environmental data

Parameter	Value	Unit
Ambient temperature	20	$^{\circ}\text{C}$
Ground resistivity	1	Km/W
Ground resistivity, horizontal layer	2	Km/W
Thermal resistivity of PE ducts	3.5	Km/W

7.5.2 Materials

As this case is a stationary thermal problem the only parameter required is the thermal conductivity of the materials.

7.5.3 Physics

The boundary conditions for this model are set as follows.

Ambient temperature at burial depth, 20 $^{\circ}\text{C}$, is set to the top ground layer. The rest of the ground boundaries are set to thermal insulation.

7.5.3.1 Heat sources

A temperature dependent heat source is placed in the DC conductor.

Losses for AC cable conductor and sheath are added as fixed losses which are not temperature dependent. Dielectric losses derived from IEC are added to the insulation layer.

Losses are added to the corresponding layer as a loss per meter value (W/m) multiplied by total length (m). This is equivalent to adding a volumetric loss (W/m^3) where the calculated losses are divided by the cross section of each layer.

7.5.4 Mesh

Mesh is created by adding a free triangular mesh at the cables cross-section. A rectangular box has been added around the AC trefoil to give a simpler interface to the upcoming mesh layers. The cable cross sections are then swept along the cable paths.

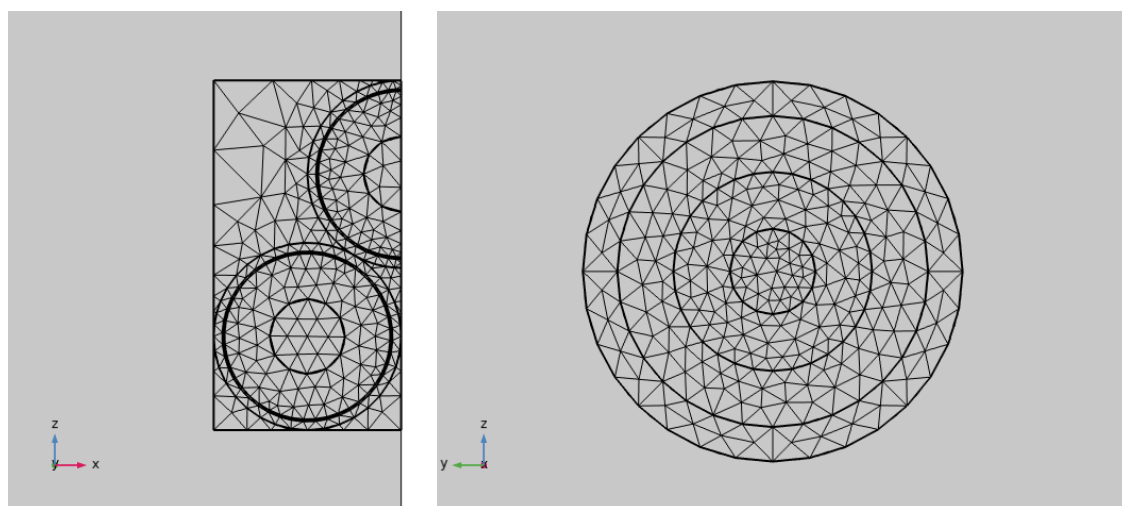


Figure 7 - 57 Mesh detail, AC (left) and DC cables (right)

The rest of the model (all soil layers) are meshed using a tetrahedral mesh.

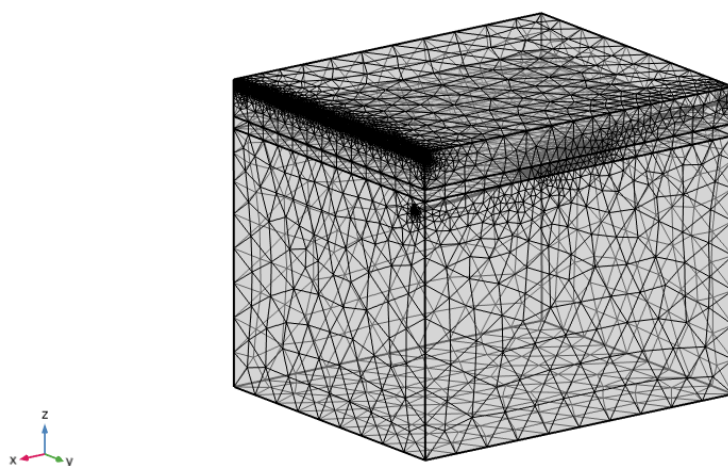


Figure 7 - 58 Final mesh of entire geometry

7.5.5 Results

The temperature profile for the HVDC-cable conductors with a current of 1800 A can be seen in Figure 7 - 59 below. The results have been mirrored around $x=0$ to provide the full, two-sided graph.

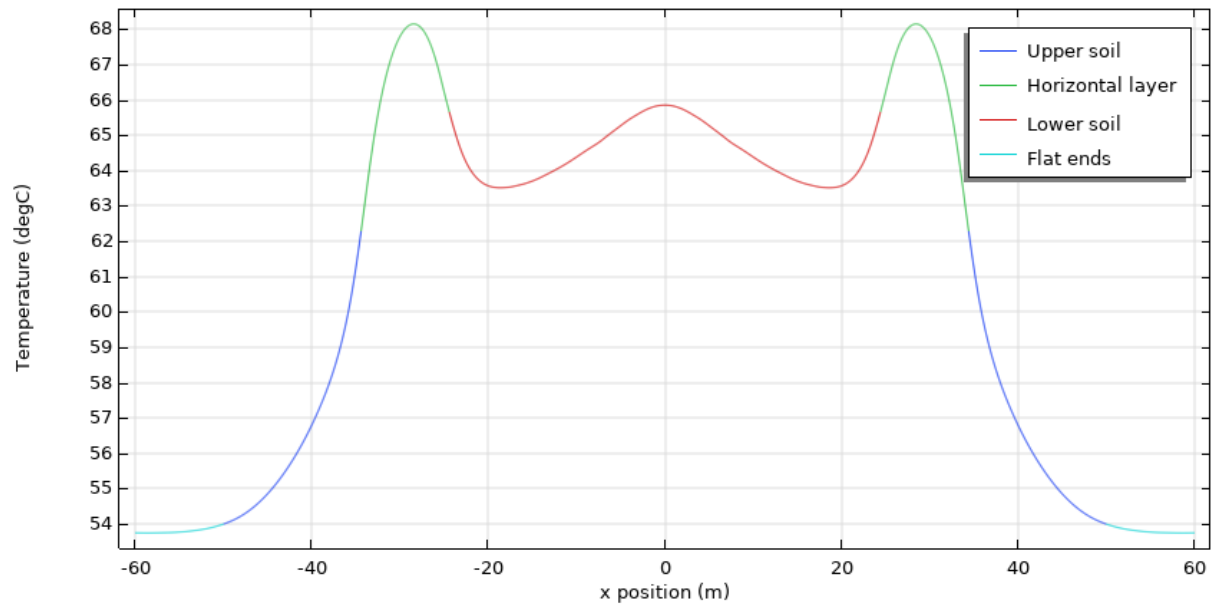


Figure 7 - 59 DC cable conductor temperature vs length along DC cables

In the graph below with burial depth on the x-axis and temperature on the y-axis, the maximum conductor temperature can be seen to occur at around 7.3 m depth.

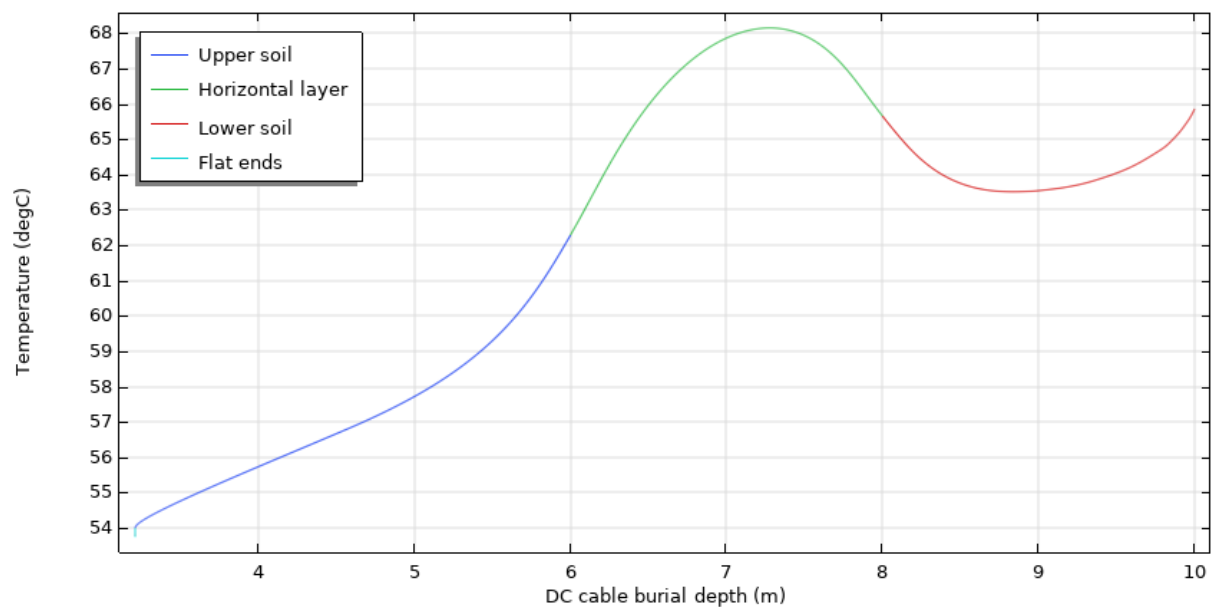


Figure 7 - 60 DC cable conductor temperature vs burial depth

The estimated AC cable conductor temperatures can be seen below.

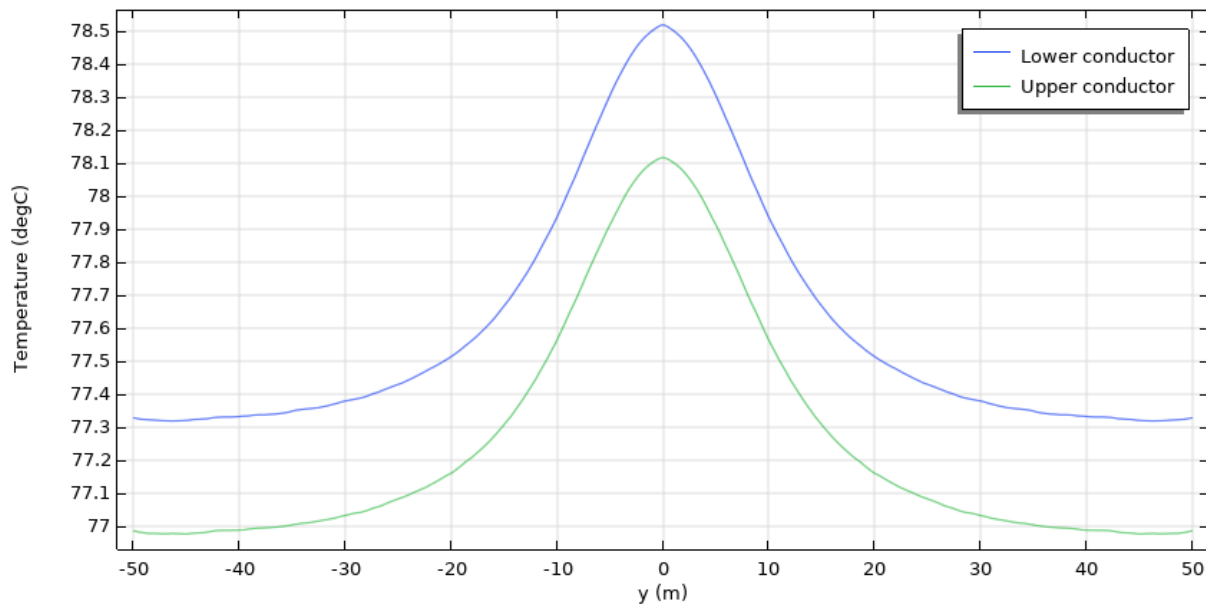


Figure 7 - 61 Estimated AC cable conductor temperatures

7.6 Case 6: Non-ventilated tunnel

This case study aims to present the thermal analysis of cables installed in non-ventilated tunnels. The heat transfer by natural convection and radiation plays a significant role in this case and thus must be accounted for. According to the literature, the IEC 60287-2-1 standard deals with certain cable installations where natural convection is considered, such as cables in free air or in air-filled troughs and suggests semi-empirical formulae to estimate the corresponding cable thermal performance. However, it fails to consider some important parameters, such as the trough aspect ratio and the surrounding soil thermal properties, which may affect the thermal behaviour of the cable. Moreover, the mutual effects of neighbouring troughs are not clearly considered and current rating calculations are often unrealistic. Some relevant work has already been published, such as [106].

Furthermore, the IEC 60287-2-3 Standard and Electra 143/144 [80] suggest methods to calculate the cable thermal rating when in ventilated tunnels. Although these methods are expected to be quite accurate at moderate and higher air velocities, they may not be so accurate for nearly-zero air velocities occurring in non-ventilated tunnels.

To overcome the above-mentioned obstacles, Finite Element Method (FEM) is introduced. An indicative 2D geometry is presented in section 7.6.1, mentioning the important boundaries; the modelling approach is briefly presented in section 7.6.2; finally, the extracted results in form of temperature 2D plots are presented in section 7.6.3.

7.6.1 Case definition

An indicative geometry of the tunnel is developed which can be seen in Figure 7 - 62. It represents the respective domains and boundaries. The soil domain which acts as a heat sink has been considered large enough so that it does not influence our region of interest. Using a sensitivity study, it was considered to be 50 m on the left, right and bottom side of the tunnel domain and the boundaries have been assigned with adiabatic boundary condition.

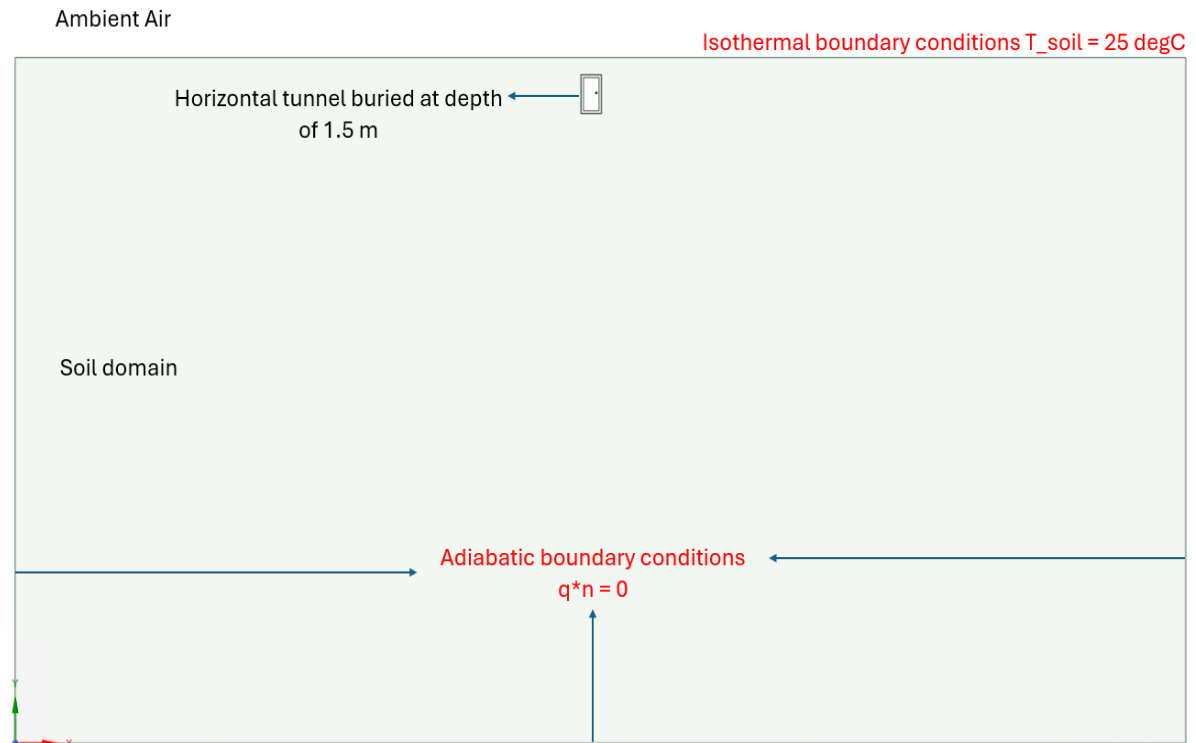


Figure 7 - 62 Full Scale Model Representation

Geometric and thermal data for the installation is listed in Table 7 - 51 and Table 7 - 52. The dimensional model is shown in Figure 7 - 63, representing the depth of the tunnel from the soil surface and the location of cables within the tunnel.

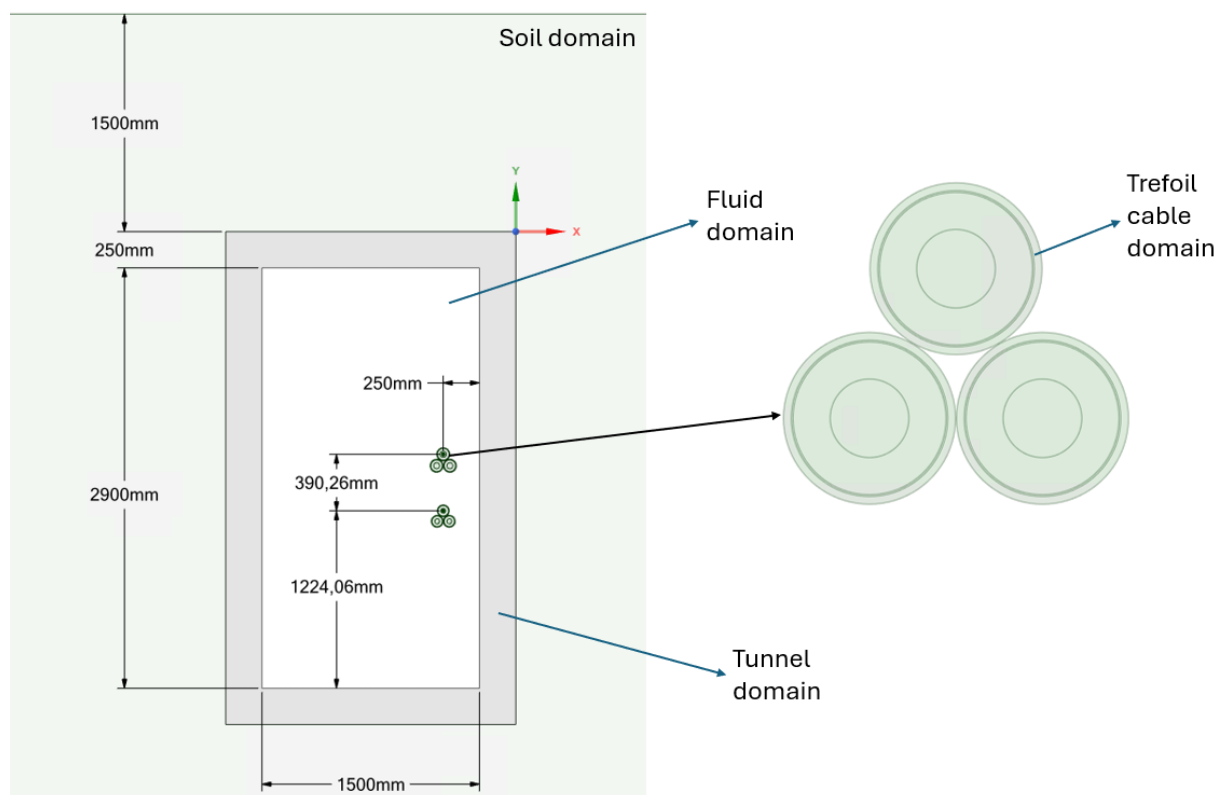


Figure 7 - 63 Dimensional model

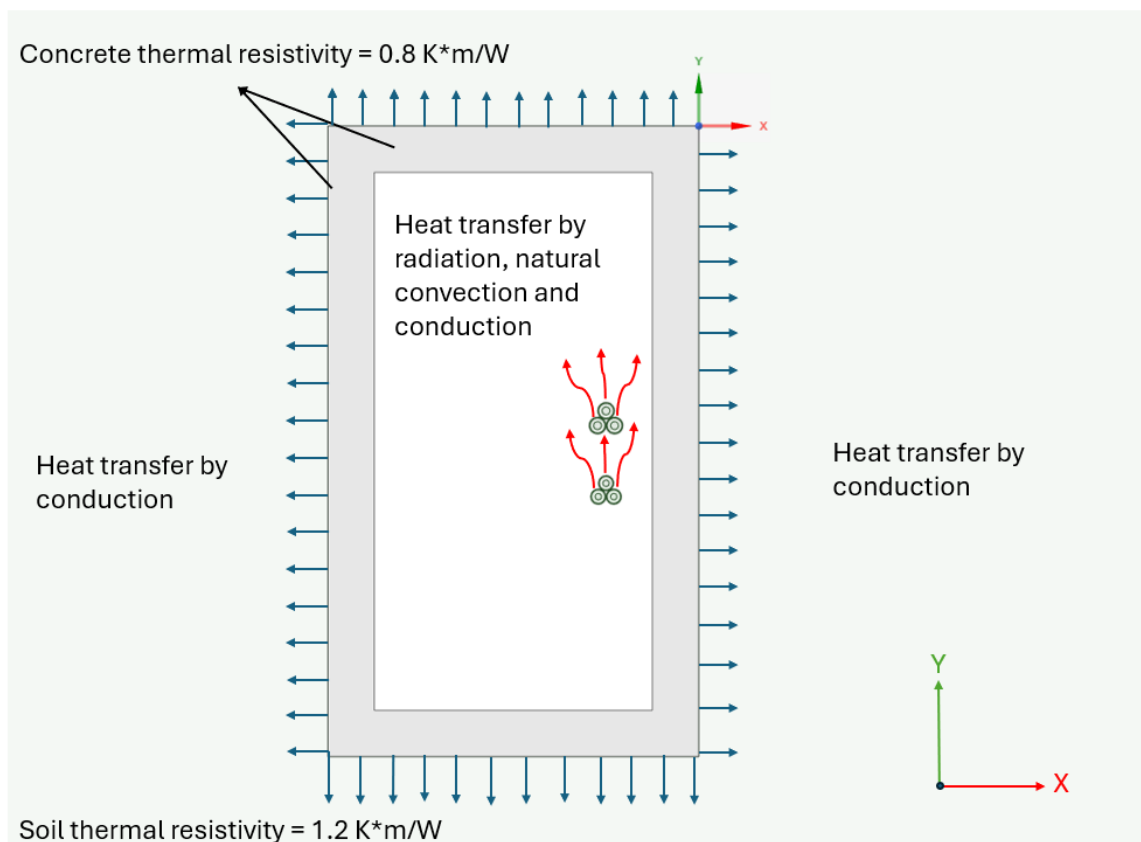


Figure 7 - 64 Modes of Heat Transfer

Table 7 - 51 Geometric data for the installation

Installation Data	Value
Tunnel burial depth to tunnel centre line	2950 mm
Thickness of tunnel walls	250 mm
Internal tunnel height	2900 mm
Internal tunnel width	1500 mm
Spacing from cable to nearest tunnel wall	250 mm
Vertical spacing between circuits	390 mm

Table 7 - 52 Thermal properties of the environment

Thermal Properties of Domains	Value
Thermal Resistivity of Soil	$1.2 \text{ Km}/\text{W}$
Thermal Resistivity of Tunnel Walls	$0.8 \text{ Km}/\text{W}$
Emissivity of Cables	0.9
Emissivity of Inner Tunnel Walls	0.9
Temperature at ground level	$25 \text{ }^{\circ}\text{C}$

To resolve and capture the high gradient temperatures and momentum around the cable, boundary layer meshing has been implemented as shown in Figure 7 - 65.

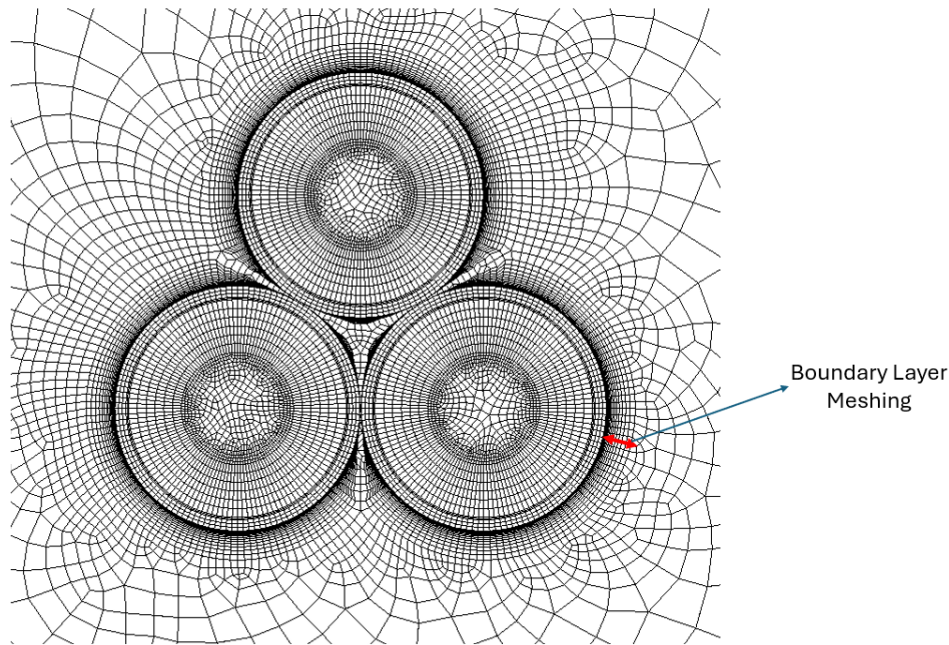


Figure 7 - 65 Detailed mesh of Fluid and Solid Domain

7.6.2 Modelling Approach

The losses in the cables are calculated as per the IEC 60287-1-1 Standard. The heat transfer within the cable takes place through conduction mechanism from the cable conductor to its outer surface. The heat transfer in the fluid domain between the cable surface and the inner tunnel surface is influenced by natural convection, surface to surface radiation and conduction. The conduction mechanism is considered where the fluid motion is very small. To address the fluid dynamics problem in this fluid domain, Navier-Stokes equations should be considered in a strictly analytical way. However, Navier-Stokes approach does not always give a solution, particularly in problems with complex geometries, such as that where two trefoil circuits are installed in tunnels. To overcome this obstacle, Computational Fluid Dynamics (CFD) may be used to represent the fluid motion inside the tunnel. Heat transfer in fluid domain is approached by introducing the heat source term in the Navier-Stokes and the energy equations as per [120].

The models developed are assumed to be in a steady-state condition. The ideal fluid properties have been considered for the fluid domain. For a high-fidelity solution, the air properties are temperature incompressible dependent. Turbulent flow is considered. To take into consideration of the buoyancy effects due to the energy equation, the gravitational acceleration $g = 9.81 \text{ m/s}^2$ has been added along the negative y-axis. The walls of the fluid domain are modelled as no-slip walls, which signifies that the velocity is zero at the interface and is monotonically increasing above the surface. In order to accurately capture these fluid-solid interactions, boundary layer meshing as shown in the Figure 7 - 65 has been performed. The soil domain around the tunnel is assumed to be a homogeneous medium with thermal resistivity of 1.2 Km/W . The material properties have been considered as per the IEC Standard 60287-2-1 and CIGRE TB 880.

Previous studies [121] conclude that the heat dissipation rate is closely related to the cable arrangement and also the thermal radiation would have significant impact on the cable temperature and air velocity distribution caused due to buoyancy.

The most critical parameters considered during modelling are the following:

- Soil Thermal Resistivity = 1.2 [Km/W]
- Concrete Tunnel Thermal Resistivity = 0.8 [Km/W]
- Load Factor for the circuits = 1
- Emissivity of the cable surface = 0.9
- Emissivity of the inner walls of tunnel = 0.9
- No moisture migration effect

7.6.2.1 Specifications of the Cables Installed

The HVAC cables are assumed to be in trefoil configuration as shown in the Figure 7 - 66. The data of the cables are summarized in Table 7 - 53, both are operating at 50 Hz. The top circuit consists of three single-core cables of 1200 mm² Aluminium conductors with 150 kV rated voltage and 740 A conductor current and the bottom circuit consists of three single-core cables of 800 mm² Aluminium conductors with 150 kV rated voltage and 600 A conductor current. The cable metallic sheaths are made of Aluminium for both circuits. The cables are laid one above the other with a spacing of 390 mm between them. Single-point bonding scheme is applied at the sheaths of both circuits and eddy losses as per the IEC 60287-1-1 Standard are taken into account.

Table 7 - 53 Environmental and system data for the case study

Property	Circuit A	Circuit B
Cross-section	1200 mm ²	800 mm ²
Conductor outer diameter	42 mm	33 mm
Insulator outer diameter	76 mm	69 mm
Sheath inner diameter	80 mm	75 mm
Sheath outer diameter	82 mm	77 mm
Overall diameter of the cable	92 mm	83 mm
Conductor loss, W_c	19.241 W/m	17.535 W/m
Dielectric loss, W_d	0.588 W/m	0.505 W/m
Sheath loss, W_{sh}	2.343 W/m	1.404 W/m
Thermal Resistivity of Conductor	0.0049 Km/W	0.0049 Km/W
Thermal Resistivity of Insulation	3.5 Km/W	3.5 Km/W
Thermal Resistivity of Sheath	0.0049 Km/W	0.0049 Km/W
Thermal Resistivity of Tapes	12 Km/W	12 Km/W
Thermal Resistivity of Jacket	3.5 Km/W	3.5 Km/W

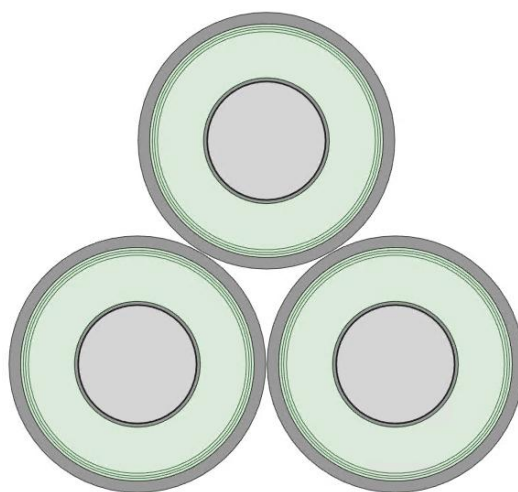


Figure 7 - 66 Trefoil Cable Geometry

7.6.3 Results

The thermal contours and plots have been studied and evaluated against different approaches. The key-assumptions and considerations are also highlighted.

7.6.3.1 Fully Coupled CFD Results

The contour plots can be seen in Figure 7 - 67 and the vector plot in the Figure 7 - 68. The vector plot represents the particle tracking of fluid particles with different velocity magnitudes. The fluid particles move upwards above the cable trefoil and then downstream close to the tunnel walls, thus illustrating the air circulation in the tunnel interior. The air velocity does not exceed 0.3 m/s, thus implying that natural convection takes place.

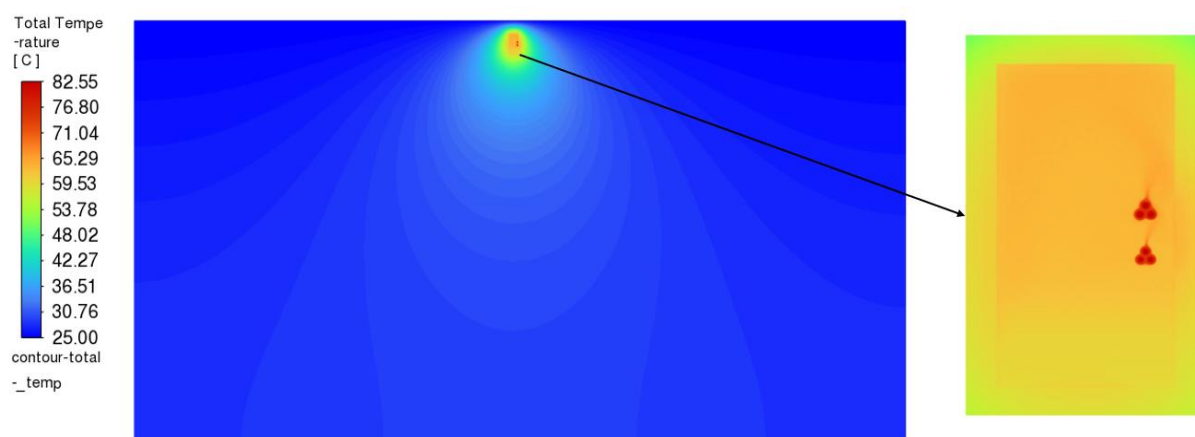


Figure 7 - 67 Temperature contour

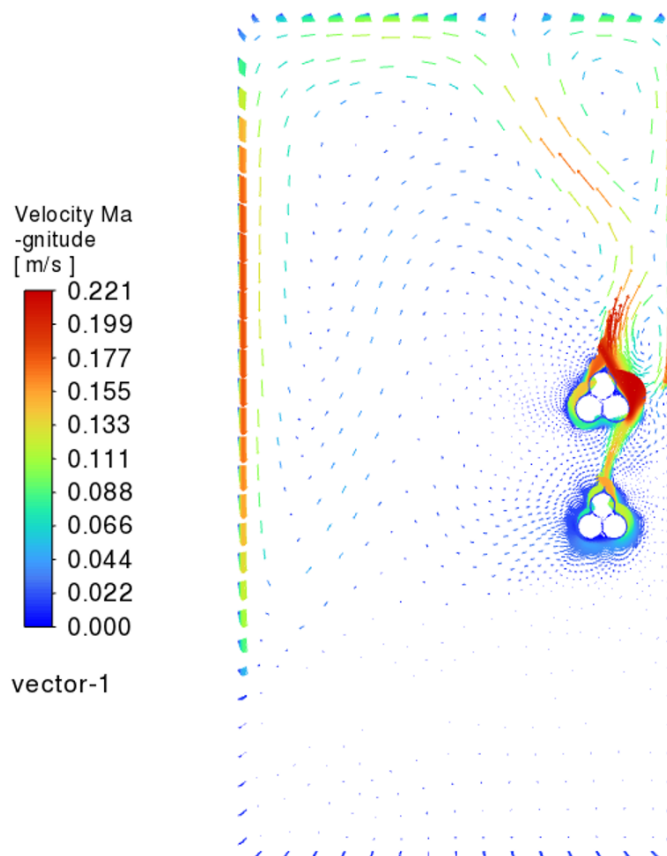


Figure 7 - 68 Velocity vector plot

7.6.3.2 Validation using the Long Horizontal Cylinder Method

This method has been extensively studied and used in various correlations. These correlations are driven by the local Nusselt's number and Rayleigh number over the entire cylinder, which strongly influence the natural convection. Churchill and Chu [122] have recommended a single correlation for a wide range of Rayleigh number up to 10^{12} .

$$\overline{Nu_D} = \left\{ 0.60 + \frac{0.387 Ra_D^{\frac{1}{6}}}{\left[1 + (0.559/Pr)^{\frac{9}{16}} \right]^{\frac{8}{27}}} \right\}^2$$

Equation 7.37

The Long Horizontal Cylinder Method has been assigned to the outermost surface formed by the three cables touching one another. Equation 7.37 refers to the heat transfer coefficient potentially applied on a cable being installed in free air. According to the Newton's law of cooling, due to the difference in temperature between the circuits and the surrounding, a proportional rate of heat will be dissipated from the circuits. For considering this surrounding temperature there are two possibilities, one being the tunnel inner wall surface temperature and one being the mean temperature between the cable surface and tunnel inner wall surface. The latter being the worst-case and is close to reality we consider the second possibility. Also, the Long Horizontal Cylinder Method does not take into account the mutual convection effect which is considered in the Fully Coupled CFD model. Hence, with the above-mentioned assumptions and considerations we expect the Long Horizontal Cylinder Method to provide some deviation from the Fully Coupled CFD Model.

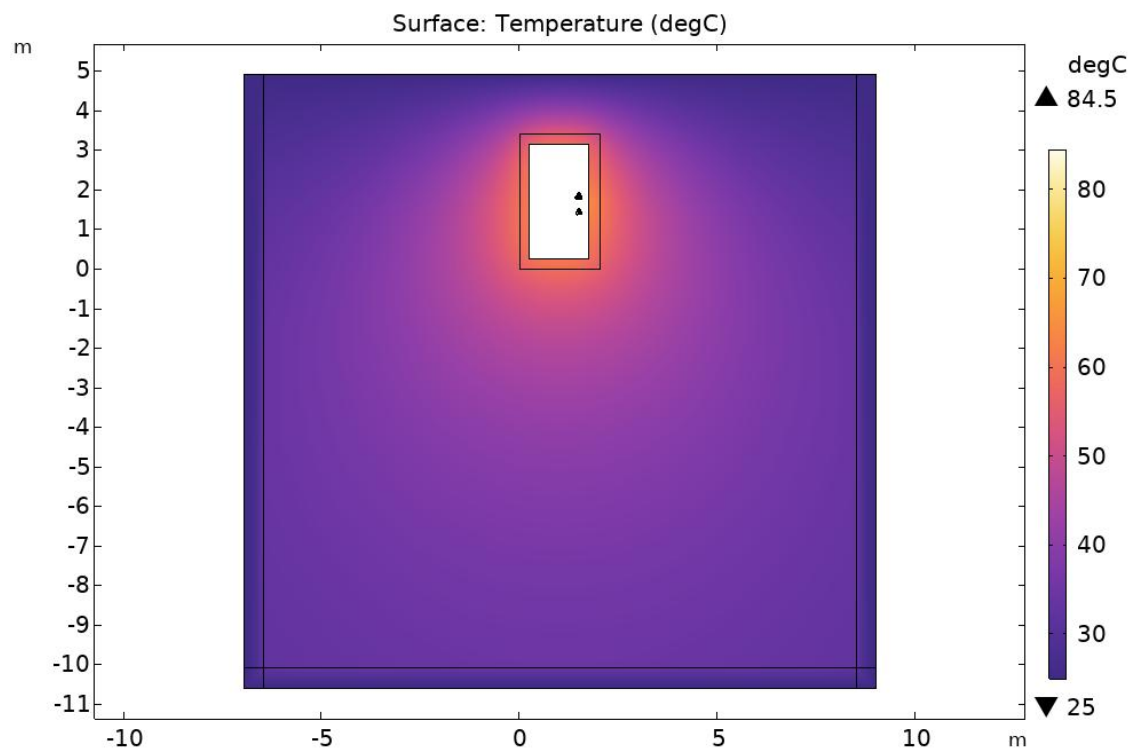


Figure 7 - 69 Temperature contour obtained by long horizontal cylinder

7.6.3.3 Validation against Electra 143/144

Although this report addresses the heat transfer in a non-ventilated tunnel, it is worth studying the heat dissipation with very low inlet velocities by applying the Electra 143/144 [80] method. Hence, we assume the velocity of the fluid is set to 0.1 m/s. This is expected to have very negligible effect on the heat dissipation. In presence of the inlet velocity the Reynold number is taken into account to address a thermal problem instead of Rayleigh number. This accounts to significant deviation in the result.

Along with this, Electra 143/144 also considers the longitudinal axis of the cable. To exclude the effect of the longitudinal heat convection along the tunnel, a short length of 10 m is considered.

The following Table 7 - 54 provides a summary of the approaches implemented with maximum temperatures. We can see that the Fully Coupled CFD Model provides us the results close to both the models.

Table 7 - 54 Summary of results with various approaches

Conductor (Top and Clockwise)	Fully Coupled CFD Model	Long Horizontal Cylinder Method	Electra 143/144 Method
Circuit A – Phase A	82.5 °C	83.4 °C	79.5 °C
Circuit A – Phase B	82.0 °C	84.5 °C	79.4 °C
Circuit A – Phase C	82.5 °C	83.9 °C	79.6 °C
Circuit B – Phase A	82.6 °C	84.5 °C	78.2 °C
Circuit B – Phase B	82.5 °C	84.2 °C	78.3 °C
Circuit B – Phase C	82.0 °C	83.6 °C	78.2 °C
Avg. Inner Tunnel Wall Temperature	59.7 °C	60.5 °C	-

7.6.3.4 Conclusion

The thermal performance of the circuits installed in unventilated tunnels are investigated in this report. Comparable results between the methods examined are obtained. The deviations of 2 K and approx. -3 K against Long Horizontal Cylinder and Electra, respectively, may be justified by the different modelling assumptions adopted by each method, such as the fact that no convective mutual effect is considered by the Long Horizontal Cylinder and Reynolds number instead of Rayleigh number is employed by the Electra 143/144 method. Although not applicable in a strict way, the methods used for validation are the only ones currently existing in literature to compare against the CFD-coupled-with-heat transfer model developed in the present study. Further validation of the developed CFD model against experimental measurements would enhance even more the existing confidence level of the adopted modelling approach.

APPENDIX A. Definitions, abbreviations and symbols

A.1. General terms

Table A - 1 Definition of general terms used in this TB

Acronym	Phrase	Definition
TB	Technical Brochure	A publication produced by CIGRÉ representing the state-of-the-art guidelines and recommendations produced by an SC WG. Hardcopy TBs can be purchased, or Individual Members, or staff of a Collective Member can download the PDF for free using their login credentials (copyright restrictions for use within their own CIGRE Membership only)
SC	Study Committee	One of the 16 technical domain groups of CIGRE
WG	Working Group	A group formed by a SC to develop a TB on a particular subject of interest

A.2. Specific terms in this TB

Table A - 2 Definition of specific terms used in this TB

Acronym	Phrase
AC	Alternating Current
API	Application Program Interface
CFD	Computational Fluid Dynamics
Client	A person or organisation who instructs / specifies an FEM ampacity study.
Contract	Refers to the agreement between the Client and the Contractor in relation to performing FEM ampacity study.
Contractor	A person or organisation who performs / presents / reports on the FEM study & calculated results to the Client.
CPS	Cable Protection System
DC	Direct Current
DDB	Distribution Duct Bank
DTS	Distributed Temperature Sensing
EPC	Engineering, Procurement and Construction
FE	Finite Element
FEA	Finite Element Analysis
FEM	Finite Element Method
HDD	Horizontal Directional Drilling
HDPE	High Density PolyEthylene
HST	Hot Spot Temperature
HV	High Voltage
IEC	International Electrotechnical Commission
LV	Low Voltage
MV	Medium Voltage
PDE	Partial Differential Equation
PE	PolyEthylene
TEE	Thermo-Electric Equivalent (circuit)
TEE	Thermo-Electric Equivalent
XLPE	Cross-linked PolyEthylene

A.3. Symbols

Table A - 3 General symbols used in this TB

Symbol	Description	Unit
σ	Conductivity of material at operating temperature	S/m
ρ_0	Electrical resistivity at reference temperature (20 °C)	$\Omega \cdot m$
α	Temperature correction parameter for resistivity	1/K
C_{cab}	Capacitance per unit length of cable	F/m
U_0	Voltage to earth	V
T_c	Conductor temperature	°C
$\tan \delta$	Loss factor of the insulation at operating frequency	
R_{AC}	IEC AC resistance at operating temperature	Ω/m
R_{sc}	Screen resistance at operating temperature	Ω/m
R_{sh}	Sheath resistance at operating temperature	Ω/m
R_a	Armour resistance at operating temperature	Ω/m
A_c	Conductor area in model, note that this is the cross section in the model and not the IEC conductor cross-section	m ²
A_{sc}	Screen area in model	m ²
A_{sh}	Sheath area in model	m ²
A_a	Armour area in model	m ²
A_d	Dielectric area in model	m ²
I	Conductor current	A
θ_c	Conductor temperature	°C
θ_{sc}	Screen temperature	°C
θ_{sh}	Sheath temperature	°C
θ_a	Armour temperature	°C
$\theta_{duct\ air}$	Temperature of air in-between cable and duct (average)	°C
θ_{amb}	Ambient temperature	°C
$\theta_{j,ext}$	Temperature of external surface of J-tube	K
$\theta_{j,inner}$	Temperature of internal surface of J-tube	K
θ_{air}	Temperature of ambient air	K
θ_{cable}	Temperature of cable surface	K
W_c	Conductor losses	W/m
W_{sc}	Screen losses	W/m
W_{sh}	Sheath losses	W/m
W_a	Armour losses	W/m
W_d	Dielectric losses	W/m
P_c	Conductor losses per volume unit (W_c/A_c)	W/m ³
P_{sc}	Screen losses per volume unit (W_{sc}/A_{sc})	W/m ³
P_{sh}	Sheath losses per volume unit (W_{sh}/A_{sh})	W/m ³
P_a	Armour losses per volume unit (W_a/A_a)	W/m ³
P_d	Dielectric losses per volume unit (W_d/A_d)	W/m ³
λ_1	Screen/sheath loss factor ($\lambda'_1 + \lambda''_1$)	
λ'_1	Screen/sheath loss factor for circulating current	
λ''_1	Screen/sheath loss factor for eddy current	

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APPENDIX C. Material parameters

This appendix provides typical material properties that have been used in the case studies in chapter 7.

C.1. Thermal resistivities

Table C - 1 Thermal resistivity

Material	Thermal resistivity (K·m/W)	Source
Insulating materials		
PE	3.5	[9]
XLPE	3.5	[9]
Semiconducting XLPE	2.5	[123]
Rubber	5.0	[9]
Swellable tapes	12.0	[123]
Protective coverings		
Compounded jute and fibrous materials	6.0	[9]
Rubber sandwich protection	6.0	[9]
PVC up to and including 35 kV cables	5.0	[9]
PVC greater than 35 kV cables	6.0	[9]
PE	3.5	[9]
Copper woven fabric tapes	6	[123]
Woven nylon tape	6	[123]
Polypropylene yarn	6	[123]
Materials for duct installations		
Concrete	1.0	[9]
PVC	6.0	[9]
PE	3.5	[9]
Metals		
Copper	1/401	[124]
Aluminium	1/237	[124]
Lead	1/35	[124]
Iron	1/80	[124]
Stainless steel	1/15	[124]
Fluids		
Air	38.0 (300 K $\approx$ 27 °C) 33.3 (350 K $\approx$ 77 °C)	[124]
Water	1.6 (300 K $\approx$ 27 °C) 1.5 (350 K $\approx$ 77 °C)	[124]

C.2. Thermal specific heat

IEC specifies volumetric specific heat [$\text{J}/(\text{m}^3\text{K})$] while FEM SW uses density [kg/m^3] and heat capacity [$\text{J}/(\text{kg}\cdot\text{K})$]

As only the density $\times$ heat capacity is used for thermal time dependent studies with solid materials, volumetric specific heat can be used as heat capacity and density should be set to 1. Important to note that this is only valid for solid materials in thermal models. For more detail see Guidance Point 61.

Table C - 2 Thermal volumetric specific heat

Material	Volumetric specific heat ($\text{J}/(\text{m}^3\cdot\text{K})$)	Source
Insulating materials		
PE	$2.4\cdot 10^6$	[10]
XLPE	$2.4\cdot 10^6$	[10]
Semiconducting XLPE	$2.4\cdot 10^6$	Assumed based on XLPE
Polyvinyl chloride:		
up to and including 3 kV cables	$1.7\cdot 10^6$	[10]
greater than 3 kV cables	$1.7\cdot 10^6$	[10]
Rubber	$2\cdot 10^6$	[10]
Protective coverings		
Compounded jute and fibrous materials	$2\cdot 10^6$	[10]
Swellable tapes	$2\cdot 10^6$	Assumed based on fibrous materials
Rubber sandwich protection	$2\cdot 10^6$	[10]
PVC up to and including 35 kV cables	$1.7\cdot 10^6$	[10]
PVC greater than 35 kV cables	$1.7\cdot 10^6$	[10]
PE	$2.4\cdot 10^6$	[10]
Materials for duct installations		
Concrete	$1.9\cdot 10^6$	[10]
PVC	$1.7\cdot 10^6$	[10]
PE	$2.0\cdot 10^6$	[10]
Metals		
Copper	$3.45\cdot 10^6$	[10]
Aluminium	$2.5\cdot 10^6$	[10]
Lead	$1.45\cdot 10^6$	[10]
Iron	$3.8\cdot 10^6$	[10]
Stainless steel	$3.8\cdot 10^6$	[10]
Fluid		
Air	$1.0\cdot 10^3$ (350 K $\approx$ 77 °C)	[124]
Water	$4.2\cdot 10^6$ (350 K $\approx$ 77 °C)	[124]

C.3. Empirical heat transfer coefficients

Empirical, analytical formulae can be used to account for heat convection in existing FEM models, which already solve for heat conduction. In those cases, the convective heat flux is applied as a condition on the external boundary of the geometry under consideration, thus confining the physical problem and overcoming the necessity to solve for convection in a domain filled with fluid. The definition of heat transfer coefficient and certain characteristic numbers involved, such as Nusselt (Nu) and Rayleigh (Ra) numbers, have to be carefully considered before use.

Particularly for simple geometries, such as horizontal or vertical cylinders or horizontal plates, there exist analytical formulae in heat transfer books, such as [124]. Equation (9.34) in [124] refers to Nu number for a longitudinal horizontal cylinder and can be employed for a cable being installed in the open air horizontally, while equation (4.126) of [125] refers to Nu number for a vertical thin cylinder and can be employed for a cable being installed in the open air vertically.

Analytical formulae are available in the existing literature also for certain slightly more complex geometries, such as horizontal concentric cylinders and vertical annuli. Equation (9.59) in [124] suggests the use of an effective conductivity accounting for the fluid motion and can be used to represent convective heat transfer in the ring gap between ducts and cables for horizontal ducted cable installations. The equations presented in Table 10.1 and Table 10.2 of [116] can be used to represent the convective heat transfer in the annulus gap formed between a cable and a riser pole. Due to the rapid development of offshore windfarms, these latter formulae have been extensively used to represent convection through air in J-Tubes, such as in the work published by Chippendale et al. [94], while they have been verified also against Computational Fluid Dynamics (CFD) analysis, such as in the works published by [126] and [127]. It is interesting to note that the upper Ra limit referred to [116] becomes no longer applicable if J-Tubes of larger gap are considered, as discussed by the authors of [127].

For rectangular enclosures, such as horizontal troughs and non-ventilated tunnels, there are no analytical formulae available. In these cases, CFD analysis is the only way to represent convection with an acceptable accuracy, see Guidance Point 22 for more information.

C.4. Radiation parameters

Radiation properties are dependent on temperature. The values presented in table below are for surfaces at about 300 K.

Table C - 3 Emissivity and absorption coefficients for radiation

Material	Absorption coefficient σ	Source	Emissivity coefficient	Source
Bitumen/jute serving	0.8	[9]	0.9	[118]
Polychloroprene	0.8	[9]		
PVC	0.6	[9]		
PE	0.4	[9]	0.9	[118]
Lead	0.6	[9]		
Paint (black)	0.98	[124]	0.98	[124]
Paint (acrylic white)	0.26	[124]	0.90	[124]
Painted metallic risers (J-tube)	0.4	[118]	0.7	[118]
Galvanized sheet metal, new	0.65	[124]	0.13	[124]
Galvanized sheet metal, weathered	0.65	[124]	0.13	[124]
Concrete	0.6	[124]	0.88	[124]

C.5. Electrical parameters

The following resistivities are suggested by IEC. For the resistance in conductors, one should use the resistance by data sheet and backward calculate the resistivity based on conductor geometry, reference to Guidance Point 6.

Table C - 4 Electrical parameters for common metals

Material	Resistivity [ohm·m]	Temperature coefficients (α_{20}) [1/K]	Source
Conductors			
Copper	$1.7241 \cdot 10^{-8}$	0.00393	[9]
Aluminium	$2.8264 \cdot 10^{-8}$	0.00493	[9]
Sheath and armour			
Copper	$1.7241 \cdot 10^{-8}$	0.00393	Assumed based on conductor value
Aluminium	$2.84 \cdot 10^{-8}$	0.00493	[9]
Lead	$21.4 \cdot 10^{-8}$	0.004	[9]
Steel	$13.8 \cdot 10^{-8}$	0.0045	[9]



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